

# Jilin University

## Ph.D. Dissertation



### Regularity Estimates for Weak Solutions of Several Classes of Elliptic Equations

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**Dedicated to Mrs. Monica Ghielmetti  
and Prof. Ugo Pietro Gianazza**



# Abstract

This dissertation is devoted to the study of regularity properties for several classes of elliptic equations. It is organized into six chapters.

Chapter 1 serves as an introduction, where we present the background of the elliptic equations under consideration, review related works in the literature, and summarize the main results, novelties, and methods of the thesis.

In Chapter 2, we investigate the local and global regularity of weak solutions of inhomogeneous linear divergence-form non-uniformly elliptic equations without drift terms in bounded domains of the  $N$ -dimensional Euclidean space. Our focus lies on the interplay between the regularity of the weak solution, the coefficient matrix, and the inhomogeneous term. Specifically, we first prove a local  $L^\infty$  estimate for a weak solution under a proper integrability condition on the inhomogeneous term. A counterexample is then constructed to show the sharpness of the integrability exponent in the  $L^\infty$  estimate. We further show that the integrability condition on the reciprocal of the smallest eigenvalue of the coefficient matrix is optimal for both the local boundedness and the Harnack inequality to hold. Moreover, we provide examples to illustrate the impact of this condition on the boundedness of weak solutions.

Building on these results and applying the Trudinger–Moser theory, we derive a Harnack inequality for the weak solution, which subsequently yields Hölder continuity. For the Dirichlet problem associated with the above equation, we prove a logarithmic upper bound for the  $L^\infty$ -norm of weak solutions, and further discuss the Alexandroff-Bakelman-Pucci maximum principle in this setting. Finally, in the case of uniformly elliptic equations with inhomogeneous terms in logarithmic Orlicz spaces, we construct a counterexample showing the existence of unbounded solutions at the critical Orlicz space.

Chapter 3 concerns linear divergence-form non-uniformly elliptic equations with

drift terms. We study the relationship between the regularity of the weak solution, the regularity of the drift term, and the coefficient matrix. In particular, we prove local boundedness result for the weak subsolution, and through two examples, show the sharpness of our assumptions on the integrability of the drift term and the reciprocal of the smallest eigenvalue. Based on these results, we then prove a Harnack inequality for weak solutions.

Chapters 4 and 5 somehow differ from the previous two, and also from the last one, since they both deal with functions in De Giorgi classes. Even though these classes were originally defined in the context of linear elliptic equations with bounded and measurable coefficients, it was quickly understood that they contain also other objects, and have a wider applicability.

In Chapter 4, we study Phragmén–Lindelöf type theorems for nonnegative functions in homogeneous De Giorgi classes. We prove that, as  $|x|$  tends to infinity, the maximal modulus of such functions grows power-like, with an order lying strictly between 0 and 1. By constructing proper counterexamples, we also show that in general the growth order cannot be expected to reach 1.

It is well-known that functions in De Giorgi classes are continuous up to the boundary, if this satisfies a geometric density condition, and the Dirichlet boundary datum is itself continuous. In Chapter 5 we give a new proof of this classical result relying on ideas that were originally proposed by Gariepy and Ziemer to prove the sufficient part of the Wiener criterion for the continuity of solutions of quasilinear elliptic equations of  $p$ -Laplace type.

Chapter 6 is devoted to nonlocal integro-differential operators of fractional Laplacian type. For the Dirichlet problem, we obtain a logarithmic upper bound for weak subsolutions and provide a counterexample to demonstrate the sharpness of the integrability assumption on the inhomogeneous term  $f$ .

**Key Words:** Non-uniformly Elliptic Equations, Bounded Solutions, Weak Harnack Inequality, De Giorgi Classes, Sublinear Growth, Phragmén–Lindelöf Theorem, Geometric density condition, Boundary Hölder continuity, Fractional Laplacian Equations, Logarithmic Upper Bound.

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# Chapter 1

## Introduction

The study of the regularity of weak solutions of elliptic partial differential equations originated from Hilbert's 19th problem, posed at the 1900 International Congress of Mathematicians in Paris, which questioned whether the solutions of Lagrangian systems are always analytic. Subsequent significant contributions by Bernstein [10], Radó [87], Schauder [93], Leray [64], Morrey [76], and Hopf [49] completed the proof that solutions can be elevated from  $C^\alpha$  to  $C^\infty$ . In the mid-20th century, De Giorgi [30] and Nash [81] independently proved the remaining part, namely  $W^{1,2} \Rightarrow C^\alpha$ , which not only developed the regularity theory of elliptic equations, but also laid the groundwork for further research on the boundedness, continuity, and Harnack inequalities of weak solutions.

This thesis aims at investigating the regularity estimates of weak solutions of several classes of elliptic partial differential equations, with a particular focus on boundedness estimates and Harnack inequalities. These issues are not only of fundamental importance in the theory of analysis but also hold significant implications in various applied fields, including geometric analysis, material science, and fluid mechanics. By studying specific types of equations, this work seeks to uncover connections between the regularity of solutions and the structure of equations, thereby enriching the theory of weak solutions and providing further mathematical support for related models.

In the following chapters, we will delve into the regularity problems of weak solutions of several specific classes of elliptic equations.

In Chapter 2 we study the boundedness and Hölder continuity of weak solutions of

the non-homogeneous linear divergence-form *non-uniformly elliptic* equation

$$-\operatorname{div}(\mathbf{A}(x)\nabla u(x)) = f(x), \quad x \in \Omega, \quad (1.1)$$

where  $\Omega \subset \mathbb{R}^N$  ( $N \geq 2$ ) is a bounded domain and the coefficient matrix  $\mathbf{A}(x) = (a_{ij}(x))_{i,j=1}^N$  satisfies

(H1) *Symmetry and measurability:*  $a_{ij} = a_{ji}$  are measurable in  $\Omega$ ;

(H2) *Non-uniform ellipticity:* there exist non-negative measurable functions  $\lambda(x), \mu(x)$  such that for every  $\xi \in \mathbb{R}^N$

$$\lambda(x)|\xi|^2 \leq \mathbf{A}(x)\xi \cdot \xi \leq \mu(x)|\xi|^2,$$

with

$$\lambda^{-1} \in L^q(\Omega), \quad \mu \in L^p(\Omega), \quad p, q > 1, \quad \frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}.$$

**Historical background.** The study of (1.1) can be traced back to the seminal works of De Giorgi [30] and Nash [81]. For the uniformly elliptic case ( $\lambda^{-1}, \mu \in L^\infty$ ) and  $f \equiv 0$ , they proved the local Hölder continuity of weak solutions. Moser [78, 79] later derived the Harnack inequality via his celebrated iteration technique. When  $f \not\equiv 0$ , Ladyzhenskaya & Ural'tseva [59] showed that for any  $f \in L^s(\Omega)$  with  $s > \frac{N}{2}$  non-negative weak solutions are locally bounded and Hölder continuous. Using rearrangement techniques, Talenti [98] proved explicit upper bounds for weak solutions of the Dirichlet problem.

Since then, a vast literature has been devoted to the local and global regularity of linear and non-linear elliptic equations; we refer to [44, 59, 71, 72, 97, 102] and the references therein.

**Degenerate/singular regimes.** If  $\lambda^{-1} \notin L^\infty$  the equation is *degenerate*; if  $\mu \notin L^\infty$  it is *singular*. For  $f \equiv 0$ , Murthy & Stampacchia [80] and Trudinger [101] proved local boundedness and the Harnack inequality under the assumptions

$$\lambda^{-1} \in L^q(\Omega), \quad \mu \in L^p(\Omega), \quad \frac{1}{p} + \frac{1}{q} < \frac{2}{N}.$$

A crucial ingredient is the weighted Sobolev inequality

$$\left( \int_{\Omega} |u|^\gamma \mu \, dx \right)^{1/\gamma} \leq C \left( \int_{\Omega} |\nabla u|^2 \lambda \, dx \right)^{1/2}, \quad \gamma > 2,$$

valid for every  $u \in H_0^1(\Omega)$ . Bella & Schäffner [7] later obtained the same conclusion under the *optimal* Lebesgue-scale condition

$$\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}, \quad N \geq 2.$$

Their key observation is to invoke Sobolev-type inequalities on  $(N-1)$ -dimensional spheres instead of the usual  $N$ -dimensional balls, yielding optimal Caccioppoli estimates for the truncations of  $u$ .

Fabes, Kenig & Serapioni [36] imposed the stronger Muckenhoupt  $A_2$  condition on the eigenvalues of  $\mathbf{A}$ , ensuring the scale-invariant Sobolev and Poincaré inequalities

$$\left( \int_B |u - u_B|^\gamma \lambda \, dx \right)^{2/\gamma} \leq Cr^2 \left( \int_B \lambda^{-1} \, dx \right)^{2/\gamma-1} \int_B |\nabla u|^2 \lambda \, dx,$$

for every ball  $B = B(y, r)$  and  $u \in H^1(B)$  with  $u_B := \int_B u \, dx$ . Further results can be found in [22, 27, 38].

**Sharper upper bounds.** Beyond the above contributions, various improvements on the size of the upper bound have been obtained. When  $\mathbf{A}$  is uniformly elliptic, Cianchi [26] used non-increasing rearrangements to prove boundedness for  $f$  in certain critical Orlicz spaces; Xu [105] derived logarithmic-type upper bounds depending on  $\|f\|_{L^s}$  by means of Moser iteration. Cruz-Uribe & Rodney [27] extended these results to degenerate equations assuming  $f \in L^A(\Omega)$  with  $A(t) = t^{\sigma'} \log(e+t)^q$  ( $q > \sigma' > 0$ ).

In another direction, Li & Wang [66] established an Alexandroff-Bakelman-Pucci (ABP) bound for divergence-form equations with uniformly elliptic  $\mathbf{A}$ . For equations in non-divergence-form, the classical ABP estimate is due to Alexandroff [3], Bakelman [6] and Pucci [86]; see also [14, 16, 17, 103].

**Motivation and open questions.** Inspired by the above works, we investigate how the regularity of weak solutions is dictated by the interplay among the integrability of  $\lambda^{-1}$ ,  $\mu$  and  $f$ . It is generally believed that the smoothness of solutions depends on the smoothness of both the coefficients and the datum. For instance, when  $f \equiv 0$ , Zhong [107] constructed discontinuous solutions of certain degenerate equations, thereby giving a partial answer to a conjecture raised by De Giorgi in 1995 [29].

Under the assumption

$$\frac{1}{p} + \frac{1}{q} < \frac{2}{N},$$

Trudinger [101, 102] proved local and global boundedness estimates and the Harnack inequality for  $f \equiv 0$  and  $f \not\equiv 0$ . Bella & Schäffner [7] later relaxed Trudinger's integrability assumption to the *sharp* condition

$$\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}, \quad N \geq 2,$$

but only for  $f \equiv 0$ . The precise regularity theory for  $f \not\equiv 0$  under this critical scaling remained unknown.

In the present chapter we give a complete answer to the following questions:

- (i) Do the boundedness results and the Harnack inequality of [102] persist under the critical condition  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$  when  $f \not\equiv 0$ ?
- (ii) If so, what is the *critical* Lebesgue integrability exponent for  $f$ ?
- (iii) Is  $L^{N/2}$  the sharp Lebesgue space for  $\lambda^{-1}$  guaranteeing Hölder continuity?

**Main contributions.** Throughout, the coefficients  $a_{ij}(x)$  are merely measurable and satisfy (H1)-(H2);  $f$  is any function in a suitable Lebesgue space.

(1) *Local  $L^\infty$  bound.* Assume  $q > \frac{N}{2}$  and  $N \geq 2$ . We prove that the local  $L^\infty$ -norm of a weak solution  $u$  of (1.1) is controlled by

$$\|u\|_{L^\gamma(B_R)} + \|f\|_{L^s(B_R)}, \quad \gamma > 0, \quad s > \frac{Nq}{2q - N}.$$

Since the usual Sobolev embedding on  $(N-1)$ -dimensional spheres precludes the classical treatment of  $f$ , we introduce an interpolation inequality

$$\begin{aligned} \left( \int_{B_\sigma} |(u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta|^{2s'} dx \right)^{\frac{1}{s'}} &\leq \varepsilon \left( \int_{B_\sigma} |(u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta|^{q_*} dx \right)^{\frac{2}{q_*}} \\ &\quad + C(\varepsilon) \left( \int_{B_\sigma} |(u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta|^{p^*} dx \right)^{\frac{2}{p^*}} \end{aligned}$$

to absorb the non-homogeneous term. Sharpness is demonstrated by explicit counterexamples.

(2) *Critical case  $q = \frac{N}{2}$ .* If  $\lambda^{-1} \in L^{N/2}(\Omega)$ , then for every  $s \in [1, \infty)$  there exists  $f \in L^s(\Omega)$  such that (1.1) possesses an *unbounded* weak solution. For  $N \geq 3$  we even have examples with  $f \in L^\infty(\Omega)$  and  $\sup_\Omega u = +\infty$ .

(3) *Sub-critical range*  $\frac{N-1}{2} < q < \frac{N}{2}$ . For  $N \geq 3$  we again produce  $f \in L^\infty(\Omega)$  for which  $\sup_\Omega u = +\infty$ .

(4) *Sharpness of the scaling*. When  $\frac{1}{p} + \frac{1}{q} \geq \frac{2}{N-1}$ , counterexamples in [8, 39] show that weak solutions may be unbounded even for  $f \equiv 0$ .

(5) *Two-dimensional limit case*. When  $N = 2$  and  $p = q = 1$  (hence  $\frac{1}{p} + \frac{1}{q} = \frac{2}{N-1}$ ), Bella & Schäffner [7] proved local boundedness for  $f \equiv 0$  via the one-dimensional maximum principle and Sobolev embeddings. We show that this result fails for  $f \not\equiv 0$  with  $f \in L^s(\Omega)$ ,  $s \in [1, \infty)$ .

**Comparison with previous results.** Surprisingly, our integrability condition on  $f$  coincides *exactly* with that in [102]. Hence, weakening the coefficient regularity from  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N}$  to  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$  *does not* require a stronger assumption on  $f$ . The reason is that the integrability exponent of  $f$  depends only on the exponent  $q$  of  $\lambda^{-1}$  and on the dimension  $N$ ; the critical value  $q = \frac{N}{2}$  is simultaneously compatible with the condition  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N}$  used in [102]. This observation is also reflected in the global  $L^\infty$ -estimates proved in Sections 6 and 7.

**Logarithmic upper bounds.** In Section 6 we establish a logarithmic-type global upper bound for weak solutions of the Dirichlet problem associated with (1.1). The result extends earlier work of Xu [105] from uniformly to non-uniformly elliptic equations and dispenses with the *a priori* assumption that  $u$  is globally bounded. The proof relies on a two-parameter approximation of test functions and is optimal in view of the counterexamples mentioned above.

**ABP-type estimates.** Section 7 establishes an ABP-type global upper bound, still under the assumption  $\sup_{\partial\Omega} u < \infty$ . We generalize the result of Li & Wang [66] to the present non-uniformly elliptic setting; the proof follows their strategy verbatim.

**Orlicz data.** Finally, Section 8 deals with the case where  $f$  belongs to an Orlicz space and  $\mathbf{A}$  is *uniformly* elliptic. When  $\mathbf{A}$  is merely non-uniformly elliptic, related results are scarce. Cruz-Uribe & Rodney [27] obtained partial boundedness results, but the Orlicz space for  $f$  is *not* critical; i.e. the integrability index of  $f$  is not optimal. A natural question is whether alternative techniques can push the integrability threshold further—this remains open.

For uniformly elliptic  $\mathbf{A}$ , the theory is essentially complete. Cianchi [26] used non-increasing rearrangements to prove boundedness when  $f \in L^A(\Omega)$  with  $A(t) = t^{N/2} \log(e + t)^s$ ,  $s > \frac{N}{2} - 1$ . Cruz-Uribe & Rodney [27] recovered the same conclusion via De Giorgi iteration and showed, by a counterexample based on the Poisson equation, that  $s < \frac{N}{2} - 1$  may lead to unbounded solutions. The *critical* case  $s = \frac{N}{2} - 1$  was left open; we close this gap by constructing, again via potential-theoretic methods, an explicit example for which the weak solution is unbounded. Hence the condition  $s > \frac{N}{2} - 1$  is indeed optimal.

In Chapter 3 we investigate the local boundedness and the Harnack inequality for weak solutions of the linear divergence-form equation

$$-\operatorname{div}(\mathbf{A}(x)\nabla u) + \operatorname{div}(u\mathbf{b}) = \operatorname{div} \mathbf{f} + f, \quad x \in \Omega, \quad (1.2)$$

where the coefficient matrix  $\mathbf{A}$  obeys the same structural assumptions (H1)-(H2) as in Chapter 2, while

$$\mathbf{b} = (b_1, \dots, b_N) \quad \text{and} \quad \mathbf{f} = (f_1, \dots, f_N)$$

are vector fields whose integrability will be specified below. Compared with the model studied in Chapter 2, the left-hand side of (1.2) contains an additional drift term  $\operatorname{div}(u\mathbf{b})$ , and the right-hand side has been augmented by the divergence term  $\operatorname{div} \mathbf{f}$ .

**Historical background.** The earliest systematic study of such problems dates back to Ladyzhenskaya & Ural'tseva [59], who considered the general divergence-form (summation over repeated indices is assumed)

$$\frac{\partial}{\partial x_i} (a_{ij} u_{x_j} + a_i u) + b_i u_{x_i} + a u = f + \frac{\partial f_i}{\partial x_i}, \quad x \in \Omega, \quad (1.3)$$

under the structural assumptions

$$\begin{cases} \nu |\xi|^2 \leq a_{ij} \xi_i \xi_j \leq \mu |\xi|^2, & \nu > 0, \\ \left\| \sum_{i=1}^N (a_i^2 + b_i^2) + |a| \right\|_{L^{s/2}(\Omega)} \leq \mu, & s > N, \end{cases} \quad (1.4)$$

and the integrability conditions

$$\left\| \sum_{i=1}^N f_i^2 + |f| \right\|_{L^{s/2}(\Omega)} \leq \mu, \quad s > N. \quad (1.5)$$

By means of De Giorgi iteration and the special function class  $\mathcal{B}_2(\Omega, M, \gamma, \delta, \frac{1}{q})$ , Ladyzhenskaya & Ural'tseva proved local boundedness and Hölder continuity of weak



solutions; examples given in [59, Ch. 1, §2] show that the above conditions are indeed sharp.

Subsequent works have sought to relax the integrability requirement on the drift term  $\mathbf{b}$ . Safonov [90] considered the non-divergence form equation (the same assumption on repeated indices holds)

$$a_{ij}D_{ij}u + b_iD_iu = f \quad \text{in } \Omega, \quad u \in W_{\text{loc}}^{2,N}(\Omega) \cap C(\bar{\Omega}),$$

with  $b_i \in L^N(\Omega)$  and  $f \in L^N(\Omega)$ , and proved the maximum principle as well as interior and boundary Harnack inequalities via the growth lemma (see [60, 61]). Later, Nazarov & Ural'tseva [82] studied the simpler model obtained from (1.3) by setting  $a_i = a = f = f_i = 0$  and assuming merely  $|\mathbf{b}| \in L^s(\Omega)$ ,  $\frac{N}{2} < s \leq N$ , together with the structural condition  $\text{div } \mathbf{b} \leq 0$  in the sense of distributions. Using Moser iteration and the De Giorgi lemma, they obtained local boundedness and Hölder continuity. The structural condition  $\text{div } \mathbf{b} \leq 0$  arises naturally in problems from fluid mechanics, most notably the Navier-Stokes equations; we refer the interested reader to [23, 53, 54, 94, 106].

**Non-uniform ellipticity.** In the non-uniformly elliptic setting, Trudinger [102] extended the results of [59] by allowing

$$\begin{cases} \lambda(x)|\xi|^2 \leq a_{ij}(x)\xi_i\xi_j \leq \mu(x)|\xi|^2, & \forall \xi \in \mathbb{R}^N, \\ \lambda^{-2} \in L^q(\Omega), \mu \in L^p(\Omega), & \frac{1}{p} + \frac{1}{q} < \frac{2}{N}, \\ [b_{ij}(a_ia_j + b_ib_j) + |a|] \in L^s(\Omega), & \frac{1}{s} + \frac{1}{q} < \frac{2}{N}, \\ \sqrt{b_{ij}f_if_j} \in L^{2s}(\Omega), f \in L^s(\Omega), & \frac{1}{s} + \frac{1}{q} < \frac{2}{N}, \end{cases} \quad (1.6)$$

where  $[b_{ij}]$  denotes the inverse of  $[a_{ij}]$ . He observed that the global boundedness estimate is independent of any upper bound on  $\mu(x)$ , whereas local regularity *does* require a restriction on the largest eigenvalue.

**Main results of Chapter 3.** Motivated by the above works and guided by the results of Chapter 2, we investigate the regularity of weak solutions of (1.2) under the critical non-uniform ellipticity assumption

$$\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}, \quad p, q > 1.$$

To be precise, we assume

$$\mathbf{b}, \mathbf{f} \in \left(L^t(\Omega)\right)^N, \quad t > \frac{Nq}{q-N}, \quad q > N.$$

Let us list the main results concerning different topics:

- (i) *Local boundedness.* By an interpolation-energy estimate combined with Moser's iteration, we prove that every non-negative weak subsolution of (1.2) is locally bounded. The integrability exponent  $t = \frac{Nq}{q-N}$  is shown to be *critical* via an explicit counterexample.
- (ii) *Harnack inequality.* Relying on the boundedness result and the Moser-Trudinger theory, we establish the Harnack inequality for weak solutions.
- (iii) *Loss of boundedness for  $q \leq N$ .* When  $q \in (\frac{N-1}{2}, N]$ , we construct a simple example showing that even for the homogeneous equation ( $f = \operatorname{div} \mathbf{f} = 0$ ) weak solutions may fail to be locally bounded.

**Comparison with Chapter 2.** The above phenomenon reveals that the drift term  $\operatorname{div}(u\mathbf{b})$  has a *stronger* influence on regularity than the inhomogeneous data  $\operatorname{div} \mathbf{f} + f$ : its mere presence forces  $\lambda^{-1}$  to possess a *higher* integrability in order to retain boundedness and the Harnack inequality.

In Chapter 4 we focus on the regularity of the *homogeneous* De Giorgi classes

$$[DG]_p^+(\mathbb{R}_+^N, \hat{\gamma}) \quad \text{and} \quad [DG]_p^+(\mathbb{R}^N, \hat{\gamma}),$$

with the aim of proving a Phragmén-Lindelöf-type theorem that describes the growth rate at infinity of the local maximum modulus. Precise definitions are given in Section 4.1.

It is known since De Giorgi [30] and Ladyzhenskaya & Ural'tseva [59] that every function in  $[DG]_p^+(\Omega, \hat{\gamma})$  is locally bounded, and every function in  $[DG]_p(\Omega, \hat{\gamma})$  is locally Hölder continuous; DiBenedetto & Trudinger [35] further showed that every non-negative member satisfies the Harnack inequality. What makes these classes attractive is their remarkable universality: they contain not only weak solutions of elliptic equations with bounded and measurable coefficients, but also minima (and even  $Q$ -minima) of functionals that need not satisfy the Euler equation. The purpose of the present chapter is to investigate whether such general functions still exhibit a Phragmén-Lindelöf behaviour.

A complete survey of all Phragmén-Lindelöf results is beyond the scope of this introduction; we therefore restrict ourselves to those directly connected with our work.

**Classical and modern precedents.**

- (i) *Sub-harmonic functions.* The first classical statement (see [83]) concerns sub-harmonic functions in the two-dimensional half-plane: if  $u \leq 0$  on the boundary and  $u$  is somewhere positive, then

$$\liminf_{r \rightarrow \infty} \frac{\mu_+(r)}{r} > 0, \quad \mu_+(r) := \sup_{(x,y) \in B_r \cap \mathbb{R}_+^2} |u(x,y)|.$$

Originally formulated for analytic functions, the result is in essence about two-variable harmonic functions and was later extended by Ahlfors to  $N$ -dimensional harmonic functions (see [2, Theorem 2.8] and the subsequent remark).

- (ii) *Conformally invariant energies.* Granlund, Lindqvist & Martio [45] studied extremals of

$$\int_{\Omega} F(x, Du) dx, \quad F(x, \xi) \approx |\xi|^N,$$

thus covering the planar harmonic case but not the classical harmonic case in  $\mathbb{R}^N$  for  $N \geq 3$ .

- (iii)  *$p$ -Laplacian.* Lindqvist [68] established Phragmén-Lindelöf theorems for non-negative subsolutions of the  $p$ -Laplacian (Theorem 4.6) and for  $N$ -harmonic functions vanishing on  $\mathbb{H}^q$  ( $1 \leq q \leq N - 1$ , Theorem 4.8). The proofs rely on the comparison principle and explicit estimates for  $N$ -harmonic measure; optimality is discussed via simple examples.

- (iv) *General  $p$ .* When  $p \neq N$  the explicit computation of the  $p$ -harmonic measure is no longer obvious. However, appropriate upper and lower bounds suffice, and Lundström [70] extended Lindqvist's Theorem 2.1 to general  $p$  (see [70, Corollary 4.3]).

**Quantitative rigidity results.** If one assumes directly that  $u$  is a non-negative solution of  $\Delta_p u = 0$  in  $\{x_N > 0\}$  vanishing continuously on  $\{x_N = 0\}$ , then the qualitative Phragmén-Lindelöf conclusion can be refined: after suitable normalisation one has  $u(x) = x_N$ . Such rigidity goes back to Loomis & Widder [69] for two-dimensional harmonic functions; Rudin [89] extended it to  $N$ -dimensional positive harmonic functions, and analogous statements for the  $p$ -Laplacian can be found in [13, 52]. For

two-dimensional non-divergence linear elliptic equations with smooth coefficients, similar rigidity results were obtained by Gilbarg [43] and Serrin [95].

**Our approach.** Inspired by [68, 70], we face two major obstructions:

- (O1) The De Giorgi classes are far too general to allow any kind of comparison principle. (In fact, whether a comparison principle holds for  $[DG]_p^\pm(\Omega, \hat{\gamma})$  remains an important open problem.)
- (O2) The weak Harnack inequality contains a constant  $\tau$  whose smallness is qualitative, raising the question of whether the exponent  $\alpha_1$  in Theorem 4.1.1 is only an artefact of the method or can be improved to the expected value  $\alpha_1 = 1$ . A similar concern arises for  $\alpha_2$ , since Lundström proved that for  $p$ -harmonic subsolutions the sharp exponent is  $\alpha_2 = \frac{p-N+M}{p-1}$ .

To overcome (O1) we develop a new strategy that exploits the weak Harnack inequality (see Lemma 4.2.3 and Proposition 4.2.8). Concerning (O2), we show by means of carefully built counterexamples that the restriction  $\alpha_1 \in (0, 1)$  is an intrinsic consequence of the broadness of the De Giorgi class, rather than a limitation of the technique.

In Chapter 5 we investigate the regularity of functions belonging to the homogeneous De Giorgi classes

$$[DG]_p^\pm(E, \gamma), \quad 1 < p \leq N,$$

focusing on their Hölder behaviour up to the boundary  $\partial E$ . In contrast to the preceding chapter, we no longer restrict ourselves to the interior, but study how the Hölder modulus behaves near the boundary.

There are at least two compelling reasons for this study. First, as already observed, these classes encompass a surprisingly broad family of functions: local solutions to uniformly elliptic equations with measurable and bounded coefficients,  $Q$ -minima of general functionals, and even functions for which no underlying Euler equation is available. Second, and more remarkably, despite their generality, every member is locally bounded and Hölder continuous ([30, 59]) and satisfies a Harnack inequality ([35]). Moreover, if the boundary satisfies a suitable density condition and the Dirichlet datum is Hölder continuous, the Hölder regularity extends up to the boundary ([77, 96]).

We give a new proof of such a classical result, based on ideas that were originally introduced by Gariepy & Ziemer [42] when studying the sufficient part of the Wiener

criterion for a wide class of quasilinear equation of  $p$ -Laplacian type. In their technique three auxiliary functions play a fundamental role, namely

$$\varphi(t) = \left(\frac{1}{1-t}\right)^\alpha, \quad \psi(t) = (1-t)^\beta, \quad \zeta(t) = \ln(1-t),$$

with  $\alpha > 0$ ,  $\beta \in (0, 1)$ , and  $t \in [0, 1)$ . The geometric density condition is needed to obtain proper estimates involving the composition of  $\varphi$ ,  $\psi$ ,  $\zeta$  with functions in De Giorgi classes; apart from that, it is immaterial. Whenever such estimates are available, the method works; therefore, this raises the natural question whether the same regularity of solutions continues to hold even when rougher boundaries are considered. This issue will not be dealt with here.

In Chapter 6 we investigate the regularity of weak solutions of non-local uniformly elliptic equations. The fractional Laplacian  $(-\Delta)^s$  originates as the infinitesimal generator of rotationally symmetric  $2s$ -stable Lévy processes; as  $s \rightarrow 1^-$  the Lévy process converges to Brownian motion and  $(-\Delta)^s$  approaches the classical Laplacian. Standard references include [5, 11, 62, 91].

In 2007, Caffarelli & Silvestre [19] introduced the *extension method*, which realises  $(-\Delta)^s$  ( $0 < s < 1$ ) as a Neumann boundary operator for a degenerate elliptic equation in one higher dimension:

$$\begin{cases} \operatorname{div} (y^{1-2s} \nabla U) = 0, & (x, y) \in \mathbb{R}^N \times (0, \infty), \\ U(x, 0) = u(x), & x \in \mathbb{R}^N, \end{cases}$$

with

$$(-\Delta)^s u(x) = -d_s \lim_{y \rightarrow 0^+} y^{1-2s} \partial_y U(x, y), \quad d_s > 0.$$

This localisation allows one to import many classical techniques from elliptic partial differential equations; see [15, 18, 20, 88].

**De Giorgi-Moser-Nash theory for non-local operators.** Kassmann [50] first extended the De Giorgi-Moser-Nash theory to a class of non-local integro-differential operators and proved that  $\mathcal{L}$ -harmonic functions are bounded from below and Hölder continuous. Subsequent works have further developed this theory; in particular, Fernández-Real & Ros-Oton [37, Chapter 2] obtained an  $L^\infty$  bound for weak solutions of the Dirichlet problem with right-hand side  $f \in L^q(\Omega)$ ,  $q > \frac{N}{2s}$ . (Our main results in this chapter will build upon this estimate.) Further developments can be found in [41]. We

emphasise that the parameters  $q$  and  $s$  in this chapter are independent of those used earlier.

**Problem setting.** We study the Dirichlet problem

$$\begin{cases} \mathcal{L}u = f, & x \in \Omega, \\ u = 0, & x \in \mathbb{R}^N \setminus \Omega, \end{cases} \quad (1.7)$$

where  $\mathcal{L}$  is a non-local operator modelled on the fractional Laplacian and defined in the Cauchy-principal-value sense

$$\mathcal{L}u(x) = \text{P.V.} \int_{\mathbb{R}^N} \frac{A(x, y)(u(x) - u(y))}{|x - y|^{N+2s}} dy,$$

with symmetric and uniformly elliptic kernel

$$0 < \lambda \leq A(x, y) = A(y, x) \leq \Lambda,$$

and with  $f \in L^q(\Omega)$ ,  $q > \frac{N}{2s}$ .

**Main results.** The goal of this chapter is to provide a *logarithmic* upper bound for non-negative weak subsolutions of (1.7), thereby improving the standard  $L^\infty$  estimate of [105]. To make the result meaningful we first show that the integrability assumption on  $f$  is *sharp*:

- (i) *Sharpness.* We construct an explicit counterexample demonstrating that when  $q = \frac{N}{2s}$  and  $N \geq 2$  the Dirichlet problem (1.7) admits an *unbounded* weak solution. Although the optimality of  $q > \frac{N}{2s}$  is often tacitly assumed, we have been unable to locate a definitive statement or counterexample in the literature. The closest result appears in Kuusi, Mingione & Sire [58, Corollary 1.3], where a similar threshold is identified for continuity. Our example also shows that the measure assumption in [58, Corollary 1.3] is optimal.
- (ii) *Logarithmic estimate.* Using exponential-type test functions and a Moser iteration adapted to the non-local setting, we prove a logarithmic upper bound for weak subsolutions. Compared with the local case [105], new difficulties arise from the non-local interactions. Unlike [37], the exponential test functions require additional techniques to absorb the exponential growth created by the iteration.

## Chapter 2

# Non-homogeneous Linear Non-uniformly Elliptic Equations without Drift

### 2.1 Main Results

#### Problem setting

We study the divergence-form second-order elliptic equation

$$-\operatorname{div}(\mathbf{A}(x)\nabla u) = f, \quad x \in \Omega, \quad (2.1)$$

where  $\Omega \subset \mathbb{R}^N$  ( $N \geq 2$ ) is a bounded domain and the coefficient matrix  $\mathbf{A}(x) = (a_{ij}(x))_{i,j=1}^N$  satisfies

(H1) *Symmetry and measurability:*  $a_{ij} = a_{ji}$  are measurable in  $\Omega$ ;

(H2) *Non-uniform ellipticity:* there exist non-negative measurable functions  $\lambda(x), \mu(x)$  such that

$$\lambda(x)|\xi|^2 \leq \mathbf{A}(x)\xi \cdot \xi \leq \mu(x)|\xi|^2, \quad \forall \xi \in \mathbb{R}^N,$$

with

$$\lambda^{-1} \in L^q(\Omega), \quad \mu \in L^p(\Omega), \quad p, q > 1, \quad \frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}.$$

## Local supremum estimates

Our first result provides an interior estimate for the local maximum modulus of  $|u|$ .

**Theorem 2.1.1** *Assume (H1)–(H2) and let  $f \in L^s(\Omega)$  for some*

$$s > \frac{Nq}{2q - N} \quad \text{with} \quad q > \frac{N}{2}.$$

*Then for every  $\gamma > 0$  and every ball  $B_R \Subset \Omega$  there exists a constant  $C = C(N, p, q, \gamma) \geq 1$  such that any weak solution  $u$  of (2.1) satisfies*

$$\sup_{B_{\theta R}} |u| \leq C \left( \Lambda(B_R) \right)^{\frac{\delta p'}{\gamma(\delta-1)}} \left[ \frac{\|u\|_{L^\gamma(B_R)}}{\left( (1-\theta)R \right)^{\frac{N}{\gamma}(m_*+1)}} + \frac{\|f\|_{L^s(B_R)}}{\left( (1-\theta)R \right)^{\frac{N}{\gamma}m_* + \frac{N}{s} - 2}} \right], \quad (2.2)$$

where  $\theta \in (0, 1)$  and

$$\begin{aligned} \theta &\in (0, 1), \quad \delta = 2 - \frac{1}{\chi}, \quad \chi := \frac{2q}{(q+1)p_*}, \quad \frac{1}{p_*} = \min \left\{ \frac{1}{2} + \frac{1}{N-1} - \frac{1}{2p}, 1 \right\}, \\ m_* &= \frac{p'\delta}{\delta-1} \max \left\{ \frac{1}{p} + \frac{1}{q}, \frac{2}{q} \right\}, \quad p' = \frac{p}{p-1}, \\ \Lambda(B_R) &:= \left( \int_{B_R} \lambda^{-q} dx \right)^{\frac{1}{q}} \left( \int_{B_R} \mu^p dx \right)^{\frac{1}{p}} + \left( \int_{B_R} \lambda^{-q} dx \right)^{\frac{2}{q}}. \end{aligned}$$

**Remark 2.1.1** As  $q \rightarrow \infty$  one has  $\frac{Nq}{2q-N} \rightarrow \frac{N}{2}$ , which recovers the classical threshold for uniformly elliptic equations.

**Remark 2.1.2** Theorem 2.1.1 shows that the integrability exponent for  $f$  depends not only on the dimension  $N$  but also on the integrability index  $q$  of  $\lambda^{-1}$ . A natural question therefore arises: *What is the critical Lebesgue space for the inhomogeneous term?*

The next example demonstrates that the exponent  $\frac{Nq}{2q-N}$  in Theorem 2.1.1 is *sharp*; details are given in Section 2.3.

**Example 2.1.1** Let  $\Omega = B_{1/4}(0)$ ,  $q > \frac{N}{2}$  and  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$ . Take

$$\mathbf{A}(x) = \text{diag}(\lambda(x), \dots, \lambda(x)), \quad \lambda(x) = \mu(x) = |x|^\beta \left( \log \frac{1}{|x|} \right)^\theta,$$



with  $\beta = \frac{N}{q} < 2$  and  $\theta = \frac{1}{q} + \frac{1}{2} - \frac{1}{N}$ . Then there exists

$$f \in \begin{cases} L^{\frac{Nq}{2q-N}}(\Omega), & N \geq 3, \\ L^s(\Omega) \ (1 \leq s < \frac{q}{q-1}), & N = 2, \end{cases}$$

such that the corresponding weak solution  $u$  of (2.1) is unbounded. Explicitly,

$$\begin{aligned} f(x) = & (N+1)|x|^{\beta-1} \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{1/2}(0)} \frac{|y|^{1-N}}{(|x|+|y|)^2 \log \frac{1}{|y|}} dy \\ & - \theta |x|^{\beta-1} \left( \log \frac{1}{|x|} \right)^{\theta-1} \int_{B_{1/2}(0)} \frac{|y|^{1-N}}{(|x|+|y|)^2 \log \frac{1}{|y|}} dy \\ & - 2|x|^\beta \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{1/2}(0)} \frac{|y|^{1-N}}{(|x|+|y|)^3 \log \frac{1}{|y|}} dy, \end{aligned}$$

and

$$u(x) = \int_{B_{1/2}(0)} \frac{|y|^{1-N}}{(|x|+|y|) \log \frac{1}{|y|}} dy.$$

**Remark 2.1.3** When  $N = 2$  and  $q > 1$  we obtain only *almost* optimality, i.e.  $s > \frac{q}{q-1}$ . We conjecture that the borderline case  $L^{\frac{q}{q-1}}$  behaves differently, but a definitive statement is still missing.

**Remark 2.1.4** Example 2.1.1 shows that if the integrability of  $\lambda^{-1}$  is kept fixed while that of  $f$  is relaxed, the solution may leave  $L^\infty(\Omega)$ .

### Critical case $q = \frac{N}{2}$

In Theorem 2.1.1 the exponent  $\frac{Nq}{2q-N}$  blows up as  $q \rightarrow \frac{N}{2}$ . One may therefore ask whether  $\|u\|_\infty$  can be controlled by  $\|f\|_\infty$  when  $q = \frac{N}{2}$  and  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$ . The interpolation inequality used in the proof breaks down in this limit ( $q_* = 2$ ). The following example shows that the answer is negative.

**Example 2.1.2** Let  $\Omega = B_{1/4}(0)$ ,  $q = \frac{N}{2}$  and  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$ . Take

$$\lambda(x) = \mu(x) = |x|^2 \left( \log \frac{1}{|x|} \right)^\theta, \quad \theta = \frac{5}{2N}.$$

Then for  $N \geq 3$  there exists  $f \in L^\infty(\Omega)$ , and for  $N = 2$  there exists  $f \in L^s(\Omega)$  for every  $s \in [1, \infty)$ , such that the weak solution  $u \in H^1(\Omega, \mathbf{A})$  of (2.1) is unbounded.

### Sub-critical integrability $\frac{N-1}{2} < q < \frac{N}{2}$

We now show that for  $N \geq 3$  and  $q \in (\frac{N-1}{2}, \frac{N}{2})$  one can still find  $f \in L^\infty(\Omega)$  producing unbounded solutions.

**Example 2.1.3** Let  $\Omega = B_{1/4}(0)$ ,  $N \geq 3$ ,  $q \in (\frac{N-1}{2}, \frac{N}{2})$  and  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$ . Choose

$$\lambda(x) = \mu(x) = |x|^\beta \left( \log \frac{1}{|x|} \right)^\theta, \quad \beta = \frac{N}{q} > 2, \quad \theta > \frac{1}{q}.$$

Then there exists  $f \in L^\infty(\Omega)$  such that  $\sup_\Omega u = +\infty$ .

**Remark 2.1.5** Examples 2.1.2 and 2.1.3 show that  $L^{\frac{N}{2}}(\Omega)$  is the critical Lebesgue space for  $\lambda^{-1}$  if one wants to guarantee  $\|u\|_\infty < \infty$ .

### Super-critical integrability $\frac{1}{p} + \frac{1}{q} \geq \frac{2}{N-1}$

For  $\frac{1}{p} + \frac{1}{q} \geq \frac{2}{N-1}$  and  $N \geq 3$  we refer to the examples in [7, 8, 39]. In particular, [39, Theorem 2] provides an unbounded weak solution for  $\frac{1}{p} + \frac{1}{q} > \frac{2}{N-1}$  with  $f \equiv 0$ . Bella & Schäffner [8] extended this to the  $p$ -Laplacian setting for  $\frac{1}{p} + \frac{1}{q} \geq \frac{m}{N-1}$ . When  $N = 2$ ,  $p = q = 1$  and  $f \equiv 0$ , they proved in [7] that  $\|u\|_\infty < \infty$ . We further show that if  $f \not\equiv 0$ ,  $q = \frac{2}{N-1}$  ( $N \geq 3$ ) and  $p = \infty$ , then there exists  $f \in L^\infty(\Omega)$  such that  $u$  is unbounded; see Remark 2.3.1.

**Remark 2.1.6** When  $N = 2$ ,  $q = 1$ ,  $p \geq 1$  and  $f \not\equiv 0$  we cannot relate  $\|u\|_\infty$  to  $\|f\|_\infty$ . The maximum principle and Moser iteration break down, and new methods or counter-examples are required.

## Harnack inequality and Hölder continuity

Building upon Theorem 2.1.1 we establish the following Harnack estimate.

**Theorem 2.1.2** Assume (H1)–(H2) and let  $f \in L^s(\Omega)$  for some  $s > \frac{Nq}{2q-N}$  with  $q > \frac{N}{2}$ . Let  $u \in H^1(\Omega, \mathbf{A})$  be a non-negative weak solution of (2.1). Then for every ball  $B_R \subset \Omega$

$$\sup_{B_{R/2}} u \leq C \left( \inf_{B_{R/2}} u + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right),$$

where  $C$  depends on  $N, p, q$  and  $\Lambda(B_R)$ .

We conclude with several standard consequences of Theorem 2.1.2; for brevity we state them without proof. In the strictly elliptic regime where  $\Lambda(B_R)$  is uniformly bounded, the Harnack inequality implies global Hölder continuity. Because the constant in (2.1.2) depends on  $\Lambda(B_R)$ , global Hölder continuity usually fails in the non-uniform case (see [7, 101]). Nevertheless, Theorem 2.1.2 yields the following local result.

**Corollary 2.1.1** *Under the hypotheses of Theorem 2.1.2, any weak solution  $u \in H^1(\Omega, \mathbf{A})$  satisfies*

$$\operatorname{osc}_{B_{r/2}} u \leq C \left( \frac{r}{R} \right)^\alpha \left[ \left( \frac{1}{R^N} \int_{B_R} |u|^\gamma dx \right)^{\frac{1}{\gamma}} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right],$$

for all  $0 < r \leq R$  with  $B_R \subset \Omega$ , where  $\alpha = \alpha(\Lambda(B_R), N, p, q, s) > 0$  and  $C = C(\Lambda(B_R), N, p, q)$ .

A direct consequence of Corollary 2.1.1 is the local Hölder continuity of  $u$ .

**Corollary 2.1.2** *Let  $u \in H^1(\Omega, \mathbf{A})$  be a weak solution of (2.1) and suppose  $f \in L^s(\Omega)$  for some  $s > \frac{Nq}{2q-N}$ . Then  $u \in C_{\text{loc}}^\alpha(\Omega)$  with  $\alpha = \alpha(\Lambda, N, p, q, s) \in (0, 1)$ . Moreover, for every  $B_R \subset \Omega$  and all  $x, y \in B_{R/8}$*

$$|u(x) - u(y)| \leq C \left( \frac{|x - y|}{R} \right)^\alpha \left[ \left( \frac{1}{R^N} \int_{B_R} |u|^\gamma dx \right)^{\frac{1}{\gamma}} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right],$$

with  $C = C(\Lambda(B_R), N, p, q) \geq 1$ .

## Improved global bounds for the Dirichlet problem

We now turn to the Dirichlet problem associated with (2.1). Below we write *problem (2.1)* to mean the Dirichlet problem

$$u \in H_0^1(\Omega, \mathbf{A}), \quad -\operatorname{div}(\mathbf{A} \nabla u) = f \text{ in } \Omega.$$

**Theorem 2.1.3 (Logarithmic upper bound)** *Assume (H1)–(H2) and let  $f \in L^{s_0}(\Omega)$  with  $s_0 = \frac{Nq}{2q-N}$  and  $q > \frac{N}{2}$ . Then for every*

$$0 < \alpha < \frac{4}{c_s^2 \|\lambda^{-1}\|_{L^q(\Omega)}},$$

the weak solution  $u \in H_0^1(\Omega, \mathbf{A})$  satisfies

$$\begin{aligned} \|u\|_{L^\infty(\Omega)} &\leq \frac{C}{\alpha} \|f\|_{L^{s_0}(\Omega)} \\ &\quad \cdot \log \left[ \left( \frac{\alpha}{2 - c_s \|\lambda^{-1}\|_{L^q(\Omega)} \sqrt{\alpha}} + 1 \right) \left( \frac{\alpha \|\lambda^{-1}\|_{L^q(\Omega)} \|f\|_{L^s(\Omega)}}{\|f\|_{L^{s_0}(\Omega)}} + 1 \right) \right], \end{aligned}$$

where  $C = C(\Omega, N, q, s) \geq 1$  and  $c_s$  is the optimal constant in the Sobolev inequality

$$\|g\|_{L^{\frac{Np}{N-p}}(\Omega)} \leq c_s \|\nabla g\|_{L^p(\Omega)}, \quad g \in W_0^{1,p}(\Omega), \quad p < N.$$

**Remark 2.1.7** Unlike the estimates in [27, 105], our proof does not require an *a priori* bound on  $u$ . The key ingredient is a refined Caccioppoli inequality for  $e^{\alpha u_+}$  obtained via truncation and approximation.

Finally, we present an improved ABP-type upper bound for weak solutions:

**Theorem 2.1.4 (ABP-type estimate)** *If (H1)–(H2) hold and  $q > \frac{N}{2}$ , then any weak subsolution  $u \in H^1(\Omega, \mathbf{A})$  satisfies*

$$\begin{aligned} \sup_{\Omega} u_+ &\leq \sup_{\partial\Omega} u_+ + C \left( \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}} \right)^{\frac{q_*}{2s'(1-\frac{q_*}{2s'})}} \left[ |\Omega|^{\frac{2}{N}+1-\frac{2s'}{q_*}} \|\lambda^{-1}\|_{L^q(\Omega)} + 1 \right] \\ &\quad \cdot \left( \int_{\{\psi=\Upsilon_\psi\} \cap \Omega} (f_+(x))^s dx \right)^{\frac{1}{s}}, \end{aligned}$$

where  $s' = \frac{s}{s-1}$ ,  $q_* = \frac{2Nq}{(q+1)N-2q}$ ,  $\psi = u - \sup_{\partial\Omega} u_+$ , and  $\Upsilon_\psi$  is the minimiser of the functional

$$J(\omega) = \int_{\Omega} a_{ij} \partial_i \omega \partial_j \omega dx$$

over the convex set

$$K_\psi = \left\{ \omega \in H_0^1(\Omega, \mathbf{A}) : \omega \geq \psi \text{ a.e. in } \Omega \right\}.$$

The constant  $C$  depends only on  $N, q, s$ .

**Remark 2.1.8** If  $\frac{1}{2} + \frac{1}{2q} < \frac{2}{N}$ , i.e.  $\frac{2q}{q-1} > \frac{Nq}{2q-N}$ , the estimate holds for every  $f \in L^s(\Omega)$  with  $s \geq \frac{2q}{q-1}$ ; otherwise it holds for  $s > \frac{Nq}{2q-N}$ .

## Organisation of the chapter

Section 2 gathers notation and auxiliary results. Section 3 constructs Example 2.1.1; Examples 2.1.2 and 2.1.3 are treated as variants thereof. Section 4 proves Theorem 2.1.1 via a Moser-type iteration under the condition  $\frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}$ , following and refining the approach of [7]. Section 5 establishes the Harnack inequality (Theorem 2.1.2) by Trudinger–Moser theory. Section 6 derives the logarithmic estimate (Theorem 2.1.3) along the lines of [105], overcoming several technical obstacles. Section 7 proves the ABP-type bound (Theorem 2.1.4) by adapting the method of [66]. Finally, Section 8 discusses recent progress on boundedness when  $f$  belongs to certain Orlicz spaces.

## 2.2 Preliminaries

### Notation

Throughout,  $\|\cdot\|_s$  ( $s \geq 1$ ) denotes the norm in  $L^s(\Omega)$ . When the centre is clear from the context we write  $B_\rho$  for  $B_\rho(0)$ . The letter  $C$  stands for a generic positive constant that may vary from line to line and depends only on the basic parameters (dimension, integrability exponents, etc.). Whenever the dependence needs to be emphasised we write, e.g.  $C(N, s)$ .

### Weighted Sobolev spaces

The spaces  $H^1(\Omega, \mathbf{A})$  and  $H_0^1(\Omega, \mathbf{A})$  are defined as the completions of  $C^\infty(\Omega)$  and  $C_0^\infty(\Omega)$ , respectively, with respect to the norm

$$\|u\|_{H^1(\Omega, \mathbf{A})} := \left( \int_{\Omega} \mathbf{A} \nabla u \cdot \nabla u \, dx \right)^{\frac{1}{2}} + \left( \int_{\Omega} |u|^{\frac{2q}{q+1}} \, dx \right)^{\frac{q+1}{2q}},$$

where  $\mathbf{A}$  satisfies (H1)–(H2) and  $q > 1$ . In particular, on  $H_0^1(\Omega, \mathbf{A})$  we use the equivalent norm

$$\|u\|_{H_0^1(\Omega, \mathbf{A})} = \left( \int_{\Omega} \mathbf{A} \nabla u \cdot \nabla u \, dx \right)^{\frac{1}{2}}.$$

For further information on these weighted Sobolev spaces we refer to [101, 102]. The spaces introduced here contain the corresponding ones defined in [102]; hence the existence results and geometric properties stated there remain valid in our setting. We only recall the following chain rule.

**Lemma 2.2.1** ([102]) *Let  $F: \mathbb{R} \rightarrow \mathbb{R}$  be uniformly Lipschitz with  $F(0) = 0$ . Then  $u \in H_0^1(\Omega, \mathbf{A})$  (or  $u \in H^1(\Omega, \mathbf{A})$ ) implies  $F(u) \in H_0^1(\Omega, \mathbf{A})$  (or  $F(u) \in H^1(\Omega, \mathbf{A})$ ), and*

$$\nabla F(u) = F'(u) \nabla u \quad \text{a.e. in } \Omega.$$

## Weak solutions

**Definition 2.2.1** *A function  $u$  is called a weak solution (weak subsolution, weak supersolution) of (2.1) in  $\Omega$  if  $u \in H^1(\Omega, \mathbf{A})$  and*

$$\int_{\Omega} \mathbf{A} \nabla u \cdot \nabla \varphi \, dx = \int_{\Omega} f \varphi \, dx \quad (\leq 0, \geq 0) \quad \forall \varphi \geq 0, \varphi \in H_0^1(\Omega, \mathbf{A}). \quad (2.3)$$

*We say that  $u$  is a local weak solution if the above identity holds for every bounded open set  $\Omega' \Subset \Omega$ .*

**Remark 2.2.1** Because  $f \in L^s(\Omega)$  with  $s > \frac{Nq}{2q-N}$ , the right-hand side of the above identity is well defined by Hölder's and Sobolev's inequalities.

## An improved Caccioppoli inequality

The following lemma first appeared in [7]. By choosing the minimiser of a suitable functional and applying the Sobolev inequality on spheres, one obtains a refined Caccioppoli estimate.

**Lemma 2.2.2** ([7]) *Fix  $N \geq 2$  and  $t \geq 1$ ; if  $N \geq 3$  assume further  $t > \frac{N-1}{2}$ . Let  $0 < \rho < \sigma < \infty$  and let  $v \in W^{1,t*}(B_{\sigma}(x_0))$  with*

$$\frac{1}{t_*} = \min \left\{ \frac{1}{2} - \frac{1}{2t} + \frac{1}{N-1}, 1 \right\}.$$

*Suppose  $g \in L^t(B_{\sigma}(x_0))$  is non-negative and  $gv^2 \in L^1(B_{\sigma}(x_0))$ . Define*

$$J(\rho, \sigma, v) = \inf \left\{ \int_{B_{\sigma}(x_0)} gv^2 |\nabla \eta|^2 \, dx \mid \eta \in C_0^1(B_{\sigma}(x_0)), 0 \leq \eta \leq 1, \right. \\ \left. \eta \equiv 1 \text{ on } B_{\rho}(x_0) \right\}.$$

*Then there exists a constant  $c = c(N, t) \geq 1$  such that*

$$J(\rho, \sigma, v) \leq c(\sigma - \rho)^{-\frac{2N}{N-1}} \|g\|_{L^t(B_{\sigma}(x_0) \setminus B_{\rho}(x_0))} \\ \cdot \left( \|\nabla v\|_{L^{t*}(B_{\sigma}(x_0) \setminus B_{\rho}(x_0))}^2 + \rho^{-2} \|v\|_{L^{t*}(B_{\sigma}(x_0) \setminus B_{\rho}(x_0))}^2 \right). \quad (2.4)$$

## Notation for the ABP estimate

For later use we set

$$J(\omega) = \int_{\Omega} a_{ij} \partial_i \omega \partial_j \omega \, dx, \quad \omega \in H^1(\Omega, \mathbf{A}).$$

Given  $\psi \in H_0^1(\Omega, \mathbf{A})$  we introduce the convex set

$$K_\psi = \left\{ \omega \in H_0^1(\Omega, \mathbf{A}) : \omega \geq \psi \text{ a.e. in } \Omega \right\}.$$

The functional  $J$  is strictly convex and coercive on  $K_\psi$ ; hence there exists a unique minimiser  $\Upsilon_\psi \in K_\psi$  such that

$$J(\Upsilon_\psi) = \inf_{\omega \in K_\psi} J(\omega).$$

## Orlicz spaces

For Section 2.8 we recall the basic definitions. A function  $A: [0, \infty) \rightarrow [0, \infty)$  is called a *Young function* if

$$A(s) = \int_0^s a(r) \, dr, \quad s \geq 0,$$

where  $a: [0, \infty) \rightarrow [0, \infty)$  is left-continuous, non-decreasing, and neither identically zero nor identically infinite on  $(0, \infty)$ .

For such an  $A$  the Orlicz space  $L^A(\Omega)$  is defined as

$$L^A(\Omega) = \left\{ f \text{ measurable} \mid \exists \tilde{\lambda} > 0, \int_{\Omega} A\left(\frac{|f|}{\tilde{\lambda}}\right) dx < \infty \right\},$$

endowed with the *Luxemburg norm*

$$\|f\|_{L^A(\Omega)} = \inf \left\{ \tilde{\lambda} > 0 \mid \int_{\Omega} A\left(\frac{|f|}{\tilde{\lambda}}\right) dx \leq 1 \right\}.$$

When  $A(t) = t^p$  ( $p \geq 1$ ) we recover the classical Lebesgue space  $L^p(\Omega)$ .

## 2.3 Proof of Example 2.1.1 – 2.1.3

**Proof of Example 2.1.1.** Let  $\Omega$  be the ball  $B = B_{\frac{1}{4}}(0)$ . Define

$$\mathbf{A}(x) = \begin{pmatrix} a_1(x) & 0 & \cdots & 0 \\ 0 & a_1(x) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_1(x) \end{pmatrix}, \quad a_1(x) = |x|^\beta \left( \log \frac{1}{|x|} \right)^\theta,$$

and

$$\begin{aligned} f(x) &= (N+1)|x|^{\beta-1} \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \\ &\quad - \theta |x|^{\beta-1} \left( \log \frac{1}{|x|} \right)^{\theta-1} \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \\ &\quad - 2|x|^\beta \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy, \end{aligned}$$

where  $\beta = \frac{N}{q} < 2$ ,  $\theta = \frac{1}{2} + \frac{1}{q} - \frac{1}{N}$ .

It is easy to observe that

$$u(x) = \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-1} \frac{1}{\log \frac{1}{|y|}} dy$$

is the weak solution of (2.1) and  $u(0) = +\infty$ . To make the argument rigorous, we exploit some technique from [27] to show that

$$\int_{\Omega} \mathbf{A}(x) \nabla u \nabla \varphi dx = \int_{\Omega} f \varphi dx, \quad \forall \varphi \geq 0 \in C_0^\infty(\Omega).$$

For each  $k \geq 2$ , let  $\chi_k$  be a  $C^\infty$ , non-negative and radial function satisfying

$$\chi_k(x) = \begin{cases} 0, & |x| \leq 2^{-k-2}, \\ 1, & 2^{-k-1} \leq |x| < \frac{1}{4}. \end{cases}$$

Define  $f_k = \chi_k f$ . Notice that each  $f_k$  is continuous, and

$$u_k = \chi_k \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-1} \frac{1}{\log \frac{1}{|y|}} dy \in C^2(\Omega)$$

is the solution of  $-\operatorname{div}(\mathbf{A}(x) \nabla u_k) = f_k$ , in the weak sense,

$$\int_{\Omega} \mathbf{A}(x) \nabla u_k \nabla \varphi dx = \int_{\Omega} f_k \varphi dx, \quad \forall \varphi \geq 0 \in C_0^\infty(\Omega).$$

The weak convergence  $u_k \rightharpoonup u$  in  $H^1(\Omega, \mathbf{A})$  implies

$$\int_{\Omega} \mathbf{A}(x) \nabla u_k \nabla \varphi dx \rightarrow \int_{\Omega} \mathbf{A}(x) \nabla u \nabla \varphi dx, \quad k \rightarrow \infty.$$

Next, we will verify that  $\lambda^{-1}(x) \in L^q(\Omega)$ . Indeed, by using a polar coordinate transformation, we obtain

$$\int_{B_{\frac{1}{2}}(0)} \left| \frac{1}{\lambda(x)} \right|^q dx = \alpha_N \int_0^{\frac{1}{2}} \left| \frac{1}{r^\beta \left( \log \frac{1}{r} \right)^\theta} \right|^q r^{N-1} dr = \alpha_N \int_0^{\frac{1}{2}} \frac{1}{\left( \log \frac{1}{r} \right)^{\theta q}} r^{N-1-\beta q} dr$$



$$\begin{aligned}
 &= \alpha_N \int_2^\infty \frac{1}{(\log t)^{\theta q}} \left(\frac{1}{t}\right)^{N+1-\beta q} dt = \alpha_N \int_2^\infty \frac{1}{(\log t)^{\theta q}} \frac{1}{t} dt \\
 &< +\infty,
 \end{aligned}$$

where  $\alpha_N$  is the surface area of an  $N$ -dimensional sphere.

To complete the proof of Example 2.1.1, we will demonstrate that  $f \in L^{\frac{Nq}{2q-N}}(\Omega)$  when  $N \geq 3$ . It is sufficient to demonstrate one term of  $f$  that belongs to  $L^{\frac{Nq}{2q-N}}(\Omega)$ , the remaining terms can be handled in a similar manner. For example, we will show that

$$|x|^\beta \left(\log \frac{1}{|x|}\right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \in L^{\frac{Nq}{2q-N}}(B_{\frac{1}{4}}(0)).$$

In fact, we know

$$\begin{aligned}
 &\int_{B_{\frac{1}{4}}(0)} \left| |x|^\beta \left(\log \frac{1}{|x|}\right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \right|^{\frac{Nq}{2q-N}} dx \\
 &= \int_{B_{\frac{1}{4}}(0)} |x|^{\frac{\beta Nq}{2q-N}} \left(\log \frac{1}{|x|}\right)^{\frac{\theta Nq}{2q-N}} \left| \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \right|^{\frac{Nq}{2q-N}} dx. \quad (2.5)
 \end{aligned}$$

Then, we will estimate

$$\int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy. \quad (2.6)$$

We decompose (2.6) into two terms

$$\begin{aligned}
 &\int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \\
 &= \int_{B_{\frac{1}{2}}(0) \cap \{|y| < |x|^{\frac{1}{2}}\}} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \\
 &\quad + \int_{B_{\frac{1}{2}}(0) \cap \{|y| \geq |x|^{\frac{1}{2}}\}} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy = I_1 + I_2.
 \end{aligned}$$

By using a polar coordinate transformation, we find  $I_1$  may be estimated as follows

$$\begin{aligned}
 I_1 &= \int_{B_{\frac{1}{2}}(0) \cap \{|y| < |x|^{\frac{1}{2}}\}} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \\
 &\leq \alpha_N \int_0^{|x|^{\frac{1}{2}}} (|x| + r)^{-3} \frac{1}{\log \frac{1}{r}} dr
 \end{aligned}$$

$$\begin{aligned}
 &\leq \frac{2\alpha_N}{\log \frac{1}{|x|}} \int_0^{|x|^{\frac{1}{2}}} (|x| + r)^{-3} dr \\
 &= \frac{\alpha_N}{\log \frac{1}{|x|}} \left( |x|^{-2} - \left( |x|^{\frac{1}{2}} + |x| \right)^{-2} \right) \leq \frac{\alpha_N}{\log \frac{1}{|x|}} |x|^{-2}.
 \end{aligned} \tag{2.7}$$

Similarly, for  $I_2$ , we also estimate

$$\begin{aligned}
 I_2 &\leq \alpha_N \int_{|x|^{\frac{1}{2}}}^{\frac{1}{2}} (|x| + r)^{-3} \frac{1}{\log \frac{1}{r}} dr \leq \frac{\alpha_N}{\log 2} \left( |x| + |x|^{\frac{1}{2}} \right)^{-3} \int_{|x|^{\frac{1}{2}}}^{\frac{1}{2}} dr \\
 &= \frac{\alpha_N}{\log 2} \left( |x| + |x|^{\frac{1}{2}} \right)^{-3} \left( \frac{1}{2} - |x|^{\frac{1}{2}} \right) \leq \frac{\alpha_N}{\log 2} |x|^{-\frac{3}{2}} \left( \frac{1}{2} - |x|^{\frac{1}{2}} \right) = \frac{\alpha_N}{2 \log 2} |x|^{-\frac{3}{2}}.
 \end{aligned} \tag{2.8}$$

By using (2.6)–(2.8), (2.5) has the following estimate

$$\begin{aligned}
 &\int_{B_{\frac{1}{4}}(0)} |x|^{\frac{\beta N q}{2q-N}} \left( \log \frac{1}{|x|} \right)^{\frac{\theta N q}{2q-N}} \left| \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \right|^{\frac{N q}{2q-N}} dx \\
 &\leq c \int_{B_{\frac{1}{4}}(0)} |x|^{\frac{\beta N q}{2q-N}} \left( \log \frac{1}{|x|} \right)^{\frac{\theta N q}{2q-N}} \left( \frac{1}{\log \frac{1}{|x|}} |x|^{-2} \right)^{\frac{N q}{2q-N}} dx \\
 &\quad + c \int_{B_{\frac{1}{4}}(0)} |x|^{\frac{\beta N q}{2q-N}} \left( \log \frac{1}{|x|} \right)^{\frac{\theta N q}{2q-N}} \left( |x|^{-\frac{3}{2}} \right)^{\frac{N q}{2q-N}} dx \\
 &= c \int_0^{\frac{1}{4}} r^{\frac{(\beta-2)Nq}{2q-N} + N - 1} \left( \log \frac{1}{r} \right)^{\frac{Nq(\theta-1)}{2q-N}} dr + c \int_0^{\frac{1}{4}} r^{\frac{(\beta-\frac{3}{2})Nq}{2q-N} + N - 1} \left( \log \frac{1}{r} \right)^{\frac{Nq\theta}{2q-N}} dr \\
 &= I_3 + I_4,
 \end{aligned}$$

where  $c$  only depends on  $N, q$ . Now, we estimate  $I_3$ . Notice

$$\begin{aligned}
 \frac{(\beta-2)Nq}{2q-N} + N - 1 &= \frac{(\frac{N}{q} - 2)Nq}{2q-N} + N - 1 = -1, \\
 \frac{Nq(\theta-1)}{2q-N} &= \frac{N-q-2Nq}{2(2q-N)} < -1,
 \end{aligned}$$

then, a simple computation shows  $I_3 < +\infty$ .

Similarly, due to

$$\frac{(\beta-\frac{3}{2})Nq}{2q-N} + N - 1 = \frac{Nq}{q-\frac{N}{2}} - 1 > -1.$$

it is not hard to prove that

$$I_4 < +\infty.$$

Thus we arrive at

$$\int_{B_{\frac{1}{4}}(0)} \left| |x|^\beta \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy \right|^{\frac{Nq}{2q-N}} dx < +\infty.$$

For the case  $N = 2$ , by a similar argument, we could also verify  $f \in L^s(B_{\frac{1}{4}}(0))$ , for  $s \in [1, \frac{q}{q-1})$ .

**Proof of Example 2.1.2.** We take  $\beta = 2$  in the proof of Example 2.1.1, and we have

$$\begin{aligned} a_1(x) &= |x|^2 \left( \log \frac{1}{|x|} \right)^\theta, \\ f(x) &= (N+1)|x| \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \\ &\quad + \theta |x| \left( \log \frac{1}{|x|} \right)^{\theta-1} \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \\ &\quad - 2|x|^2 \left( \log \frac{1}{|x|} \right)^\theta \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy, \end{aligned}$$

where  $\theta = \frac{5}{2N}$ . We observe

$$u(x) = \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-1} \frac{1}{\log \frac{1}{|y|}} dy$$

is the weak solution of (2.1), and  $u(0) = \infty$ . The rest of procedure is similar to Example 2.1.1. Finally, we find  $f \in L^\infty(B_{\frac{1}{4}}(0))$  when  $N \geq 3$  and  $f \in L^s(B_{\frac{1}{4}}(0))$ ,  $s \in [1, +\infty)$  when  $N = 2$ .

**Proof of Example 2.1.3.** We set  $\beta = \frac{N}{q} \in (2, \frac{2N}{N-1})$ ,  $\theta = 1$  in Example 2.1.1. Let

$$\begin{aligned} a_1(x) &= |x|^{\frac{N}{q}} \left( \log \frac{1}{|x|} \right)^{\frac{2}{q}}, \\ f(x) &= (N+1)|x|^{\frac{N}{q}-1} \left( \log \frac{1}{|x|} \right) \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \\ &\quad - |x|^{\frac{N}{q}-1} \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-2} \frac{1}{\log \frac{1}{|y|}} dy \end{aligned}$$

$$-2|x|^{\frac{N}{q}} \left( \log \frac{1}{|x|} \right) \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-3} \frac{1}{\log \frac{1}{|y|}} dy,$$

and  $f \in L^\infty(B_{\frac{1}{4}}(0))$ . Then we take

$$u(x) = \int_{B_{\frac{1}{2}}(0)} |y|^{1-N} (|x| + |y|)^{-1} \frac{1}{\log \frac{1}{|y|}} dy,$$

by a similar argument as above, we could verify  $u$  is the weak solution of (2.1).

**Remark 2.3.1** It's interesting to see that Example 2.1.3 could be extended directly to the case of  $\frac{1}{p} + \frac{1}{q} = \frac{2}{N-1}$ ,  $N \geq 3$ . Indeed, when we take  $q = \frac{2}{N-1}$ ,  $p = +\infty$ , we find that there exists  $f \in L^\infty(\Omega)$ , which makes the weak solution  $u$  unbounded.

## 2.4 Proof of Theorem 2.1.1

To better understand the outline of our proof, we will divide this proof into three steps.

**Step 1. Establish the estimate of subsolution for  $\theta = \frac{1}{4}$ ,  $R = 1$  and  $\gamma = 2p'$ , where  $p' = \frac{p}{p-1}$ .** i.e., if  $u$  is subsolution of (2.1) for  $\gamma = 2p'$ , then we claim that

$$\|u_+\|_{L^\infty(B_{\frac{1}{4}})} \leq C (\Lambda(B_1))^{\frac{\delta}{2(\delta-1)}} \left[ \left( \int_{B_1} |u_+|^{2p'} dx \right)^{\frac{1}{2p'}} + \|f\|_{L^s(B_1)} \right]. \quad (2.9)$$

To do this, we first introduce the definition of some test function. For  $\beta \geq 1$  and  $m \in (0, +\infty)$ , we define

$$\phi_m = \begin{cases} (u_+ + k)^\beta - k^\beta, & u \leq m, \\ \beta(m + k)^{\beta-1}(u - m) + (m + k)^\beta - k^\beta, & u > m, \end{cases}$$

where  $k := \|f\|_{L^s(B_1)}$ . Set  $\phi = \phi_m \eta^2$ , where  $\eta(x) \geq 0$ ,  $\eta \in C_0^\infty(B_1)$ , with  $\eta \equiv 1$  for  $x \in B_\rho$ ;  $\eta \equiv 0$  for  $x \in B_1 \setminus B_\sigma$ , where  $\frac{1}{4} \leq \rho < \sigma \leq \frac{1}{2}$ . Using (2.3), we obtain

$$\int_{\Omega} \mathbf{A}(x) \nabla u \nabla (\phi_m \eta^2) dx \leq \int_{\Omega} f \eta^2 \phi_m dx. \quad (2.10)$$

Then, by means of (2.10), the conditions (H1)–(H2), Young's inequality, and convexity of  $\phi_m$  in the form of  $\phi_m \leq (u_+ + k) \phi'_m$ , we get

$$\int_{\Omega} (\mathbf{A}(x) \nabla u_+ \nabla u_+) \phi'_m \eta^2 dx \leq \frac{1}{2} \int_{\Omega} \mathbf{A}(x) \nabla u_+ \nabla u_+ \phi'_m \eta^2 dx$$

$$+ 8 \int_{\Omega} \mu |\nabla \eta|^2 (u_+ + k)^2 \phi'_m dx + \int_{\Omega} f \eta^2 \phi_m dx. \quad (2.11)$$

Once again, using the elliptic condition  $\mathbf{A}(x) \nabla u_+ \nabla u_+ \geq \lambda(x) |\nabla u_+|^2$ , it follows from (2.11) that

$$\int_{\Omega} \lambda |\nabla u_+|^2 \phi'_m \eta^2 dx \leq 16 \int_{\Omega} \mu |\nabla \eta|^2 (u_+ + k)^2 \phi'_m dx + 2 \int_{\Omega} f \eta^2 \phi_m dx. \quad (2.12)$$

Imposing Hölder's inequality on the last term of (2.12), we obtain

$$\int_{B_\sigma} f \phi_m \eta^2 dx \leq \int_{B_\sigma} \frac{|f|}{k} (u_+ + k) \phi_m \eta^2 dx \leq \left( \int_{B_\sigma} \left| (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right|^{2s'} dx \right)^{\frac{1}{s'}}, \quad (2.13)$$

where  $s' = \frac{s}{s-1}$ .

Next, we estimate the right-hand side of (2.13). Actually,  $q > \frac{N}{2}$  implies  $p_* < 2s' = \frac{2s}{s-1} < q^* := \frac{2Nq}{(q+1)N-2q}$ . And then to combine Sobolev's inequality and interpolation inequality yields the following inequalities

$$\begin{aligned} \left( \int_{B_\sigma} \left| (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right|^{2s'} dx \right)^{\frac{1}{s'}} &\leq \varepsilon \left( \int_{B_\sigma} \left| (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right|^{q^*} dx \right)^{\frac{2}{q^*}} \\ &\quad + \frac{1}{\varepsilon} \left( \int_{B_\sigma} \left| (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right|^{p_*} dx \right)^{\frac{2}{p_*}} \\ &\leq \varepsilon \left( \int_{B_\sigma} \left| \nabla \left( (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \\ &\quad + \frac{1}{\varepsilon} \left( \int_{B_\sigma} \left| (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right|^{p_*} dx \right)^{\frac{2}{p_*}}, \end{aligned} \quad (2.14)$$

for sufficiently small  $\varepsilon$  (which will be determined later). Furthermore, keeping Hölder's inequality and Cauchy-Schwarz inequality in mind, we estimate the first term of the right-hand side of (2.14) as follows

$$\begin{aligned} &\left( \int_{B_\sigma} \left| \nabla \left( (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \\ &\leq \left\| \lambda^{-1} \right\|_{L^q(B_\sigma)} \left( \int_{B_\sigma} \lambda \left| \nabla \left( (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \eta \right) \right|^2 dx \right) \\ &\leq \left\| \lambda^{-1} \right\|_{L^q(B_\sigma)} \left( 2 \int_{B_\sigma} \lambda \left| \nabla \left( (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \right) \right|^2 \eta^2 dx \right. \\ &\quad \left. + 2 \int_{B_\sigma} \lambda |\nabla \eta|^2 (u_+ + k) \phi_m dx \right) \\ &\leq \left\| \lambda^{-1} \right\|_{L^q(B_\sigma)} \left( 2 \int_{B_\sigma} \lambda \left| \nabla \left( (u_+ + k)^{\frac{1}{2}} \phi_m^{\frac{1}{2}} \right) \right|^2 \eta^2 dx \right. \end{aligned}$$

$$+ 2 \int_{B_\sigma} \mu |\nabla \eta|^2 (u_+ + k) \phi_m dx \Big). \quad (2.15)$$

By applying Hölder's inequality and substituting (2.13)–(2.15) into (2.12), we conclude

$$\begin{aligned} \left( \int_{B_\rho} |\eta \nabla G_m(u)|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} &\leq \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx \\ &\leq C \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \mu |\nabla \eta|^2 (u_+ + k)^2 \phi'_m dx \\ &\quad + C \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \left( \int_{B_\sigma} |(u_+ + k)^2 \phi'_m \eta|^{\frac{p^*}{2}} dx \right)^{\frac{2}{p^*}}, \end{aligned} \quad (2.16)$$

where  $G_m(u) = \int_0^{u_+} |\phi'_m(s)|^{\frac{1}{2}} ds$ .

To optimize the first term on the right-hand side of (2.16) with respect to  $\eta$ , we need to verify that  $\mu(u_+ + k)^2 \phi'_m$  and  $(u_+ + k) (\phi'_m)^{\frac{1}{2}}$  satisfy the conditions of Lemma 2.2.2. Indeed, it is not hard to find  $\mu(u_+ + k)^2 \phi'_m \in L^1(B_\sigma)$  and  $(u_+ + k) (\phi'_m)^{\frac{1}{2}} \in W^{1,p^*}(B_\sigma)$  by virtue of  $u \in H^1(\Omega, \mathbf{A})$ ,  $p^* < \frac{2q}{q+1}$  as well as the properties of  $\phi'_m$ . Further Lemma 2.2.2 shows

$$\begin{aligned} &\inf_{\eta} \left\{ \int_{B_\sigma} \mu |\nabla \eta|^2 (u_+ + k)^2 \phi'_m dx \right\} \\ &\leq C(\sigma - \rho)^{-\frac{2N}{N-1}} \|\mu\|_{L^p(B_\sigma)} \left\| (u_+ + k) (\phi'_m)^{\frac{1}{2}} \right\|_{W^{1,p^*}(B_\sigma)}^2. \end{aligned} \quad (2.17)$$

On the other hand, when  $m$  goes to infinity, we have

$$\begin{aligned} \lim_{m \rightarrow +\infty} |\nabla G_m(u)|^2 &= \lim_{m \rightarrow +\infty} |G'_m(u)|^2 |\nabla u_+|^2 \geq \beta(u_+ + k)^{\beta-1} |\nabla u_+|^2 \text{ a.e.}, \\ (u_+ + k)^2 \phi'_m &\leq \beta(u_+ + k)^{\beta+1}, \quad \forall m \in [1, +\infty). \end{aligned}$$

Obviously, to use (2.16), (2.17) and  $\eta \equiv 1$  in  $B_\rho$  derives the following inequality

$$\begin{aligned} \left( \int_{B_\rho} \left| \nabla \left( (u_+ + k)^{\frac{\beta+1}{2}} \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} &\leq C(\beta + 1)(\sigma - \rho)^{\frac{-N}{N-1}} (\Lambda(B_\sigma))^{\frac{1}{2}} \\ &\quad \cdot \left\| (u_+ + k)^{\frac{\beta+1}{2}} \right\|_{W^{1,p^*}(B_\sigma)}. \end{aligned} \quad (2.18)$$

To make Moser iteration argument work, we further prove that there exists  $\delta > 1$  such that the term  $\left\| (u_+ + k)^{\frac{(\beta+1)\delta}{2}} \right\|_{W^{1,p^*}(B_\rho)}$  may be dominated by  $\left\| (u_+ + k)^{\frac{\beta+1}{2}} \right\|_{W^{1,p^*}(B_\sigma)}$ . Indeed, on the one hand, choose  $\delta = 2 - \frac{1}{\chi} > 1$ , with  $\chi = \frac{2q}{(q+1)p^*} > 1$ , and apply Hölder's inequality with exponent  $\frac{2q}{(q+1)p^*}$  to have

$$\left( \int_{B_\rho} \left| \nabla \left( (u_+ + k)^{\frac{\beta+1}{2} \delta} \right) \right|^{p^*} dx \right)^{\frac{1}{p^*}}$$

$$\begin{aligned}
 &\leq \left( \int_{B_\rho} \left| \delta(u_+ + k)^{\frac{\beta+1}{2}(\delta-1)} \nabla \left( (u_+ + k)^{\frac{\beta+1}{2}} \right)^{p_*} dx \right|^{\frac{1}{p_*}} \right. \\
 &\leq \delta \left( \int_{B_\rho} \left| \nabla \left( (u_+ + k)^{\frac{\beta+1}{2}} \right)^{\frac{2q}{q+1}} dx \right|^{\frac{q+1}{2q}} \right. \\
 &\quad \cdot \left. \left. \left( \int_{B_\rho} \left| (u_+ + k)^{\frac{\beta+1}{2}} \right|^{p_*} dx \right)^{\frac{\delta-1}{p_*}} \right. \right.
 \end{aligned} \tag{2.19}$$

On the other hand, with the aid of Sobolev's inequality  $W^{1,p_*}(B_\rho) \hookrightarrow L^t(B_\rho)$  where  $1 \leq t \leq \frac{Np_*}{N-p_*}$ , we have

$$\left( \int_{B_\rho} (u_+ + k)^{\frac{\beta+1}{2}\delta p_*} dx \right)^{\frac{1}{\delta p_*}} \leq C \left\| (u_+ + k)^{\frac{\beta+1}{2}} \right\|_{W^{1,p_*}(B_\rho)}. \tag{2.20}$$

Thus, combining (2.18), (2.19) with (2.20), we have

$$\begin{aligned}
 \left\| (u_+ + k)^{\frac{\beta+1}{2}\delta} \right\|_{W^{1,p_*}(B_\rho)} &\leq C(\beta+1) (\Lambda(B_1))^{\frac{1}{2}} \\
 &\quad \cdot (\sigma - \rho)^{\frac{-N}{N-1}} \left\| (u_+ + k)^{\frac{\beta+1}{2}} \right\|_{W^{1,p_*}(B_\sigma)}^\delta.
 \end{aligned} \tag{2.21}$$

Set  $\alpha = \frac{\beta+1}{2} \geq 1$ , then (2.21) can be rewritten as

$$\left\| (u_+ + k)^{\alpha\delta} \right\|_{W^{1,p_*}(B_\rho)}^{\frac{1}{\delta\alpha}} \leq \left( C\alpha (\Lambda(B_1))^{\frac{1}{2}} (\sigma - \rho)^{\frac{-N}{N-1}} \right)^{\frac{1}{\delta\alpha}} \left\| (u_+ + k)^\alpha \right\|_{W^{1,p_*}(B_\sigma)}^{\frac{1}{\alpha}}. \tag{2.22}$$

We further choose  $\alpha = \delta^m$ ,  $\rho_m = \frac{1}{4} + \left(\frac{1}{4}\right)^{m+1}$ ,  $m \in \mathbb{N}$ , and deduce the following recurrence relations for the function  $u_+ + k$

$$\begin{aligned}
 \left\| (u_+ + k)^{\delta^{m+1}} \right\|_{W^{1,p_*}(B_{\rho_{m+1}})}^{\frac{1}{\delta^{m+1}}} &\leq ((\Lambda(B_1)))^{\frac{1}{2\delta^{m+1}}} \delta^{\frac{m}{\delta^{m+1}}} \left( C4^{\frac{N}{N-1}} \right)^{\frac{m+1}{\delta^{m+1}}} \\
 &\quad \cdot \left\| (u_+ + k)^{\delta^m} \right\|_{W^{1,p_*}(B_{\rho_m})}^{\frac{1}{\delta^m}}.
 \end{aligned} \tag{2.23}$$

By iteration, we find

$$\|u_+ + k\|_{L^\infty(B_{\frac{1}{4}})} \leq C [(\Lambda(B_1))]^{\sum_{i=0}^{\infty} \frac{1}{2\delta^{i+1}}} \left( 4^{\frac{-N}{N-1}} \delta \right)^{\sum_{i=0}^{\infty} \frac{i+1}{\delta^{i+1}}} \|u_+ + k\|_{W^{1,p_*}(B_{\frac{1}{2}})}. \tag{2.24}$$

Subsequently, we estimate the right-hand side of (2.24). Using (2.16) with  $\beta = 1$ ,  $\rho = \frac{1}{2}$ ,  $\sigma = 1$ , the fact  $p^* < \frac{2q}{q+1} \leq 2 \leq 2p'$ , and  $\|\nabla \eta\|_{L^\infty(B_\sigma)} \leq c(\sigma - \rho)^{-1}$ , we obtain

$$\begin{aligned}
 \left( \int_{B_{\frac{1}{2}}} |\nabla(u_+ + k)|^{p_*} dx \right)^{\frac{1}{p_*}} &\leq \left( \int_{B_{\frac{1}{2}}} |\nabla(u_+ + k)|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\
 &\leq (\Lambda(B_1))^{\frac{1}{2}} \left[ \left( \int_{B_1} |u_+ + k|^{2p'} dx \right)^{\frac{1}{2p'}} \right]
 \end{aligned}$$

$$+ \left( \int_{B_1} |u_+ + k|^{p^*} dx \right)^{\frac{1}{p^*}} \Bigg].$$

And then by Hölder's inequality, we get

$$\left( \int_{B_1} |u_+ + k|^{p^*} dx \right)^{\frac{1}{p^*}} \leq C \left( \int_{B_1} |u_+ + k|^{2p'} dx \right)^{\frac{1}{2p'}}.$$

In turn, combining the above inequalities with (2.24), we obtain

$$\begin{aligned} \|u_+\|_{L^\infty(B_{\frac{1}{4}})} &\leq C (\Lambda(B_1))^{\frac{\delta}{2(\delta-1)}} \left( \int_{B_1} |u_+ + k|^{2p'} dx \right)^{\frac{1}{2p'}} \\ &\leq C (\Lambda(B_1))^{\frac{\delta}{2(\delta-1)}} \left[ \left( \int_{B_1} |u_+|^{2p'} dx \right)^{\frac{1}{2p'}} + \|f\|_{L^s(B_1)} \right]. \end{aligned} \quad (2.25)$$

which proves the claim.

**Step 2. The general case.** Although the procedure is standard, we also give a sketch of our proof to make our article self-contained. For more details, we may refer to [46]. First, let

$$\tilde{\mathbf{A}}(y) = \mathbf{A}(Ry), \quad \tilde{\lambda}(y) = \lambda(Ry), \quad \tilde{\mu}(y) = \mu(Ry), \quad \tilde{u}(y) = u(Ry),$$

$$\tilde{f}(y) = R^2 f(Ry), \quad \tilde{k} = \|\tilde{f}\|_{L^s(B_1)},$$

where  $y \in B_1$ . We apply what we just proved to  $\tilde{u}$  in  $B_1$  and rewrite the result in terms of  $u$ , then for any  $\gamma \geq 2p'$ , we derive

$$\|u_+\|_{L^\infty(B_{\frac{1}{4}R})} \leq C (\Lambda(B_R))^{\frac{p'\delta}{\gamma(\delta-1)}} \left( R^{-\frac{N}{\gamma}} \left( \int_{B_R} |u_+|^\gamma dx \right)^{\frac{1}{\gamma}} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right), \quad (2.26)$$

where  $C = C(p, q, N, \gamma) \geq 1$ . Again, by the same argument as above, (2.26) clearly also holds for  $B_R$  replaced by  $B_{(1-\theta)R}(x_0)$ , where  $x_0 \in B_{\theta R}$ , we arrive at

$$\begin{aligned} &\|u_+\|_{L^\infty(B_{\frac{1}{4}(1-\theta)R}(x_0))} \\ &\leq C \left[ \Lambda(B_{(1-\theta)R}(x_0)) \right]^{\frac{p'\delta}{\gamma(\delta-1)}} \left\{ [(1-\theta)R]^{-\frac{N}{\gamma}} \left( \int_{B_{(1-\theta)R}(x_0)} |u_+|^\gamma dx \right)^{\frac{1}{\gamma}} \right. \\ &\quad \left. + [(1-\theta)R]^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right\} \\ &\leq C (\Lambda(B_R))^{\frac{p'\delta}{\gamma(\delta-1)}} \left[ \frac{1}{(1-\theta)^{\frac{N}{\gamma}(m_*+1)}} R^{-\frac{N}{\gamma}} \left( \int_{B_R} |u_+|^\gamma dx \right)^{\frac{1}{\gamma}} \right. \end{aligned}$$



$$+ \frac{1}{(1-\theta)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \Big], \quad (2.27)$$

where  $m_* = \frac{p'\delta}{\delta-1} \max \left\{ \frac{1}{p} + \frac{1}{q}, \frac{2}{q} \right\}$ ,  $C = C(N, p, q, \gamma)$ .

Next, we will be in position to prove (2.2) for  $\gamma \in (0, 2p')$ . First, we observe that

$$\left( \int_{B_R} |u_+|^{2p'} dx \right)^{\frac{1}{2p'}} \leq \|u_+\|_{L^\infty(B_R)}^{\frac{2p'-\gamma}{2p'}} \|u_+\|_{L^\gamma(B_R)}^{\frac{\gamma}{2p'}}. \quad (2.28)$$

In addition, applying the Young's inequality to (2.28), and combining this with (2.27) for  $\gamma = 2p'$ , we obtain

$$\begin{aligned} & \|u_+\|_{L^\infty(B_{\theta R})} \\ & \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C (\Lambda(B_R))^{\frac{\delta p'}{\gamma(\delta-1)}} \frac{1}{(1-\theta)^{\frac{N}{\gamma}(m_*+1)}} R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_R)} \\ & \quad + (\Lambda(B_R))^{\frac{\delta}{2(\delta-1)}} \frac{1}{(1-\theta)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)}, \end{aligned} \quad (2.29)$$

where  $C = C(N, p, q, \gamma)$ . Then, set  $r := \theta R$  ( $\theta R < R \leq 1$ ,  $\theta \in (0, 1)$ ) and apply (2.29) to get

$$\begin{aligned} \|u_+\|_{L^\infty(B_r)} & \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C (\Lambda(B_R))^{\frac{\delta p'}{\gamma(\delta-1)}} \frac{1}{\left(1 - \frac{r}{R}\right)^{\frac{N}{\gamma}(m_*+1)}} R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_1)} \\ & \quad + C (\Lambda(B_R))^{\frac{\delta}{2(\delta-1)}} \frac{1}{\left(1 - \frac{r}{R}\right)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} R^{2-\frac{N}{s}} \|f\|_{L^s(B_1)} \\ & \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C (\Lambda(B_1))^{\frac{\delta p'}{\gamma(\delta-1)}} \frac{1}{(R-r)^{\frac{N}{\gamma}(m_*+1)}} \|u_+\|_{L^\gamma(B_1)} \\ & \quad + C (\Lambda(B_1))^{\frac{\delta}{2(\delta-1)}} \frac{1}{(R-r)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} \|f\|_{L^s(B_1)} \\ & \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C (\Lambda(B_1))^{\frac{\delta p'}{\gamma(\delta-1)}} \\ & \quad \cdot \left( \frac{1}{(R-r)^{\frac{N}{\gamma}(m_*+1)}} \|u_+\|_{L^\gamma(B_1)} + \frac{1}{(R-r)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} \|f\|_{L^s(B_1)} \right), \end{aligned} \quad (2.30)$$

where  $C = C(N, p, q, \gamma)$ . Let  $h(r) := \|u_+\|_{L^\infty(B_r)}$ , then (2.30) becomes

$$\begin{aligned} h(r) & \leq \frac{1}{2} h(R) + C (\Lambda(B_1))^{\frac{\delta p'}{\gamma(\delta-1)}} \left( \frac{1}{(R-r)^{\frac{N}{\gamma}(m_*+1)}} \|u_+\|_{L^\gamma(B_1)} \right. \\ & \quad \left. + \frac{1}{(R-r)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} \|f\|_{L^s(B_1)} \right). \end{aligned}$$

Proceeding as a proof of in [46, Lemma 4.3], we may derive

$$\begin{aligned} \|u_+\|_{L^\infty(B_\theta)} &\leq C (\Lambda(B_1))^{\frac{\delta p'}{\gamma(\delta-1)}} \\ &\cdot \left( \frac{1}{(1-\theta)^{\frac{N}{\gamma}(m_*+1)}} \|u_+\|_{L^\gamma(B_1)} + \frac{1}{(1-\theta)^{\frac{N}{\gamma}m_* + \frac{N}{s}-2}} \|f\|_{L^s(B_1)} \right), \end{aligned} \quad (2.31)$$

where  $C = C(N, p, q, \gamma)$ .

**Step 3. Consider**  $\|(-u)_+\|_{L^\infty(B_\theta)}$ . We consider  $u$  is supersolution of (2.1), and rewrite the above result in terms of  $-u$ , then we could prove the claim.

The proof of Theorem 2.1.1 is complete.

**Remark 2.4.1** The proof of Theorem 2.1.1 clearly shows that the reduction in the integrability of  $\lambda^{-1}$  leads to an improvement in the integrability index of  $f$  compared to the uniformly elliptic case. However, this does not mean that the integrability assumption on  $\mu$  is redundant. In fact, since  $p_* < \frac{2q}{q+1}$ , we also get  $p > \frac{1}{\frac{2}{N-1} - \frac{1}{q}}$ . Therefore, a sufficiently strong integrability assumption on  $\mu$  is essential for the continuation of the proof.

## 2.5 Proof of Theorem 2.1.2

To end the proof of Theorem 2.1.2, we first give a key lemma, namely, weak Harnack inequality.

**Lemma 2.5.1** (*Weak Harnack inequality*) Assume that (H1)-(H2) hold,  $q > \frac{N}{2}$  and  $f \in L^s(\Omega)$  for some  $s > \frac{Nq}{2q-N}$ . Let  $u \in H^1(\Omega, \mathbf{A})$  be a nonnegative supersolution in  $\Omega$ . Then for any  $B_R \subset \Omega$ ,  $\gamma \in (0, \frac{q_*}{2})$  and any  $0 < \theta < \tau < 1$ , there exists a constant  $C$  such that

$$\left( \frac{1}{R^N} \int_{B_{\tau R}} u^\gamma dx \right)^{\frac{1}{\gamma}} \leq C \left( \inf_{B_{\theta R}} u + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right),$$

where  $q_* = \frac{2Nq}{N(q+1)-2q}$ ,  $C \leq c_1 e^{c_2 \Lambda(B_R)}$ ,  $c_1 = c_1(\gamma, N, p, q, s, \tau, \theta) \geq 1$ ,  $c_2 = c_2(\gamma, N, p, q) > 0$ .

**Remark 2.5.1** Theorem 2.1.2 may be concluded by combining Theorem 2.1.1 with Lemma 2.5.1.

The prove of Lemma 2.5.1 is based on the strategy of [7, Theorem 4.1]. Even though experts might already anticipate how to adapt the argument of [7], for the completeness of the article, we give a detailed proof.

**Proof of Lemma 2.5.1.** To better comprehend the structure of our proof, we will divide our proof into three steps. And without loss of generality, we set  $R = 1$  here.

**Step 1.** We claim that

$$\exp \left( \int_{B_\tau} \log(u+h) dx \right) \leq C \left( \inf_{B_\theta} u + \|f\|_{L^s(B_1)} \right), \quad (2.32)$$

where  $C = C(N, \tau, \theta, \Lambda(B_1), p, q, s)$ .

First, we take  $\phi = \frac{\varphi}{u+h}$  in (2.3), where  $h := \|f\|_{L^s(B_1)}$  and  $\varphi \geq 0 \in H_0^1(B_1, \mathbf{A})$ , then we have

$$\int_{B_1} \left( \frac{1}{u+h} \mathbf{A}(x) \nabla u \nabla \varphi - \frac{\varphi}{(u+h)^2} \mathbf{A}(x) \nabla u \nabla u \right) dx \geq \int_{B_1} \frac{f\varphi}{u+h} dx. \quad (2.33)$$

Due to  $\frac{\varphi}{(u+h)^2} \mathbf{A}(x) \nabla u \nabla u \geq 0$ , (2.33) becomes

$$\int_{B_1} \mathbf{A}(x) \nabla \left( \log \frac{k}{u+h} \right) \nabla \varphi dx \leq - \int_{B_1} \frac{f\varphi}{u+h} dx. \quad (2.34)$$

Set  $\log \frac{k}{u+h} = v$  and  $\frac{f}{u+h} = \tilde{f}$ , and rewrite (2.34) as

$$\int_{B_1} \mathbf{A}(x) \nabla v \nabla \varphi dx \leq - \int_{B_1} \tilde{f} \varphi dx, \quad (2.35)$$

where  $\|\tilde{f}\|_{L^s(B_1)} \leq 1$ . Then (2.35) implies that  $v$  is the weak subsolution of  $-\operatorname{div}(\mathbf{A}(x) \nabla v) = -\tilde{f}$ .

To give a priori estimate for  $v$ , we choose  $\varphi = \varphi_m \eta^2$  in (2.35), where

$$\varphi_m = \begin{cases} (v_+ + 1)^\beta - 1, & v < m, \\ \beta(m+1)^{\beta-1}(v-m) + (m+1)^\beta - 1, & v \geq m, \end{cases}$$

and  $\eta \in C_0^\infty(B_\tau)$ ,  $\eta \equiv 1$  in  $B_\rho$ ;  $\eta \equiv 0$  in  $B_\tau \setminus B_\sigma$  and  $0 < \theta \leq \rho < \sigma \leq \tau \leq 1$ , by the same procedure of Theorem 2.1.1, we have

$$\sup_{B_\theta} v \leq C (\Lambda(B_1))^{\frac{\delta p'}{q_*(\delta-1)}} \left( \frac{1}{(\tau-\theta)^{\frac{N}{q_*(m_*+1)}}} \|v\|_{L^{q_*}(B_\tau)} + \frac{1}{(\tau-\theta)^{\frac{Nm_*}{q_*} + \frac{N}{s} - 2}} \right), \quad (2.36)$$

where  $q_* = \frac{2Nq}{N(q+1)-2q}$ . In the sequel, we will estimate the term  $\|v\|_{L^{q_*}(B_\tau)}$  in (2.36). Let  $\phi = \frac{\eta_1^2}{u+h}$  in (2.3), where  $\eta_1 \in C_0^\infty(B_1)$  and  $\eta_1 \equiv 1$  in  $B_\tau$ ;  $\eta_1 \equiv 0$  in  $B_1 \setminus B_{\frac{1+\tau}{2}}$ . After direct calculation, we have

$$\int_{B_1} \eta_1^2 \mathbf{A}(x) \nabla v \nabla v dx + \int_{B_1} 2\eta_1 \mathbf{A}(x) \nabla v \nabla \eta_1 dx \leq - \int_{B_1} \tilde{f} \eta_1^2 dx.$$

and then using the Young's inequality and Hölder's inequality yields

$$\begin{aligned} \int_{B_1} \lambda |\nabla v|^2 \eta_1^2 dx &\leq C \int_{B_1} \mu |\nabla \eta_1|^2 dx + |B_1|^{\frac{1}{s'}} \\ &\leq C \frac{\|\mu\|_{L^p(B_1)}}{(1-\tau)^2} |B_1|^{\frac{1}{p'}} + |B_1|^{\frac{1}{s'}}. \end{aligned} \quad (2.37)$$

Subsequently, we let  $k = \exp \left( \int_{B_\tau} \log(u+h) dx \right)$  satisfying  $\int_{B_\tau} v dx = 0$ , and apply Poincaré's inequality to get

$$\|v\|_{L^{q^*}(B_\tau)} \leq C \|\nabla v\|_{L^{\frac{2q}{q+1}}(B_\tau)} \leq \|\lambda^{-1}\|_{L^q(B_\tau)}^{\frac{1}{2}} \|\lambda^{\frac{1}{2}} \nabla v\|_{L^2(B_\tau)}. \quad (2.38)$$

Substituting (2.37) into (2.38) and it is not hard to verify

$$\sup_{B_\theta} v \leq C (\Lambda(B_1))^{\frac{\delta p'}{q^*(\delta-1)} + \frac{1}{2}} \left( \frac{1}{(\tau-\theta)^{\frac{N}{q^*}(m_*+1)}(1-\tau)} + \frac{1}{(\tau-\theta)^{\frac{Nm_*}{q^*} + \frac{N}{s} - 2}} \right).$$

Finally, (2.32) follows from the definitions of  $v$  and  $k$ .

**Step 2.** We claim that there exist  $p_0 = p_0(\tau, \theta, \Lambda(B_1), p, s, N, q) > 0$  and  $C = C(\tau, \theta, \Lambda(B_1), p, s, N, q) > 0$  such that

$$\left( \int_{B_{\frac{3\tau+\theta}{4}}} |u+h|^{p_0} dx \right)^{\frac{1}{p_0}} \leq C \exp \left( \int_{B_\tau} \log(u+h) dx \right). \quad (2.39)$$

First of all, we take  $\phi = \eta^2 \frac{1}{u+h} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right)$  in (2.3), where  $\omega = \log \frac{u+h}{k} = -v$ ,  $\beta \geq 1$ ,  $\eta \equiv 1$  in  $B_\rho$ ;  $\eta_1 \equiv 0$  in  $B_\tau \setminus B_\sigma$  and  $\eta \in C_0^\infty(B_\sigma)$ , with  $\frac{3\tau+\theta}{4} \leq \rho < \sigma \leq \tau < 1$ , and notice

$$\begin{aligned} \nabla \phi &= \frac{2\eta}{u+h} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \nabla \eta - \frac{\eta^2}{(u+h)^2} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \nabla u \\ &\quad + \frac{\beta \eta^2}{u+h} (\omega_+ + 1)^{\beta-1} \nabla (\omega_+ + 1). \end{aligned}$$

Then, we obtain

$$\begin{aligned} &\int_{B_\sigma} \frac{\eta^2}{(u+h)^2} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \mathbf{A}(x) \nabla u \nabla u dx \\ &\quad - \beta \int_{B_\sigma} \frac{\eta^2}{u+h} (\omega_+ + 1)^{\beta-1} \mathbf{A}(x) \nabla u \nabla (\omega_+ + 1) dx \\ &\leq \int_{B_\sigma} \frac{2\eta}{u+h} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \mathbf{A}(x) \nabla u \nabla \eta dx \\ &\quad - \int_{B_\sigma} f \frac{\eta^2}{u+h} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) dx. \end{aligned} \quad (2.40)$$

Subsequently, using Young's inequality, we can estimate the first term on the right-hand side of (2.40) as follows

$$\begin{aligned}
 & \int_{B_\sigma} \frac{2\eta}{u+h} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \mathbf{A}(x) \nabla u \nabla \eta dx \\
 & \leq \frac{1}{4} \int_{B_\sigma} \frac{\eta^2 (\omega_+ + 1)^{\beta-1}}{(u+h)^2} \mathbf{A}(x) \nabla u \nabla u dx + 8 \int_{B_\sigma} (\omega_+ + 1)^{\beta+1} \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & \quad + \frac{1}{4} \int_{B_\sigma} \frac{\eta^2}{(u+h)^2} (2\beta)^\beta \mathbf{A}(x) \nabla u \nabla u dx + 8 \int_{B_\sigma} (2\beta)^\beta \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & \leq \frac{1}{4} \int_{B_\sigma} \frac{\eta^2 (\omega_+ + 1)^\beta}{(u+h)^2} \mathbf{A}(x) \nabla u \nabla u dx + 8 \int_{B_\sigma} (\omega_+ + 1)^{\beta+1} \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & \quad + \frac{1}{4} \int_{B_\sigma} \frac{\eta^2}{(u+h)^2} (2\beta)^\beta \mathbf{A}(x) \nabla u \nabla u dx + 8 \int_{B_\sigma} (2\beta)^\beta \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & = \frac{1}{4} \int_{B_\sigma} \frac{\eta^2}{(u+h)^2} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) \mathbf{A}(x) \nabla u \nabla u dx \\
 & \quad + 8 \int_{B_\sigma} (\omega_+ + 1)^{\beta+1} \mathbf{A}(x) \nabla \eta \nabla \eta dx + 8 \int_{B_\sigma} (2\beta)^\beta \mathbf{A}(x) \nabla \eta \nabla \eta dx. \tag{2.41}
 \end{aligned}$$

By definition of  $\omega_+$ , we observe that the first term on the right-hand side in (2.41) can be absorbed into the first term on the left-hand side of (2.40). This leads to the following inequality

$$\begin{aligned}
 & \int_{B_\sigma} \eta^2 \left[ \frac{3}{4} \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) - \beta (\omega_+ + 1)^{\beta-1} \right] \mathbf{A}(x) \nabla (\omega_+ + 1) \nabla (\omega_+ + 1) dx \\
 & \leq 8 \int_{B_\sigma} (\omega_+ + 1)^{\beta+1} \mathbf{A}(x) \nabla \eta \nabla \eta dx + 8 \int_{B_\sigma} (2\beta)^\beta \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & \quad + \int_{B_\sigma} |\tilde{f}| \eta^2 \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) dx.
 \end{aligned}$$

Owing to  $(\omega_+ + 1)^\beta + (2\beta)^\beta \geq 2\beta (\omega_+ + 1)^{\beta-1}$ , the previous inequality reduces

$$\begin{aligned}
 & \int_{B_\sigma} \beta \eta^2 (\omega_+ + 1)^{\beta-1} \mathbf{A}(x) \nabla (\omega_+ + 1) \nabla (\omega_+ + 1) dx \\
 & \leq 16 \int_{B_\sigma} (\omega_+ + 1)^{\beta+1} \mathbf{A}(x) \nabla \eta \nabla \eta dx + 16 \int_{B_\sigma} (2\beta)^\beta \mathbf{A}(x) \nabla \eta \nabla \eta dx \\
 & \quad + 2 \int_{B_\sigma} |\tilde{f}| \eta^2 \left( (\omega_+ + 1)^\beta + (2\beta)^\beta \right) dx. \tag{2.42}
 \end{aligned}$$

With the help of (H2) and Hölder's inequality in (2.42), we obtain

$$\begin{aligned}
 & \frac{4\beta}{(\beta+1)^2} \int_{B_\sigma} \lambda \left| \nabla (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^2 \eta^2 dx \\
 & \leq 16 \int_{B_\sigma} \mu |\nabla \eta|^2 (\omega_+ + 1)^{\beta+1} dx
 \end{aligned}$$

$$\begin{aligned}
 & + (2\beta)^\beta \left( 16 \|\mu\|_{L^p(B_\sigma)} |B_1|^{\frac{1}{p'}} \frac{1}{(\sigma - \rho)^2} + 2 |B_1|^{\frac{1}{s'}} \right) \\
 & + 2 \int_{B_\sigma} |\tilde{f}| \eta^2 (\omega_+ + 1)^{\beta+1} dx.
 \end{aligned} \tag{2.43}$$

Hence, following the lines of proof of (2.13)-(2.15), we conclude

$$\begin{aligned}
 \int_{B_\sigma} |\tilde{f}| \eta^2 (\omega_+ + 1)^\beta dx & \leq \left( \int_{B_\sigma} \left| \eta (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^{2s'} dx \right)^{\frac{1}{s'}} \\
 & \leq \varepsilon \left( \int_{B_\sigma} \left| \nabla \left( (\omega_+ + 1)^{\frac{\beta+1}{2}} \eta \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \\
 & \quad + \frac{1}{\varepsilon} \left( \int_{B_\sigma} \left| (\omega_+ + 1)^{\frac{\beta+1}{2}} \eta \right|^{p^*} dx \right)^{\frac{2}{p^*}} \\
 & \leq 2\varepsilon \|\lambda^{-1}\|_q \left[ \int_{B_\sigma} \lambda \left| \nabla (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^2 \eta^2 dx \right. \\
 & \quad \left. + \int_{B_\sigma} \mu |\nabla \eta|^2 (\omega_+ + 1)^{\beta+1} dx \right] \\
 & \quad + \frac{1}{\varepsilon} \left( \int_{B_\sigma} \left| (\omega_+ + 1)^{\frac{\beta+1}{2}} \eta \right|^{p^*} dx \right)^{\frac{2}{p^*}}.
 \end{aligned} \tag{2.44}$$

By setting  $\varepsilon = \frac{\beta}{2(\beta+1)^2 \|\lambda^{-1}\|_{L^q(B_\sigma)}}$  and absorbing the first term on the right-hand side of (2.44) into the left-hand side of (2.43), it is easy to calculate that

$$\begin{aligned}
 \int_{B_\sigma} \lambda \left| \nabla (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^2 \eta^2 dx & \leq \frac{16(\beta+1)^2}{\beta} \int_{B_\sigma} \mu |\nabla \eta|^2 (\omega_+ + 1)^{\beta+1} dx \\
 & \quad + \frac{2\|\lambda^{-1}\|_q (\beta+1)^4}{\beta^2} \left( \int_{B_\sigma} \left| (\omega_+ + 1)^{\frac{\beta+1}{2}} \eta \right|^{p^*} dx \right)^{\frac{2}{p^*}} \\
 & \quad + 32(\beta+1)^2 (2\beta)^{\beta-1} \|\mu\|_{L^p(B_\sigma)} \frac{1}{(\sigma - \rho)^2}.
 \end{aligned} \tag{2.45}$$

Furthermore, minimizing the right hand of (2.45) for  $\eta(|x|) \in C_0^\infty(B_\tau)$ , with  $\eta(|x|) \equiv 1$  in  $B_\rho$ ;  $\eta(|x|) \equiv 0$  in  $B_\tau \setminus B_\sigma$ , and using Lemma 2.2.2, we obtain

$$\begin{aligned}
 & \int_{B_\sigma} \lambda \left| \nabla (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^2 \eta^2 dx \\
 & \leq \left[ \frac{16(\beta+1)^2}{\beta(\sigma - \rho)^{\frac{2N}{N-1}}} \|\mu\|_{L^p(B_\sigma)} + 2\|\lambda^{-1}\|_{L^q(B_\sigma)} \frac{(\beta+1)^4}{\beta^2} \right] \\
 & \quad \cdot \left\| (\omega_+ + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p^*}(B_\sigma)}^2 + 32(\beta+1)^2 (2\beta)^{\beta-1} \|\mu\|_{L^p(B_\sigma)} \frac{1}{(\sigma - \rho)^2}.
 \end{aligned} \tag{2.46}$$

Then, we replace  $u$  by  $\omega$  in inequalities (2.18)–(2.20) and use (2.46) to get

$$\left\| (\omega_+ + 1)^{\frac{\beta+1}{2}} \delta \right\|_{W^{1,p^*}(B_\rho)}$$

$$\begin{aligned}
 &\leq C\Lambda(B_1)^{\frac{1}{2}} \left( \frac{\beta+1}{(\sigma-\rho)^{\frac{N}{N-1}}\sqrt{\beta}} + \frac{(\beta+1)^2}{\beta} \right) \left\| (\omega_+ + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p*}(B_\sigma)}^\delta \\
 &\quad + (2\beta)^{\frac{\beta-1}{2}} (\beta+1) \Lambda(B_1)^{\frac{1}{2}} \frac{1}{\sigma-\rho} \left( \int_{B_\rho} \left| (\omega_+ + 1)^{\frac{\beta+1}{2}} \right|^{p*} dx \right)^{\frac{\delta-1}{p*}}. \quad (2.47)
 \end{aligned}$$

Keeping Young's inequality in mind, we find that the second term on the right-hand side of (2.47) has the following estimate

$$\begin{aligned}
 &\left\| (\omega_+ + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p*}(B_\rho)}^\delta \\
 &\leq C\Lambda(B_1)^{\frac{\delta}{2(\delta-1)}} \left( \frac{\beta+1}{(\sigma-\rho)^{\frac{N}{N-1}}\sqrt{\beta}} + \frac{(\beta+1)^2}{\beta} + (\sigma-\rho)^{\frac{-\delta}{\delta-1}} \right) \\
 &\quad \cdot \left\| (\omega_+ + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p*}(B_\sigma)}^\delta + \left[ (2\beta)^{\frac{\beta-1}{2}} (\beta+1) \right]^\delta. \quad (2.48)
 \end{aligned}$$

Set  $\alpha = \frac{\beta+1}{2} \geq 1$ , then, (2.48) is rewritten as

$$\begin{aligned}
 &\left\| (\omega_+ + 1)^{\alpha\delta} \right\|_{W^{1,p*}(B_\rho)}^{\frac{1}{\alpha\delta}} \\
 &\leq C^{\frac{1}{\alpha\delta}} \Lambda(B_1)^{\frac{1}{2\alpha(\delta-1)}} \left( \frac{\alpha}{(\sigma-\rho)^{\frac{N}{N-1}}\sqrt{2\alpha-1}} + \frac{\alpha^2}{2\alpha-1} + (\sigma-\rho)^{\frac{\delta}{\delta-1}} \right)^{\frac{1}{\delta\alpha}} \\
 &\quad \cdot \left\| (\omega_+ + 1)^\alpha \right\|_{W^{1,p*}(B_\sigma)}^{\frac{1}{\alpha}} + 4\alpha.
 \end{aligned}$$

Therefore, we set  $\alpha = \delta^m$ ,  $\rho_m = \frac{3\tau+\theta}{4} + \frac{1}{2^{m+2}}(\tau-\theta)$ ,  $m \in \mathbb{N}$ , and build up the following iterative process

$$\begin{aligned}
 \left\| (\omega_+ + 1)^{\delta^{m+1}} \right\|_{W^{1,p*}(B_{\rho_{m+1}})}^{\frac{1}{\delta^{m+1}}} &\leq C^{\frac{1}{\delta^{m+1}}} \Lambda(B_1)^{\frac{1}{2(\delta-1)\delta^m}} \left( 2^{\frac{\delta}{\delta-1} + \frac{N}{N-1}} \delta \right)^{\frac{m+1}{\delta^m}} \\
 &\quad \cdot \left\| (\omega_+ + 1)^{\delta^m} \right\|_{W^{1,p*}(B_{\rho_m})}^{\frac{1}{\delta^m}} + 4\delta^m. \quad (2.49)
 \end{aligned}$$

which implies

$$\begin{aligned}
 \left\| (\omega_+ + 1)^{\delta^{m+1}} \right\|_{W^{1,p*}(B_{\rho_{m+1}})}^{\frac{1}{\delta^{m+1}}} &\leq C\Lambda(B_1)^{\frac{\delta}{2(\delta-1)^2}} \left( 2^{\frac{\delta}{\delta-1} + \frac{N}{N-1}} \delta \right)^{\frac{\delta}{(\delta-1)^2}} \\
 &\quad \cdot \left( \left\| \omega_+ + 1 \right\|_{W^{1,p*}(B_{\rho_0})} + \delta^m \right), \quad (2.50)
 \end{aligned}$$

where  $C$  depends on  $N, p, q, \tau, \theta$ .

Finally, recall that  $\omega = -v$ , together with (2.37) and (2.38), and we get

$$\left\| \omega \right\|_{W^{1,p*}(B_\tau)} = \left\| v \right\|_{W^{1,p*}(B_\tau)} \leq C \left\| v \right\|_{W^{1, \frac{2q}{q+1}}(B_\tau)} \leq C(\tau, \Lambda(B_1), p, q, s).$$

Then for any  $i > p_*$ , there exists  $m_0 \in \mathbb{N}$  such that  $\delta^{m_0} p_* \leq i \leq \delta^{m_0+1} p_*$ , (2.50) can be rewritten as

$$\|\omega_+\|_{L^i(B_{\frac{3\tau+\theta}{4}})} \leq C_1 \Lambda(B_1)^{C_2} i,$$

where  $C_1 = C_1(\tau, p, q, s, N, \theta)$  and  $C_2 = C_2(N, p, q)$ . We further let  $p_0 = (2C_1 \Lambda(B_1)^{C_2} e)^{-1}$  satisfying

$$\frac{p_0^i \|\omega_+\|_{L^i(B_{\frac{3\tau+\theta}{4}})}^i}{i!} \leq \frac{1}{2^i}.$$

thus we have

$$\begin{aligned} \left( \int_{B_{\frac{3\tau+\theta}{4}}} \left( \frac{u+h}{k} \right)^{p_0} dx \right)^{\frac{1}{p_0}} &\leq \left( \int_{B_{\frac{3\tau+\theta}{4}}} e^{p_0 \omega_+} dx + |B_1| \right)^{\frac{1}{p_0}} \\ &\leq \left( \sum_{i=0}^{\infty} \frac{p_0^i \|\omega_+\|_{L^i(B_{\frac{3\tau+\theta}{4}})}^i}{i!} + |B_1| \right)^{\frac{1}{p_0}} \leq (2 + |B_1|)^{\frac{1}{p_0}}. \end{aligned}$$

The proof of the inequality (2.39) is complete.

**Step 3.** We claim that for any  $\gamma \in [p_0, \frac{q_*}{2})$ , there exists a constant  $C = C(p, q, \Lambda(B_1), \gamma, N) \geq 1$  such that

$$\left( \int_{B_\theta} (u+h)^\gamma dx \right)^{\frac{1}{\gamma}} \leq C \left( \int_{B_{\frac{3\tau+\theta}{4}}} (u+h)^{p_0} dx \right)^{\frac{1}{p_0}}, \quad (2.51)$$

where  $C \leq C_3 \Lambda(B_1)^{C_4(\frac{1}{p_0} - \frac{1}{\gamma})}$ ,  $C_3 = C_3(\gamma, N, p, q, \tau, \theta) \in [1, +\infty)$ ,  $C_4 = C_4(\gamma, N, p, q) \in [1, +\infty)$ .

To complete the proof of (2.51), we take  $\phi = \eta^2 (u+h)^\beta$  with  $\beta \in (-1, 0)$  in (2.3), and  $0 < \frac{\tau+3\theta}{4} \leq \rho < \sigma \leq \frac{\tau+\theta}{2} < 1$ ,  $\eta \equiv 1$  in  $B_\rho$ ;  $\eta_1 \equiv 0$  in  $B_{\frac{\tau+\theta}{2}} \setminus B_\sigma$ , we obtain

$$\begin{aligned} \beta \int_{B_\sigma} (u+h)^{\beta-1} \eta^2 \mathbf{A}(x) \nabla u \nabla (u+h) dx &+ 2 \int_{B_\sigma} \eta (u+h)^\beta \mathbf{A}(x) \nabla u \nabla \eta dx \\ &\geq \int_{B_\sigma} f \eta^2 (u+h)^\beta dx. \end{aligned}$$

then by Young's inequality, we get

$$-\beta \int_{B_\sigma} (u+h)^{\beta-1} \eta^2 \mathbf{A}(x) \nabla u \nabla (u+h) dx$$



$$\begin{aligned}
 &\leq - \int_{B_\sigma} f \eta^2 (u+h)^\beta dx + 2 \int_{B_\sigma} \eta (u+h)^\beta \mathbf{A}(x) \nabla u \nabla \eta dx \\
 &\leq - \frac{\beta}{2} \int_{B_\sigma} (u+h)^{\beta-1} \eta^2 \mathbf{A}(x) \nabla u \nabla (u+h) dx \\
 &\quad - \int_{B_\sigma} f \eta^2 (u+h)^\beta dx - \frac{8}{\beta} \int_{B_\sigma} \mu |\nabla \eta|^2 (u+h)^{\beta+1} dx.
 \end{aligned} \tag{2.52}$$

Absorbing the first term on the right-hand side into the left-hand side of (2.52), we obtain

$$\begin{aligned}
 - \frac{4\beta}{(\beta+1)^2} \int_{B_\sigma} \lambda \left| \nabla (u+h)^{\frac{\beta+1}{2}} \right|^2 dx &\leq \int_{B_\sigma} |f| \eta^2 (u+h)^\beta dx \\
 &\quad - \frac{8}{\beta} \int_{B_\sigma} \mu |\nabla \eta|^2 (u+h)^{\beta+1} dx.
 \end{aligned}$$

Next, recalling (2.44) and replacing  $\omega_+ + 1$  by  $u+h$ , the above inequality may be rewritten as

$$\begin{aligned}
 \int_{B_\sigma} \lambda \left| \nabla (u+h)^{\frac{\beta+1}{2}} \right|^2 dx &\leq \frac{16(\beta+1)^2}{|\beta|} \int_{B_\sigma} \mu |\nabla \eta|^2 (u+h)^{\beta+1} dx \\
 &\quad + \frac{2\|\lambda^{-1}\|_q (\beta+1)^4}{\beta^2} \left( \int_{B_\sigma} (u+h)^{\frac{(\beta+1)p_*}{2}} dx \right)^{\frac{2}{p_*}}.
 \end{aligned} \tag{2.53}$$

In turn, keeping Lemma 2.2.2 and Hölder's inequality in mind, we get

$$\begin{aligned}
 \left\| \nabla (u+h)^{\frac{\beta+1}{2}} \right\|_{L^{\frac{2q}{q+1}}(B_\rho)} &\leq C \left( \frac{\beta+1}{\sqrt{\beta}} + \frac{(\beta+1)^2}{|\beta|} \right) \frac{\Lambda(B_1)^{\frac{1}{2}}}{(\sigma-\rho)^{\frac{N}{N-1}}} \\
 &\quad \cdot \left\| (u+h)^{\frac{\beta+1}{2}} \right\|_{W^{1,p_*}(B_\sigma)}.
 \end{aligned} \tag{2.54}$$

Set  $\alpha = \frac{\beta+1}{2} \in (0, \frac{1}{2})$ , then (2.19), (2.20) and (2.54) imply

$$\begin{aligned}
 &\left\| (u+h)^{\alpha\delta} \right\|_{W^{1,p_*}(B_\rho)}^{\frac{1}{\alpha\delta}} \\
 &\leq \frac{(C\Lambda(B_1)^{\frac{1}{2}})^{\frac{1}{\alpha\delta}}}{(\sigma-\rho)^{\frac{N}{(N-1)\alpha\delta}}} \left( \frac{2\alpha}{\sqrt{1-2\alpha}} + \frac{4\alpha^2}{1-2\alpha} \right)^{\frac{1}{\alpha\delta}} \left\| (u+h)^\alpha \right\|_{W^{1,p_*}(B_\sigma)}^{\frac{1}{\alpha}}.
 \end{aligned}$$

Taking  $\alpha = \frac{\alpha_0}{\delta^m}$  where  $\alpha_0 \in (0, \frac{1}{2})$ , and  $\rho_m = \frac{\tau+\theta}{2} + \frac{1}{2^m} \frac{(\theta-\tau)}{4}$ , we get

$$\begin{aligned}
 &\left\| (u+h)^{\frac{\alpha_0}{\delta^{m-1}}} \right\|_{W^{1,p_*}(B_{\rho_{m-1}})}^{\frac{\delta^{m-1}}{\alpha_0}} \\
 &\leq \left( \frac{C\Lambda(B_1)^{\frac{1}{2}}}{(\tau-\theta)} \right)^{\frac{\delta^{m-1}}{\alpha_0}} \left( \frac{\alpha_0}{\delta^m \sqrt{1-\frac{2\alpha_0}{\delta^m}}} + \frac{4\alpha_0^2}{\delta^{2m} (1-\frac{2\alpha_0}{\delta^m})} \right)^{\frac{\delta^{m-1}}{\alpha_0}}
 \end{aligned}$$

$$\cdot 2^{\frac{mN\delta^{m-1}}{\alpha_0(N-1)}} \left\| (u+h)^{\frac{\alpha_0}{\delta^m}} \right\|_{W^{1,p*}(B_{\rho_m})}^{\frac{\delta^m}{\alpha_0}}.$$

Then by iteration, we obtain

$$\begin{aligned} & \left\| (u+h)^{\alpha_0} \right\|_{W^{1,p*}(B_{\rho_0})}^{\frac{1}{\alpha_0}} \\ & \leq \left( \frac{C\Lambda(B_1)^{\frac{1}{2}}\alpha_0^2}{(\tau-\theta)\left(1-\frac{2\alpha_0}{\delta}\right)} \right)^{\frac{\delta^m-1}{(\delta-1)\alpha_0}} \left( \frac{2^{\frac{N}{N-1}}}{\delta} \right)^{\frac{1}{\alpha_0} \left( \frac{(m+1)}{\delta-1} \delta^{m+1} - \frac{\delta^{m+1}-1}{(\delta-1)^2} \right)} \\ & \cdot \left\| (u+h)^{\frac{\alpha_0}{\delta^m}} \right\|_{W^{1,p*}(B_{\rho_m})}^{\frac{\delta^m}{\alpha_0}}. \end{aligned} \quad (2.55)$$

Subsequently, we fix  $m \in \mathbb{N}$  satisfying  $\frac{2p\alpha_0}{\delta^m(p-1)} \leq p_0 \leq \frac{2p\alpha_0}{\delta^{m-1}(p-1)}$ , and apply Hölder's inequality to (2.53) to estimate the right-hand of (2.55) as follows

$$\begin{aligned} \left\| \nabla (u+h)^{\frac{\alpha_0}{\delta^m}} \right\|_{L^{p*}(B_{\rho_m})} & \leq C \left\| \nabla (u+h)^{\frac{\alpha_0}{\delta^m}} \right\|_{L^{\frac{2q}{q+1}}(B_{\frac{\tau+\theta}{2}})} \\ & \leq C_5 \left\| (u+h)^{\frac{\alpha_0}{\delta^m}} \right\|_{L^{\frac{2p}{p-1}}(B_{\frac{3\tau+\theta}{4}})} \leq C_6 \|u+h\|_{L^{p_0}(B_{\frac{3\tau+\theta}{4}})}^{\frac{\alpha_0}{\delta^m}}, \end{aligned}$$

where  $C_5, C_6$  depend on  $\tau, \theta, \alpha_0, \Lambda(B_1), N, p, q, s$ . Further, inserting the above inequality into (2.55), one has

$$\left\| (u+h)^{\alpha_0} \right\|_{W^{1,p*}(B_{\rho_0})}^{\frac{1}{\alpha_0}} \leq C\Lambda(B_1)^{\frac{p\delta}{(p-1)p_0(\delta-1)} - \frac{1}{2\alpha_0(\delta-1)}} \|u+h\|_{L^{p_0}(B_{\frac{3\tau+\theta}{4}})}. \quad (2.56)$$

Once again, by Sobolev's inequality, (2.53) and (2.54), we find

$$\|(u+h)^{\alpha_0}\|_{L^{q*}(B_\theta)} \leq C \|(u+h)^{\alpha_0}\|_{W^{1,\frac{2q}{q+1}}(B_\theta)} \leq C \|(u+h)^{\alpha_0}\|_{W^{1,p*}(B_{\frac{\tau+3\theta}{4}})}, \quad (2.57)$$

where  $C = C(\tau, \theta, N, \Lambda(B_1), q, p, \alpha_0)$ . Then a combination of (2.56) and (2.57) yields the desired claim (2.51).

This completes the proof of Lemma 2.5.1.

## 2.6 Proof of Theorem 2.1.3

In this section, without loss of generality, we can assume that  $\|f\|_{\frac{Nq}{2q-N}} = 1$  for  $f \neq 0$ . We shall begin with the Lemma 2.6.1. This lemma is mainly based on [27, Lemma 2.22] and [105, Lemma B], but in comparison, we do not assume that  $u$  is bounded. Our idea of proof is motivated by [21, 26].

**Lemma 2.6.1** *Assume that (H1)-(H2) hold,  $q > \frac{N}{2}$  and  $f \in L^s(\Omega)$  for some  $s > \frac{Nq}{2q-N}$ . Let  $u$  be the non-negative, weak subsolution of (2.1) in  $\Omega$  with zero Dirichlet boundary value. Then for any  $\alpha \in \left(0, \frac{4}{c_s^2 \|\lambda^{-1}\|_q}\right)$ , where  $c_s$  is the sharp constant of the Sobolev inequality<sup>1</sup>, there exists  $C = C(c_s, \Omega, N, q)$  such that*

$$\left( \int_{\Omega} e^{\frac{\alpha q_* u}{2}} dx \right)^{\frac{1}{q_*}} \leq \frac{2\alpha |\Omega|^{1 - \frac{(q_*)'}{s_0}}}{\left(2 - c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \sqrt{\alpha}\right)^2} + |\Omega|^{\frac{1}{q_*}},$$

where  $q_* = \frac{2Nq}{N(q+1)-2q}$ .

**Proof.** First we fix  $\alpha \in \left(0, \frac{4}{c_s^2 \|\lambda^{-1}\|_q}\right)$  and define

$$\varphi(u) = \phi_M^\alpha(u) = \begin{cases} e^{\alpha u} - 1, & u \leq M, \\ \alpha e^{\alpha M} u + (1 - \alpha M)e^{\alpha M} - 1, & u > M. \end{cases}$$

Using Lemma 2.2.1, we have  $\phi_M^\alpha(u) \in H_0^1(\Omega, \mathbf{A})$ . Substituting  $\phi_M^\alpha(u)$  into (2.3) yields

$$\begin{aligned} \int_{\Omega} \mathbf{A}(x) \nabla u \nabla \phi_M^\alpha dx &= \int_{\Omega} \mathbf{A}(x) \nabla u \nabla u (\phi_M^\alpha)' dx \\ &\leq \int_{\Omega} |f| \phi_M^\alpha dx \leq \int_{\Omega} |f| \left[ \left| e^{\frac{\alpha u}{2}} - 1 \right|^2 + 2 \left| e^{\frac{\alpha u}{2}} - 1 \right| \right] dx, \quad \forall M \geq 1. \end{aligned} \quad (2.58)$$

Then from (H1)-(H2), Lemma 2.2.1 and (2.58), it infers

$$\begin{aligned} \int_{\Omega} \lambda(x) |\nabla F_M^\alpha(u)|^2 dx &\leq \int_{\Omega} \mathbf{A}(x) \nabla u \nabla \phi_M^\alpha dx \\ &\leq \int_{\Omega} |f| \left| e^{\frac{\alpha u}{2}} - 1 \right|^2 dx + 2 \int_{\Omega} |f| \left| e^{\frac{\alpha u}{2}} - 1 \right| dx, \end{aligned} \quad (2.59)$$

where  $F_M^\alpha = \int_0^u \left| (\phi_M^\alpha)'(s) \right|^{\frac{1}{2}} ds$ . In addition, according to the definition of  $\phi_M^\alpha$ , we know

$$\lim_{M \rightarrow \infty} |\nabla F_M^\alpha(u)| \geq \sqrt{\alpha} e^{\frac{\alpha u}{2}} |\nabla u|. \quad (2.60)$$

Therefore, together with Hölder's inequality, (2.60) and (2.59), we have

$$\begin{aligned} \left( \int_{\Omega} \left| \nabla \left( e^{\frac{\alpha u}{2}} - 1 \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} &= \left( \int_{\Omega} \left| \nabla e^{\frac{\alpha u}{2}} \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\ &\leq \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \left( \int_{\Omega} \lambda(x) \left| \nabla e^{\frac{\alpha u}{2}} \right|^2 dx \right)^{\frac{1}{2}} \end{aligned}$$

---

<sup>1</sup>Here the Sobolev inequality is  $\|u\|_{p^*} \leq c_s \|\nabla u\|_p$ , where  $p^* = \frac{Np}{N-p}$ ,  $u \in W_0^{1,p}(\Omega)$ .

$$\begin{aligned}
 &\leq \|\lambda^{-1}\|_q^{\frac{1}{2}} \left( \frac{\alpha}{4} \int_{\Omega} \lambda(x) |\nabla F_M^\alpha(u)|^2 dx \right)^{\frac{1}{2}} \\
 &\leq \|\lambda^{-1}\|_q^{\frac{1}{2}} \left[ \frac{\alpha}{4} \int_{\Omega} |f| \left| e^{\frac{\alpha u}{2}} - 1 \right|^2 dx \right. \\
 &\quad \left. + \frac{\alpha}{2} \int_{\Omega} |f| \left| e^{\frac{\alpha u}{2}} - 1 \right| dx \right]^{\frac{1}{2}}. \tag{2.61}
 \end{aligned}$$

Once again, we apply Sobolev's inequality and (2.61) to observe that the following inequality holds

$$\begin{aligned}
 \left( \int_{\Omega} \left| e^{\frac{\alpha u}{2}} - 1 \right|^{q_*} dx \right)^{\frac{1}{q_*}} &\leq c_s \left( \int_{\Omega} \left| \nabla \left( e^{\frac{\alpha u}{2}} - 1 \right) \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\
 &\leq c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \left[ \frac{\alpha}{4} \left( \int_{\Omega} \left| e^{\frac{\alpha u}{2}} - 1 \right|^{q_*} dx \right)^{\frac{2}{q_*}} \right. \\
 &\quad \left. + \frac{\alpha}{2} \|f\|_{(q_*)'} \left( \int_{\Omega} \left| e^{\frac{\alpha u}{2}} - 1 \right|^{q_*} dx \right)^{\frac{1}{q_*}} \right]^{\frac{1}{2}}, \tag{2.62}
 \end{aligned}$$

where  $q_* = \frac{2Nq}{N(q+1)-2q}$ . Meanwhile, we notice that

$$\|f\|_{(q_*)'} \leq |\Omega|^{1-\frac{(q_*)'}{s_0}},$$

and  $1 - \frac{c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \sqrt{\alpha}}{2} > 0$ , so, it follows from (2.62) that

$$\left( \int_{\Omega} \left| e^{\frac{\alpha u}{2}} - 1 \right|^{q_*} dx \right)^{\frac{1}{q_*}} \leq \frac{2\alpha \|f\|_{(p_*)'}}{(2 - c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \sqrt{\alpha})^2} \leq \frac{2\alpha |\Omega|^{1-\frac{(q_*)'}{s_0}}}{(2 - c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \sqrt{\alpha})^2}. \tag{2.63}$$

By Minkowski's inequality and (2.63), we get

$$\begin{aligned}
 \left( \int_{\Omega} e^{\frac{\alpha q_* u}{2}} dx \right)^{\frac{1}{q_*}} &\leq \left( \int_{\Omega} \left| e^{\frac{\alpha u}{2}} - 1 \right|^{q_*} dx \right)^{\frac{1}{q_*}} + |\Omega|^{\frac{1}{q_*}} \\
 &\leq \frac{2\alpha |\Omega|^{1-\frac{(q_*)'}{s_0}}}{(2 - c_s \|\lambda^{-1}\|_q^{\frac{1}{2}} \sqrt{\alpha})^2} + |\Omega|^{\frac{1}{q_*}}. \tag{2.64}
 \end{aligned}$$

This completes the proof of Lemma 2.6.1.  $\square$

**Proof of Theorem 2.1.3.** Without any loss of generality, we assume  $\sup_{\Omega} u = \|u\|_{\infty}$ , otherwise we consider  $-u$ . Let  $\varphi = \widetilde{\phi_M^\alpha}(u)\tilde{\eta}$ , where

$$\widetilde{\phi_M^\alpha}(u) = \begin{cases} e^{\frac{\alpha}{2}u_+}, & u \leq M, \\ \frac{\alpha}{2}e^{\frac{\alpha}{2}M}(u - M) + e^{\frac{\alpha}{2}M}, & u > M, \end{cases} \quad \text{and } \tilde{\eta} \geq 0 \in C_0^\infty(\Omega). \tag{2.65}$$

We choose  $\varphi$  as a test function in (2.3) and obtain

$$\int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \widetilde{\phi_M^\alpha}) \tilde{\eta} dx + \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \tilde{\eta}) \widetilde{\phi_M^\alpha} dx = \int_{\Omega} f \widetilde{\phi_M^\alpha} \tilde{\eta} dx.$$

In view of  $\int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \widetilde{\phi_M^\alpha}) \tilde{\eta} dx \geq 0$ , we have

$$\int_{\Omega} \widetilde{\phi_M^\alpha} \mathbf{A}(x) \nabla u \nabla \tilde{\eta} dx \leq \int_{\Omega} f \widetilde{\phi_M^\alpha} \tilde{\eta} dx. \quad (2.66)$$

Since  $C_0^\infty(\Omega)$  is dense in  $H_0^1(\Omega, \mathbf{A})$ , it is obvious that (2.66) holds for any  $\tilde{\eta} \in H_0^1(\Omega, \mathbf{A})$ .

Therefore, we may choose  $\tilde{\eta}$  satisfying

$$\tilde{\eta} = \eta_M^\beta(u) = \begin{cases} \left(e^{\frac{\alpha}{2}u_+}\right)^\beta - 1, & u \leq M, \\ \frac{\alpha\beta}{2}e^{\frac{\alpha\beta}{2}M}(u - M) + e^{\frac{\alpha\beta}{2}M} - 1, & u > M, \end{cases}$$

where  $\beta \geq 1$  as a test function. Hence, substituting  $\eta_M^\beta$  into (2.66), we have

$$\int_{\Omega} (\mathbf{A}(x) \nabla u_+ \nabla u_+) (\eta_M^\beta)' \widetilde{\phi_M^\alpha} dx \leq \int_{\Omega} f \widetilde{\phi_M^\alpha} \eta_M^\beta dx. \quad (2.67)$$

Using Lemma 2.2.1 and (H1), we rewrite (2.67) as

$$\int_{\Omega} \lambda |\nabla \tilde{G}(u)|^2 dx \leq \int_{\Omega} f \widetilde{\phi_M^\alpha} \eta_M^\beta dx,$$

where  $\tilde{G}(u) = \int_0^u \left[ (\eta_M^\beta)'(s) \widetilde{\phi_M^\alpha}(s) \right]^{\frac{1}{2}} ds$ . A simple computation shows

$$\lim_{M \rightarrow \infty} |\nabla \tilde{G}(u)|^2 \geq \frac{8\beta}{(\beta+1)^2} \left| \nabla \left( e^{\frac{\alpha}{2}u_+} \right)^{\frac{\beta+1}{2}} \right|^2, \quad (2.68)$$

$$\widetilde{\phi_M^\alpha} \eta_M^\beta \leq \left( e^{\frac{\alpha}{2}u_+} \right)^{\beta+1}, \quad \forall M \geq 1. \quad (2.69)$$

Let  $\omega = e^{\frac{\alpha}{2}u_+}$ , then together with (2.67)–(2.69), we obtain

$$\int_{\Omega} \lambda \left| \nabla \omega^{\frac{\beta+1}{2}} \right|^2 dx \leq \frac{\alpha(\beta+1)^2}{8\beta} \int_{\Omega} f \omega^{\beta+1} dx. \quad (2.70)$$

With the aid of the Sobolev inequality and (2.70), we deduce

$$\begin{aligned} \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{q_*} dx \right)^{\frac{1}{q_*}} &\leq \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} - 1 \right)^{q_*} dx \right)^{\frac{1}{q_*}} + |\Omega|^{\frac{1}{q_*}} \\ &\leq c_s \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \left( \int_{\Omega} \lambda \left| \nabla \omega^{\frac{\beta+1}{2}} \right|^2 dx \right)^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}} \\ &\leq c_s \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \frac{\sqrt{\alpha}(\beta+1)}{2\sqrt{2}\beta} \|f\|_{\frac{1}{s}}^{\frac{1}{2}} \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{2s'} dx \right)^{\frac{1}{2s'}} \end{aligned}$$

$$\begin{aligned}
 & + |\Omega|^{\frac{1}{q_*} - \frac{1}{2s'}} \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{2s'} dx \right)^{\frac{1}{2s'}} \\
 & = \left[ c_s \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \frac{\sqrt{\alpha}(\beta+1)}{2\sqrt{2}\beta} \|f\|_s^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*} - \frac{1}{2s'}} \right] \\
 & \quad \cdot \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{2s'} dx \right)^{\frac{1}{2s'}}, \tag{2.71}
 \end{aligned}$$

where  $2s' < q_*$  due to  $s > s_0 = \left(\frac{q_*}{2}\right)'$ . Because of  $\frac{\beta+1}{2\sqrt{\beta}} \geq 1$ , (2.71) may be rewritten as the following form

$$\begin{aligned}
 \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{q_*} dx \right)^{\frac{2}{q_*(\beta+1)}} & \leq \left[ c_s \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \frac{\sqrt{\alpha}(\beta+1)}{2\sqrt{2}\beta} \|f\|_s^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*} - \frac{1}{2s'}} \right]^{\frac{2}{\beta+1}} \\
 & \quad \cdot \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{2s'} dx \right)^{\frac{1}{s'(\beta+1)}} \\
 & \leq \left[ C \frac{(\beta+1)}{\sqrt{\beta}} \left( \sqrt{\alpha} \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \|f\|_s^{\frac{1}{2}} + 1 \right) \right]^{\frac{2}{\beta+1}} \\
 & \quad \cdot \left( \int_{\Omega} \left( \omega^{\frac{\beta+1}{2}} \right)^{2s'} dx \right)^{\frac{1}{s'(\beta+1)}}, \tag{2.72}
 \end{aligned}$$

Here  $C = C(N, p, q, s, \Omega) \geq 1$ . Now, if we set  $\frac{\beta+1}{2} = \chi^i$ ,  $i \in \mathbb{N}$  in (2.72), where  $\chi = \frac{q_*}{2s'} > 1$ , then (2.72) becomes

$$\begin{aligned}
 \|\omega\|_{\chi^i q_*} & \leq \left[ C \left( \sqrt{\alpha} \left\| \lambda^{-1} \right\|_q^{\frac{1}{2}} \|f\|_s^{\frac{1}{2}} + 1 \right) \right]^{\frac{1}{\chi^i}} \chi^{\frac{i}{\chi^i}} \|\omega\|_{2\chi^i s'} \\
 & \leq \left[ C \left( \alpha \left\| \lambda^{-1} \right\|_q \|f\|_s + 1 \right) \right]^{\sum_{j=0}^i \frac{1}{2\chi^j}} \chi^{\sum_{j=0}^i \frac{j}{\chi^j}} \|\omega\|_{2s'}. \tag{2.73}
 \end{aligned}$$

And then, taking  $i \rightarrow +\infty$  yields

$$\|\omega\|_{\infty} \leq \left[ C \left( \alpha \left\| \lambda^{-1} \right\|_q \|f\|_s + 1 \right) \right]^{\frac{\chi}{2(\chi-1)}} \chi^{\frac{\chi}{(\chi-1)^2}} \|\omega\|_{2s'}, \tag{2.74}$$

where  $C = C(N, p, q, s, \Omega)$ . Using Lemma 2.6.1 and Hölder's inequality, (2.74) can be rewritten as

$$\begin{aligned}
 \|\omega\|_{\infty} & \leq C \left[ \left( \alpha \left\| \lambda^{-1} \right\|_q \|f\|_s + 1 \right) \right]^{\frac{\chi}{2(\chi-1)}} \left( \int_{\Omega} \omega^{2s'} dx \right)^{\frac{1}{2s'}} \\
 & \leq C \left[ \left( \alpha \left\| \lambda^{-1} \right\|_q \|f\|_s + 1 \right) \right]^{\frac{\chi}{2(\chi-1)}} \left( \left( \int_{\Omega} e^{\frac{\alpha}{2} q_* u} dx \right)^{\frac{1}{q_*}} + |\Omega|^{\frac{1}{q_*}} \right) \\
 & \leq C \left[ \left( \alpha \left\| \lambda^{-1} \right\|_q \|f\|_s + 1 \right) \right]^{\frac{\chi}{2(\chi-1)}} \left( \frac{\alpha |\Omega|^{1 - \frac{(q_*)'}{s_0}}}{\left( 2 - c_s \left\| \lambda^{-1} \right\|_q \sqrt{\alpha} \right)^2} + |\Omega|^{\frac{1}{q_*}} \right).
 \end{aligned}$$

By the definition of  $\omega$ , we further obtain

$$e^{\frac{\alpha}{2}\|u_+\|} \leq C \left[ \left( \alpha \|\lambda^{-1}\|_q \|f\|_s + 1 \right) \right]^{\frac{\chi}{2(\chi-1)}} \left( \frac{\alpha |\Omega|^{1-\frac{(q^*)'}{s_0}}}{(2 - c_s \|\lambda^{-1}\|_q \sqrt{\alpha})^2} + |\Omega|^{\frac{1}{q^*}} \right). \quad (2.75)$$

Obviously, (2.75) is equivalent to

$$\|u\|_\infty \leq \frac{C}{\alpha} \log \left[ \left( \frac{\alpha}{2 - c_s \|\lambda^{-1}\|_q \sqrt{\alpha}} + 1 \right) \left( \alpha \|\lambda^{-1}\|_q \|f\|_s + 1 \right) \right], \quad (2.76)$$

where  $C = C(N, p, q, s, \Omega) \geq 1$ . This completes the proof.

**Remark 2.6.1** We need to point out some differences below. Firstly, in the proof of Lemma 2.6.1, we construct a truncation function that satisfies the condition of Lemma 2.2.1 to bypass some difficulties from the global boundedness of  $u$ , as described in [27, 105]. Secondly, in the proof of Theorem 2.1.3, we construct a Caccioppoli inequality for  $e^{\alpha u_+}$  to avoid the transformation in [27, 105] which essentially depends on the bounded condition.

## 2.7 Proof of Theorem 2.1.4

**Lemma 2.7.1** (1) *There exists a unique  $\Upsilon_\psi \in K_\psi$  such that*

$$J(\Upsilon_\psi) = \inf_{\omega \in K_\psi} J(\omega).$$

(2) *There exists a unique  $\tilde{\Upsilon}_\psi$  such that*

$$\int_{B_{2d}} a_{ij} \partial \omega \partial_j (\omega - \tilde{\Upsilon}_\psi) dx \geq 0, \quad \forall \omega \in K_\psi.$$

(3) *There exists a unique  $\hat{\Upsilon}_\psi$  such that*

$$\int_{B_{2d}} a_{ij} \partial_i \hat{\Upsilon}_\psi \partial_j (\omega - \hat{\Upsilon}_\psi) dx \geq 0, \quad \forall \omega \in K_\psi.$$

(4)  $\Upsilon_\psi = \tilde{\Upsilon}_\psi = \hat{\Upsilon}_\psi$ .

**Remark 2.7.1** Lemma 2.7.1 is analogous to the Lemma 1.2 in [66]. And the proof follows a similar strategy. The difference lies in the fact that the coefficient matrix  $\mathbf{A}(x)$  is not uniformly elliptic matrix, but rather a non-uniformly elliptic matrix.

Next lemma is a type of comparison lemma, and its proof mainly uses the assumption (H2) and Hölder's inequality, and we will put the details of the proof behind the lemma for the convenience of the reader.

**Lemma 2.7.2** *Let  $\omega \in H_0^1(\Omega, \mathbf{A})$  be a supersolution to (2.1) in  $\Omega$ , and let  $f \in L^{\frac{2q}{q-1}}(\Omega)$ . Suppose that*

$$\omega(x) \leq 0 \quad \text{a.e. in } \Omega \quad \Rightarrow \quad f(x) \geq 0 \quad \text{a.e. in } \Omega.$$

*Then we have  $\omega \geq 0$  a.e. in  $\Omega$ .*

**Proof.** First, we define  $\omega_- = \max\{0, -\omega\} \in H_0^1(\Omega, \mathbf{A})$ , from the assumption in Lemma 2.7.2 and taking test function  $\phi = \omega_-$ , we have

$$\int_{\Omega} a_{ij} \partial_i \omega \partial_j \omega_- dx \geq \int_{\Omega} f \omega_- dx = \int_{\{x \in \Omega | \omega \leq 0\}} f \omega_- dx \geq 0.$$

And then

$$\int_{\Omega} \lambda(x) |\nabla \omega_-|^2 dx \leq \int_{\Omega} a_{ij} \partial_i \omega_- \partial_j \omega_- dx = \int_{\Omega} a_{ij} \partial_i (-\omega) \partial_j \omega_- dx \leq 0.$$

Using Poincaré's inequality and Hölder's inequality, we obtain

$$\begin{aligned} \left( \int_{\Omega} |\omega_-|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} &\leq C \left( \int_{\Omega} |\nabla \omega_-|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \\ &\leq C \|\lambda^{-1}\|_{L^q} \left( \int_{\Omega} \lambda(x) |\nabla \omega_-|^2 dx \right) \leq 0, \end{aligned}$$

where  $C$  is a Poincaré constant. Consequently, we derive  $\omega_- = 0$  a.e. in  $\Omega$ , which implies that  $\omega \geq 0$  a.e. in  $\Omega$ .  $\square$

In the following, we will give an existence lemma to a specific nonlinear elliptic equation.

**Lemma 2.7.3** *Suppose  $f \in L^{\frac{2q}{q-1}}(\Omega)$  and  $\theta$  is a Lipschitz function defined in  $(-\infty, +\infty)$  such that  $0 \leq \theta(x) \leq 1$ , then the nonlinear equation*

$$-\partial_i (a_{ij} \partial_j u(x)) = f_+(x) \cdot \theta(u - \psi) \tag{2.77}$$

*has a solution in  $H_0^1(\Omega, \mathbf{A})$  for any  $\psi \in L^{\frac{2q}{q+1}}(\Omega)$ .*

**Proof.** First, by the Lax-Milgram Theorem, we could get, for any fixed  $\omega \in L^{\frac{2q}{q+1}}(\Omega)$ , the linear equation

$$-\partial_i (a_{ij} \partial_j u(x)) = f_+(x) \cdot \theta(\omega - \psi) \tag{2.78}$$



has a unique weak solution  $u \in H_0^1(\Omega, \mathbf{A})$ , and we define the operator  $T$ , which satisfies  $u = T\omega$ . Since  $0 \leq \theta \leq 1$ , by the Hölder inequality and simple computation, we could get

$$\begin{aligned}
 \int_{\Omega} a_{ij} \partial_j T \omega \partial_i T \omega dx &= \int_{\Omega} f_+(x) \cdot \theta(\omega - \psi) \cdot T \omega dx \\
 &\leq \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \left( \int_{\Omega} |T \omega|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\
 &\leq C \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \left( \int_{\Omega} |\nabla T \omega|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\
 &\leq C \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \left( \int_{\Omega} \lambda(x) |\nabla T \omega|^2 dx \right)^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} \\
 &\leq C \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \left( \int_{\Omega} a_{ij} \partial_j T \omega \partial_i T \omega dx \right)^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}},
 \end{aligned}$$

where  $C$  is Poincaré constant. Thus, we have

$$\|T \omega\|_{H_0^1(\Omega, \mathbf{A})} \leq C \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}}. \quad (2.79)$$

Second, we define a closed, convex and nonempty set

$$D := \left\{ \omega \in L^{\frac{2q}{q+1}}(\Omega) : \|\omega\|_{L^{\frac{2q}{q+1}}(\Omega)} \leq C^2 \|\lambda^{-1}\|_{L^q(\Omega)} \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \right\},$$

where  $C$  is the same constant as in (2.79). It is not hard to observe  $T(D) \subset T(L^{\frac{2q}{q+1}}(\Omega)) \subset D$ , and then for a sequence  $\{\omega_j\}_{j=1}^{\infty} \subset D$ ,  $\{T(\omega_j)\}_{j=1}^{\infty} \subset H_0^1(\Omega, \mathbf{A}) \cap D$ , and because  $H_0^1(\Omega, \mathbf{A})$  is compactly embedded in  $L^{\frac{2q}{q+1}}(\Omega)$ , there exists a subsequence  $\{\omega_{j_k}\}_{k=1}^{\infty}$ , such that  $\{T(\omega_{j_k})\}_{k=1}^{\infty}$  converges strongly to a point in  $D$  (we call it  $\omega^*$ ). Therefore the operator  $T: D \rightarrow D$  is compact.

Third, we will prove that  $T$  is also a continuous operator. We might as well use the sequence  $\{\omega_j\}_{j=1}^{\infty}$  from the previous step, assuming that they converge strongly to  $\omega_0$  in  $D$ . Because  $\{T \omega_j\}_{j=1}^{\infty}$  has a uniform bound, it makes  $\{T \omega_{j_k}\}_{k=1}^{\infty}$  weakly converge to  $\omega^*$  in  $H_0^1(\Omega, \mathbf{A}) \cap D$ . Since the embedding from  $H_0^1(\Omega, \mathbf{A})$  to  $L^{\frac{2q}{q+1}}(\Omega)$  is compact, there exists subsequence of  $\{T \omega_{j_k}\}_{k=1}^{\infty}$ , which we still denote as  $\{T \omega_{j_k}\}_{k=1}^{\infty}$ , which strongly converges to  $\omega^*$  in  $D$ . Then it suffices to prove  $\omega^* = T(\omega_0)$ , that is,

$$\int_{\Omega} a_{ij}(x) \partial_i \omega^* \partial_j \varphi dx = \int_{\Omega} f_+(x) \cdot \theta(\omega_0 - \psi) \cdot \varphi dx, \quad \forall \varphi \in C_0^{\infty}(\Omega).$$

From the definition of the operator  $T$ , we know

$$\int_{\Omega} a_{ij}(x) \partial_i T \omega_{j_k} \partial_j \varphi dx = \int_{\Omega} f_+(x) \cdot \theta(\omega_{j_k} - \psi) \cdot \varphi dx, \quad \forall \varphi \in C_0^{\infty}(\Omega).$$

Indeed, since  $T\omega_{j_k} \rightarrow \omega^*$  weakly in  $H_0^1(\Omega, \mathbf{A})$ , it follows

$$\int_{\Omega} a_{ij}(x) \partial_i T\omega_{j_k} \partial_j \varphi dx \rightarrow \int_{\Omega} a_{ij}(x) \partial_i \omega^* \partial_j \varphi dx, \text{ as } k \rightarrow +\infty.$$

On the other hand, because  $\theta(x)$  is a Lipschitz function, we get

$$\begin{aligned} & \int_{\Omega} f_+(x) |\theta(\omega_{j_k} - \psi) - \theta(\omega_0 - \psi)| \varphi dx \\ & \leq \|\varphi\|_{L^\infty(\Omega)} \int_{\Omega} f_+(x) |\theta(\omega_{j_k} - \psi) - \theta(\omega_0 - \psi)| dx \\ & \leq L \|\varphi\|_{L^\infty(\Omega)} \int_{\Omega} f_+(x) |\omega_{j_k} - \omega_0| dx \\ & \leq L \|\varphi\|_{L^\infty(\Omega)} \|f_+\|_{L^{\frac{2q}{q-1}}(\Omega)} \|\omega_{j_k} - \omega_0\|_{L^{\frac{2q}{q+1}}(\Omega)} \rightarrow 0, \text{ as } k \rightarrow +\infty, \end{aligned}$$

where  $L$  is the Lipschitz constant of  $\theta(x)$ . Thus, we obtain

$$\int_{\Omega} f_+(x) \cdot \theta(\omega_{j_k} - \psi) \cdot \varphi dx \rightarrow \int_{\Omega} f_+(x) \cdot \theta(\omega_0 - \psi) \cdot \varphi dx.$$

Therefore, the operator  $T : D \rightarrow D$  is continuous.

Consequently, according to the Schauder fixed point theory, there exists  $u \in D$ , such that  $u = Tu \in H_0^1(\Omega, \mathbf{A})$ .  $\square$

Now, we are going to prove another important lemma in this section, the proof is also similar to Theorem 2.1 in [66]. For the completeness of the article, we write it below.

**Lemma 2.7.4** *We suppose  $f \in L^{\frac{2q}{q-1}}(\Omega)$  and  $\psi \in H_0^1(\Omega, \mathbf{A})$  is the weak subsolution of (2.1) in  $\Omega$ . Then, we derive that  $\Upsilon_\psi(x)$  is the weak subsolution to the equation*

$$-\partial_i(a_{ij}(x) \partial_j \Upsilon_\psi(x)) = f_+(x) \cdot \chi_E(x) \quad (2.80)$$

in  $\Omega$ , where  $\chi_E(x)$  represents the characteristic function of the set  $E$  with  $E = \{x \in \Omega; \psi(x) = \Upsilon_\psi(x)\}$ .

**Proof.** First, we consider a sequence of Lipschitz functions  $\{\theta_m(t)\}_{m=1}^\infty$ , where

$$\theta_m(t) = \begin{cases} 1, & t \leq \frac{1}{2m}, \\ 2 - 2mt, & \frac{1}{2m} \leq t \leq \frac{1}{m}, \\ 0, & t > \frac{1}{m}. \end{cases}$$

Then, from Lemma 2.7.3, there exists  $u_m \in H_0^1(\Omega, \mathbf{A})$  satisfying (2.77) with  $\theta(t) = \theta_m(t)$ .

Next, we will prove  $\{u_m\}_{m=1}^\infty \subset K_\psi$ . In fact, since  $\psi$  is the weak subsolution to (2.1) and the equation (2.77) holds with  $\theta$  and  $u$  replaced by  $\theta_m$  and  $u_m$ , respectively, we obtain

$$-\partial_i(a_{ij}\partial_j(u_m - \psi)) \geq f_+(x) \cdot \theta_m(u_m - \psi) - f, \quad x \in \Omega,$$

in the weak sense. Because  $u_m - \psi \leq 0$  implies  $f_+(x) \cdot \theta_m(u_m - \psi) - f \geq 0$  for  $x \in \Omega$  a.e., by Lemma 2.7.2, we have  $u_m \geq \psi$  a.e. in  $\Omega$  for  $m \geq 1$ .

Third, since  $\{u_m\}_{m=1}^{+\infty}$  has a uniform bound in  $H_0^1(\Omega, \mathbf{A})$ , there exists  $\tilde{u} \in H_0^1(\Omega, \mathbf{A})$ , satisfying  $u_m \rightarrow \tilde{u}$  weakly in  $H_0^1(\Omega, \mathbf{A})$ , and strongly in  $L^{\frac{2q}{q+1}}(\Omega)$ . Then, we claim that

$$\lim_{m \rightarrow +\infty} \sup_{k \geq m} \theta_k(u_k - \psi) \leq \chi_{\tilde{E}} \text{ a.e. in } \Omega, \quad (2.81)$$

where  $\tilde{E} = \{x \in \Omega; \tilde{u} = \psi(x)\}$ .

Indeed, we could find a set  $K$  with  $|K| = 0$ , such that  $u_m(x) \rightarrow \tilde{u}(x)$  and  $u_m(x) \geq \psi(x)$  for every  $x \in \Omega \setminus K$ ,  $m \geq 1$ . Now, we could see if  $x \in \tilde{E}$ , the inequality (2.81) holds clearly. When  $x \in \Omega \setminus \{\tilde{E} \cup K\}$ , we have  $\tilde{u}(x) > \psi(x)$ . Since  $u_m(x) \rightarrow \tilde{u}(x)$ , it follows that for any fixed  $x_0 \in \Omega \setminus \{\tilde{E} \cup K\}$ , there exists an integer  $m_0 = m_0(x_0)$  such that for all  $m \geq m_0(x_0)$ , we have  $u_m(x_0) > \tilde{u}(x_0) - \frac{\tilde{u}(x_0) - \psi(x_0)}{2} = \psi(x_0) + \frac{\tilde{u}(x_0) - \psi(x_0)}{2}$ . Now define  $m_1 = m_1(x_0) = \max\{m_0, \frac{2}{\tilde{u}(x_0) - \psi(x_0)}\}$ , then for all  $m \geq m_1$ , it follows that  $u_m(x_0) - \psi(x_0) > \frac{1}{m}$ , and hence  $\theta_m(u_m(x_0) - \psi(x_0)) = 0$  for  $m \geq m_1$ . This completes the proof of the claim.

Since  $u_m$  is the weak solution of (2.77) with  $\theta = \theta_m$ , together with (2.81) and  $u_m \rightarrow \tilde{u}$  weakly in  $H_0^1(\Omega, \mathbf{A})$ , we obtain

$$\int_{\Omega} a_{ij}\partial_i\tilde{u}\partial_j\varphi dx \leq \int_{\Omega} f_+(x) \cdot \chi_{\tilde{E}} \cdot \varphi dx, \quad \forall \varphi \geq 0 \in H_0^1(\Omega, \mathbf{A}).$$

Therefore, the remaining part is to show that  $\tilde{u} = \Upsilon_\psi$ . Because  $\theta_m(t)$  is decreasing function, we deduce that for any  $v \in H_0^1(\Omega, \mathbf{A})$ ,

$$\begin{aligned} & \int_{\Omega} (a_{ij}\partial_i(v - u_m)\partial_j(v - u_m) \\ & - f_+(x) \cdot (\theta_m(v - \psi) - \theta_m(u_m - \psi)) \cdot (v - u_m)) dx \geq 0. \end{aligned}$$

Together with (2.77), where  $\theta$  and  $u$  replaced by  $\theta_m$  and  $u_m$ , respectively, we get

$$\int_{\Omega} (a_{ij}\partial_i v \partial_j(v - u_m) - f_+(x) \cdot \theta_m(v - \psi) \cdot (v - u_m)) dx \geq 0. \quad (2.82)$$

For any fixed integer  $m_2 \geq 1$ , we select  $v - \psi > \frac{1}{m_2}$  a.e. in  $\Omega$ . Therefore, for  $m \geq m_2$ , we have  $\theta_m(v - \psi) = 0$ , then (2.82) becomes

$$\int_{\Omega} a_{ij} \partial_i v \partial_j (v - u_m) dx \geq 0.$$

Since  $u_m \rightarrow \tilde{u}$  weakly in  $H_0^1(\Omega, \mathbf{A})$ , we have

$$\int_{\Omega} a_{ij} \partial_i v \partial_j (v - \tilde{u}) dx \geq 0. \quad (2.83)$$

In view of the arbitrariness of  $m_2$ , we deduce that (2.83) holds for any  $v \in K_{\psi}$ . Then, from Lemma 2.7.1, we have  $\tilde{u} = \tilde{\Upsilon}_{\psi} = \Upsilon_{\psi}$ .  $\square$

Next, we will prove a global  $L^{\infty}$  result for weak subsolution of (2.1).

**Lemma 2.7.5** *Let  $u$  be the weak subsolution to (2.1) in  $\Omega$ , and assume (H1)-(H2) hold, then there exists constant  $C$ , such that*

$$\begin{aligned} \|u\|_{L^{\infty}(\Omega)} &\leq \sup_{\partial\Omega} u + C \left( \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}} \right)^{\frac{q_*}{1 - \frac{2s'}{2s'}}} \\ &\quad \cdot \left[ |\Omega|^{\frac{2}{N} + 1 - \frac{2s'}{q_*}} \|\lambda^{-1}\|_{L^q(\Omega)} + 1 \right] \|f\|_{L^s(\Omega)}, \end{aligned} \quad (2.84)$$

where  $q_* = \frac{2Nq}{(q+1)N-2q}$ ,  $s' = \frac{s}{s-1}$ ,  $s > \frac{Nq}{2q-N}$ ,  $q > \frac{N}{2}$  and  $C$  depends on  $N$ ,  $q$ .

**Proof.** First, without loss of generality, we just take test function  $\varphi(x) = [(u - \sup_{\partial\Omega} u)_+ + k]^{\beta} - k^{\beta}$ , where  $k = \|f\|_{L^s(\Omega)}$ . Subsequently, we derive

$$\begin{aligned} &\frac{4\beta}{(\beta+4)^2} \int_{\Omega} \lambda(x) \left| \nabla \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} \right|^2 dx \\ &\leq \frac{4\beta}{(\beta+1)^2} \int_{\Omega} a(x) \nabla \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} \nabla \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} dx \\ &\leq \int_{\Omega} f \left\{ \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\beta} - k^{\beta} \right\} dx \\ &\leq \left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{(\beta+1)s'} dx \right)^{\frac{1}{s'}}, \end{aligned} \quad (2.85)$$

where  $s' = \frac{s}{s-1}$ ,  $s > \frac{Nq}{2q-N}$ .

Combining Sobolev's embedding inequality and Hölder's inequality with previous inequality, we have

$$\left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2} q_*} dx \right)^{\frac{1}{q_*}}$$

$$\begin{aligned}
 &\leq \left( \int_{\Omega} \left| \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} - k^{\frac{\beta+1}{2}} \right|^{q_*} dx \right)^{\frac{1}{q_*}} + k^{\frac{\beta+1}{2}} |\Omega|^{\frac{1}{q_*}} \\
 &\leq C \left( \int_{\Omega} \left| \nabla \left\{ \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} - k^{\frac{\beta+1}{2}} \right\} \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} + k^{\frac{\beta+1}{2}} |\Omega|^{\frac{1}{q_*}} \\
 &\leq C \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} \left( \int_{\Omega} \lambda(x) \left| \nabla \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\frac{\beta+1}{2}} \right|^2 dx \right)^{\frac{1}{2}} + k^{\frac{\beta+1}{2}} |\Omega|^{\frac{1}{q_*}} \\
 &\leq \left( C \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} \frac{(\beta+1)}{2\sqrt{\beta}} + |\Omega|^{\frac{1}{q_*}} \right) \left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{(\beta+1)s'} dx \right)^{\frac{1}{2s'}},
 \end{aligned}$$

where  $q_* = \frac{2Nq}{(q+1)N-2q}$  and  $C$  depends on  $N, q$ .

Since  $q_* > 2s'$ , we take  $\frac{\beta+1}{2} = \left(\frac{q_*}{2s'}\right)^m$  with  $m \in \mathbb{N}_+$ , then we have

$$\begin{aligned}
 &\left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\left(\frac{q_*}{2s'}\right)^m q_*} dx \right)^{\frac{1}{\left(\frac{q_*}{2s'}\right)^m q_*}} \\
 &\leq \left( C \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} \frac{(\beta+1)}{2\sqrt{\beta}} + |\Omega|^{\frac{1}{q_*}} \right)^{\frac{1}{\left(\frac{q_*}{2s'}\right)^m}} \\
 &\quad \cdot \left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ + k \right]^{\left(\frac{q_*}{2s'}\right)^{m-1} q_*} dx \right)^{\frac{1}{\left(\frac{q_*}{2s'}\right)^{m-1} q_*}}.
 \end{aligned}$$

When  $m$  goes to infinity, we have

$$\begin{aligned}
 \sup_{\Omega} u - \sup_{\partial\Omega} u &\leq C \left( \|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}} \right)^{\frac{\frac{q_*}{2s'}}{1-\frac{q_*}{2s'}}} \\
 &\quad \cdot \left[ \left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ \right]^{q_*} dx \right)^{\frac{1}{q_*}} + k \right],
 \end{aligned}$$

where  $C$  depends on  $N, q, s$ .

In the sequel, we keep on estimating the term  $\left( \int_{\Omega} [(u - \sup_{\partial\Omega} u)_+]^{q_*} dx \right)^{\frac{1}{q_*}}$ . Taking the test function  $\varphi = (u - \sup_{\partial\Omega} u)_+ \in H_0^1(\Omega, \mathbf{A})$ , then, by Poincaré's inequality, Sobolev's embedding inequality and Hölder's inequality, we obtain

$$\begin{aligned}
 &\left( \int_{\Omega} \left[ (u - \sup_{\partial\Omega} u)_+ \right]^{q_*} dx \right)^{\frac{1}{q_*}} \\
 &\leq C |\Omega|^{\frac{1}{N}} \left( \int_{\Omega} \left| \nabla \left[ (u - \sup_{\partial\Omega} u)_+ \right] \right|^{q_*} dx \right)^{\frac{1}{q_*}} \\
 &\leq C |\Omega|^{\frac{1}{N}} \left( \int_{\Omega} \left| \nabla \left[ (u - \sup_{\partial\Omega} u)_+ \right] \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}}
 \end{aligned}$$

$$\leq C|\Omega|^{\frac{1}{N}+\frac{1}{2}-\frac{s'}{q_*}}\|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}}\|f\|_{L^s(\Omega)}^{\frac{1}{2}}\left(\int_{\Omega}\left[(u-\sup_{\partial\Omega}u)_+\right]^{q_*}dx\right)^{\frac{1}{2q_*}},$$

where  $C$  depends on  $N, q$ . Then, if  $\left(\int_{\Omega}[(u-\sup_{\partial\Omega}u)_+]^{q_*}dx\right)^{\frac{1}{q_*}} \neq 0$ , we get

$$\left(\int_{\Omega}\left[(u-\sup_{\partial\Omega}u)_+\right]^{q_*}dx\right)^{\frac{1}{q_*}} \leq C|\Omega|^{\frac{2}{N}+1-\frac{2s'}{q_*}}\|\lambda^{-1}\|_{L^q(\Omega)}\|f\|_{L^s(\Omega)}.$$

Thus, we have

$$\begin{aligned} \sup_{\Omega}u - \sup_{\partial\Omega}u &\leq C\left(\|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}}\right)^{\frac{\frac{q_*}{2s'}}{1-\frac{q_*}{2s'}}} \\ &\quad \cdot \left(|\Omega|^{\frac{2}{N}+1-\frac{2s'}{q_*}}\|\lambda^{-1}\|_{L^q(\Omega)}\|f\|_{L^s(\Omega)} + \|f\|_{L^s(\Omega)}\right), \end{aligned}$$

where  $C$  depends on  $N, q$ . □

**The proof of Theorem 2.1.4:** Recall that we assume  $\psi = u - \sup_{\partial\Omega}u_+$ ,  $f \in L^s(\Omega)$  with  $s \in [\frac{2q}{q-1}, +\infty] \cap (\frac{nq}{2q-n}, +\infty]$ , because  $\psi$  is the weak subsolution to (2.1), according to Lemma 2.7.4, then  $\Upsilon_{\psi}(x)$  is the subsolution of the equation

$$-\partial_i(a_{ij}(x)\partial_j u) = F(x)$$

where  $F(x) = f_+(x) \cdot \chi_E(x)$  and  $E = \{x \in \Omega; \psi(x) = \Upsilon_{\psi}(x)\}$ .

Then, by Lemma 2.7.5 with  $f = F(x)$ , we get

$$\begin{aligned} \sup_{\Omega}\Upsilon_{\psi} &\leq \sup_{\partial\Omega}\Upsilon_{\psi} + C\left(\|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}}\right)^{\frac{\frac{q_*}{2s'}}{1-\frac{q_*}{2s'}}} \\ &\quad \cdot \left[|\Omega|^{\frac{2}{N}+1-\frac{2s'}{q_*}}\|\lambda^{-1}\|_{L^q(\Omega)} + 1\right]\|F\|_{L^s(\Omega)}, \end{aligned} \tag{2.86}$$

where  $C$  depends on  $N, q$ .

Since  $\Upsilon_{\psi}(x) \geq \psi(x)$  and  $\sup_{\partial\Omega}\Upsilon_{\psi} = 0$ , we obtain

$$\begin{aligned} \sup_{\Omega}u - \sup_{\partial\Omega}u_+ &= \sup_{\Omega}\psi(x) \leq \sup_{\Omega}\Upsilon_{\psi}(x) \\ &\leq C\left(\|\lambda^{-1}\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{1}{q_*}}\right)^{\frac{\frac{q_*}{2s'}}{1-\frac{q_*}{2s'}}} \\ &\quad \cdot \left(|\Omega|^{\frac{2}{N}+1-\frac{2s'}{q_*}}\|\lambda^{-1}\|_{L^q(\Omega)} + 1\right)\|F\|_{L^s(\Omega)}, \end{aligned}$$

where  $s \in [\frac{2q}{q-1}, +\infty] \cap (\frac{Nq}{2q-N}, +\infty]$  and  $C$  depends on  $N, q$ .

## 2.8 Boundedness Theory When $f$ Belongs to a Logarithmic Orlicz Space

This section deals with the boundedness of weak solutions  $u$  to Problem (2.1) when the inhomogeneous term  $f$  lies in a *critical* Orlicz space that contains an additional logarithmic factor and is strictly larger than the usual Lebesgue scale. As already mentioned in Chapter 1, if the coefficient matrix  $\mathbf{A}(x)$  is non-uniformly elliptic and merely satisfies hypotheses (H1)–(H2), the optimal integrability threshold on  $f$  in such Orlicz spaces for which  $u$  remains bounded is still unknown. For uniformly elliptic equations the picture is more complete, yet the *borderline* case where  $f$  attains the critical integrability has not been settled. Building on the works of [26, 27], the main purpose of this section is to exhibit an explicit counter-example showing that, exactly at this critical level, a uniform maximum-norm estimate for  $u$  may fail. This completes the existing theory by marking the precise barrier beyond which boundedness is no longer guaranteed.

Without loss of generality we illustrate the phenomenon for the Dirichlet problem of the Poisson equation (i.e.  $\mathbf{A} = \mathbf{I}$ ). For general uniformly elliptic operators the construction follows the same lines.

**Example 2.8.1** *Let  $N \geq 3$  and  $\Omega = B_\tau(0)$ , where  $\tau = \frac{1}{e^e}$ . Then there exists*

$$f(x) = |x|^{-2} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} \in L^A(\Omega),$$

*where  $A = A(t) = t^{\frac{N}{2}} (\log(e + t))^{\frac{N}{2}-1}$ , such that (2.1) has an unbounded solution.*

**Remark 2.8.1** Although the preceding example is stated only for bounded domains, it is worth emphasizing that Theorem 5 in [26] already guarantees an  $L^\infty$  estimate for the solution  $u$  whenever  $\Omega \subseteq \mathbb{R}^N$  is an open set—bounded or unbounded—and the exponent satisfies  $s > \frac{N-2}{2}$ . For unbounded domains, a minor adaptation of our argument shows that the same threshold  $s = \frac{N-2}{2}$  remains critical for the validity of the maximum-norm estimate.

**Proof.** Firstly, with the definition of  $f$  at hand, a simple computation shows

$$u(0) = \frac{1}{(N-2)\omega_N} \int_{B_\tau(0)} |x|^{-N} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} dx$$

$$\begin{aligned}
 & - \frac{1}{(N-2)\omega_N\tau^{N-2}} \int_{B_\tau(0)} |x|^{-2} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} dx \\
 & = \infty,
 \end{aligned} \tag{2.87}$$

where  $\omega_N := \int_{|\omega|=1} d\omega$  denotes the surface area of the  $N$ -dimensional unit ball.

However, for the sake of completeness, we will provide the rigorous arguments, which are motivated by [27, Section 5]. Indeed, for  $k \in \mathbb{N}_+$ , we define  $f_k(x) = \chi_k(x)f(x)$  as follows

$$\chi_k(x) = \begin{cases} 0, & \text{if } |x| < \frac{\tau}{2^{k+1}}, \\ 1, & \text{if } \frac{\tau}{2^k} \leq x < \tau, \\ \frac{2^{k+1}}{\tau}|x| - 1, & \text{if } \frac{\tau}{2^{k+1}} \leq |x| < \frac{\tau}{2^k}. \end{cases}$$

Obviously,  $f_k$  is continuous. Then it is well known that the Dirichlet problem

$$\begin{cases} -\Delta u_k = f_k, & x \in B_\tau(0), \\ u_k = 0, & x \in \partial B_\tau(0) \end{cases} \tag{2.88}$$

has one classical solution  $u_k \in C^2(B_\tau(0))$ . Further, the potential theory (see for example [44]) ensures

$$\begin{aligned}
 u_k(0) &= \frac{1}{(N-2)\omega_N} \int_{B_\tau(0)} |x|^{2-N} f_k(x) dx - \frac{1}{(N-2)\omega_N\tau^{N-2}} \int_{B_\tau(0)} f_k(x) dx \\
 &\geq \frac{1}{(N-2)\omega_N} \int_{\frac{\tau}{2^k} \leq |x| < \tau} \frac{1}{|x|^N} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} dx \\
 &\quad - \frac{1}{(N-2)\omega_N\tau^{N-2}} \int_{\frac{\tau}{2^k} \leq |x| < \tau} |x|^{-2} \left( \log(e + |x|^{-1}) \right)^{-1} \\
 &\quad \cdot \left( \log \log(e + |x|^{-1}) \right)^{-1} dx.
 \end{aligned}$$

Then, a direct computation shows that (2.89) may be rewritten as

$$\begin{aligned}
 & \int_{\frac{\tau}{2^k} \leq |x| < \tau} |x|^{-N} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} dx \\
 &= \omega_N \int_{\frac{\tau}{2^k}}^{\tau} r^{-1} \left( \log(e + r^{-1}) \right)^{-1} \left( \log \log(e + r^{-1}) \right)^{-1} dr \\
 &\geq \omega_N \int_{\frac{\tau}{2^k}}^{\tau} (1 + er)^{-1} r^{-1} \left( \log(e + r^{-1}) \right)^{-1} \left( \log \log(e + r^{-1}) \right)^{-1} dr \\
 &= \omega_N \left( \log \log \log(e + \frac{2^k}{\tau}) - \log \log \log(e + \frac{1}{\tau}) \right).
 \end{aligned}$$



In addition, it is not hard to verify that

$$\begin{aligned}
 & \int_{\frac{\tau}{2^k} \leq |x| < \tau} |x|^{-2} \left( \log(e + |x|^{-1}) \right)^{-1} \left( \log \log(e + |x|^{-1}) \right)^{-1} dx \\
 &= \omega_N \int_{\frac{\tau}{2^k}}^{\tau} |r|^{N-3} \left( \log(e + r^{-1}) \right)^{-1} \left( \log \log(e + r^{-1}) \right)^{-1} dr \\
 &\leq \omega_N \tau^{N-3} \left( \log(e + \tau^{-1}) \right)^{-1} \left( \log \log(e + \tau^{-1}) \right)^{-1} < +\infty.
 \end{aligned}$$

Obviously, we have  $u_k(0) \rightarrow +\infty$ , as  $k \rightarrow +\infty$ .

To end our proof, we need to verify  $f \in L^A(B_\tau(0))$ . In fact, on the one hand, we observe

$$\begin{aligned}
 A(|f|) &= |x|^{-N} \left( \log(e + |x|^{-1}) \right)^{-\frac{N}{2}} \left( \log \log(e + |x|^{-1}) \right)^{-\frac{N}{2}} (\log(e + |f|))^s \\
 &\leq 2s|x|^{-N} \left( \log(e + |x|^{-1}) \right)^{-\frac{N}{2}+s} \left( \log \log(e + |x|^{-1}) \right)^{-\frac{N}{2}}.
 \end{aligned} \tag{2.89}$$

which, together with some simple computations, implies

$$\begin{aligned}
 \int_{B_\tau(0)} |A(|f|)| dx &\leq 2s\omega_N \int_0^\tau \frac{1}{r} \frac{1}{\left( \log(e + \frac{1}{r}) \right)^{\frac{N}{2}-s}} \frac{1}{\left( \log \log(e + \frac{1}{r}) \right)^{\frac{N}{2}}} dr \\
 &\leq 2q\omega_N \int_0^\tau \frac{1}{r} \frac{1}{\left( \log(\frac{1}{r}) \right)^{\frac{N}{2}-s}} \frac{1}{\left( \log \log(\frac{1}{r}) \right)^{\frac{N}{2}}} dr \\
 &= 2q\omega_N \int_{\log \frac{1}{\tau}}^{+\infty} \frac{1}{t^{\frac{N}{2}-s}} \frac{1}{(\log t)^{\frac{N}{2}}} dt.
 \end{aligned} \tag{2.90}$$

On the other hand, we know

$$A(|f|) \geq |x|^{-N} \left( \log(e + |x|^{-1}) \right)^{-\frac{N}{2}+s} \left( \log \log(e + |x|^{-1}) \right)^{-\frac{N}{2}}. \tag{2.91}$$

Similar to (2.90), we have

$$\begin{aligned}
 \int_{B_\tau(0)} |A(|f|)| dx &\geq \omega_N \int_0^\tau \frac{1}{r} \frac{1}{\left( \log(e + \frac{1}{r}) \right)^{\frac{N}{2}-s}} \frac{1}{\left( \log \log(e + \frac{1}{r}) \right)^{\frac{N}{2}}} dr \\
 &\geq \omega_N \int_0^\tau \frac{1}{r(er + 1)} \frac{1}{\left( \log(e + \frac{1}{r}) \right)^{\frac{N}{2}-s}} \frac{1}{\left( \log \log(e + \frac{1}{r}) \right)^{\frac{N}{2}}} dr \\
 &= \omega_N \int_{\log(e + \frac{1}{\tau})}^{+\infty} \frac{1}{t^{\frac{N}{2}-s}} \frac{1}{(\log t)^{\frac{N}{2}}} dt.
 \end{aligned} \tag{2.92}$$

Combining with (2.90) and (2.92), we find the fact that  $A(f) \in L^1(B_1(0))$  if and only if  $\frac{N}{2} - s \geq 1$ . The proof is finished.  $\square$



## Chapter 3

# Non-homogeneous Linear Non-uniformly Elliptic Equations with Drift

### 3.1 Main Results

#### Problem setting

We consider the linear non-uniformly elliptic equation with drift

$$-\operatorname{div}(\mathbf{A}(x)\nabla u) + \operatorname{div}(u\mathbf{b}) = \operatorname{div}\mathbf{f} + f, \quad x \in \Omega, \quad (3.1)$$

under the following structural assumptions:

(H1)  $\Omega \subset \mathbb{R}^N$  ( $N \geq 2$ ) is a bounded domain.

(H2)  $\mathbf{A}(x)$  is symmetric and satisfies

$$\lambda(x)|\xi|^2 \leq a_{ij}(x)\xi_i\xi_j \leq \mu(x)|\xi|^2, \quad \forall \xi \in \mathbb{R}^N,$$

with

$$\lambda^{-1} \in L^q(\Omega), \quad \mu \in L^p(\Omega), \quad \frac{1}{p} + \frac{1}{q} < \frac{2}{N-1}, \quad p, q > 1.$$

(H3)  $f \in L^s(\Omega)$  with  $s > \frac{Nq}{2q-N}$  and  $\mathbf{f}, \mathbf{b} \in \left(L^t(\Omega)\right)^N$  with

$$t > \frac{Nq}{q-N}, \quad q > N.$$

## Local boundedness

Our first result gives an interior  $L^\infty$  estimate for non-negative weak subsolutions.

**Theorem 3.1.1** *Assume (H1)–(H3). Then for any  $\gamma > 0$ , any ball  $B_{2R}(x_0) \subset \Omega$  and any  $r < R$ , any weak subsolution  $u$  of (3.1) satisfies*

$$\begin{aligned} & \|u_+\|_{L^\infty(B_r(x_0))} \\ & \leq C_0 \left(1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R(x_0))}\right)^{\frac{4p'\delta}{(\delta-1)\gamma}} \Lambda(B_R(x_0))^{\frac{2p'\delta}{\gamma(\delta-1)}} \left(1 - \frac{r}{R}\right)^{-\frac{N}{\gamma} - \frac{4Np'\delta s_*}{\gamma(\delta-1)}} \\ & \quad \cdot \left(R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_R(x_0))} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R(x_0))} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R(x_0))}\right), \end{aligned} \quad (3.2)$$

where the constant  $C_0$  depends on  $N, p, q, t$ .  $\delta = 2 - \frac{1}{\chi} > 1$ ,  $\chi = \frac{2q}{(q+1)p_t^*} > 1$ ,  $p_t^* = \max\{p_*, t_*\}$ ,  $\frac{1}{t_*} = \min\{\frac{1}{2} - \frac{1}{2t} + \frac{1}{N-1}, 1\}$ ,  $\frac{1}{p_*} = \min\{\frac{1}{2} - \frac{1}{2p} + \frac{1}{N-1}, 1\}$ ,  $p' = \frac{p}{p-1}$ ,  $s_* = (\frac{1}{2p} + \frac{1}{2q} + \frac{1}{q} + \frac{2}{q})$ ,

$$\begin{aligned} \Lambda(B_R(x_0)) &= \left(\int_{B_R(x_0)} \mu^p dx\right)^{\frac{1}{2p}} \left(\int_{B_R(x_0)} |\lambda|^{-q} dx\right)^{\frac{1}{2q}} \\ &\quad + \left(\int_{B_R(x_0)} |\lambda|^{-q} dx\right)^{\frac{1}{q}} + \left(\int_{B_R(x_0)} |\lambda|^{-q} dx\right)^{\frac{2}{q}}. \end{aligned}$$

**Remark 3.1.1** The integrability indices for  $|\mathbf{b}|$  and  $|\mathbf{f}|$  can be chosen independently; the same comment applies to Theorem 3.1.2.

A natural question is whether the ranges of  $t$  and  $q$  in Theorem 3.1.1 are optimal. The following two examples give an affirmative answer.

**Example 3.1.1 (Optimality of  $\lambda^{-1}$ )** Let  $N \geq 3$ ,  $q = N$  and  $\Omega = B_{1/e}(0)$ . Choose  $\mathbf{A}(x) = \lambda(x)\mathbf{I}$  with

$$\lambda(x) = \mu(x) = |x| \left(\log \frac{1}{|x|}\right)^\beta, \quad \frac{1}{N} < \beta < 1,$$

and set

$$\mathbf{b}(x) = -\frac{x}{|x|} \frac{\left(\log \frac{1}{|x|}\right)^{\beta-1}}{\log \log \frac{1}{|x|}}, \quad f = f_i = 0.$$

Then (3.1) possesses an unbounded weak solution.

**Example 3.1.2 (Optimality of b)** Let  $N \geq 3$  and  $t = \frac{Nq}{q-N}$  with  $q > N$ . Take

$$\mathbf{A}(x) = |x|^{\frac{N}{q}} \left( \log \frac{1}{|x|} \right)^{1-\beta} \log \log \frac{1}{|x|} \mathbf{I}, \quad \mathbf{b}(x) = -\frac{x}{|x|^{2-\frac{N}{q}} \left( \log \frac{1}{|x|} \right)^\beta},$$

with  $\beta \in (\frac{1}{N} - \frac{1}{q}, 1 - \frac{1}{q}]$ , and let  $f = \mathbf{f} = 0$ . Then  $u(x) = \log \log \frac{1}{|x|}$  is an unbounded weak solution of (3.1).

## Harnack inequality

Based on Theorem 3.1.1 we deduce a weak Harnack estimate.

**Theorem 3.1.2** Under (H1)–(H3), let  $u$  be a non-negative weak supersolution of (3.1) in  $B_{5R}(x_0) \subset \Omega$ . Then for every

$$\gamma \in \left(0, \frac{q_*}{2}\right), \quad q_* = \frac{2Nq}{(q+1)N-2q}, \quad \tau, \theta \in (0, 1),$$

we have

$$\begin{aligned} & \left( \frac{1}{R^N} \int_{B_{\theta R}(x_0)} u^\gamma dx \right)^{\frac{1}{\gamma}} \\ & \leq C_1 \left( \inf_{B_{\tau R}(x_0)} u + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R(x_0))} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R(x_0))} \right), \end{aligned}$$

where

$$\begin{aligned} C_1 &= C_2 \left( 1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R(x_0))} \right)^{\frac{4\delta-2}{\alpha_1(\delta-1)} + \frac{2\gamma}{q_*}} \Lambda(B_R(x_0))^{\frac{2\delta-1}{\alpha_1(\delta-1)} + \frac{\gamma}{q_*}}, \\ \alpha_1 &= (2p' C_3 e)^{-1}, \\ C_3 &= C_4 \left( R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R(x_0))} + 1 \right)^{\frac{2\delta}{(\delta-1)^2} + \frac{1}{2}} \Lambda(B_R(x_0))^{\frac{\delta}{(\delta-1)^2} + \frac{1}{2}}, \end{aligned}$$

and the constants  $C_2, C_4$  depend on  $N, p, q, t, \theta, \tau$ ; the exponent  $\delta$  is the same as in Theorem 3.1.1.

Combining Theorems 3.1.1 and 3.1.2 we obtain the Harnack inequality.

**Theorem 3.1.3 (Harnack inequality)** Let  $u$  be a non-negative weak solution of (3.1) in  $B_{5R}(x_0) \subset \Omega$  under (H1)–(H3). Then

$$\sup_{B_{R/2}(x_0)} u \leq C_5 \left( \inf_{B_{R/2}(x_0)} u + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R(x_0))} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R(x_0))} \right),$$

where  $C_5$  depends on

$$\left( R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R(x_0))} + 1 \right), \quad \Lambda(B_R(x_0)), \quad N, p, q, t.$$

## Organisation of the chapter

Section 3.2 introduces the precise notion of weak solution. Section 3.3 proves Examples 3.1.1 and 3.1.2. Section 3.4 is devoted to the proof of Theorem 3.1.1. Finally, Section 3.5 proves Theorem 3.1.2.

## 3.2 Preliminaries

### Function spaces

The Banach spaces  $H^1(\Omega, \mathbf{A})$  and  $H_0^1(\Omega, \mathbf{A})$  employed in this chapter coincide with those introduced in Section 2.2.

### Weak solutions

**Definition 3.2.1** *A function  $u \in H^1(\Omega, \mathbf{A})$  is called a weak solution (weak subsolution, weak supersolution) of (3.1) in  $\Omega$  if*

$$\int_{\Omega} \mathbf{A}(x) \nabla u \cdot \nabla \varphi \, dx - \int_{\Omega} u \mathbf{b} \cdot \nabla \varphi \, dx \, (=, \leq, \geq) \, - \int_{\Omega} \mathbf{f} \cdot \nabla \varphi \, dx + \int_{\Omega} f \varphi \, dx,$$

*for every non-negative test function  $\varphi \in H_0^1(\Omega, \mathbf{A})$ .*

## 3.3 Proof of Example 3.1.1–3.1.2

**The Proof of Example 3.1.1.** We could verify  $u(x) = \log \log \frac{1}{|x|}$  is the weak solution of (3.1), because  $u \mathbf{b} = \lambda(x) \nabla u$ . And it is easy to verify that  $\mathbf{b} \in (L^\infty(B_{\frac{1}{e^e}}(0)))^n$ ,  $\lambda^{-1} \in L^n(B_{\frac{1}{e^e}}(0))$  and  $\mu \in L^\infty(B_{\frac{1}{e^e}}(0))$ .

**The Proof of Example 3.1.2.** First, we could verify

$$\mathbf{A}(x) \nabla u = u \mathbf{b},$$

and  $\mathbf{b} \in L^t(B_{\frac{1}{e^e}}(0))$ . By direct computation, we have

$$\int_{B_{\frac{1}{e^e}}(0)} \frac{1}{\lambda^q(x)} \, dx = C(n) \int_0^{\frac{1}{e^e}} \frac{1}{r(\log \frac{1}{r})^{(1-\beta)q}} \frac{1}{(\log \log \frac{1}{r})^q} \, dr < +\infty,$$

$$\mu(x) \in L^\infty(B_{\frac{1}{e^e}}(0)).$$

### 3.4 Proof of Theorem 3.1.1

**Proof.** Without loss of generality, we assume  $x_0 = 0$ .

**Step 1. The case for  $\gamma = 2p'$ .** We begin by introducing the test function used in the subsequent discussion. For  $\beta \geq 1$  and  $m \in (0, +\infty)$ , we define

$$\phi_m(u) = \begin{cases} (u_+ + k)^\beta - k^\beta, & u \leq m, \\ \beta(m + k)^{\beta-1}(u - m) + (m + k)^\beta - k^\beta, & u > m, \end{cases}$$

where  $k = \|f\|_{L^s(B_1)} + \|\mathbf{f}\|_{L^t(B_1)}$ . Set  $\varphi = \phi_m \eta^2$ , where  $\eta(x) \geq 0$ ,  $\eta \in C_0^\infty(B_1)$ , with  $\eta \equiv 1$  for  $x \in B_\rho$ ;  $\eta \equiv 0$  for  $x \in B_1 \setminus B_\sigma$ , where  $\frac{1}{4} \leq \rho < \sigma \leq \frac{1}{2}$ . Then from (3.2.1), we get

$$\begin{aligned} & \int_{\Omega} \mathbf{A}(x) \nabla u \left( 2\eta \phi_m(u) \nabla \eta + \phi'_m(u) \eta^2 \nabla u \right) \\ & \leq \int_{\Omega} 2\eta \phi_m(u) |\nabla \eta| (u_+ + k) \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \\ & \quad + \int_{\Omega} \phi'_m(u) \eta^2 |\nabla u| (u_+ + k) \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) dx + \int_{\Omega} f \eta^2 \phi_m dx. \end{aligned} \quad (3.3)$$

Next we will deal with the first term on the right side of the inequality (3.3), by Young's inequality, we get

$$\begin{aligned} \int_{\Omega} \eta \phi_m(u) |\nabla \eta| (u_+ + k) \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) & \leq \frac{1}{\varepsilon} \int_{B_\sigma} \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) |\nabla \eta|^2 |G_m(u)|^2 dx \\ & \quad + \varepsilon \int_{B_\sigma} \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) |G_m(u)|^2 \eta^2 dx, \end{aligned} \quad (3.4)$$

where  $G_m(u) = [(\phi_m(u) + k^\beta)(u_+ + k)]^{\frac{1}{2}}$ . Applying Lemma 2.2.2 to the first term on the right-hand side of (3.4) and taking  $g = \frac{|\mathbf{f}|}{k} + |\mathbf{b}|$ ,  $v = G_m(u)$ , we obtain

$$\begin{aligned} & \inf_{\eta(x)} \int_{B_\sigma} \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) |\nabla \eta|^2 |G_m(u)|^2 dx \\ & \leq \frac{C(N, t)}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma \setminus B_\rho)} \|G_m(u)\|_{W^{1,t*}(B_\sigma \setminus B_\rho)}^2, \end{aligned}$$

where  $\frac{1}{t_*} = \min\{\frac{1}{2} - \frac{1}{2t} + \frac{1}{N-1}, 1\}$ . Similarly, we have

$$\inf_{\eta(x)} \int_{B_\sigma} \mu |\nabla \eta|^2 |G_m(u)|^2 dx \leq \frac{C(N, p)}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \|\mu\|_{L^p(B_\sigma \setminus B_\rho)} \|G_m(u)\|_{W^{1,p*}(B_\sigma \setminus B_\rho)}^2,$$

where  $\frac{1}{p_*} = \min\{\frac{1}{2} - \frac{1}{2p} + \frac{1}{N-1}, 1\}$ . Subsequently, we take  $p_t^* = \max\{t_*, p_*\}$ , then we obtain

$$\begin{aligned} & \max\{\|G_m(u)\|_{W^{1,t_*}(B_\sigma \setminus B_\rho)}, \|G_m(u)\|_{W^{1,p_*}(B_\sigma \setminus B_\rho)}\} \\ & \leq \left(|B_\sigma|^{1-\frac{t_*}{p_t^*}} + |B_\sigma|^{1-\frac{p_*}{p_t^*}}\right) \|G_m(u)\|_{W^{1,p_t^*}(B_\sigma \setminus B_\rho)}. \end{aligned}$$

Therefore, we deduce

$$\begin{aligned} & \inf_{\eta(x)} \int_{B_\sigma} \left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right) |\nabla \eta|^2 |G_m(u)|^2 dx \\ & \leq \frac{C(N, t, p)}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma \setminus B_\rho)} \|G_m(u)\|_{W^{1,p_t^*}(B_\sigma \setminus B_\rho)}^2. \end{aligned}$$

We proceed to estimate the first term on the right side of (3.4), by Hölder's inequality, Sobolev's inequality, we get

$$\begin{aligned} & \int_{B_\sigma} \left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right) |G_m(u)|^2 \eta^2 dx \\ & \leq |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \|\eta G_m(u)\|_{L^{q_*}(B_\sigma)}^2 \\ & \leq C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \|\nabla(\eta G_m(u))\|_{L^{\frac{2q}{q+1}}(B_\sigma)}^2 \\ & \leq C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla(\eta G_m(u))|^2 dx \\ & \leq C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \\ & \quad \cdot \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \int_{B_\sigma} \lambda |\nabla \eta|^2 |G_m(u)|^2 dx + \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx \right), \end{aligned}$$

where  $q_* = \frac{2Nq}{(q+1)N-2q}$ . Return to (3.3), we estimate the second term on the right side, by Young's inequality, we have

$$\begin{aligned} & \int_{\Omega} \phi'_m(u) \eta^2 |\nabla u| (u_+ + k) \left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right) dx \\ & \leq \frac{1}{4} \int_{B_\sigma} \lambda |\nabla u|^2 \phi'_m(u) \eta^2 dx + 4 \int_{B_\sigma} \frac{\left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right)^2}{\lambda} \phi'_m(u) (u_+ + k)^2 \eta^2 dx. \end{aligned} \quad (3.5)$$

Next, we proceed to estimate the second term on the right side of (3.5), by Hölder's inequality and interpolation inequality due to  $p_* < 2 < \frac{2}{1-\frac{2}{t}-\frac{1}{q}} < q_*$ , we have

$$4 \int_{B_\sigma} \frac{\left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right)^2}{\lambda} \phi'_m(u) (u_+ + k)^2 \eta^2 dx$$



$$\begin{aligned}
 &\leq 4\beta \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \|G_m(u)\eta\|_{L^{1-\frac{2}{t}-\frac{1}{q}}(B_\sigma)}^2 \\
 &\leq 4\beta \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \\
 &\quad \cdot \left( \varepsilon \|G_m(u)\eta\|_{L^{q*}(B_\sigma)}^2 + \frac{1}{\varepsilon} \|G_m(u)\eta\|_{L^{p*}(B_\sigma)}^2 \right).
 \end{aligned}$$

by Sobolev's inequality and Hölder's inequality, we have

$$\begin{aligned}
 \|G_m(u)\eta\|_{L^{q*}(B_\sigma)}^2 &\leq C(N, q) \|\nabla(\eta G_m(u))\|_{L^{\frac{2q}{q+1}}(B_\sigma)}^2 \\
 &\leq C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla(\eta G_m(u))|^2 dx \\
 &\leq C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla\eta|^2 |G_m(u)|^2 dx \\
 &\quad + C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx \\
 &\leq C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \mu |\nabla\eta|^2 |G_m(u)|^2 dx \\
 &\quad + C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx. \tag{3.6}
 \end{aligned}$$

Using Hölder's inequality and the interpolation inequality again, we could deal with the term  $\int_\Omega f \eta^2 \phi_m dx$ , namely

$$\begin{aligned}
 \int_\Omega f \eta^2 \phi_m dx &\leq \int_{B_\sigma} \eta^2 \frac{|f|}{k} |G_m(u)|^2 dx \\
 &\leq \|G_m(u)\eta\|_{L^{\frac{2s}{s-1}}(B_\sigma)}^2 \\
 &\leq \left( \varepsilon \|G_m(u)\eta\|_{L^{q*}(B_\sigma)}^2 + \frac{1}{\varepsilon} \|G_m(u)\eta\|_{L^{p*}(B_\sigma)}^2 \right).
 \end{aligned}$$

Now, we return to (3.3), and keep simplifying it, then (3.3) becomes

$$\begin{aligned}
 &\inf_{\eta(x)} \int_{B_\sigma} \lambda |G_m(u)|^2 \eta^2 \\
 &\leq 32\beta \frac{C(N, p)}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \|\mu\|_{L^p(B_\sigma \setminus B_\rho)} \|G_m(u)\|_{W^{1, p*}(B_\sigma \setminus B_\rho)}^2 \\
 &\quad + \frac{2}{\varepsilon_1} \frac{C(N, t, p)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma \setminus B_\rho)} \|G_m(u)\|_{W^{1, p_t^*}(B_\sigma \setminus B_\rho)}^2 \\
 &\quad + 2\varepsilon_1 C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \\
 &\quad \cdot \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \inf_{\eta(x)} \int_{B_\sigma} \mu |\nabla\eta|^2 |G_m(u)|^2 dx + \inf_{\eta(x)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx \right)
 \end{aligned}$$

$$\begin{aligned}
 & + 8\beta \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \\
 & \cdot \left( \varepsilon_2 C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta(x)} \int_{B_\sigma} \mu |\nabla \eta|^2 |G_m(u)|^2 dx \right. \\
 & \quad \left. + \varepsilon_2 C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta(x)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx + \frac{1}{\varepsilon_2} \|G_m(u)\eta\|_{L^{p^*}(B_\sigma)}^2 \right) \\
 & + 2\varepsilon_3 C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta(x)} \int_{B_\sigma} \mu |\nabla \eta|^2 |G_m(u)|^2 dx \\
 & + 2\varepsilon_3 C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta(x)} \int_{B_\sigma} \lambda |\nabla G_m(u)|^2 \eta^2 dx + \frac{2}{\varepsilon_3} \|G_m(u)\eta\|_{L^{p^*}(B_\sigma)}^2.
 \end{aligned}$$

and note that  $\left\| \frac{|\mathbf{f}|}{k} \right\|_{L^t(B_\sigma)} \leq 1$ , then we have

$$\begin{aligned}
 \varepsilon_1 &= \frac{1}{8C(N, t) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} (1 + \|\mathbf{b}\|_{L^t(B_\sigma)}) \|\lambda^{-1}\|_{L^q(B_\sigma)}}, \\
 \varepsilon_2 &= \frac{1}{32\beta (1 + \|\mathbf{b}\|_{L^t(B_\sigma)})^2 \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 C(N, q)}, \quad \varepsilon_3 = \frac{1}{8C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)}}.
 \end{aligned}$$

Consequently, the previous inequality becomes

$$\begin{aligned}
 & \frac{1}{4} \int_{B_\rho} \lambda |G_m(u)|^2 dx \\
 & \leq \left( \left( 32\beta + \frac{3}{4} \right) \frac{C(N, p)}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \|\mu\|_{L^p(B_\sigma \setminus B_\rho)} \right. \\
 & \quad + 16C(N, t, p, q) (1 + \|\mathbf{b}\|_{L^t(B_\sigma)})^2 \frac{1}{(\sigma - \rho)^{\frac{2N}{N-1}} \rho^2} \|\lambda^{-1}\|_{L^q(B_\sigma)} \\
 & \quad \left. + 256\beta^2 (1 + \|\mathbf{b}\|_{L^t(B_\sigma)})^4 \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 C(N, q) + 8C(N, q) \|\lambda^{-1}\|_{L^q(B_\sigma)} \right) \\
 & \quad \cdot \|G_m(u)\|_{W^{1,p_t^*}(B_\sigma)}^2 \\
 & \leq \left[ \frac{\beta^2 C}{(\sigma - \rho)^{\frac{2N}{N-1}}} (1 + \|\mathbf{b}\|_{L^t(B_\sigma)})^4 \left( \|\mu\|_{L^p(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 \right) \right] \\
 & \quad \cdot \|G_m(u)\|_{W^{1,p_t^*}(B_\sigma)}^2, \tag{3.7}
 \end{aligned}$$

where  $C$  depends on  $N, q, p, t$ . By applying Hölder's inequality, we conclude

$$\left( \int_{B_\rho} |\nabla G_m(u)|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \leq \|\lambda^{-1}\|_{L^q(B_\rho)} \int_{B_\rho} \lambda |\nabla G_m(u)|^2 dx. \tag{3.8}$$

When  $m$  goes to infinity, we have

$$\lim_{m \rightarrow +\infty} |\nabla G_m(u)|^{\frac{2q}{q+1}} = \left( |\nabla(u_+ + k)^{\frac{\beta+1}{2}}| \right)^{\frac{2q}{q+1}},$$

$$G_m(u) \leq (u_+ + k)^{\frac{\beta+1}{2}}, \quad \forall m \in [1, +\infty).$$

By Fatou's lemma, we have

$$\begin{aligned} \int_{B_\rho} |\nabla(u_+ + k)|^{\frac{\beta+1}{2}} dx &= \int_{B_\rho} \lim_{m \rightarrow +\infty} |\nabla G_m(u)|^{\frac{2q}{q+1}} dx \\ &\leq \lim_{m \rightarrow +\infty} \int_{B_\rho} |\nabla G_m(u)|^{\frac{2q}{q+1}} dx. \end{aligned} \quad (3.9)$$

Combining (3.7), (3.8) and (3.9), and setting  $v = (u_+ + k)^{\frac{\beta+1}{2}}$ , we have

$$\begin{aligned} \|v\|_{L^{\frac{2q}{q+1}}(B_\rho)}^2 &\leq \frac{\beta^2 C}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left(1 + \|\mathbf{b}\|_{L^t(B_\sigma)}\right)^4 \\ &\quad \cdot \left(\|\mu\|_{L^p(B_\sigma)} \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 + \|\lambda^{-1}\|_{L^q(B_\sigma)}^4\right) \\ &\quad \cdot \|v\|_{W^{1,p_t}^*(B_\sigma)}^2. \end{aligned} \quad (3.10)$$

To make Moser iteration argument work, we further demonstrate that there exists  $\delta > 1$  such that the term  $\|v^\delta\|_{W^{1,p_t}^*(B_\rho)}$  may be dominated by  $\|v\|_{W^{1,p_t}^*(B_\sigma)}^\delta$ . Indeed, on the one hand, choose  $\delta = 2 - \frac{1}{\chi} > 1$  with  $\chi = \frac{2q}{(q+1)p_t^*} > 1$ , and apply Hölder's inequality with exponent  $\frac{2q}{(q+1)p_t^*}$  to get

$$\begin{aligned} \left(\int_{B_\rho} |\nabla(v^\delta)|^{p_t^*} dx\right)^{\frac{1}{p_t^*}} &\leq \left(\int_{B_\rho} |\delta v^{\delta-1} \nabla v|^{p_t^*} dx\right)^{\frac{1}{p_t^*}} \\ &\leq \delta \left(\int_{B_\rho} |\nabla v|^{\frac{2q}{q+1}} dx\right)^{\frac{q+1}{2q}} \left(\int_{B_\rho} |v|^{p_t^*} dx\right)^{\frac{\delta-1}{p_t^*}}. \end{aligned} \quad (3.11)$$

On the other hand, with the aid of the Sobolev inequality  $W^{1,p_t}^*(B_\rho) \hookrightarrow L^t(B_\rho)$  where  $1 \leq t \leq \frac{Np_t^*}{N-p_t^*}$ , we have

$$\left(\int_{B_\rho} v^{\delta p_t^*} dx\right)^{\frac{1}{\delta p_t^*}} \leq C \|v\|_{W^{1,p_t}^*(B_\rho)}. \quad (3.12)$$

Then, combining (3.10), (3.11) and (3.12), we obtain the iterative inequality

$$\begin{aligned} \|v^\delta\|_{W^{1,p_t}^*(B_\rho)} &\leq \frac{\beta C}{(\sigma - \rho)^{\frac{N}{N-1}}} \left(1 + \|\mathbf{b}\|_{L^t(B_\sigma)}\right)^2 \|v\|_{W^{1,p_t}^*(B_\sigma)}^\delta \\ &\quad \cdot \left(\|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2\right), \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ . Let us raise both sides to the  $\frac{\delta(\beta+1)}{2}$  power, and take  $\frac{\beta+1}{2} = \delta^{m-1}$ , and  $\rho = \rho_{N+1} = \frac{1}{4} + \left(\frac{1}{4}\right)^{m+1}$ ,  $\sigma = \rho_N = \frac{1}{4} + \left(\frac{1}{4}\right)^m$ , where  $m \in \mathbb{N}_+$ , we have

$$\|(u_+ + k)^{\delta^m}\|_{W^{1,p_t}^*(B_{\rho_{m+1}})}^{\frac{1}{\delta^m}}$$

$$\begin{aligned} &\leq \delta^{\frac{m-1}{\delta m}} 4^{\frac{mn}{(n-1)\delta m}} \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_\sigma)} \right)^2 \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} \right. \right. \\ &\quad \left. \left. + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right) \right]^{\frac{1}{\delta m}} \|(u_+ + k)^{\delta m - 1}\|_{W^{1,p_t^*}(B_{\rho_m})}^{\frac{1}{\delta m - 1}}. \end{aligned}$$

Taking  $m \rightarrow \infty$ , we get

$$\begin{aligned} \|u_+ + k\|_{L^\infty(B_{\frac{1}{4}})} &\leq C \left( 1 + \|\mathbf{b}\|_{L^t(B_\sigma)} \right)^{\frac{2}{\delta - 1}} \|u_+ + k\|_{W^{1,p_t^*}(B_{\frac{1}{2}})} \\ &\quad \cdot \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right)^{\frac{1}{\delta - 1}}, \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ . Now we deal with the term  $\|u_+ + k\|_{W^{1,p_t^*}(B_{\frac{1}{2}})}$ . We choose test function  $\varphi = u_+ \eta^2$ , where  $\eta \equiv 1$  in  $B_{\frac{1}{2}}$ ;  $\eta \equiv 0$  in  $B_1 \setminus B_{\frac{2}{3}}$ . Following a similar procedure as above, and using the fact that  $p_t^* < \frac{2q}{q+1} \leq 2 \leq 2p'$  and  $\|\nabla \eta\|_{L^\infty(B_\sigma)} \leq c(\sigma - \rho)^{-1}$ , we obtain

$$\begin{aligned} &\left( \int_{B_{\frac{1}{2}}} |\nabla(u_+ + k)|^{p_t^*} dx \right)^{\frac{1}{p_t^*}} \\ &\leq C \left( \int_{B_{\frac{1}{2}}} |\nabla(u_+ + k)|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{2q}} \\ &\leq \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_1)} \right)^2 \left( \|\mu\|_{L^p(B_1)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_1)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_1)} + \|\lambda^{-1}\|_{L^q(B_1)}^2 \right) \right] \\ &\quad \cdot \|u_+ + k\|_{L^{2p'}(B_1)}. \end{aligned}$$

And then by Hölder's inequality, we get

$$\left( \int_{B_1} |u_+ + k|^{p_t^*} dx \right)^{\frac{1}{p_t^*}} \leq C \left( \int_{B_1} |u_+ + k|^{2p'} dx \right)^{\frac{1}{2p'}}.$$

Consequently, we obtain

$$\begin{aligned} \|u_+ + k\|_{L^\infty(B_{\frac{1}{4}})} &\leq C \left( 1 + \|\mathbf{b}\|_{L^t(B_1)} \right)^{\frac{2\delta}{\delta - 1}} \|u_+ + k\|_{L^{2p'}(B_1)} \\ &\quad \cdot \left( \|\mu\|_{L^p(B_1)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_1)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_1)} + \|\lambda^{-1}\|_{L^q(B_1)}^2 \right)^{\frac{\delta}{\delta - 1}}, \quad (3.13) \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ .

**Step 2. The general case.** Although the procedure is standard, we also give a sketch of our proof to make our article self-contained. For more details, we may refer to [46]. First, let

$$\tilde{\mathbf{A}}(y) = \mathbf{A}(Ry), \quad \tilde{\lambda}(y) = \lambda(Ry), \quad \tilde{\mu}(y) = \mu(Ry), \quad \tilde{u}(y) = u(Ry),$$

$$\tilde{f}(y) = R^2 f(Ry), \quad \tilde{\mathbf{b}}(y) = R\mathbf{b}(Ry), \quad \tilde{\mathbf{f}}(y) = R\mathbf{f}(Ry), \quad \tilde{k} = \|\tilde{f}\|_{L^s(B_1)} + \|\tilde{\mathbf{f}}\|_{L^t(B_1)},$$

where  $y \in B_1$ . We apply what we just proved to  $\tilde{u}$  in  $B_1$  and rewrite the result in terms of  $u$ , then for any  $\gamma \geq 2p'$ , we derive

$$\begin{aligned} & \|u_+\|_{L^\infty(B_{\frac{1}{4}R})} \\ & \leq C \left(1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R)}\right)^{\frac{2\delta}{\delta-1}} \Lambda(B_R)^{\frac{\delta}{\delta-1}} \left(R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_R)} + \bar{k}\right), \end{aligned} \quad (3.14)$$

where

$$\Lambda(B_R) = \left(\int_{B_R} \mu^p dx\right)^{\frac{1}{2p}} \left(\int_{B_R} |\lambda|^{-q} dx\right)^{\frac{1}{2q}} + \left(\int_{B_R} |\lambda|^{-q} dx\right)^{\frac{1}{q}} + \left(\int_{B_R} |\lambda|^{-q} dx\right)^{\frac{2}{q}},$$

$\bar{k} = R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R)}$ ,  $C = C(p, q, N, \gamma) \geq 1$ , and (3.14) also holds for  $B_R$  replaced by  $B_{(1-\theta)R}(\tilde{x}_0)$ , for any  $\tilde{x}_0 \in B_{\theta R}$ , we arrive at

$$\begin{aligned} & \|u_+\|_{L^\infty(B_{\frac{1}{4}(1-\theta)R}(\tilde{x}_0))} \\ & \leq C \left(1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_{(1-\theta)R}(\tilde{x}_0))}\right)^{\frac{2\delta}{\delta-1}} \Lambda(B_{(1-\theta)R}(\tilde{x}_0))^{\frac{\delta}{\delta-1}} \\ & \quad \cdot \left( ((1-\theta)R)^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_{(1-\theta)R}(\tilde{x}_0))} + ((1-\theta)R)^{2-\frac{N}{s}} \|f\|_{L^s(B_R(\tilde{x}_0))} \right. \\ & \quad \left. + ((1-\theta)R)^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R(\tilde{x}_0))} \right), \end{aligned} \quad (3.15)$$

where  $C = C(N, p, q, \gamma)$ .

Next, we will be in position to prove (3.2) for  $\gamma \in (0, 2p')$ . First, we observe that

$$\left(\int_{B_R} |u_+|^{2p'} dx\right)^{\frac{1}{2p'}} \leq \|u_+\|_{L^\infty(B_R)}^{\frac{2p'-\gamma}{2p'}} \|u_+\|_{L^\gamma(B_R)}^{\frac{\gamma}{2p'}}. \quad (3.16)$$

In addition, applying the Young's inequality to (3.16), and combining this with (3.15) for  $\gamma = 2p'$ , we obtain

$$\begin{aligned} \|u_+\|_{L^\infty(B_{\theta R})} & \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C \left(1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R)}\right)^{\frac{4p'\delta}{(\delta-1)\gamma}} \Lambda(B_{(1-\theta)R})^{\frac{2p'\delta}{\gamma(\delta-1)}} \\ & \quad \cdot \left( ((1-\theta)R)^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_{(1-\theta)R})} + ((1-\theta)R)^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} \right. \\ & \quad \left. + ((1-\theta)R)^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R)} \right), \end{aligned} \quad (3.17)$$

where  $C = C(N, p, q, \gamma)$ . Then, set  $r := \theta R$  ( $\theta R < R \leq 1$ ,  $\theta \in (0, 1)$ ) and apply (3.17) to get

$$\|u_+\|_{L^\infty(B_r)} \leq \frac{1}{2} \|u_+\|_{L^\infty(B_R)} + C \left(1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R)}\right)^{\frac{4p'\delta}{(\delta-1)\gamma}} \Lambda(B_R)^{\frac{2p'\delta}{\gamma(\delta-1)}}$$

$$\begin{aligned}
 & \cdot \left( R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_R)} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R)} \right) \\
 & \cdot \left( 1 - \frac{r}{R} \right)^{-\frac{N}{\gamma} - \frac{4Np'\delta s_*}{\gamma(\delta-1)}}, \tag{3.18}
 \end{aligned}$$

where  $C = C(N, p, q, \gamma)$ ,  $s_* = (\frac{1}{2p} + \frac{1}{2q} + \frac{1}{q} + \frac{2}{q})$ . Let  $h(r) := \|u_+\|_{L^\infty(B_r)}$ , then (3.18) becomes

$$\begin{aligned}
 h(r) & \leq \frac{1}{2}h(R) + C \left( 1 + R^{1-\frac{N}{t}} \|\mathbf{b}\|_{L^t(B_R)} \right)^{\frac{4p'\delta}{(\delta-1)\gamma}} \Lambda(B_R)^{\frac{2p'\delta}{\gamma(\delta-1)}} \left( 1 - \frac{r}{R} \right)^{-\frac{N}{\gamma} - \frac{4Np'\delta s_*}{\gamma(\delta-1)}} \\
 & \cdot \left( R^{-\frac{N}{\gamma}} \|u_+\|_{L^\gamma(B_R)} + R^{2-\frac{N}{s}} \|f\|_{L^s(B_R)} + R^{1-\frac{N}{t}} \|\mathbf{f}\|_{L^t(B_R)} \right).
 \end{aligned}$$

Proceeding as a proof of Lemma 4.3 in [46], we may derive

$$\begin{aligned}
 \|u_+\|_{L^\infty(B_\theta)} & \leq C \left( 1 + \|\mathbf{b}\|_{L^t(B_1)} \right)^{\frac{4p'\delta}{(\delta-1)\gamma}} \Lambda(B_1)^{\frac{2p'\delta}{\gamma(\delta-1)}} (1 - \theta)^{-\frac{N}{\gamma} - \frac{4Np'\delta s_*}{\gamma(\delta-1)}} \\
 & \cdot \left( \|u_+\|_{L^\gamma(B_1)} + \|f\|_{L^s(B_1)} + \|\mathbf{f}\|_{L^t(B_1)} \right), \tag{3.19}
 \end{aligned}$$

where  $C = C(N, p, q, \gamma)$ . □

### 3.5 Proof of Theorem 3.1.2

**Proof.** Without loss of generality, we assume that  $u > 0$  and  $x_0 = 0$ .

**Step 1.** We claim that  $k + \inf_{B_{\frac{1}{4}}} u \geq \frac{C}{\|\frac{1}{u}\|_{L^{2p'\alpha_1(B_{\frac{2}{3}})}}}$  with  $k := \|f\|_{L^s(B_1)} + \|\mathbf{f}\|_{L^t(B_1)}$ .

We take  $\varphi = \psi(u)\eta^2$ , where  $\psi(u) = (u + k)^\beta$  with  $\beta < -1$ ,  $\eta \equiv 1$  in  $B_\rho$  and  $\eta \equiv 0$  in  $B_{\frac{1}{2}} \setminus B_\sigma$ ,  $\frac{1}{4} \leq \rho < \sigma \leq \frac{1}{2}$ . Then we have

$$\begin{aligned}
 & - \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla u) \psi'(u) \eta^2 dx \\
 & \leq 2 \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \eta) \psi(u) \eta dx - \int_{\Omega} u \mathbf{b} (2\eta \psi(u) \nabla \eta + \psi'(u) \eta^2 \nabla u) dx \\
 & \quad + \int_{\Omega} \mathbf{f} (2\eta \psi(u) \nabla \eta + \psi'(u) \eta^2 \nabla u) dx - \int_{\Omega} f(x) \psi(u) \eta^2 dx \\
 & \leq 2 \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \eta) \psi(u) \eta dx + \int_{\Omega} |f(x)| \psi(u) \eta^2 dx \\
 & \quad + \int_{\Omega} \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) (u + k) (2\eta \psi(u) |\nabla \eta| + \psi'(u) \eta^2 |\nabla u|) dx. \tag{3.20}
 \end{aligned}$$

Now, we deal with the terms on the right side of (3.20),

First, by Young's inequality, we have

$$2 \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla \eta) \psi(u) \eta dx$$

$$\leq 2\varepsilon_0 \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla u) (u+k)^{\beta-1} \eta^2 dx + \frac{2}{\varepsilon_0} \int_{\Omega} (\mathbf{A}(x) \nabla \eta \nabla \eta) (u+k)^{\beta+1} dx.$$

Second, by the interpolation inequality and Sobolev inequality, we obtain

$$\begin{aligned} \int_{\Omega} f \eta^2 \psi(u) dx &\leq \int_{B_{\sigma}} \eta^2 \frac{|f|}{k} v^2 dx \leq \|v\eta\|_{L^{\frac{2s}{s-1}}(B_{\sigma})}^2 \\ &\leq \left( \varepsilon_3 \|v\eta\|_{L^{q^*}(B_{\sigma})}^2 + \frac{1}{\varepsilon_3} \|v\eta\|_{L^{p^*}(B_{\sigma})}^2 \right) \\ &\leq \varepsilon_3 C(N) \|\lambda^{-1}\|_{L^q(B_{\sigma})} \int_{B_{\sigma}} \mu |\nabla \eta|^2 v^2 dx + \frac{1}{\varepsilon_3} \|v\eta\|_{L^{p^*}(B_{\sigma})}^2 \\ &\quad + \varepsilon_3 C(N) \|\lambda^{-1}\|_{L^q(B_{\sigma})} \int_{B_{\sigma}} \lambda |\nabla v|^2 \eta^2 dx. \end{aligned}$$

Third, by Young's inequality, we obtain

$$\begin{aligned} &2 \int_{\Omega} \left( \frac{|f|}{k} + |\mathbf{b}| \right) (u+k) \eta \psi(u) |\nabla \eta| dx \\ &\leq \frac{1}{\varepsilon_1} \int_{B_{\sigma}} \left( \frac{|f|}{k} + |\mathbf{b}| \right) v^2 |\nabla \eta|^2 dx + \varepsilon_1 \int_{B_{\sigma}} \left( \frac{|f|}{k} + |\mathbf{b}| \right) v^2 \eta^2 dx, \end{aligned}$$

where  $v = (u+k)^{\frac{\beta+1}{2}}$ . Then, from Lemma 2.2.2, we have

$$\begin{aligned} &\inf_{\eta(x)} \int_{B_{\sigma}} \left( \frac{|f|}{k} + |\mathbf{b}| \right) |\nabla \eta|^2 v^2 dx \\ &\leq \frac{C(N, t, p)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left\| \frac{|f|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\sigma} \setminus B_{\rho})} \|v\|_{W^{1, p_t^*}(B_{\sigma} \setminus B_{\rho})}^2, \end{aligned}$$

where  $p_t^* = \max\{p_*, t_*\}$ . Then, by Hölder's inequality and Sobolev's inequality, we get

$$\begin{aligned} &\int_{B_{\sigma}} \left( \frac{|f|}{k} + |\mathbf{b}| \right) v^2 \eta^2 dx \\ &\leq |B_{\sigma}|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|f|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\sigma})} \|\eta v\|_{L^{q_*}(B_{\sigma})}^2 \\ &\leq C(N, q) |B_{\sigma}|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|f|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\sigma})} \|\nabla(\eta v)\|_{L^{\frac{2q}{q+1}}(B_{\sigma})}^2 \\ &\leq C(N, q) |B_{\sigma}|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|f|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\sigma})} \|\lambda^{-1}\|_{L^q(B_{\sigma})} \int_{B_{\sigma}} \lambda |\nabla(\eta v)|^2 dx \\ &\leq C(N, q) |B_{\sigma}|^{1-\frac{1}{t}-\frac{2}{q_*}} \left\| \frac{|f|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\sigma})} \\ &\quad \cdot \|\lambda^{-1}\|_{L^q(B_{\sigma})} \left( \int_{B_{\sigma}} \mu |\nabla \eta|^2 v^2 dx + \int_{B_{\sigma}} \lambda |\nabla v|^2 \eta^2 dx \right), \end{aligned}$$

where  $q_* = \frac{2Nq}{(q+1)N-2q}$ . Fourth, we derive

$$\int_{\Omega} \psi'(u) \eta^2 |\nabla u| (u_+ + k) \left( \frac{|f|}{k} + |\mathbf{b}| \right) dx$$

$$\leq \frac{|\beta|}{4} \int_{B_\sigma} \lambda |\nabla u|^2 (u+k)^{\beta-1} \eta^2 dx + 4|\beta| \int_{B_\sigma} \frac{\left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right)^2}{\lambda} v^2 \eta^2 dx,$$

where

$$\begin{aligned} & 4|\beta| \int_{B_\sigma} \frac{\left(\frac{|\mathbf{f}|}{k} + |\mathbf{b}|\right)^2}{\lambda} v^2 \eta^2 dx \\ & \leq 4|\beta| \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \|v\eta\|_{L^{1-\frac{2}{t}-\frac{1}{q}}(B_\sigma)}^2 \\ & \leq 4|\beta| \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \varepsilon_2 \|v\eta\|_{L^{q^*}(B_\sigma)}^2 + \frac{1}{\varepsilon_2} \|v\eta\|_{L^{p^*}(B_\sigma)}^2 \right) \\ & \leq 4|\beta| \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \varepsilon_2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \mu |\nabla \eta|^2 v^2 dx \right. \\ & \quad \left. + \varepsilon_2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla v|^2 \eta^2 dx + \frac{1}{\varepsilon_2} \|v\eta\|_{L^{p^*}(B_\sigma)}^2 \right). \end{aligned}$$

Together with the above estimates, we obtain

$$\begin{aligned} & |\beta| \inf_{\eta} \int_{B_\sigma} (\mathbf{A}(x) \nabla u \nabla u) (u+k)^{\beta-1} \eta^2 dx \\ & \leq 2\varepsilon_0 \inf_{\eta} \int_{\Omega} (\mathbf{A}(x) \nabla u \nabla u) (u+k)^{\beta-1} \eta^2 dx + \frac{2}{\varepsilon_0} \inf_{\eta} \int_{\Omega} (\mathbf{A}(x) \nabla \eta \nabla \eta) v^2 dx \\ & \quad + \frac{1}{\varepsilon_1} \frac{C(N, p, t)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma \setminus B_\rho)} \|v\|_{W^{1,p_t}^*(B_\sigma \setminus B_\rho)}^2 \\ & \quad + \varepsilon_1 C(N) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \|\lambda^{-1}\|_{L^q(B_\sigma)} \\ & \quad \cdot \left( \inf_{\eta} \int_{B_\sigma} \lambda |\nabla \eta|^2 v^2 dx + \inf_{\eta} \int_{B_\sigma} \lambda |\nabla v|^2 \eta^2 dx \right) \\ & \quad + \frac{|\beta|}{4} \inf_{\eta} \int_{B_\sigma} \lambda |\nabla u|^2 (u+k)^{\beta-1} \eta^2 dx \\ & \quad + 4|\beta| \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)} \\ & \quad \times \left( \varepsilon_2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta} \int_{B_\sigma} \mu |\nabla \eta|^2 v^2 dx \right. \\ & \quad \left. + \varepsilon_2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta} \int_{B_\sigma} \lambda |\nabla v|^2 \eta^2 dx + \frac{1}{\varepsilon_2} \|v\eta\|_{L^{p^*}(B_\sigma)}^2 \right) \\ & \quad + \varepsilon_3 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta} \int_{B_\sigma} \mu |\nabla \eta|^2 v^2 dx \\ & \quad + \varepsilon_3 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)} \inf_{\eta} \int_{B_\sigma} \lambda |\nabla v|^2 \eta^2 dx + \frac{1}{\varepsilon_3} \|v\eta\|_{L^{p^*}(B_\sigma)}^2. \end{aligned}$$



We take

$$\begin{aligned}\varepsilon_0 &= \frac{|\beta|}{32}, \quad \varepsilon_1 = \frac{|\beta|}{4(\beta+1)^2 C(N) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma)} \|\lambda^{-1}\|_{L^q(B_\sigma)}}, \\ \varepsilon_2 &= \frac{1}{16(\beta+1)^2 \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 C(N)}, \\ \varepsilon_3 &= \frac{|\beta|}{4(\beta+1)^2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)}}.\end{aligned}$$

Then, it is not hard to get

$$\begin{aligned}& \left( \frac{4|\beta|}{(\beta+1)^2} - \frac{|\beta|}{4(\beta+1)^2} - \frac{|\beta|}{4(\beta+1)^2} - \frac{|\beta|}{(\beta+1)^2} - \frac{|\beta|}{4(\beta+1)^2} - \frac{|\beta|}{4(\beta+1)^2} \right) \\& \cdot \inf_{\eta} \int_{\Omega} \lambda(x) |\nabla v|^2 \eta^2 dx \\& \leq \frac{64}{|\beta|} \inf_{\eta} \int_{B_\sigma} \mu |\nabla \eta|^2 v^2 dx + \frac{3|\beta|}{4(\beta+1)^2} \inf_{\eta} \int_{B_\sigma} \mu |\nabla \eta|^2 v^2 dx \\& \quad + \frac{4(\beta+1)^2 C(N, q, t, p) \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_\sigma \setminus B_\rho)}^2 \|\lambda^{-1}\|_{L^q(B_\sigma)}}{(\sigma - \rho)^{\frac{2N}{N-1}}} \|v\|_{W^{1,p_t^*}(B_\sigma \setminus B_\rho)}^2 \\& \quad + 64|\beta|(\beta+1)^2 C(N) \left\| \left( \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right) \right\|_{L^t(B_\sigma)}^4 \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 \|v\eta\|_{L^{p^*}(B_\sigma)}^2 \\& \quad + \frac{4(\beta+1)^2 C(N) \|\lambda^{-1}\|_{L^q(B_\sigma)}}{|\beta|} \|v\eta\|_{L^{p^*}(B_\sigma)}^2.\end{aligned}$$

By direct computation, we get

$$\begin{aligned}& \inf_{\eta} \int_{\Omega} \lambda(x) |\nabla v|^2 \eta^2 dx \\& \leq \left[ \frac{|\beta|^4 C(N, q, p, t) \|1 + |\mathbf{b}|\|_{L^t(B_\sigma)}^4 \left( \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 + \|\mu\|_{L^p(B_\sigma)} \right)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \right] \\& \quad \cdot \|v\|_{W^{1,p_t^*}(B_\sigma)}^2.\end{aligned}\tag{3.21}$$

By the same calculation as (3.11) and (3.12), we have

$$\begin{aligned}\|v^\delta\|_{W^{1,p_t^*}(B_\rho)} & \leq \frac{|\beta|^2 C}{(\sigma - \rho)^{\frac{N}{N-1}}} \left( 1 + \| |\mathbf{b}| \|_{L^t(B_\sigma)} \right)^2 \|v\|_{W^{1,p_t^*}(B_\sigma)}^\delta \\& \quad \cdot \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right),\end{aligned}$$

where  $C$  depends on  $p, q, t, N$ . Let  $\frac{\beta+1}{2} = -\alpha_1 \delta^{m-1}$ , where  $\alpha_1 > 0$  to be determined in Step 2, and define  $\rho = \rho_{m+1} = \frac{1}{4} + \left(\frac{1}{4}\right)^{m+1}$ ,  $\sigma = \rho_m = \frac{1}{4} + \left(\frac{1}{4}\right)^m$ ,  $m \in \mathbb{N}_+$ , we have

$$\|(u+k)^{-\alpha_1 \delta^m}\|_{W^{1,p_t^*}(B_{\rho_{m+1}})}^{\frac{1}{\alpha_1 \delta^m}}$$

$$\begin{aligned}
 &\leq \delta^{\frac{2m-2}{\alpha_1 \delta^m}} 4^{\frac{mN}{\alpha_1(N-1)\delta^m}} \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_\sigma)} \right)^2 \right. \\
 &\quad \cdot \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right) \Big]^{\frac{1}{\alpha_1 \delta^m}} \\
 &\quad \cdot \|(u+k)^{-\alpha_1 \delta^{m-1}}\|_{W^{1,p_t^*}(B_{\rho_m})}^{\frac{1}{\alpha_1 \delta^{m-1}}},
 \end{aligned}$$

where  $C$  depends on  $\alpha_1, N, p, q, t$ . Then by iteration, we get

$$\begin{aligned}
 \left\| \frac{1}{u+k} \right\|_{L^\infty(B_{\frac{1}{4}})} &\leq C \left( 1 + \|\mathbf{b}\|_{L^t(B_{\frac{1}{2}})} \right)^{\frac{2\delta}{\alpha_1(\delta-1)}} \\
 &\quad \cdot \left( \|\mu\|_{L^p(B_{\frac{1}{2}})}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_{\frac{1}{2}})}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_{\frac{1}{2}})} + \|\lambda^{-1}\|_{L^q(B_{\frac{1}{2}})}^2 \right)^{\frac{\delta}{\alpha_1(\delta-1)}} \\
 &\quad \cdot \left\| \frac{1}{(u+k)^{\alpha_1}} \right\|_{W^{1,p_t^*}(B_{\frac{1}{2}})}^{\frac{1}{\alpha_1}},
 \end{aligned}$$

where  $C$  depends on  $p, q, t, N, \alpha_1$ .

We proceed to estimate  $\left\| \frac{1}{(u+k)^{\alpha_1}} \right\|_{W^{1,p_t^*}(B_{\frac{1}{2}})}^{\frac{1}{\alpha_1}}$ , take  $\eta_1 \equiv 1$  in  $B_{\frac{1}{2}}$ ,  $\eta_1 \equiv 0$  in  $B_1 \setminus B_{\frac{2}{3}}$  and set  $v_0 = \frac{1}{(u+k)^{\alpha_1}}$ , by Hölder's inequality, we have

$$\begin{aligned}
 \left( \int_{B_{\frac{1}{2}}} |\nabla v_0|^{p_t^*} dx \right)^{\frac{1}{\alpha_1 p_t^*}} &\leq |B_{\frac{1}{2}}|^{\frac{1}{\alpha_1 p^*}(1-\frac{1}{\chi})} \|\nabla(v_0)\|_{L^{\frac{2q}{q+1}}(B_{\frac{1}{2}})}^{\frac{1}{\alpha_1}} \\
 &\leq |B_{\frac{1}{2}}|^{\frac{1}{\alpha_1 p^*}(1-\frac{1}{\chi})} \left( \|\lambda^{-1}\|_{L^q} \left( \int_{B_{\frac{2}{3}}} \eta_1^2 \lambda |\nabla v_0|^2 dx \right)^{\frac{1}{2}} \right)^{\frac{1}{\alpha_1}}.
 \end{aligned}$$

In view of (3.21) with  $\beta = -1 - 2\alpha_1$ , we obtain

$$\begin{aligned}
 &\left( \int_{B_{\frac{2}{3}}} \eta_1^2 \lambda |\nabla v_0|^2 dx \right)^{\frac{1}{2\alpha_1}} \\
 &\leq C \left( \|\mu\|_{L^p(B_{\frac{2}{3}})} + \|\lambda^{-1}\|_{L^q}^3 \left\| \frac{|\mathbf{f}|}{k} + |\mathbf{b}| \right\|_{L^t(B_{\frac{2}{3}})}^4 \right)^{\frac{1}{2\alpha_1}} \|v_0\|_{L^{2p'}(B_{\frac{2}{3}})}^{\frac{1}{\alpha_1}},
 \end{aligned}$$

where  $C$  depends on  $\alpha_1, N, p, q, t$ . Combining the previous inequalities, we have

$$\begin{aligned}
 &\left\| \frac{1}{u+k} \right\|_{L^\infty(B_{\frac{1}{4}})} \\
 &\leq C \left( 1 + \|\mathbf{b}\|_{L^t(B_{\frac{2}{3}})} \right)^{\frac{4\delta-2}{\alpha_1(\delta-1)}} \\
 &\quad \cdot \left( \|\mu\|_{L^p(B_{\frac{2}{3}})}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})} + \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})}^2 \right)^{\frac{2\delta-1}{\alpha_1(\delta-1)}}
 \end{aligned}$$

$$\cdot \left\| \frac{1}{(u+k)} \right\|_{L^{2\alpha_1 p'}(B_{\frac{2}{3}})},$$

where  $C$  depends on  $N, p, q, t, \alpha_1$ .

**Step 2.** We aim to show

$$\|u+k\|_{L^{2p'\alpha_1}(B_{\frac{2}{3}})} \left\| \frac{1}{u+k} \right\|_{L^{2p'\alpha_1}(B_{\frac{2}{3}})} \leq C.$$

To this end, it suffices to prove that

$$\int_{B_{\frac{2}{3}}} e^{\alpha_1 2p' |\log \frac{(u+k)}{h}|} dx \leq C,$$

$$\text{where } h = e^{\frac{1}{|B_{\frac{2}{3}}|} \int_{B_{\frac{2}{3}}} \log(u+k) dx}.$$

We take the test function  $\varphi = \eta^2 \frac{1}{u+k} \phi(|\omega|)$ , where  $\phi(|\omega|) = (|\omega| + 1)^\beta + (2\beta)^\beta$  with  $\beta \geq 1$ , and  $\omega = \log \frac{u+k}{h}$ ,  $\eta \equiv 1$ , in  $B_\rho$ ;  $\eta \equiv 0$  in  $B_1 \setminus B_\sigma$ ,  $\frac{2}{3} \leq \rho < \sigma \leq 1$ . Then, we have

$$\nabla \varphi(x) = 2\eta \frac{\phi(|\omega|)}{u+k} \nabla \eta - \eta^2 \frac{\phi(|\omega|)}{(u+k)^2} \nabla u + \frac{\eta^2}{(u+k)^2} \phi'(|\omega|) (\text{sgn}(\omega)) \nabla u.$$

Substituting it into (3.2.1), we obtain

$$\begin{aligned} \int_{\Omega} \mathbf{A}(x) \nabla u \nabla \varphi &= 2 \int_{B_\sigma} \eta \phi(|\omega|) \mathbf{A}(x) \nabla \omega \nabla \eta dx - \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \phi(|\omega|) \eta^2 dx \\ &\quad + \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \eta^2 \phi'(|\omega|) (\text{sgn} \omega) dx, \end{aligned}$$

$$\begin{aligned} \int_{\Omega} u \mathbf{b} \nabla \varphi dx &= 2 \int_{B_\sigma} (\tilde{\mathbf{b}} \cdot \nabla \eta) \eta \phi(|\omega|) dx - \int_{B_\sigma} (\tilde{\mathbf{b}} \cdot \nabla \omega) \eta^2 \phi(\omega) dx \\ &\quad + \int_{B_\sigma} (\tilde{\mathbf{b}} \cdot \nabla \omega) \eta^2 \phi'(|\omega|) (\text{sgn} \omega) dx, \end{aligned}$$

$$\begin{aligned} \int_{\Omega} \mathbf{f} \nabla \varphi dx &= 2 \int_{B_\sigma} (\tilde{\mathbf{f}} \cdot \nabla \eta) \eta \phi(|\omega|) dx - \int_{B_\sigma} (\tilde{\mathbf{f}} \cdot \nabla \omega) \eta^2 \phi(\omega) dx \\ &\quad + \int_{B_\sigma} (\tilde{\mathbf{f}} \cdot \nabla \omega) \eta^2 \phi'(|\omega|) (\text{sgn} \omega) dx, \end{aligned}$$

$$\int_{\Omega} f \eta^2 \frac{\phi(|\omega|)}{u+k} dx = \int_{B_\sigma} \tilde{f} \eta^2 \phi(|\omega|) dx,$$

where  $\tilde{\mathbf{f}} = \frac{\mathbf{f}}{u+k}$ ,  $\tilde{\mathbf{b}} = \frac{u\mathbf{b}}{u+k}$ ,  $\tilde{f} = \frac{f}{u+k}$  and note that  $\|\tilde{\mathbf{f}}\|_{L^t} \leq 1$ ,  $\|\tilde{\mathbf{b}}\|_{L^t} \leq \|\mathbf{b}\|_{L^t}$ ,  $\|\tilde{f}\|_{L^s} \leq 1$ .

Then we have

$$\int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \phi(|\omega|) \eta^2 dx - \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \eta^2 \phi'(\omega) (\text{sgn} \omega)$$

$$\begin{aligned}
 &\leq 2 \int_{B_\sigma} \eta \phi(|\omega|) \mathbf{A}(x) \nabla \omega \nabla \eta dx + 2 \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \eta) \eta \phi(|\omega|) \\
 &\quad - \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \omega) \eta^2 \phi(\omega) dx + \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \omega) \eta^2 \phi'(|\omega|) (\operatorname{sgn} \omega) dx \\
 &\quad - \int_{B_\sigma} \tilde{f} \eta^2 \phi(|\omega|) dx.
 \end{aligned} \tag{3.22}$$

We estimate the first term on the right side of (3.22), by Young's inequality, we have

$$2 \int_{B_\sigma} \eta \phi(|\omega|) \mathbf{A}(x) \nabla \omega \nabla \eta dx \tag{3.23}$$

$$\begin{aligned}
 &\leq 2\varepsilon_0 \int_{B_\sigma} \eta^2 (\mathbf{A}(x) \nabla \omega \nabla \omega) \left[ (|\omega| + 1)^{\beta-1} + (2\beta)^\beta \right] dx \\
 &\quad + \frac{2}{\varepsilon_0} \int_{B_\sigma} \mathbf{A}(x) \nabla \eta \nabla \eta \left[ (|\omega| + 1)^{\beta+1} + (2\beta)^\beta \right] dx.
 \end{aligned} \tag{3.24}$$

Next, we estimate the second term on the right side of (3.22), by Young's inequality, we get

$$\begin{aligned}
 &2 \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \eta) \eta \phi(|\omega|) \\
 &= 2 \int_{B_\sigma} (|\omega| + 1)^\beta ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \eta) \eta dx + (2\beta)^\beta \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \eta) \eta dx \\
 &\leq \varepsilon_1 \int_{B_\sigma} |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| \eta^2 (|\omega| + 1)^{\beta+1} dx + \frac{1}{\varepsilon_1} \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta-1} dx \\
 &\quad + (2\beta)^\beta \frac{C}{\sigma - \rho} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t} |B_\sigma|^{1-\frac{1}{t}} \\
 &\leq \varepsilon_1 |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t(B_\sigma)} \|(|\omega| + 1)^{\frac{\beta+1}{2}}\|_{L^{q^*}(B_\sigma)}^2 \\
 &\quad + \frac{1}{\varepsilon_1} \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta+1} dx + (2\beta)^\beta \frac{C}{\sigma - \rho} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t} |B_\sigma|^{1-\frac{1}{t}} \\
 &\leq \left[ \varepsilon_1 C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} (\|\mathbf{b}\|_{L^t} + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \int_{B_\sigma} \lambda |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right. \right. \\
 &\quad \left. \left. + \int_{B_\sigma} \lambda |\nabla (|\omega| + 1)^{\frac{\beta+1}{2}}|^2 \eta^2 dx \right) \right] + \frac{1}{\varepsilon_1} \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta+1} dx \\
 &\quad + (2\beta)^\beta \frac{C}{\sigma - \rho} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t} |B_\sigma|^{1-\frac{1}{t}} \\
 &\leq \left[ \varepsilon_1 C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} (\|\mathbf{b}\|_{L^t} + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \left( \int_{B_\sigma} \lambda |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right. \right. \\
 &\quad \left. \left. + \frac{(\beta+1)^2}{4} \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) (|\omega| + 1)^{\beta-1} \eta^2 dx \right) \right] \\
 &\quad + \frac{1}{\varepsilon_1} \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta+1} dx + (2\beta)^\beta \frac{C}{\sigma - \rho} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t} |B_\sigma|^{1-\frac{1}{t}}.
 \end{aligned}$$

Now, we will deal with the third term on the right side of (3.22)

$$\begin{aligned}
 & \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \omega) \eta^2 \phi(\omega) dx \\
 & \leq \int_{B_\sigma} (|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| \cdot |\nabla \omega|) \eta^2 (|\omega| + 1)^\beta dx + \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \omega) \eta^2 (2\beta)^\beta dx \\
 & = \text{I} + \text{II},
 \end{aligned}$$

where

$$\begin{aligned}
 \text{I} & \leq \int_{B_\sigma} |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| |\nabla \omega| \eta^2 (|\omega| + 1)^\beta dx \\
 & \leq \varepsilon_2 \int_{B_\sigma} \lambda |\nabla \omega|^2 (|\omega| + 1)^{\beta-1} \eta^2 dx + \frac{1}{\varepsilon_2} \int_{B_\sigma} \frac{|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}|^2}{\lambda} (|\omega| + 1)^{\beta+1} \eta^2 dx \\
 & \leq \varepsilon_2 \int_{B_\sigma} \lambda |\nabla \omega|^2 (|\omega| + 1)^{\beta-1} \eta^2 dx \\
 & \quad + \frac{1}{\varepsilon_2} \left[ (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{\frac{2}{1-\frac{2}{t}-\frac{1}{q}}}(B_\sigma)}^2 \right] \\
 & \leq \varepsilon_2 \int_{B_\sigma} \lambda |\nabla \omega|^2 (|\omega| + 1)^{\beta-1} \eta^2 dx + \frac{1}{\varepsilon_2} \left[ (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \right. \\
 & \quad \cdot \left( \varepsilon_3 \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{q*}(B_\sigma)}^2 + \frac{1}{\varepsilon_3} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p*}(B_\sigma)}^2 \right) \Big] \\
 & \leq \varepsilon_2 \int_{B_\sigma} \lambda |\nabla \omega|^2 (|\omega| + 1)^{\beta-1} \eta^2 dx + \frac{1}{\varepsilon_2} \left[ (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \right. \\
 & \quad \cdot \left( \varepsilon_3 C(N, q) \|\lambda^{-1}\|_{L^q} \left( \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right. \right. \\
 & \quad \left. \left. + \int_{B_\sigma} \lambda |\nabla (|\omega| + 1)^{\frac{\beta+1}{2}}|^2 \eta^2 dx \right) + \frac{1}{\varepsilon_3} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p*}(B_\sigma)}^2 \right) \Big] \\
 & \leq \varepsilon_2 \int_{B_\sigma} \lambda |\nabla \omega|^2 (|\omega| + 1)^{\beta-1} \eta^2 dx + \frac{1}{\varepsilon_2} \left[ (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \right. \\
 & \quad \cdot \left( \varepsilon_3 C(N, q) \|\lambda^{-1}\|_{L^q} \left( \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right. \right. \\
 & \quad \left. \left. + \frac{(\beta+1)^2}{4} \int_{B_\sigma} (a(x) \nabla \omega \nabla \omega) (|\omega| + 1)^{\beta-1} |\eta|^2 dx \right) \right. \\
 & \quad \left. \left. + \frac{1}{\varepsilon_3} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p*}(B_\sigma)}^2 \right) \right],
 \end{aligned}$$

and

$$\begin{aligned}
 \text{II} & \leq (2\beta)^\beta \int_{B_\sigma} |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| |\nabla \omega| \eta^2 dx \\
 & \leq \varepsilon_4 (2\beta)^\beta \int_{B_\sigma} \lambda |\nabla \omega|^2 \eta^2 dx + \frac{(2\beta)^\beta}{\varepsilon_4} \int_{B_\sigma} \frac{|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}|^2}{\lambda} \eta^2 dx
 \end{aligned}$$

$$\leq \varepsilon_4 (2\beta)^\beta \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \eta^2 dx + \frac{(2\beta)^\beta}{\varepsilon_4} (\|\mathbf{b}\|_{L^t} + 1)^2 \|\lambda^{-1}\|_{L^q} |B_\sigma|^{1-\frac{2}{t}-\frac{1}{q}}.$$

Next, we estimate the fourth term on the right side of (3.22)

$$\begin{aligned} \int_{B_\sigma} ((\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \cdot \nabla \omega) \eta^2 \phi'(|\omega|) (\operatorname{sgn} \omega) dx &\leq \beta \int_{B_\sigma} |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| |\nabla \omega| \eta^2 (|\omega| + 1)^{\beta-1} dx \\ &\leq \beta \int_{B_\sigma} |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| |\nabla \omega| \eta^2 (|\omega| + 1)^\beta dx \\ &= \beta \mathbf{I}. \end{aligned}$$

Then, we estimate the last term on the right side of (3.22)

$$\begin{aligned} &\int_{B_\sigma} \tilde{f} \eta^2 \phi(|\omega|) dx \\ &= \int_{B_\sigma} \tilde{f} \eta^2 (|\omega| + 1)^\beta dx + (2\beta)^\beta \int_{B_\sigma} \tilde{f} \eta^2 dx \\ &\leq \|\tilde{f}\|_{L^s(B_\sigma)} \left( \int_{B_\sigma} |\eta| (|\omega| + 1)^{\frac{\beta+1}{2}} |2s'| dx \right)^{\frac{1}{s'}} + (2\beta)^\beta \|\tilde{f}\|_{L^s(B_\sigma)} |B_\sigma|^{1-\frac{1}{s'}} \\ &\leq \varepsilon_5 C(N, q) \|\lambda^{-1}\|_{L^q} \left[ \int_{B_\sigma} \lambda |\nabla (|\omega| + 1)^{\frac{\beta+1}{2}}|^2 \eta^2 dx + \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right] \\ &\quad + \frac{1}{\varepsilon_5} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p^*}(B_\sigma)}^2 + (2\beta)^\beta |B_\sigma|^{1-\frac{1}{s'}}, \end{aligned}$$

where

$$\begin{aligned} \varepsilon_0 &= \frac{1}{32}, \quad \varepsilon_1 = \frac{1}{4(\beta+1)^2 C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} (\|\mathbf{b}\|_{L^t} + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)}}, \quad \varepsilon_2 = \frac{1}{16}, \\ \varepsilon_3 &= \frac{1}{64(\beta+1)^2 (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 C(N, q)}, \quad \varepsilon_4 = \frac{1}{16}, \\ \varepsilon_5 &= \frac{1}{4(\beta+1)^2 C(N, q) \|\lambda^{-1}\|_{L^q}}. \end{aligned}$$

Let us substitute the above estimates into (3.22), it is not hard to get

$$\begin{aligned} &\int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) \eta^2 \left[ (|\omega| + 1)^\beta + (2\beta)^\beta - \beta (|\omega| + 1)^{\beta-1} \right. \\ &\quad \left. - \frac{1}{16} ((|\omega| + 1)^{\beta-1} + (2\beta)^\beta) - \frac{4}{16} (|\omega| + 1)^{\beta-1} - \frac{1}{16} (2\beta)^\beta \right] dx \\ &\leq 64 \int_{B_\sigma} \mathbf{A}(x) \nabla \eta \nabla \eta \left[ (|\omega| + 1)^{\beta+1} + (2\beta)^\beta \right] dx \\ &\quad + \left[ \frac{1}{4(\beta+1)^2} \left( \int_{B_\sigma} \lambda |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right) \right] \\ &\quad + 4(\beta+1)^2 C(N, q) |B_\sigma|^{1-\frac{1}{t}-\frac{2}{q^*}} (\|\mathbf{b}\|_{L^t} + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \end{aligned}$$

$$\begin{aligned}
 & \cdot \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta+1} dx \\
 & + (2\beta)^\beta \frac{C}{\sigma - \rho} \|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}\|_{L^t} |B_\sigma|^{1-\frac{1}{t}} \\
 & + \frac{1}{4(\beta+1)} \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \\
 & + 1024(\beta+1)^3 (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1)^2 \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 C(N, q) \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p^*}(B_\sigma)}^2 \\
 & + 16(2\beta)^\beta (\|\mathbf{b}\|_{L^t} + 1)^2 \|\lambda^{-1}\|_{L^q} |B_\sigma|^{1-\frac{2}{t}-\frac{1}{q}} \\
 & + \frac{1}{4(\beta+1)^2} \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \\
 & + 4(\beta+1)^2 C(N, q) \|\lambda^{-1}\|_{L^q} \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p^*}(B_\sigma)}^2 + (2\beta)^\beta |B_\sigma|^{1-\frac{1}{s'}}.
 \end{aligned}$$

Notice that

$$\begin{aligned}
 & (|\omega| + 1)^\beta + \frac{7}{8}(2\beta)^\beta - \frac{21\beta}{16}(|\omega| + 1)^{\beta-1} \\
 & \geq \frac{7}{8} \left( (|\omega| + 1)^\beta + (2\beta)^\beta \right) - \frac{21\beta}{16}(|\omega| + 1)^{\beta-1} \\
 & \geq \frac{7\beta}{4}(|\omega| + 1)^{\beta-1} - \frac{21\beta}{16}(|\omega| + 1)^{\beta-1} = \frac{7}{16}\beta(|\omega| + 1)^{\beta-1},
 \end{aligned}$$

the above inequality becomes

$$\begin{aligned}
 & \int_{B_\sigma} (\mathbf{A}(x) \nabla \omega \nabla \omega) (|\omega| + 1)^{\beta-1} \eta^2 dx \\
 & \leq C \left( \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx \right) \\
 & \quad + 4\beta C(N) (\|\mathbf{b}\|_{L^t} + 1) \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} |\nabla \eta|^2 |\tilde{\mathbf{f}} - \tilde{\mathbf{b}}| (|\omega| + 1)^{\beta+1} dx \\
 & \quad + \frac{C(\|\mathbf{b}\|_{L^t} + 1)^2 (\|\mu\|_{L^p} + \|\lambda^{-1}\|_{L^q} + 1)}{(\sigma - \rho)^2} (2\beta)^\beta \\
 & \quad + (\beta+1)^2 (\|\mathbf{b}\|_{L^t(B_\sigma)}^2 + 1)^2 \left( \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 + \|\lambda^{-1}\|_{L^q(B_\sigma)} \right) \\
 & \quad \times \|(|\omega| + 1)^{\frac{\beta+1}{2}} \eta\|_{L^{p^*}(B_\sigma)}^2. \tag{3.25}
 \end{aligned}$$

From Lemma 2.2.2, we have

$$\begin{aligned}
 & \inf_{\eta(x)} \int_{B_\sigma} (|\tilde{\mathbf{f}} - \tilde{\mathbf{b}}|) |\nabla \eta|^2 (|\omega| + 1)^{\frac{\beta+1}{2}} dx \\
 & \leq \frac{C(N, p, t)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left\| \tilde{\mathbf{f}} - \tilde{\mathbf{b}} \right\|_{L^t(B_\sigma \setminus B_\rho)} \left\| (|\omega| + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p^*}_t(B_\sigma \setminus B_\rho)}^2,
 \end{aligned}$$

$$\inf_{\eta(x)} \int_{B_\sigma} \mu |\nabla \eta|^2 (|\omega| + 1)^{\beta+1} dx$$

$$\leq \frac{C(N, p, t)}{(\sigma - \rho)^{\frac{2N}{N-1}}} \|\mu\|_{L^p(B_\sigma \setminus B_\rho)} \left\| (|\omega| + 1)^{\frac{\beta+1}{2}} \right\|_{W^{1,p_t}^*(B_\sigma \setminus B_\rho)}^2.$$

Then, by simple computation, one can obtain

$$\begin{aligned} & \int_{B_\sigma} \lambda |\nabla(|\omega| + 1)^{\frac{\beta+1}{2}}|^2 \eta^2 dx \\ & \leq C(\beta + 1)^2 \frac{(\|\mathbf{b}\|_{L^t(B_\sigma)} + 1)^4}{(\sigma - \rho)^{\frac{2N}{N-1}}} \left( \|\lambda^{-1}\|_{L^q(B_\sigma)}^3 + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\mu\|_{L^p(B_\sigma)} \right) \\ & \quad \cdot \left\| (|\omega| + 1)^{\frac{\beta+1}{2}} \eta \right\|_{W^{1,p_t}^*(B_\sigma)}^2 + \frac{C(\|\mathbf{b}\|_{L^t} + 1)^2 (\|\mu\|_{L^p} + \|\lambda^{-1}\|_{L^q} + 1)}{(\sigma - \rho)^2} (2\beta)^\beta, \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ . And applying Hölder's inequality yields

$$\left( \int_{B_\rho} \left| \nabla(|\omega| + 1)^{\frac{\beta+1}{2}} \right|^{\frac{2q}{q+1}} dx \right)^{\frac{q+1}{q}} \leq \|\lambda^{-1}\|_{L^q(B_\sigma)} \int_{B_\sigma} \lambda |\nabla(|\omega| + 1)^{\frac{\beta+1}{2}}|^2 \eta^2 dx. \quad (3.26)$$

In order to get the iterative inequality, replacing  $v$  with  $(|\omega| + 1)^{\frac{\beta+1}{2}}$  in (3.11), (3.12), and together with (3.25)–(3.26), and Young's inequality, we have

$$\begin{aligned} & \left\| (|\omega| + 1)^{\frac{\delta(\beta+1)}{2}} \right\|_{W^{1,p_t}^*(B_\rho)} \\ & \leq C(\beta + 1) \frac{(\|\mathbf{b}\|_{L^t(B_\sigma)} + 1)^{\frac{2\delta}{\delta-1}}}{(\sigma - \rho)^{\frac{\delta N}{(\delta-1)(N-1)}}} \left\| (|\omega| + 1)^{\frac{\beta+1}{2}} \eta \right\|_{W^{1,p_t}^*(B_\sigma)}^\delta \\ & \quad \cdot \left( \|\lambda^{-1}\|_{L^q(B_\sigma)}^4 + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 + \|\lambda^{-1}\|_{L^q(B_\sigma)} \|\mu\|_{L^p(B_\sigma)} \right)^{\frac{\delta}{(\delta-1)^2}} + \left( \frac{\beta + 1}{2} \right)^{\frac{\delta(\beta+1)}{2}}, \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ . We set  $\frac{\beta+1}{2} = \delta^{m-1}$ ,  $\rho = \frac{2}{3} + \frac{1}{3^{m+1}}$ ,  $\sigma = \frac{2}{3} + \frac{1}{3^m}$ ,  $m \in \mathbb{N}_+$ , by iteration, we have

$$\begin{aligned} & \left\| (|\omega| + 1)^{\delta^m} \right\|_{W^{1,p_t}^*(B_{\frac{2}{3}})}^{\frac{1}{\delta^m}} \\ & \leq C(\|\mathbf{b}\|_{L^t(B_1)} + 1)^{\frac{2\delta}{(\delta-1)^2}} \left( \left\| (|\omega| + 1) \right\|_{W^{1,p_t}^*(B_1)} + \delta^m \right) \\ & \quad \cdot \left( \|\lambda^{-1}\|_{L^q(B_1)}^4 + \|\lambda^{-1}\|_{L^q(B_1)}^2 + \|\lambda^{-1}\|_{L^q(B_\sigma)} \|\mu\|_{L^p(B_1)} \right)^{\frac{\delta}{2(\delta-1)^2}}, \quad (3.27) \end{aligned}$$

Then, we take  $\varphi = \eta^2 \frac{1}{u+k}$  in (3.2.1), with  $\eta \equiv 1$ , in  $B_1$  and  $\eta \equiv 0$  in  $B_3 \setminus B_2$ , we have

$$\begin{aligned} \int_{B_2} \eta^2 \mathbf{A}(x) \nabla \omega \nabla \omega dx & \leq 2 \int_{B_2} \eta \cdot (\tilde{\mathbf{f}} - \tilde{\mathbf{b}}) \nabla \eta dx + \int_{B_2} (\tilde{\mathbf{b}} - \tilde{\mathbf{f}}) \nabla \omega \eta^2 dx \\ & \quad + 2 \int_{B_2} \eta \mathbf{A}(x) \nabla \omega \nabla \eta dx - \int_{B_2} \tilde{f} \eta^2 dx, \end{aligned}$$

where  $\tilde{\mathbf{f}} = \frac{\mathbf{f}}{u+k}$ ,  $\tilde{\mathbf{b}} = \frac{u\mathbf{b}}{u+k}$ ,  $\tilde{f} = \frac{f}{u+k}$  and recall that  $\|\tilde{\mathbf{f}}\|_{L^t} \leq 1$ ,  $\|\tilde{\mathbf{b}}\|_{L^t} \leq \|\mathbf{b}\|_{L^t}$ ,  $\|f\|_{L^s} \leq 1$ .

By Hölder's inequality, we get

$$\int_{B_2} (\tilde{\mathbf{b}} - \tilde{\mathbf{f}}) \nabla \omega \eta^2 dx$$



$$\begin{aligned}
 &\leq \frac{1}{4} \int_{B_2} \lambda |\nabla \omega|^2 \eta^2 dx + 4 \int_{B_2} \frac{|\tilde{\mathbf{b}} - \tilde{\mathbf{f}}|}{\lambda} \eta^2 dx \\
 &\leq \frac{1}{4} \int_{B_2} \lambda |\nabla \omega|^2 \eta^2 dx + 4(\|\mathbf{b}\|_{L^t(B_2)} + 1) \|\lambda^{-1}\|_{L^q(B_2)} |B_2|^{1-\frac{1}{t}-\frac{1}{q}}.
 \end{aligned}$$

Consequently, we have

$$\begin{aligned}
 &\|(|\omega| + 1)\|_{W^{1,p_t^*}(B_1)} \\
 &\leq |B_1|^{1-\frac{(q+1)p_t^*}{2q}} \|(|\omega| + 1)\|_{L^{\frac{2q}{q+1}}(B_1)} \\
 &\leq |B_1|^{1-\frac{(q+1)p_t^*}{2q}} \|\lambda^{-1}\|_{L^q(B_1)}^{\frac{1}{2}} \left( \int_{B_2} \lambda |\nabla \omega|^2 \eta^2 dx \right)^{\frac{1}{2}} \\
 &\leq |B_1|^{1-\frac{(q+1)p_t^*}{2q}} \|\lambda^{-1}\|_{L^q(B_1)}^{\frac{1}{2}} \left( 16 \left\| 1 + |\tilde{\mathbf{b}}| \right\|_{L^t(B_2)}^{\frac{1}{2}} |B_2|^{\frac{1}{2}-\frac{1}{2t}} \right. \\
 &\quad \left. + 4 \left\| \tilde{\mathbf{b}} \right\|_{L^t(B_2)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_2)} |B_2|^{\frac{1}{2}-\frac{1}{2t}-\frac{1}{2q}} + 4 \|\mu\|_{L^p}^{\frac{1}{2}} + |B_2|^{\frac{1}{2}-\frac{1}{2s}} \right).
 \end{aligned}$$

Then, we deduce

$$\begin{aligned}
 &\|(|\omega| + 1)^{\delta^m}\|_{W^{1,p_t^*}(B_{\frac{2}{3}})}^{\frac{1}{\delta^m}} \\
 &\leq C(\|\mathbf{b}\|_{L^t(B_2)} + 1)^{\frac{2\delta}{(\delta-1)^2} + \frac{1}{2}} \\
 &\quad \cdot \left( \|\lambda^{-1}\|_{L^q(B_2)}^2 + \|\lambda^{-1}\|_{L^q(B_2)} + \|\lambda^{-1}\|_{L^q(B_2)}^{\frac{1}{2}} \|\mu\|_{L^p(B_2)}^{\frac{1}{2}} \right)^{\frac{\delta}{(\delta-1)^2} + \frac{1}{2}} \delta^m,
 \end{aligned}$$

where  $C$  depends on  $N, p, q, t$ . Then for any  $i > p_t^* \delta$ , there exists  $m_0 \in \mathbb{N}_+$  such that  $\delta^{m_0} p_t^* \leq i \leq \delta^{m_0+1} p_t^*$ , the previous inequality can be rewritten as

$$\| |\omega| + 1 \|_{L^i(B_{\frac{2}{3}})} \leq \tilde{C} i,$$

where  $C$  depends on  $N, t, p, q$ , and

$$\begin{aligned}
 \tilde{C} &= C(\|\mathbf{b}\|_{L^t(B_2)} + 1)^{\frac{2\delta}{(\delta-1)^2} + \frac{1}{2}} \\
 &\quad \cdot \left( \|\lambda^{-1}\|_{L^q(B_2)}^2 + \|\lambda^{-1}\|_{L^q(B_2)} + \|\lambda^{-1}\|_{L^q(B_2)}^{\frac{1}{2}} \|\mu\|_{L^p(B_2)}^{\frac{1}{2}} \right)^{\frac{\delta}{(\delta-1)^2} + \frac{1}{2}}.
 \end{aligned}$$

We further let  $\alpha_1 = (2p'\tilde{C})^{-1}$  satisfy

$$\frac{(2\alpha_1 p')^i \| |\omega| \|_{L^i(B_{\frac{2}{3}})}^i}{i!} \leq \frac{1}{2^i},$$

thus we have

$$\int_{B_{\frac{2}{3}}} e^{\alpha_1 2p' |\omega|} dx \leq \sum_{i=0}^{\infty} \frac{(2\alpha_1 p')^i \| |\omega| \|_{L^i(B_{\frac{2}{3}})}^i}{i!} \leq 2.$$

**Step 3.** We prove a reverse Hölder's inequality, we claim that for any  $\gamma \in [\alpha_1 2p', \frac{q_*}{2})$ , there exist constants  $\bar{C} \geq 1$ ,  $\theta \in (0, \frac{2}{3})$ , such that

$$\left( \int_{B_\theta} (u+k)^\gamma dx \right)^{\frac{1}{\gamma}} \leq \bar{C} \left( \int_{B_{\frac{2}{3}}} (u+k)^{\alpha_1 2p'} dx \right)^{\frac{1}{\alpha_1 2p'}}, \quad (3.28)$$

where

$$\bar{C} = \left[ \frac{C \|1 + |\mathbf{b}|\|_{L^t(B_{\rho_0})}^2 \Lambda(B_{\rho_0})}{(\frac{1}{3} - \frac{\theta}{2})^{\frac{N}{N-1}}} \right]^{\frac{\gamma}{q_*}}$$

with  $C$  depending on  $N, q, p, t, \theta$  and

$$\Lambda(B_{\rho_0}) = \left( \|\lambda^{-1}\|_{L^q(B_{\rho_0})} + \|\lambda^{-1}\|_{L^q(B_{\rho_0})}^2 + \|\mu\|_{L^p(B_{\rho_0})}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_{\rho_0})}^{\frac{1}{2}} \right)$$

.

To complete the proof of (3.28), we take  $\varphi = \eta^2 (u+k)^\beta$  with  $\beta \in (-1, 0)$  in (3.2.1), and  $0 < \frac{1}{3} + \frac{\theta}{2} \leq \rho < \sigma \leq \frac{2}{3}$ ,  $\eta \equiv 1$  in  $B_\rho$ ;  $\eta_1 \equiv 0$  in  $B_{\frac{2}{3}} \setminus B_\sigma$ , we continue to use the previous computation in Step 1, we derive

$$\begin{aligned} \|v^\delta\|_{W^{1,p_t^*}(B_\rho)} &\leq \frac{(|\beta|+1)^2 C}{(\sigma-\rho)^{\frac{N}{N-1}}} \left(1 + \|\mathbf{b}\|_{L^t(B_\sigma)}\right)^2 \|v\|_{W^{1,p_t^*}(B_\sigma)}^\delta \\ &\quad \cdot \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right), \end{aligned}$$

where  $v = (u+k)^{\frac{\beta+1}{2}}$ ,  $C$  depends on  $p, q, N, t$ . Taking  $\frac{\beta+1}{2} = \frac{\alpha_2}{\delta^m}$ ,  $\alpha_2 \in (0, \frac{1}{2})$ , and  $\rho_m = \frac{2}{3} - \frac{(\frac{1}{3}-\frac{\theta}{2})}{2^m}$ ,  $m \in \mathbb{N}$ , we get

$$\begin{aligned} &\left\| (u+k)^{\frac{\alpha_2}{\delta^{m-1}}} \right\|_{W^{1,p_t^*}(B_{\rho_{m-1}})}^{\frac{\delta^{m-1}}{\alpha_2}} \\ &\leq \left[ C 2^{\frac{mN}{N-1}} \left(1 + \|\mathbf{b}\|_{L^t(B_\sigma)}\right)^2 \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} \right. \right. \\ &\quad \left. \left. + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right) \right]^{\frac{\delta^{m-1}}{\alpha_2}} \left\| (u+k)^{\frac{\alpha_2}{\delta^m}} \right\|_{W^{1,p_t^*}(B_{\rho_m})}^{\frac{\delta^m}{\alpha_2}}. \end{aligned}$$

Then by iteration, we obtain

$$\begin{aligned} &\left\| (u+k)^{\alpha_2} \right\|_{W^{1,p_t^*}(B_{\rho_0})}^{\frac{1}{\alpha_2}} \\ &\leq 2^{\frac{(m-1)^2 \delta^m N}{(\delta-1)^2 \alpha_2 (N-1)}} \left[ C \left(1 + \|\mathbf{b}\|_{L^t(B_\sigma)}\right)^2 \right. \\ &\quad \left. \cdot \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right) \right]^{\frac{\delta^m - 1}{(\delta-1)\alpha_2}} \end{aligned}$$

$$\cdot \left\| (u+k)^{\frac{\alpha_2}{\delta^m}} \right\|_{W^{1,p_t^*}(B_{\rho_m})}^{\frac{\delta^m}{\alpha_2}}. \quad (3.29)$$

Subsequently, we fix  $m \in \mathbb{N}$ , such that  $\frac{\alpha_2}{\delta^m} \leq \alpha_1 \leq \frac{\alpha_2}{\delta^{m-1}}$ , and apply Hölder's inequality to estimate the right-hand of (3.29) as follows

$$\begin{aligned} & \left\| \nabla (u+k)^{\frac{\alpha_2}{\delta^m}} \right\|_{L^{p_t^*}(B_{\rho_m})} \\ & \leq |B_{\frac{2}{3}}|^{1-\frac{(q+1)p^*}{2q}} \left\| \nabla (u+k)^{\frac{\alpha_2}{\delta^m}} \right\|_{L^{\frac{2q}{q+1}}(B_{\rho_m})} \\ & \leq \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_{\frac{2}{3}})} \right)^2 \Lambda(B_{\frac{2}{3}}) \right] \\ & \quad \cdot \left\| (u+k)^{\frac{\alpha_2}{\delta^m}} \right\|_{L^{2p'}(B_{\frac{2}{3}})} \\ & \leq \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_{\frac{2}{3}})} \right)^2 \Lambda(B_{\frac{2}{3}}) \right] \\ & \quad \cdot \|u+k\|_{L^{\alpha_1 2p'}(B_{\frac{2}{3}})}^{\frac{\alpha_2}{\delta^m}}, \end{aligned}$$

where  $C$  depend on  $\alpha_2, N, p, q, t$  and

$$\Lambda(B_{\frac{2}{3}}) = \left( \|\mu\|_{L^p(B_{\frac{2}{3}})}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})} + \|\lambda^{-1}\|_{L^q(B_{\frac{2}{3}})}^2 \right).$$

Subsequently, one has

$$\begin{aligned} & \left\| (u+k)^{\alpha_2} \right\|_{W^{1,p_t^*}(B_{\rho_0})}^{\frac{1}{\alpha_2}} \\ & \leq \left[ C \left( 1 + \|\mathbf{b}\|_{L^t(B_\sigma)} \right)^2 \left( \|\mu\|_{L^p(B_\sigma)}^{\frac{1}{2}} \|\lambda^{-1}\|_{L^q(B_\sigma)}^{\frac{1}{2}} + \|\lambda^{-1}\|_{L^q(B_\sigma)} \right. \right. \\ & \quad \left. \left. + \|\lambda^{-1}\|_{L^q(B_\sigma)}^2 \right) \right]^{\frac{\delta \alpha_2 - \alpha_1}{(\delta-1)\alpha_2 \alpha_1} + 1} \|u+k\|_{L^{\alpha_1 2p'}(B_{\frac{2}{3}})}. \end{aligned} \quad (3.30)$$

The proof of Theorem 3.1.2 is completed.  $\square$



## Chapter 4

# Phragmén-Lindelöf-type Results for Functions in Homogeneous De Giorgi Classes

### 4.1 Main Results

Throughout this chapter  $\Omega \subset \mathbb{R}^N$  denotes an open set. For  $x_0 \in \mathbb{R}^N$  we write  $B_\rho(x_0)$  for the open ball centred at  $x_0$  with radius  $\rho > 0$ .

#### Definitions

**Definition 4.1.1** *Let  $\Omega \subseteq \mathbb{R}^N$  be open. A function  $u \in W_{\text{loc}}^{1,p}(\Omega)$  belongs to the De Giorgi class  $[DG]_p^\pm(\Omega, \hat{\gamma})$ ,  $1 < p \leq N$ , if there exists a constant  $\hat{\gamma} > 1$  such that for every  $\sigma \in (0, 1)$ ,  $\rho > 0$ ,  $x_0 \in \Omega$ ,  $k \in \mathbb{R}$  and every ball  $B_\rho(x_0) \subset \Omega$ ,*

$$\int_{B_{\sigma\rho}(x_0)} |D[(u - k)_\pm]|^p dx \leq \frac{\hat{\gamma}}{[(1 - \sigma)\rho]^p} \int_{B_\rho(x_0)} |(u - k)_\pm|^p dx. \quad (4.1)$$

We further set

$$[DG]_p(\Omega, \hat{\gamma}) := [DG]_p^+(\Omega, \hat{\gamma}) \cap [DG]_p^-(\Omega, \hat{\gamma}).$$

**Definition 4.1.2** *Let  $h \in W^{1,p}(\mathbb{R}^N) \cap C(\mathbb{R}^N)$ . For an open set  $\Omega \subset \mathbb{R}^N$  the De Giorgi class on the closure  $\bar{\Omega}$  is defined as follows. A function  $u \in [DG]_p^+(\Omega, \hat{\gamma})$  belongs to  $[DG]_p^+(\bar{\Omega}, h, \hat{\gamma})$  if  $(u - h)_+ \in W_0^{1,p}(\Omega)$  and, for every  $y \in \partial\Omega$ , every ball  $B_\rho(y)$ , every*

$\sigma \in (0, 1)$  and every  $k \geq \sup_{B_\rho(y) \cap \partial\Omega} h$ ,

$$\int_{B_{\sigma\rho}(y) \cap \Omega} |D[(u - k)_+]|^p dx \leq \frac{\hat{\gamma}}{[(1 - \sigma)\rho]^p} \int_{B_\rho(y) \cap \Omega} |(u - k)_+|^p dx.$$

The class  $[DG]_p^-(\bar{\Omega}, h, \hat{\gamma})$  is defined analogously by replacing  $(u - k)_+$  with  $(u - k)_-$  and assuming  $k \leq \inf_{B_\rho(y) \cap \partial\Omega} h$ . We set

$$[DG]_p(\bar{\Omega}, h, \hat{\gamma}) := [DG]_p^+(\bar{\Omega}, h, \hat{\gamma}) \cap [DG]_p^-(\bar{\Omega}, h, \hat{\gamma}).$$

**Remark 4.1.1** In Definitions 4.1.1–4.1.2 we used balls  $B_\rho$ ; equivalent definitions can be given with cubes  $K_\rho(y)$  of side-length  $2\rho$  centred at  $y$ . Whenever the centre is the origin we simply write  $B_\rho$  or  $K_\rho$ .

## Main theorems

**Theorem 4.1.1** Let  $1 < p \leq N$  and let  $u \in [DG]_p^+(\mathbb{R}_+^N, \hat{\gamma})$  be non-negative and not identically zero. Assume

$$\lim_{\mathbb{R}_+^N \ni y \rightarrow x \in \partial\mathbb{R}_+^N} u(y) = 0 \tag{4.2}$$

in the sense of Definition 4.1.2 with  $h \equiv 0$ . Then there exists  $0 < \alpha_1 < 1$ , depending only on  $N, p, \hat{\gamma}$ , such that

$$0 < \liminf_{r \rightarrow \infty} \frac{\mu_+(r)}{r^{\alpha_1}}, \quad \mu_+(r) := \sup_{B_r \cap \mathbb{R}_+^N} u.$$

**Theorem 4.1.2** Let  $1 < p \leq N$  and let  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$  be non-negative and not identically zero. Suppose  $u$  vanishes on an  $M$ -dimensional hyper-plane through the origin,

$$\mathbb{H}^M := \{(x_1, \dots, x_M, 0, \dots, 0)\}, \quad 1 \leq M \leq N - 1.$$

Then, if  $N - M < p \leq N$ , there exists  $0 < \alpha_2 < 1$ , depending only on  $N, p, M, \hat{\gamma}$ , such that

$$0 < \liminf_{r \rightarrow \infty} \frac{\mu_+(r)}{r^{\alpha_2}}, \quad \mu_+(r) := \sup_{B_r} u.$$

**Remark 4.1.2** Since  $u \in W^{1,p}(\Omega)$  is defined up to sets of  $p$ -capacity zero, the assumption that  $u$  vanishes on  $\mathbb{H}^M$  is meaningful; see Section 4.2.2 for the relevant definitions.

## Counter-examples

### First counter-example (Meyers)

The following example is due to Meyers [74]. In  $\mathbb{R}^2$  with coordinates  $(x, y)$  consider

$$\operatorname{div}(\mathbf{A} Du) = 0, \quad (4.3)$$

where

$$\mathbf{A} = \begin{bmatrix} 1 - (1 - \mu^2) \frac{y^2}{x^2 + y^2} & (1 - \mu^2) \frac{xy}{x^2 + y^2} \\ (1 - \mu^2) \frac{xy}{x^2 + y^2} & 1 - (1 - \mu^2) \frac{x^2}{x^2 + y^2} \end{bmatrix}, \quad 0 < \mu < 1.$$

The eigenvalues of  $\mathbf{A}$  are  $\mu^2$  and 1, hence the equation is uniformly elliptic. The function

$$u(x, y) = (x^2 + y^2)^{\frac{\mu-1}{2}} x \quad (4.4)$$

is a weak solution of (4.3) and vanishes on the line  $x = 0$ . Moreover,  $Du \in L^p_{\text{loc}}(\mathbb{R}^2)$  for  $1 < p < \frac{2}{1-\mu}$ . Thus, for a suitable  $\hat{\gamma}$ ,  $u$  belongs to  $[DG]_p^+(\mathbb{R}_+^2, \hat{\gamma})$ , showing that  $\alpha_1 = 1$  is impossible in Theorem 4.1.1.

The example extends to arbitrary dimension. Writing  $(x, y, z) \in \mathbb{R}^N$  with  $z \in \mathbb{R}^{N-2}$ , set  $u(x, y, z) = u(x, y)$  and extend  $\mathbf{A}$  trivially. Then

$$L_{xy}u + \Delta_z u = 0$$

has the same ellipticity constants, and the growth order  $\alpha_1 < 1$  persists.

### Second counter-example (extension of the previous idea)

We can generalise the preceding example in a different way. When  $N = 3$ , the function

$$u(x_1, x_2, x_3) = x_1 |x|^\alpha, \quad \alpha \in (-1, 0),$$

solves

$$\operatorname{div}(\mathbf{A} Du) = 0$$

with

$$\mathbf{A} = \left( \delta_{ij} + C(\alpha) Q_{ij}(x) \right), \quad C(\alpha) = -\frac{\alpha(\alpha + 3)}{2},$$

and  $Q_{ij}(x) = \frac{x_i x_j}{|x|^2} - \delta_{ij}$ . The eigenvalues are  $1, 1 - C, 1 - C$ , so the equation is uniformly elliptic. As before,  $\alpha_1 < 1$ .

### Third counter-example

For  $N = 4$  consider the elliptic 4-Laplacian

$$-\operatorname{div}(\mathbf{A}(x)|Du|^2 Du) = 0,$$

where  $\mathbf{A}(x)$  is the  $4 \times 4$  matrix with entries

$$a_{ij}(x) = \delta_{ij} - C(\alpha) \left( \delta_{ij} - \frac{x_i x_j}{|x|^2} \right), \quad C(\alpha) = 1 - \frac{(2 - 3\alpha)^2}{8}, \quad \alpha \in (0, 2/3).$$

The eigenvalues are  $1, 1 - C, 1 - C, 1 - C$ , hence the matrix is uniformly elliptic. The function

$$u(x) = \frac{(x_1^2 + x_2^2)^{\frac{1}{3}}}{|x|^\alpha}$$

is a subsolution. A direct computation shows

$$\sum_{i=1}^4 \frac{\partial}{\partial x_i} \left( |\nabla u|^2 [\mathbf{A} \nabla u]_i \right) = \frac{(2 - 3\alpha)^4}{54} |x|^{-3\alpha-2} > 0,$$

while  $u \in W_{\text{loc}}^{1,4}(\mathbb{R}^4)$ . Thus  $u \in [DG]_4^+(\mathbb{R}^4, \hat{\gamma})$  for some  $\hat{\gamma}$ , and  $u$  vanishes on  $\mathbb{H}^2 = \{x_1 = x_2 = 0\}$ . Since

$$\mu_+(r) \approx r^{\frac{2}{3}-\alpha} \quad \text{and} \quad \frac{2}{3} - \alpha < \frac{M}{N-1},$$

the exponent  $\alpha_2$  in Theorem 4.1.2 cannot reach the optimal value  $\frac{M}{N-1}$  obtained for the  $N$ -Laplacian in [68, Theorem 4.8]. Similar examples can be constructed for any  $p$  and any admissible  $M$ .

## Structure of the chapter

In Section 4.2 we first introduce the notation and discuss various notions of capacity and their mutual relations. We then collect the known properties of functions in the De Giorgi class, and in particular we establish a key logarithmic estimate. Based on the structural characteristics of the zero set of  $u$ , we present a new proof of the weak Harnack inequality. Sections 4.3 and 4.4 are devoted to the proofs of the main results for the cases  $1 < p < N$  and  $p = N$ , respectively. Finally, we provide a remark that briefly summarizes the techniques employed in the proofs of Theorems 4.1.1–4.1.2, as well as some applications of the main results of this chapter.



## 4.2 Preliminaries

### 4.2.1 Notation and Main Tools

A first simplification is the extension of  $u$  to  $\mathbb{R}^N$ . Indeed, for  $u \in [DG]_p^+(\mathbb{R}_+^N, \hat{\gamma})$ ,  $u$  has an upper-semicontinuous representative

$$u^*(x) = \begin{cases} \lim_{\rho \downarrow 0} \sup_{B_\rho(x)} u = \lim_{\rho \downarrow 0} \int_{B_\rho(x)} u(y) dy, & \text{for } x \in \mathcal{L}(u, \mathbb{R}_+^N), \\ 0 & \text{otherwise,} \end{cases}$$

where  $\mathcal{L}(u, \mathbb{R}_+^N)$  is the set of the Lebesgue points of  $u$ . Moreover, if  $u$  satisfies (4.2), then the zero-extension

$$\tilde{u}(x) = \begin{cases} u^*(x), & x \in \mathbb{R}_+^N, \\ 0, & x \in \mathbb{R}^N \setminus \mathbb{R}_+^N, \end{cases}$$

satisfies  $\tilde{u} \in W_{\text{loc}}^{1,p}(\mathbb{R}^N)$  and (4.1.2) with the integration extended over the full-balls. This allows us to work locally in  $\mathbb{R}^N$ . The same is valid when prescribing Dirichlet conditions at the boundary of  $\mathbb{H}^M$ , or a more general set  $E$  whose complement is  $p$  fat (see Remark 1.3). In order to keep the notation simple, in the sequel we will continue to denote by  $u$  its zero-extension.

In the following  $\omega_N$  stands for the  $(N-1)$ -dimensional measure of the surface of the unit ball in  $\mathbb{R}^N$ .

Given a measurable set  $\Omega \subset \mathbb{R}^N$ , for a function  $f \in L^1(\Omega)$  we let

$$\int_{\Omega} f(x) dx \equiv |\Omega|^{-1} \int_{\Omega} f(x) dx.$$

Finally, when we say that a constant  $C$  depends on the *data*, we mean that  $C$  depends on  $p, N$ .

### 4.2.2 Variational Capacity

Let  $K$  be a compact subset of  $\mathbb{R}^N$  and  $Q$  an open subset of  $\mathbb{R}^N$  containing  $K$ : for  $p \geq 1$ , the  $p$ -capacity of the set  $K$  with respect to the set  $Q$  is defined by

$$\text{cap}_p(K, Q) = \inf \left\{ \int_Q |D\varphi|^p dx : \varphi \in C_0^\infty(Q), \varphi \geq 1 \text{ on } K \right\}. \quad (4.5)$$

If  $U \subset Q$  is open, we let

$$\text{cap}_p(U, Q) = \sup_{K \subset U, \text{compact}} \text{cap}_p(K, Q),$$

and for an arbitrary set  $V \subset Q$  we define

$$\text{cap}_p(V, Q) = \inf_{V \subset U \subset Q, U \text{ open}} \text{cap}_p(U, Q).$$

Let us state the main properties of  $\text{cap}_p$ . For their proofs, we refer to [40], or to [47, Chapter 2].

**Proposition 4.2.1** *Let  $1 < p \leq N$ .*

1.  $\text{cap}_p(\emptyset, Q) = 0$ ;
2. *Given two sets  $K_1 \subset K_2 \subset\subset Q$ , we have  $\text{cap}_p(K_1, Q) \leq \text{cap}_p(K_2, Q)$ ;*
3. *Given two compact sets  $K_1, K_2 \subset\subset Q$ , we have*

$$\text{cap}_p(K_1 \cap K_2, Q) + \text{cap}_p(K_1 \cup K_2, Q) \leq \text{cap}_p(K_1, Q) + \text{cap}_p(K_2, Q);$$

4. *For every sequence  $K_j$  of compact subsets of  $Q$  such that  $K_1 \supset \dots \supset K_j \supset K_{j+1} \dots$ , we have*

$$\text{cap}_p(\cap_{j=1}^{+\infty} K_j, Q) = \lim_{j \rightarrow \infty} \text{cap}_p(K_j, Q);$$

5. *For every sequence  $K_j$  of subsets of  $Q$  such that  $K_1 \subset \dots \subset K_j \subset K_{j+1} \dots$ , we have*

$$\text{cap}_p(\cup_{j=1}^{+\infty} K_j, Q) = \lim_{j \rightarrow \infty} \text{cap}_p(K_j, Q);$$

6. *If  $V = \cup_i V_i \subset Q$ , we have  $\text{cap}_p(V, Q) \leq \sum_i \text{cap}_p(V_i, Q)$ ;*

7. *Let  $B_r(x_o), B_R(x_o) \subseteq \mathbb{R}^N$  for any  $x_o \in \mathbb{R}^N$ ,  $0 < r < R$ . Then, when  $1 < p < N$ , we have*

$$\text{cap}_p(B_r(x_o), B_R(x_o)) = \omega_N |q|^{p-1} |R^q - r^q|^{1-p}, \quad q = \frac{p-N}{p-1};$$

*when  $p = N$ , we have*

$$\text{cap}_p(B_r(x_o), B_R(x_o)) = \omega_N \left( \log \frac{R}{r} \right)^{1-N};$$

8. Given a compact set  $K \subset Q$  and  $\lambda > 0$ , we have

$$\text{cap}_p(\lambda K, \lambda Q) = \lambda^{N-p} \text{cap}_p(K, Q),$$

for any  $1 < p < N$ ;

9. Given a set  $A \subset Q$ , we have  $|A| \leq C[\text{cap}_p(A, Q)]^{\frac{N}{N-p}}$  for any  $1 < p < N$  and for some constant  $C$  depending only on  $p$  and  $N$ ;

10. Given a set  $A \subset Q$ , we have  $\text{cap}_p(A, Q) \leq \mathcal{H}^{N-p}(A)$  for any  $1 < p < N$ , where  $\mathcal{H}^s$  is the  $s$ -dimensional Hausdorff measure; the constant  $C$  depends only on  $p$  and  $N$ .

The last statement can be refined. Once more, we refer to [40] for the proofs.

**Proposition 4.2.2** *Let  $1 < p < N$  and  $A \subset\subset Q$ . If  $\mathcal{H}^{N-p}(A) < \infty$ , then  $\text{cap}_p(A, Q) = 0$ . On the other hand,  $1 < p \leq N$  and  $\text{cap}_p(A, Q) = 0$ , the  $\mathcal{H}^s(A) = 0$  for all  $s > N - p$ .*

### 4.2.3 Fat Sets

We come to a notion that plays a fundamental role in the following.

**Definition 4.2.1** *Given a set  $E \subset \mathbb{R}^N$ , for  $1 < p < N$  we say that  $E$  is  $p$ -locally uniformly fat if there exist  $r_o > 0$  and  $c_o > 0$  such that for any  $0 < r < r_o$  and for every  $x \in E$  we have*

$$\frac{\text{cap}_p(E \cap B_r(x), B_{2r}(x))}{\text{cap}_p(B_r(x), B_{2r}(x))} \geq c_o. \quad (4.6)$$

If  $p = N$ , then (4.6) is replaced by

$$\text{cap}_N(E \cap B_r(x), B_{2r}(x)) \geq c_o.$$

In order to understand the role played by such a notion in our context, let us first consider the case of  $E = \mathbb{R}_-^N$ .

It is easy to see that  $E$  is  $p$ -locally uniformly fat for any  $1 < p \leq N$  with  $r_o = +\infty$ . First of all, it is apparent that it suffices to consider  $x$  which belongs to the hyperplane  $x_N = 0$ .

If we let  $B_r^-(x) \stackrel{\text{def}}{=} B_r(x) \cap \mathbb{R}_-^N$ , for any  $p \in (1, N)$ , by (7) and (9) of Proposition 4.2.1 we have

$$\text{cap}_p(E \cap B_r(x), B_{2r}(x)) = \text{cap}_p(B_r^-(x), B_{2r}(x)) \geq C(N, p)r^{N-p}$$

$$\Rightarrow \frac{\text{cap}_p(E \cap B_r(x), B_{2r}(x))}{\text{cap}_p(B_r(x), B_{2r}(x))} \geq C(N, p).$$

If  $p = N$ , we have

$$\text{cap}_N(E \cap B_r(x), B_{2r}(x)) = \text{cap}_N(B_r^-(x), B_{2r}(x)) \geq \text{cap}_N(J_r(x), B_{2r}(x)),$$

where  $J_r = \{(0, 0, \dots, 0, x_N) : -r < x_N < 0\}$ ; by [67] we have

$$\text{cap}_N(J_r(x), B_{2r}(x)) > \frac{\omega_{N-2}}{\kappa_N^{N-1}} \ln 3,$$

where  $\kappa_N = \int_0^{\frac{\pi}{2}} (\sin t)^{\frac{2-N}{N-1}} dt$ , whence the conclusion immediately follows.

Now, let us consider  $E = \mathbb{H}^M$ , the  $M$ -dimensional hyperplane through the origin, that is

$$\mathbb{H}^M = \{(x_1, \dots, x_M, 0, 0, \dots, 0)\} \quad \text{with } 1 \leq M \leq N-1.$$

Again, we have that  $E$  is  $p$ -locally uniformly fat with  $r_o = \infty$  for any  $1 \leq M \leq N-1$ , provided that  $N-M < p \leq N$ . This is proved in [70, Lemma 3.2].

Moreover, when  $p = N$ , it is a straightforward consequence of the following estimate proven in [4]:

$$\text{cap}_N(E \cap B_r(x), B_{2r}(x)) \geq \omega_{q-1} \omega_{N-q-1} 2^{1-q} \frac{\left[ \int_1^3 \frac{(t^2-1)^{q-1}}{t^q} dt \right]}{\left[ \int_0^{\frac{\pi}{2}} (\sin t)^{\frac{1+q-N}{N-1}} dt \right]^{N-1}},$$

where  $x \in \mathbb{H}^M$ , and  $q = 1, 2, \dots, N-1$ .

**Remark 4.2.1** It is worth pointing out that the explicit value of the 2-capacity of a disk  $D_r$  of radius  $r$  or of a semi-ball  $B_r^-$  of the same radius with respect to  $\mathbb{R}^3$  are given in [62, Chapter II, § 3, No. 14]. We have

$$\text{cap}_2(D_r, \mathbb{R}^3) = \frac{2r}{\pi^2}, \quad \text{cap}_2(B_r^-, \mathbb{R}^3) = \frac{2r}{\pi} \left( 1 - \frac{1}{\sqrt{3}} \right).$$

Finally, concerning  $p$ -locally uniformly fat sets, we have the following result.

**Proposition 4.2.3 (Lewis, [65])** *Given  $1 < p \leq N$ , suppose  $E \subset \mathbb{R}^N$  is closed and  $p$ -locally uniformly fat with constants  $c_o$  and  $r_o$ . Then, there exist  $\varepsilon_o \in (0, 1)$  and  $c_1 > 0$  depending only on  $p, N, c_o$ , such that whenever  $x \in E$ ,  $0 < r < r_o$ , and  $p - \varepsilon_o < q < p$ , the set  $E$  is  $q$ -locally uniformly fat with constants  $c_1$  and  $r_o$ , i.e.*

$$\frac{\text{cap}_q(E \cap B_r(x), B_{2r}(x))}{\text{cap}_q(B_r(x), B_{2r}(x))} \geq c_1.$$

The original Lewis' proof does not give an explicit functional dependence of  $c_1$  on  $c_o$ . However, the proof provided in [75], later extended to metric spaces in [12] (see also [63]), allows to conclude that

$$c_1 = C(N, p)c_o.$$

#### 4.2.4 A Different Notion of Capacity for $1 < p < N$

An equivalent notion of capacity can be given, assuming  $Q \equiv \mathbb{R}^N$ . Let us first consider the case where  $1 < p < N$ . We do not go into details here and limit ourselves to underlining the link between capacity in this new framework and nonlinear potential theory; for more information, we refer the interested reader to [73]. For simplicity, we will write  $\text{cap}_p(K)$  instead of  $\text{cap}_p(K, \mathbb{R}^N)$ .

There exists a Radon measure  $\mu_K$  supported in  $K$  such that

$$\text{cap}_p(K) = \int_K d\mu_K. \quad (4.7)$$

Such a measure, called *capacitary distribution of  $K$* , can be identified with an element of  $[W^{1,p}(\mathbb{R}^N)]^*$ , and generates a potential  $\mathcal{U}_K$  by the formula

$$\mathcal{U}_K(x) \equiv \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left\{ \frac{d\mu_K(z)}{|y-z|^{N-1}} \right\}^{\frac{1}{p-1}} \frac{dy}{|x-y|^{N-1}}. \quad (4.8)$$

**Remark 4.2.2** If  $p = 2$ , by the Riesz composition formula (4.8) implies

$$\mathcal{U}_K(x) \equiv \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-2}},$$

which is, up to a normalizing constant, the classical newtonian potential generated by  $\mu_K$ .

By (2.10) of [73], the energy associated with the potential  $\mathcal{U}_K$  is

$$\mathcal{E}(\mu_K) \equiv \|\mathcal{U}_K\|_{W^{1,p}(\mathbb{R}^N)}^p = \int_{\mathbb{R}^N} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx.$$

The connection between these quantities is that the infimum in the definition of capacity (4.5) is achieved when  $\varphi \equiv \mathcal{U}_K$ . Therefore,

$$\text{cap}_p(K) = \int_{\mathbb{R}^N} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx. \quad (4.9)$$

Let us now state and prove a Poincaré-type inequality.

**Proposition 4.2.4** *Let  $v \in W^{1,1}(B_R)$  and assume that  $v$  vanishes on a subset  $E_0$  of  $B_R$ . Let  $K$  be any compact subset of  $E_0$  and let  $\mu_K$  denote its capacitary distribution. There exists a constant  $\gamma$  depending only upon  $N$  such that for  $1 < p < N$ ,*

$$\text{cap}_p(K) \int_{B_R} |v| dx \leq \gamma R^N \int_{B_R} |Dv(x)| \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-1}} \right\} dx. \quad (4.10)$$

**Proof.** Without loss of generality, we may assume that  $v$  is continuous and non-negative. Let  $z \in K \subset E_0$  and  $x \in B_R \setminus E_0$ . Then

$$v(x) - v(z) = \int_0^{|x-z|} \frac{d}{dr} v(z + r\nu) dr \leq \int_0^{|x-z|} |Dv(y)| dr,$$

where  $\nu \equiv \frac{x-z}{|x-z|}$ ,  $y = z + r\nu$ .

Let us integrate this with respect to the Lebesgue measure  $dx$  over all  $x \in B_R \setminus E_0$  and then in  $d\mu_K(z)$  over all  $z \in K$ . Recalling that  $v$  vanishes in  $K$  and taking into account (4.7) we obtain

$$\text{cap}_p(K) \int_{B_R} v dx \leq \int_K d\mu_K(z) \int_{B_R \setminus E_0} dx \left\{ \int_0^{|x-z|} |Dv(y)| dr \right\}. \quad (4.11)$$

We estimate the last two integrals for  $z \in K$  fixed, by expressing them in polar coordinates with pole at  $z$ . If  $\tilde{x} \equiv \tilde{x}(z, \nu)$  is the polar representation of  $\partial B_R$  with pole at  $z$ , we have

$$\begin{aligned} & \int_{B_R \setminus E_0} dx \left\{ \int_0^{|x-z|} |Dv(y)| dr \right\} \leq \int_{B_R} dx \left\{ \int_0^{|x-z|} |Dv(y)| dr \right\} \\ &= \int_0^{\tilde{x}(z, \nu)} |x-z|^{N-1} d|x-z| \int_{|\nu|=1} d\nu \left\{ \int_0^{|x-z|} \frac{|Dv(y)|}{|y-z|^{N-1}} |y-z|^{N-1} d|y-z| \right\} \\ &\leq \int_0^{2R} |x-z|^{N-1} d|x-z| \left\{ \int_{|\nu|=1} \int_0^{|x-z|} \frac{|Dv(y)|}{|y-z|^{N-1}} |y-z|^{N-1} d|y-z| d\nu \right\} \\ &\leq \gamma(N) R^N \int_{B_R} \frac{|Dv(y)|}{|y-z|^{N-1}} dy. \end{aligned}$$

We substitute this in (4.11) and interchange the order of integration with the aid of Fubini's theorem to obtain (4.10). ■

**Remark 4.2.3** Proposition 4.2.4 continues to hold if  $B_R$  is replaced by any convex set  $\mathcal{C}$ .

In the following we will employ Proposition 4.2.4 in a special form. Indeed, let  $u \in$

$W^{1,1}(B_{2R})$  and let  $l$  and  $k$  be any two numbers satisfying  $l > k$ . Define

$$v = \begin{cases} l - k, & \text{if } u > l, \\ u - k, & \text{if } k \leq u \leq l, \\ 0, & \text{if } u < k, \end{cases}$$

and set

$$A_{l,R} \equiv \{x \in B_R : u(x) > l\}, \quad R > 0.$$

If  $A_{k,R}$  is defined analogously, from (4.10) applied to  $v$  we deduce

$$\text{cap}_p(K) \int_{B_R} v \, dx \leq CR^N \int_{A_{k,\frac{3}{2}R} \setminus A_{l,\frac{3}{2}R}} |Du(x)| \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-1}} \right\} dx, \quad (4.12)$$

for every compact set  $K \subset \{x \in B_R : u(x) < k\}$ . From (4.12) we finally obtain

$$(l - k) \text{cap}_p(K) |A_{l,R}| \leq cR^N \int_{A_{k,\frac{3}{2}R} \setminus A_{l,\frac{3}{2}R}} |Du(x)| \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|x-z|^{N-1}} \right\} dx, \quad (4.13)$$

for every compact set  $K \subset \{x \in B_R : u(x) < k\}$ .

As a direct further consequence of Proposition 4.2.4 we have the following.

**Corollary 4.2.1** *Let  $v \in W^{1,p}(B_R)$ ,  $1 < p < N$ , satisfy the assumptions of Proposition 4.2.4 and let*

$$\delta(K; R) \equiv \frac{\text{cap}_p(K)}{R^{N-p}}.$$

*There exists a constant  $\gamma$  depending only upon  $N$  such that*

$$[\delta(K; R)]^{1/p} \int_{B_R} |v| \, dx \leq \gamma R \left( \int_{B_R} |Dv(x)|^p \, dx \right)^{1/p}. \quad (4.14)$$

**Proof.** From (4.10) by the Hölder inequality

$$\begin{aligned} \text{cap}_p(K) \int_{B_R} |v| \, dx \\ \leq \gamma R^N \left( \int_{B_R} |Dv(x)|^p \, dx \right)^{1/p} \left( \int_{B_R} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_K(z)}{|y-z|^{N-1}} \right\}^{\frac{p}{p-1}} dy \right)^{\frac{p-1}{p}}. \end{aligned}$$

Now (4.14) follows from (4.9) and the definition of  $\delta(K; R)$ . ■

### 4.2.5 A Different Notion of Capacity for $p = N$

When  $p = N$  things are more delicate. Estimates (4.7)–(4.8) and (4.9) continue to hold (see [73], but also [1, Chapter 2]), but they need to be written not in terms of the Riesz kernel

$$I_1 \stackrel{\text{def}}{=} \frac{\gamma_1}{|x|^{N-1}}, \quad \gamma_1 = \frac{\Gamma\left(\frac{N-1}{2}\right)}{2\pi^{\frac{N+1}{2}}},$$

but in terms of the Bessel kernel

$$G_1 \stackrel{\text{def}}{=} \mathcal{F}^{-1} \left\{ \frac{1}{(1 + |\xi|^2)^{\frac{1}{2}}} \right\} \equiv \frac{1}{(2\pi^N)} \int_{\mathbb{R}^N} \frac{e^{ix \cdot \xi}}{(1 + |\xi|^2)^{\frac{1}{2}}} d\xi,$$

or also

$$G_1 = \frac{1}{(2\pi)^{\frac{N}{2}}} |x|^{-\frac{N-2}{2}} \int_0^\infty \frac{t^{\frac{N}{2}}}{(1 + t^2)^{\frac{1}{2}}} J_{\frac{N-2}{2}}(|x|t) dt,$$

where  $J_\nu$  denotes the Bessel function of order  $\nu$ . It is worth pointing out that

$$G_1(x) \sim I_1(x), \quad \text{as } |x| \rightarrow 0,$$

and that for any  $c \in (0, 1)$

$$G_1(x) = O(e^{-c|x|}) \quad \text{as } |x| \rightarrow \infty.$$

Coming to Proposition 4.2.4, the estimate

$$\int_{B_R \setminus E_0} dx \left\{ \int_0^{|x-z|} |Dv(y)| dr \right\} \leq \gamma(N) R^N \int_{B_R} \frac{|Dv(y)|}{|y-z|^{N-1}} dy$$

obviously continues to hold. When we substitute it into (4.11), take into account that  $|y-z| \leq 2R$ , and interchange the order of integration with the aid of Fubini's Theorem, we can observe that

$$\begin{aligned} \int_K \frac{d\mu_K(z)}{|y-z|^{N-1}} &= \int_K e^{\frac{|y-z|}{2R}} e^{-\frac{|y-z|}{2R}} \frac{d\mu_K(z)}{|y-z|^{N-1}} \\ &\leq \gamma \int_K e^{-\frac{|y-z|}{2R}} \frac{d\mu_K(z)}{|y-z|^{N-1}} \\ &\leq \gamma \int_{\mathbb{R}^N} G_1(|x-z|) d\mu_K(z). \end{aligned}$$

Hence, (4.10) becomes

$$\text{cap}_N(K) \int_{B_R} |v| dx \leq \gamma R^N \int_{B_R} |Dv(x)| \left\{ \int_{\mathbb{R}^N} G_1(|x-z|) d\mu_K(z) \right\} dx.$$

Once we recall that for  $p = N$

$$\text{cap}_N(K) = \int_{\mathbb{R}^N} \left\{ \int_{\mathbb{R}^N} G_1(|x-z|) d\mu_K(z) \right\}^{\frac{N}{N-1}} dx,$$

it is apparent that Corollary 2.1 remains the same.



## 4.2.6 A Final Comment About Capacities

Due to the equivalences among the different notions of capacity discussed in Sections 4.2.2, 4.2.4, 4.2.5, in the following we will use them in an interchangeable way; in particular, given two quantities  $A$  and  $B$ ,  $A \approx B$  will mean that their quotient is bounded from above and from below by two constant that depend only on the data.

## 4.2.7 The De Giorgi Lemma

The following lemma is well-known; for its proof we refer to [59]. We do not state it in its full generality, we rather concentrate on a specific case tailored on the problem we are working on.

**Lemma 4.2.1** *Let  $\Omega \subset \mathbb{R}^N$  be an open domain, take  $s \geq 2$ , assume  $u \in [DG]_p^+(\Omega, \hat{\gamma})$ , let  $R > 0$  and  $x_o \in \Omega$  be such that  $B_{2R}(x_o) \subset \Omega$ , and define  $\mu_+ = \sup_{B_{2R}(x_o)} u$ . There exists a number  $\theta_o \in (0, 1)$ , depending only upon  $\hat{\gamma}$  such that if*

$$\left| \left\{ x \in B_R(x_o) : u(x) > \mu_+ \left( 1 - \frac{1}{2^s} \right) \right\} \right| < \theta_o |B_R|,$$

then

$$u(x) \leq \mu_+ \left( 1 - \frac{1}{2^{s+1}} \right), \quad \forall x \in B_{\frac{5}{6}R}(x_o).$$

**Remark 4.2.4** The value of  $\theta_o$  forces the reduction of the oscillation in the sense that we can accept a larger (namely closer to 1) value of  $\theta_o$  if we can take  $s+l$  with  $l \gg 1$  instead of  $s+1$  in Lemma 4.2.1. For the convenience of the proof below, we provide a specific form of  $\theta_o$

$$\theta_o = \min \left\{ \frac{5^{N+1} \omega_N}{N 6^{N+1}}, \frac{\omega_N^N}{N^N 6^{4N+2N^2} \hat{\gamma}^{\frac{N}{p}} \beta^N} \right\},$$

$\beta$  is a positive constant, identical to the one in Lemma 3.5, Chapter 2 of [59].

## 4.2.8 Boundedness and Weak Harnack Inequality

A function  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$  is locally bounded above, and it satisfies the following local estimate.

**Lemma 4.2.2 (Thm. 1 in [35])** *Let  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ . Then, for any  $\sigma \in (0, 1)$ ,  $q > 0$ , there exists  $C > 0$  depending only on the data  $\{N, p\}$ , and  $q$  such that for all*

fixed  $\rho > 0$ ,

$$\sup_{B_{\sigma\rho}} u_+ \leq \left( \frac{\hat{\gamma}^{\frac{N}{p}} C}{(1-\sigma)^N} \int_{B_\rho} u_+^q dx \right)^{\frac{1}{q}}.$$

Another fundamental tool is the Weak Harnack inequality for function in the “complementary” De Giorgi class.

**Lemma 4.2.3 (Theorem 2 in [35])** *Let  $v \in [DG]_p^-(\mathbb{R}^N, \hat{\gamma})$  be non-negative and let  $\rho > 0$ . Then, there exists  $\tau > 0$  depending only on the data  $\{p, N, \hat{\gamma}\}$  with the following property: for every  $\sigma, \eta \in (0, 1)$  there exists a constant  $C = C(\tau, \sigma, \eta, p, N, \hat{\gamma}) > 0$  such that*

$$\left( \int_{B_{\sigma\rho}} v^\tau dx \right)^{\frac{1}{\tau}} \leq C(\tau, \sigma, \eta, p, N, \hat{\gamma}) \inf_{B_{\eta\rho}} v. \quad (4.15)$$

**Remark 4.2.5** In Lemma 4.2.2 there is no need to assume  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ ; it suffices to have  $v \in [DG]_p^+(\Omega, \hat{\gamma})$ , where  $\Omega \subset \mathbb{R}^N$  is a bounded, open subset, such that  $B_{2\rho} \subset \Omega$ . Analogous remark holds for Lemma 4.2.3.

**Remark 4.2.6** It is a matter of straightforward computations to check that if  $u \in W_{\text{loc}}^{1,p}(\Omega)$  is bounded above,  $\mu_+ = \sup_\Omega u$ , and  $u \in [DG]_p^+(\Omega, \hat{\gamma})$ , then  $\mu_+ - u \in [DG]_p^-(\Omega, \hat{\gamma})$ .

**Remark 4.2.7** The proof of Lemma 4.2.3 relies in a fundamental way on a delicate covering argument, originally due to Krylov & Safonov [57]. The natural question arises, whether in the particular framework we are dealing with now a simpler proof is actually possible, in the spirit of the original Moser’s approach. The answer is in the affirmative, as we will show in § 4.2.12.

#### 4.2.9 A Logarithmic Estimate for $1 < p < N$

Let  $\Omega$  be either  $\mathbb{R}_-^N$  or  $\mathbb{H}^M$  for  $1 \leq M \leq N-1$  and  $N-M < p < N$ ; take  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ , assume that  $u \geq 0$ ,  $u \not\equiv 0$ , and vanishes in  $\Omega$ . For simplicity, in the sequel we refer to balls centered at the origin.

For  $R > 0$  fixed, let

$$Q(R) \stackrel{\text{def}}{=} \Omega \cap \overline{B_R}, \quad \delta(R) \stackrel{\text{def}}{=} \frac{\text{cap}_p(Q(R))}{R^{N-p}}.$$

We have the following.

**Proposition 4.2.5** *Under the previous assumptions, for any  $\sigma \in (0, \frac{p-1}{p})$  there exists  $C > 1$  which depends on  $N, p, M$ , and  $\sigma$  such that*

$$\int_{B_R} \left[ \ln \left( \frac{\mu_+}{\mu_+ - u} \right) \right]^\sigma dx \leq \frac{C \hat{\gamma}^{\frac{N+1}{p}}}{[\delta(R)]^{\frac{1}{p}}} |B_R|,$$

where  $\mu_+ = \sup_{B_{2R}} u$ .

**Proof.**  $Q(R)$  is a compact set and let  $\mu_{Q(R)}$  be its capacitary distribution. For simplicity of notation define

$$\mu_+ = \sup_{B_{2R}} u, \quad A_{s,R} \equiv \left\{ x \in B_R : u(x) > \mu_+ \left( 1 - \frac{1}{2^s} \right) \right\}, \quad s = 4, 5, \dots,$$

and notice that  $Q(R) \subset \left\{ x \in B_R : u(x) < \mu_+ \left( 1 - \frac{1}{2^s} \right) \right\}$ ,  $s = 4, 5, \dots$

Applying (4.13) with  $l = \mu_+ \left( 1 - \frac{1}{2^{s+1}} \right)$ ,  $k = \mu_+ \left( 1 - \frac{1}{2^s} \right)$ , we obtain

$$\begin{aligned} & \frac{\mu_+}{2^{s+1}} \text{cap}_p(Q(R)) |A_{s+1,R}| \\ & \leq cR^N \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} |Du(x)| \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\} dx \\ & \leq cR^N \left( \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left| D \left( u - \mu_+ \left( 1 - \frac{1}{2^s} \right) \right)_+ \right|^p dx \right)^{\frac{1}{p}} \\ & \quad \cdot \left( \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \right)^{\frac{p-1}{p}} \\ & \leq cR^N \left( \int_{B_{\frac{3}{2}R}} \left| D \left( u - \mu_+ \left( 1 - \frac{1}{2^s} \right) \right)_+ \right|^p dx \right)^{\frac{1}{p}} \\ & \quad \cdot \left( \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \right)^{\frac{p-1}{p}}. \end{aligned}$$

On the other hand,

$$\begin{aligned} \int_{B_{\frac{3}{2}R}} \left| D \left( u - \mu_+ \left( 1 - \frac{1}{2^s} \right) \right)_+ \right|^p dx & \leq \frac{\hat{\gamma}}{R^p} \int_{B_{2R}} \left| \left( u - \mu_+ \left( 1 - \frac{1}{2^s} \right) \right)_+ \right|^p dx, \\ \int_{B_{2R}} \left| \left( u - \mu_+ \left( 1 - \frac{1}{2^s} \right) \right)_+ \right|^p dx & \leq c \left( \frac{\mu_+}{2^s} \right)^p R^N. \end{aligned}$$

Collecting all the previous estimates yields

$$\begin{aligned} & \text{cap}_p(Q(R)) |A_{s+1,R}| \\ & \leq c\hat{\gamma}^{\frac{1}{p}} R^{\frac{Np+N-p}{p}} \cdot \left( \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \right)^{\frac{p-1}{p}}, \end{aligned}$$

and raising both sides to the  $\frac{p}{p-1}$  power we obtain

$$\begin{aligned} & \left[ \text{cap}_p(Q(R)) \right]^{\frac{p}{p-1}} |A_{s+1,R}|^{\frac{p}{p-1}} \\ & \leq c\hat{\gamma}^{\frac{1}{p-1}} R^{\frac{Np+N-p}{p-1}} \left( \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \right). \end{aligned}$$

We add these inequalities for  $s = 4, 5, \dots, s^* - 2$ , where  $s^*$  is a positive integer which can be chosen arbitrarily large; since  $A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}$  are pairwise disjoint sets, the right-hand side can be majorized with a convergent series. We have

$$\begin{aligned} & (s^* - 5) \left[ \text{cap}_p(Q(R)) \right]^{\frac{p}{p-1}} |A_{s^*-1,R}|^{\frac{p}{p-1}} \\ & \leq cR^{\frac{Np+N-p}{p-1}} \sum_{s=1}^{\infty} \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx, \end{aligned} \quad (4.16)$$

and recalling (4.9) we have

$$\begin{aligned} & \sum_{s=1}^{\infty} \int_{A_{s,\frac{3}{2}R} \setminus A_{s+1,\frac{3}{2}R}} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \leq \int_{\mathbb{R}^N} \left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}} dx \\ & = \text{cap}_p(Q(R)). \end{aligned}$$

From (4.16) we deduce

$$\begin{aligned} & (s^* - 5) \left[ \text{cap}_p(Q(R)) |A_{s^*-1,R}| \right]^{\frac{p}{p-1}} \leq cR^{\frac{Np}{p-1}} R^{\frac{N-p}{p-1}} \text{cap}_p(Q(R)), \\ & |A_{s^*-1,R}| \leq \frac{c\hat{\gamma}^{\frac{1}{p}} R^N}{(s^* - 5)^{\frac{p-1}{p}}} \left[ \frac{\text{cap}_p(Q(R))}{R^{N-p}} \right]^{-\frac{1}{p}}. \end{aligned}$$

The previous inequality can be rewritten as

$$\left| \left[ u > \mu_+ \left( 1 - \frac{1}{2^s} \right) \right] \cap B_R \right| \leq \frac{c}{(s-4)^{\frac{p-1}{p}} [\delta(R)]^{\frac{1}{p}}} |B_R|, \quad s \geq 5. \quad (4.17)$$

If we define

$$v \stackrel{\text{def}}{=} \ln \left( \frac{\mu_+}{\mu_+ - u} \right),$$

taking also into account the change of base in the logarithm, (4.17) yields

$$\begin{aligned} |[v > s] \cap B_R| &= \left| \left[ u > \mu_+ \left( 1 - \frac{1}{e^s} \right) \right] \cap B_R \right| \leq \left| \left[ u > \mu_+ \left( 1 - \frac{1}{2^s} \right) \right] \cap B_R \right| \\ &\leq \frac{c \hat{\gamma}^{\frac{N+1}{p}}}{s^{\frac{p-1}{p}} [\delta(R)]^{\frac{1}{p}}} |B_R|. \end{aligned}$$

If we now pick any  $\sigma \in (0, \frac{p-1}{p})$ , we have

$$\begin{aligned} \int_{B_R} v^\sigma dx &= \int_0^{+\infty} s^{\sigma-1} |[v > s] \cap B_R| ds \\ &= \int_0^5 s^{\sigma-1} |[v > s] \cap B_R| ds + \int_5^{+\infty} s^{\sigma-1} |[v > s] \cap B_R| ds \\ &\leq \frac{1}{\sigma} |B_R| + c \hat{\gamma}^{\frac{N+1}{p}} |B_R| \int_5^\infty \frac{s^{\sigma-1}}{(s-4)^{\frac{p-1}{p}} [\delta(R)]^{\frac{1}{p}}} ds \\ &\leq \left[ \frac{1}{\sigma} + \frac{c \hat{\gamma}^{\frac{N+1}{p}}}{\left( \frac{p-1}{p} - \sigma \right) [\delta(R)]^{\frac{1}{p}}} \right] |B_R| = \frac{C(\sigma) \hat{\gamma}^{\frac{N+1}{p}}}{[\delta(R)]^{\frac{1}{p}}} |B_R|, \end{aligned}$$

and we conclude. ■

#### 4.2.10 A Logarithmic Estimate for $p = N$

As before, we let  $\Omega$  be either  $\mathbb{R}_-^N$  or  $\mathbb{H}^M$  for  $1 \leq M \leq N-1$ ; take  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ , assume that  $u \geq 0$ , and vanishes in  $\Omega$ .

**Proposition 4.2.6** *Under the previous assumptions, for any  $\sigma \in (0, \frac{N-1}{N})$  there exists  $C > 1$  which depends on  $N$ ,  $M$ , and  $\sigma$  such that*

$$\int_{B_R} \left[ \ln \left( \frac{\mu_+}{\mu_+ - u} \right) \right]^\sigma dx \leq \frac{C \hat{\gamma}^{\frac{N+1}{N}}}{[\delta(R)]^{\frac{1}{N}}} |B_R|,$$

where  $\mu_+ = \sup_{B_{2R}} u$ .

The proof is largely similar to the one of Proposition 4.2.5; indeed, it suffices to substitute

$$\left\{ \int_{\mathbb{R}^N} \frac{d\mu_{Q(R)}(z)}{|x-z|^{N-1}} \right\}^{\frac{p}{p-1}}$$

with

$$\left\{ \int_{\mathbb{R}^N} G_1(|x-z|) d\mu_{Q(R)}(z) \right\}^{\frac{N}{N-1}},$$

$\text{cap}_p(Q(R))$  with  $\text{cap}_N(Q(R))$ , and recall that for  $p = N$  we have  $\delta(R) = \text{cap}_N(Q(R))$ .

### 4.2.11 De Giorgi Classes $[DG]_p^\pm(\Omega, \gamma)$ and Sub(Super)-Harmonic Functions

For any open set  $\Omega \subset \mathbb{R}^N$ , the *generalized De Giorgi classes*  $[GDG]_p^\pm(\Omega, \gamma)$  are the collection of functions  $u \in W_{\text{loc}}^{1,p}(\Omega)$ , for some  $p > 1$ , satisfying

$$\int_{B_\rho(y)} |D(u - k)_\pm|^p dx \leq \frac{\gamma}{(R - \rho)^p} \left( \frac{R}{R - \rho} \right)^{Np} \int_{B_R(y)} |(u - k)_\pm|^p dx,$$

for all balls  $B_\rho(y) \subset B_R(y) \subset \Omega$ , and all  $k \in \mathbb{R}$ , for a given positive constant  $\gamma$ . An equivalent definition can be given in terms of cubes, instead of balls.

Convex, monotone, non-decreasing functions of sub-harmonic functions are sub-harmonic. Similarly, concave, non-decreasing, functions of super-harmonic functions are super-harmonic. Similar statements hold for weak, sub(super)-solutions of linear elliptic equations with measurable coefficients [78]. The next lemma establishes analogous properties for functions  $u \in [DG]_p^\pm(\Omega, \hat{\gamma})$ .

**Lemma 4.2.4 (Lemma 2.1 in [34])** *Let  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  be convex and non-decreasing, and let  $u \in [DG]_p^+(\Omega, \hat{\gamma})$ . There exists a positive constant  $\bar{\gamma}$  depending only on the data,  $\hat{\gamma}$ , and independent of  $u$ , such that  $\varphi(u) \in [GDG]_p^+(\Omega, \bar{\gamma})$ .*

*Likewise let  $\psi : \mathbb{R} \rightarrow \mathbb{R}$  be concave and non-decreasing, and let  $u \in [DG]_p^-(\Omega, \hat{\gamma})$ . There exist a positive constant  $\bar{\gamma}$  depending only on the data,  $\hat{\gamma}$ , and independent of  $u$ , such that  $\psi(u) \in [GDG]_p^-(\Omega, \bar{\gamma})$ .*

We also have the following results.

**Lemma 4.2.5** *Let  $u \in [DG]_p^+(\Omega, \hat{\gamma})$ , assume  $0 \leq u < 1$ , and take  $\varphi(t) := \frac{1}{1-t}$ ,  $\psi(t) = \ln \frac{1}{1-t}$  with  $t \in [0, 1)$ . Then for all  $\tau > 0$  there exists a constant  $C_\tau$  depending only upon the data  $\{N, p\}$  and  $\tau$ , such that*

$$\sup_{B_\rho(y)} \frac{1}{1-u} \leq C_\tau (1 + \hat{\gamma})^{\frac{(N+1)N}{p\tau}} \left( \frac{R}{R-\rho} \right)^{\frac{N}{\tau}} \left( \int_{B_R(y)} \left( \frac{1}{1-u} \right)^\tau dx \right)^{\frac{1}{\tau}}, \quad (4.18)$$

$$\sup_{B_\rho(y)} \ln \left( \frac{1}{1-u} \right) \leq C_\tau (1 + \hat{\gamma})^{\frac{(N+1)N}{p\tau}} \left( \frac{R}{R-\rho} \right)^{\frac{N}{\tau}} \left\{ \int_{B_R(y)} \left[ \ln \left( \frac{1}{1-u} \right) \right]^\tau dx \right\}^{\frac{1}{\tau}}, \quad (4.19)$$

for any pair of balls  $B_\rho(y) \subset B_R(y) \subset \Omega$ .

**Proof.** Since both  $\varphi$  and  $\psi$  are convex, strictly increasing functions, (4.18)–(4.19) are straightforward consequences of Lemma 4.2.4 and of Lemma 4.2.2. ■

**Proposition 4.2.7 (Proposition 3.3 in [34])** *Let  $u \in [DG]_p^-(\Omega, \hat{\gamma})$  be non-negative and upper bounded by some positive constant  $M$ . Then*

$$\int_{B_\rho(y)} |D \ln u|^p dx \leq \frac{\hat{\gamma} p}{(R - \rho)^p} \int_{B_R(y)} \ln \frac{M}{u} dx,$$

for any pair of balls  $B_\rho(y) \subset B_R(y) \subset \Omega$ .

**Remark 4.2.8** An analogous statement holds on cubes as well.

### 4.2.12 A Weak Harnack Inequality

We have all the tools to prove a weak Harnack inequality using a Moser-like approach.

**Proposition 4.2.8** *Let  $u \in [DG]_p^+(B_R, \hat{\gamma})$  for some  $R > 0$  and  $1 < p \leq N$ , let  $\mu_+ \stackrel{\text{def}}{=} \sup_{B_R} u$ , assume that  $u \geq 0$  and vanishes either in  $\mathbb{R}_-^N \cap B_R$  or in  $\mathbb{H}^N \cap B_R$  with  $1 \leq M \leq N - 1$  and  $N - M < p \leq N$ . Let  $\rho \in (0, \frac{R}{2})$ ; then, there exists  $\tau_o \in (0, 1)$ , such that for any  $\tau \in (0, \tau_o)$  and  $s \in (0, \frac{1}{4})$  we have*

$$\int_{B_{2s\rho}} (\mu_+ - u)^\tau dx \leq C (1 + \hat{\gamma})^{\frac{(N+1)N}{p}} \inf_{B_{s\rho}} (\mu_+ - u)^\tau,$$

where  $C \geq 1$  depends on the data, and  $\tau$ .

**Proof.** Without loss of generality, we may assume  $\mu_+ \equiv 1$ : indeed, since De Giorgi classes are homogeneous with respect to  $u$ , by a simple normalization such a condition is always satisfied. By Lemma 4.2.5, for any  $\sigma > 0$  we have

$$\begin{aligned} \int_{B_\rho} \ln \left( \frac{1}{1-u} \right) dx &\leq C_N \rho^N \sup_{B_\rho} \ln \left( \frac{1}{1-u} \right) \\ &\leq C_N \rho^N C_\sigma (1 + \hat{\gamma})^{\frac{N(N+1)}{p\sigma}} 2^{\frac{N}{\sigma}} \left[ \int_{B_{2\rho}} \left[ \ln \left( \frac{1}{1-u} \right) \right]^\sigma dx \right]^{\frac{1}{\sigma}}, \end{aligned}$$

that is

$$\left[ \int_{B_\rho} \ln \left( \frac{1}{1-u} \right) dx \right]^\sigma \leq C_1 (1 + \hat{\gamma})^{\frac{N(N+1)}{p}} \int_{B_{2\rho}} \left[ \ln \left( \frac{1}{1-u} \right) \right]^\sigma dx,$$

where  $C_1$  depends on the data and  $\sigma$ . On the other hand, by Propositions 4.2.5–4.2.6, provided we choose  $\sigma \in (0, \frac{p-1}{p})$ , we have

$$\int_{B_{2\rho}} \left[ \ln \left( \frac{1}{1-u} \right) \right]^\sigma dx \leq \frac{C (1 + \hat{\gamma})^{\frac{N+1}{p}}}{[\delta(2\rho)]^{\frac{1}{p}}} |B_{2\rho}|,$$

and by the argument of § 4.2.3 there exists  $c_o$  which depends only on the data, such that  $\delta(2\rho) \approx \frac{\text{cap}_p(Q(2\rho), B_{4\rho})}{(2\rho)^{N-p}} \geq c_o$  for any  $\rho \in (0, \frac{R}{2})$ . Thus, for any  $\sigma \in (0, \frac{p-1}{p})$ , for any  $R > 0$ , and for any  $\rho \in (0, \frac{R}{2})$ ,

$$\int_{B_\rho} \ln \left( \frac{1}{1-u} \right) dx \leq C_2 (1 + \hat{\gamma})^{\frac{(1+N)^2}{p\sigma}},$$

where  $C_2$  depends on the data and  $\sigma$ ; moreover, since  $u \in [DG]_p^+(B_R, \hat{\gamma})$ , by Remark 4.2.6 we have  $1-u \in [DG]_p^-(B_R, \hat{\gamma})$  and by Proposition 4.2.7

$$\int_{B_{s\rho}} |D \ln(1-u)|^p dx \leq \frac{C_3 \hat{\gamma} (1 + \hat{\gamma})^{\frac{(1+N)^2}{p\sigma}}}{(1-s)^p} \rho^{N-p}, \quad s \in (0, \frac{1}{4}).$$

Working as in [100, Lemma 1.2] we conclude that there exist  $C_4 = 4$  and

$$\tau_o = \frac{\ln 2}{2^{N+1} C_3 \hat{\gamma}^{\frac{1}{p}} (1 + \hat{\gamma})^{\frac{(1+N)^2}{p^2\sigma}}} \in (0, 1),$$

such that for all  $\tau \in (0, \tau_o)$  and for all  $s \in (0, \frac{1}{4})$

$$\begin{aligned} \left( \int_{B_{2s\rho}} (1-u)^{-\tau} dx \right) \left( \int_{B_{2s\rho}} (1-u)^{\tau} dx \right) &\leq C_4 \\ \Leftrightarrow \int_{B_{2s\rho}} (1-u)^{\tau} dx &\leq \frac{C_4}{\int_{B_{2s\rho}} \frac{1}{(1-u)^{\tau}} dx}. \end{aligned}$$

On the other hand, (4.18) can be rewritten as

$$\frac{1}{\inf_{B_{s\rho}} (1-u)^{\tau}} \leq C_5 (1 + \hat{\gamma})^{\frac{(N+1)N}{p}} \int_{B_{2s\rho}} \frac{1}{(1-u)^{\tau}} dx,$$

where  $C_5$  depends on the data and  $\tau$ . Combining the previous two inequalities, we conclude. ■

### 4.3 Proof of Theorems 4.1.1–4.1.2 for $1 < p < N$

We give a single proof of both statements. Without loss of generality, from here on we assume that  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ ,  $u \geq 0$ ,  $u \not\equiv 0$  in  $\mathbb{R}^N$ , and  $u$  vanishes either in  $\Omega \stackrel{\text{def}}{=} \mathbb{R}_-^N$  or in  $\Omega \stackrel{\text{def}}{=} \mathbb{H}^M$  with  $1 \leq M \leq N-1$  and  $N-M < p < N$ . As discussed in § 4.2.3, in both instances the vanishing set  $\Omega$  is a  $p$ -locally uniformly fat set.



It is a matter of straightforward computations to show that any  $u \in W^{1,s}(B_{2R})$ ,  $s \in [\frac{Np}{N+p}, p]$  satisfies the following Sobolev-Poincaré inequality

$$\left( \int_{B_{\frac{3}{2}R}} |u|^p dx \right)^{\frac{1}{p}} \leq C(p, s) \frac{R^{\frac{N}{p}}}{\left[ \text{cap}_s \left( Q(\frac{3}{2}R), B_{2R} \right) \right]^{\frac{1}{s}}} \left( \int_{B_{2R}} |Du|^s dx \right)^{\frac{1}{s}},$$

where  $Q(\frac{3}{2}R) = \left\{ x \in B_{\frac{3}{2}R} \cap \Omega \text{ s.t. } u(x) = 0 \right\}$ . See, for example [59], or [40, Theorem 2.9]. It is apparent that  $Q(\frac{3}{2}R) \subset \Omega$ .

Taking into account the previous results, we conclude that

$$\left( \int_{B_{\frac{3}{2}R}} |u|^p dx \right)^{\frac{1}{p}} \leq C_1(N, p, q) \frac{R^{\frac{N}{p}}}{\left[ \text{cap}_q \left( Q(\frac{3}{2}R), B_{2R} \right) \right]^{\frac{1}{q}}} \left( \int_{B_{2R}} |Du|^q dx \right)^{\frac{1}{q}},$$

for any  $p - \varepsilon_o < q < p$ , where  $\varepsilon_o$  is the quantity introduced in Proposition 4.2.3.

By the same proposition, in particular, relying both on the dependence of the  $q$ -fatness on the original  $p$ -fatness, and on the explicit expression of the  $q$ -capacity of a ball yields

$$\left( \int_{B_{\frac{3}{2}R}} |u|^p dx \right)^{\frac{1}{p}} \leq C_2 \frac{R^{\frac{N}{p}}}{R^{\frac{N-q}{q}}} \left( \frac{1}{c_o} \right)^{\frac{1}{q}} \left( \int_{B_{2R}} |Du|^q dx \right)^{\frac{1}{q}}. \quad (4.20)$$

Finally, since  $u \in [DG]_p^+(\mathbb{R}^N, \hat{\gamma})$ , and  $u \geq 0$ , by Definition 4.1.1 with  $k = 0$  we have

$$\int_{B_R} |Du|^p dx \leq \frac{\hat{\gamma}}{(1 - \frac{2}{3})^p} \frac{1}{R^p} \int_{B_{\frac{3}{2}R}} u^p dx,$$

and we can conclude that

$$\left( \int_{B_R} |Du|^p dx \right)^{\frac{1}{p}} \leq C_3 \hat{\gamma}^{\frac{1}{p}} R^{\frac{N}{p} - \frac{N}{q}} \left( \frac{1}{c_o} \right)^{\frac{1}{q}} \left( \int_{B_{2R}} |Du|^q dx \right)^{\frac{1}{q}},$$

even though we are not going to use this result. Therefore, we have proved a reverse Holder inequality, much as already done by Kilpeläinen & Koskela ([51]).

We let  $\mu_+(r) \stackrel{\text{def}}{=} \sup_{B_r} u$ , choose  $\ell = \frac{7}{8}\mu_+(4R)$ , and define

$$v := \begin{cases} u, & \text{if } 0 < u \leq \ell, \\ \ell, & \text{if } u > \ell. \end{cases}$$

Then, (4.20) yields

$$\left( \int_{B_{\frac{3}{2}R}} v^p dx \right)^{\frac{1}{p}} \leq C_2 \frac{R^{\frac{N}{p}}}{R^{\frac{N-q}{q}}} \left( \frac{1}{c_o} \right)^{\frac{1}{q}} \left( \int_{B_{2R}} |Dv|^q dx \right)^{\frac{1}{q}},$$

and also

$$\begin{aligned} \ell \left| A_{\ell, \frac{3}{2}R} \right|^{\frac{1}{p}} &\leq C_2 R^{1+\frac{N}{p}-\frac{N}{q}} \left( \int_{B_{2R}} |Du|^q dx \right)^{\frac{1}{q}} \\ \ell^q \left| A_{\ell, \frac{3}{2}R} \right|^{\frac{q}{p}} &\leq C_2^q R^{q+\frac{q}{p}N-N} \int_{B_{2R}} |Du|^q dx \\ &\leq C_2^q R^{q+\frac{q}{p}N-N} \int_{B_{2R}} \frac{|Du|^q}{(\mu_+(4R) - u)^{(1+\alpha)q}} (\mu_+(4R) - u)^{(1+\alpha)q} dx, \end{aligned}$$

where  $\alpha > 0$  is to be chosen. Since

$$D \frac{1}{(\mu_+(4R) - u)^\alpha} = \alpha \frac{1}{(\mu_+(4R) - u)^{1+\alpha}} Du,$$

we can rewrite

$$\begin{aligned} [\mu_+(4R)]^q \left| A_{\ell, \frac{3}{2}R} \right|^{\frac{q}{p}} &\leq C_4 R^{q+\frac{qN}{p}-N} \int_{B_{2R}} \left| D \frac{1}{(\mu_+(4R) - u)^\alpha} \right|^q (\mu_+(4R) - u)^{(1+\alpha)q} dx, \end{aligned}$$

where  $C_4 = C_4 \left( \frac{7}{8}, c_o, N, p, \alpha \right)$ . We stipulate that

$$\left| A_{\ell, \frac{3}{2}R} \right| \geq \theta_o \left| B_{\frac{3}{2}R} \right|,$$

where  $\theta_o$  is the quantity introduced in Lemma 4.2.1; we will later consider what happens when such a stipulation does not hold. Then, for a constant  $C_5$  which depends on the original choice of  $\ell$ , and on  $c_o, N, p, \alpha$ , we have

$$\begin{aligned} &[\mu_+(4R)]^q R^{\frac{qN}{p}} \\ &\leq C_5 \hat{\gamma}^{\frac{Nq}{p^2}} R^{q+\frac{qN}{p}-N} \int_{B_{2R}} \left| D \frac{(\mu_+(4R))^\alpha}{(\mu_+(4R) - u)^\alpha} \right|^q \frac{(\mu_+(4R) - u)^{(1+\alpha)q}}{(\mu_+(4R))^{\alpha q}} dx \\ &\leq \frac{C_5 \hat{\gamma}^{\frac{Nq}{p^2}} R^{q+\frac{qN}{p}-N}}{(\mu(4R)_+)^{\alpha q}} \left( \int_{B_{2R}} \left| D \frac{(\mu_+(4R))^\alpha}{(\mu_+(4R) - u)^\alpha} \right|^p dx \right)^{\frac{q}{p}} \\ &\quad \cdot \left( \int_{B_{2R}} (\mu_+(4R) - u)^{(1+\alpha)\frac{qp}{p-q}} dx \right)^{\frac{p-q}{p}}, \end{aligned}$$

that is, taking Lemma 4.2.4 into account, and choosing  $\sigma_o \in (0, \tau_o)$  where  $\tau_o$  is the quantity stipulated in Proposition 4.2.8

$$\begin{aligned} [\mu_+(4R)]^{(1+\alpha)q} &\leq C_6 (1 + \hat{\gamma})^{\frac{Nq+pq(N+1)}{p^2}} \frac{1}{R^N} \left( \int_{B_{3R}} \left| \frac{\mu_+(4R)}{\mu_+(4R) - u} \right|^{\alpha p} dx \right)^{\frac{q}{p}} \\ &\quad \cdot \left( \int_{B_{3R}} [\mu_+(4R) - u]^{\sigma_o} dx \right)^{\frac{p-q}{p}} [\mu_+(4R)]^{(1+\alpha)q - \sigma_o \frac{p-q}{p}}; \end{aligned}$$

by Proposition 4.2.8 this yields

$$\begin{aligned} [\mu_+(4R)]^{\sigma_o \frac{p-q}{p}} &\leq C_7 (1 + \hat{\gamma})^{\frac{(p-q)N^2 + pN + pq(N+1)}{p^2}} \frac{[\mu_+(4R)]^{\alpha q}}{[\mu_+(4R) - \mu(3R)]^{\alpha q}} \\ &\quad \cdot [\mu_+(4R) - \mu(3R)]^{\sigma_o \frac{p-q}{p}} \\ [\mu_+(4R)]^{\sigma_o \frac{p-q}{p} - \alpha q} &\leq C_7 (1 + \hat{\gamma})^{\frac{(p-q)N^2 + pN + pq(N+1)}{p^2}} [\mu_+(4R) - \mu(3R)]^{\sigma_o \frac{p-q}{p} - \alpha q}, \end{aligned}$$

and we may finally choose  $\alpha$  small enough, so that

$$\sigma_o \frac{p-q}{p} - \alpha q > 0 \Rightarrow \alpha < \sigma_o \frac{p-q}{pq} \Rightarrow \alpha < \sigma_o \left( \frac{1}{q} - \frac{1}{p} \right).$$

Correspondingly, we obtain

$$\mu_+(4R) \leq C_8 (1 + \hat{\gamma})^{\frac{(p-q)N^2 + pN + pq(N+1)}{p(\sigma_o(p-q) - \alpha pq)}} [\mu_+(4R) - \mu_+(3R)],$$

and also

$$\mu_+(3R) \leq \left( 1 - \frac{1}{C_9} \right) \mu_+(4R),$$

where  $C_9 = C_8 (1 + \hat{\gamma})^{\frac{(p-q)N^2 + pN + pq(N+1)}{p(\sigma_o(p-q) - \alpha pq)}}$ ,  $C_8$  depends on the data,  $q$ ,  $\alpha$ ,  $\sigma_o$ ,  $c_o$ .

On the other hand, if the previous hypothesis is violated, that is if  $|A_{\ell, \frac{3}{2}R}| < \theta_0 |B_{\frac{3}{2}R}|$ , then by Lemma 4.2.1, we have

$$\mu_+(R) \leq \frac{15}{16} \mu_+(2R).$$

Since  $\mu_+(3R) \geq \mu_+(R)$ , and  $\mu_+(2R) \leq \mu_+(4R)$ , if we let

$$\eta = \max \left\{ \frac{15}{16}, 1 - \frac{1}{C_9} \right\},$$

we conclude that

$$\mu_+(R) \leq \eta \mu_+(4R).$$

The constants  $C_i$  depend on the set where  $u$  vanishes, but the structure of the proof is independent of the size of  $Q(R)$ .

Let us define the sequence of radii

$$R_k = 4^k, \quad k \in \mathbb{N},$$

Then, for any  $r \geq 4$ , there exists  $k \in \mathbb{N}$ , such that

$$R_k \leq r \leq R_{k+1} = 4^{k+1},$$

which means  $k \geq \frac{\ln r}{\ln 4} - 1$ . Therefore, we derive

$$\begin{aligned} \mu_+(r) &\geq \mu_+(R_k) \geq \frac{1}{\eta} \mu_+(R_{k-1}) \\ &\vdots \\ &\geq \frac{1}{\eta^k} \mu_+(R_0) \geq \frac{1}{\eta^{\frac{\ln r}{\ln 4} - 1}} \mu_+(R_0) \\ &= \mu_+(R_0) \eta \exp \left( \ln r \cdot \frac{\ln \frac{1}{\eta}}{\ln 4} \right) \\ &= \mu_+(R_0) \eta r^{\frac{\ln \frac{1}{\eta}}{\ln 4}} = \frac{\mu_+(R_0)}{R_0^{\frac{\ln \frac{1}{\eta}}{\ln 4}}} \eta r^{\frac{\ln \frac{1}{\eta}}{\ln 4}}. \end{aligned}$$

Consequently, we obtain

$$\frac{\mu_+(r)}{r^{\frac{\ln \frac{1}{\eta}}{\ln 4}}} \geq \frac{\mu_+(R_0)}{R_0^{\frac{\ln \frac{1}{\eta}}{\ln 4}}} \eta,$$

and we finish the proof, once we assume  $\alpha_i = \frac{\ln \frac{1}{\eta}}{\ln 4}$ . ■

**Remark 4.3.1** When  $u$  vanishes in  $\mathbb{R}_-^N$ , a simpler proof can be given. For any fixed  $R > 0$ , let us consider the function

$$0 \leq (\mu_+(R) - u) \in [DG]_p^-(B_R, \hat{\gamma}).$$

Applying Proposition 4.2.8 and using that  $u$  vanishes in  $B_{2s\rho}^- = B_{2s\rho} \cap \mathbb{R}_-^N$ , we obtain

$$\begin{aligned} \frac{\mu_+(R)}{2^{1/\tau}} &= \left( \frac{\mu_+(R)^\tau}{|B_{2s\rho}|} \left( |B_{2s\rho}|/2 \right) \right)^{1/\tau} = \left( \frac{1}{|B_{2s\rho}|} \int_{B_{2s\rho}^-} (\mu_+(R))^\tau dx \right)^{1/\tau} \\ &\leq \left( \int_{B_{2s\rho}} (\mu_+(R) - u(x))^\tau dx \right)^{1/\tau} \\ &\leq C (1 + \hat{\gamma})^{\frac{(N+1)N}{\tau p}} \inf_{B_{s\rho}} \left( \mu_+(R) - u(x) \right) \\ &= C (1 + \hat{\gamma})^{\frac{(N+1)N}{p\tau}} \left( \mu_+(R) - \mu_+(s\rho) \right), \end{aligned}$$

where the parameters  $s$ ,  $\rho$  and  $\tau$  are the same as those in Proposition 4.2.8, and  $C$  depends on the data and  $\tau$ . Hence, we have

$$\frac{\mu_+(R)}{2^{1/\tau}} \leq C(1 + \hat{\gamma})^{\frac{(N+1)N}{p\tau}} \left( \mu_+(R) - \mu_+(s\rho) \right),$$

which is

$$\eta \mu_+(R) \geq \mu_+(s\rho), \quad \text{with} \quad \eta = \left( 1 - \frac{1}{2^{1/\tau} C (1 + \hat{\gamma})^{\frac{(N+1)N}{p\tau}}} \right).$$

Then, we set  $R_k = \left( \frac{R}{s\rho} \right)^k$ ,  $k \in \mathbb{N}$ . Through a similar iterative process as in the final part of the proof above, we derive

$$\frac{\mu_+(r)}{r^{\alpha_1}} \geq \frac{\mu_+(R_0)}{R_0^{\alpha_1}} \eta, \quad \text{where} \quad \alpha_1 = \frac{\ln \frac{1}{\eta}}{\ln \frac{R}{s\rho}}.$$

## 4.4 Proof of Theorems 4.1.1–4.1.2 for $p = N$

When we deal with  $p = N$ , the gist of the argument remains essentially the same as discussed in the previous section. We only need to substitute  $p$  with  $N$  and take  $q \in (N - \varepsilon_o, N)$ , instead of  $q \in (p - \varepsilon_o, p)$ .

## 4.5 Final Remarks

The weak Harnack inequality of Proposition 4.2.8 does not depend on the special choices  $\Omega = \mathbb{R}_-^N$  or  $\Omega = \mathbb{H}^M$ , and it can be formulated for any  $0 \leq u \in [DG]_p^+(\mathbb{R}^N, \gamma)$  that vanishes in of a  $p$ -fat set  $\Omega$ .

Moreover, the proof of Theorems 4.1.1-4.1.2 relies on three different features:

- the zero-extension of  $u$  in  $\Omega$ ;
- the weak Harnack inequality for  $p$ -fat sets;
- the possibility to repeat the procedure for arbitrarily big radii, i.e. for  $\Omega$  unbounded.

Hence, our result is valid also for the other examples given in [61, Chapter I, § 6], i.e. the slab  $S$ , where

$$S := \{(x_1, x_2, \dots, x_N) : |x_N| < h\},$$

the double circular cone, the one-edged cone, and many others (we refrain from giving complete definitions of these objects, and we refer the interested reader to [61]).

Furthermore, the weak Harnack inequality for  $p$ -fat sets may be useful in the future also to attack properties of singular solutions, such as in [47, Theorem 7.40]. Here a similar statement seems reasonable to be valid with  $\alpha < 0$ .

## Chapter 5

# Boundary Regularity for Functions in Homogeneous De Giorgi Classes

### 5.1 Main Results

Let  $E \subset \mathbb{R}^N$  be an open set, and for  $x_o \in \mathbb{R}^N$ , let  $B_\rho(x_o)$  denote the ball of radius  $\rho$  centered at  $x_o$ . When  $x_o = 0$ , we omit the reference to the center.

#### Definitions

We have the following well-known definitions.

**Definition 5.1.1** *Given an open subset  $E \subseteq \mathbb{R}^N$ , we say that a function  $u \in W_{\text{loc}}^{1,p}(E)$  belongs to the De Giorgi class  $[DG]_p^\pm(E, \gamma)$  for  $1 < p \leq N$  if there exists a constant  $\gamma > 1$  such that for all  $\sigma \in (0, 1)$ ,  $\rho > 0$ ,  $x_o \in E$ ,  $k \in \mathbb{R}$ , and  $B_\rho(x_o) \subset E$ , the following estimate is satisfied:*

$$\int_{B_{\sigma\rho}(x_o)} |\nabla[(u - k)_\pm]|^p dx \leq \frac{\gamma}{[(1 - \sigma)\rho]^p} \int_{B_\rho(x_o)} |(u - k)_\pm|^p dx. \quad (5.1)$$

Furthermore, we let

$$[DG]_p(E, \gamma) \equiv [DG]_p^+(E, \gamma) \cap [DG]_p^-(E, \gamma).$$

**Definition 5.1.2** *Let  $h \in W^{1,p}(\mathbb{R}^N) \cap C(\mathbb{R}^N)$ . Given an open subset  $E \subset \mathbb{R}^N$ , the De Giorgi classes  $[DG]_p^+(\bar{E}, h, \gamma)$  in the closure of  $E$  are defined to consist of functions*

$u \in [DG]_p^+(E, \gamma)$  such that  $(u - h)_+ \in W_o^{1,p}(E)$  and satisfying in addition inequalities of the type

$$\forall x_o \in \partial E, \quad \forall B_\rho(x_o), \quad \forall \sigma \in (0, 1), \quad \forall k \geq \sup_{B_\rho(x_o) \cap \partial E} h,$$

$$\int_{B_{\sigma\rho}(x_o) \cap E} |\nabla[(u - k)_+]|^p dx \leq \frac{\gamma}{[(1 - \sigma)\rho]^p} \int_{B_\rho(x_o) \cap E} |(u - k)_+|^p dx.$$

The various constants are as in (5.1). The classes  $[DG]_p^-(\bar{E}, h, \gamma)$  are defined analogously by replacing  $(u - k)_+$  with  $(u - k)_-$  provided that  $k \leq \inf_{B_\rho(x_o) \cap \partial E} h$ . Furthermore, we let

$$[DG]_p(\bar{E}, h, \gamma) \equiv [DG]_p^+(\bar{E}, h, \gamma) \cap [DG]_p^-(\bar{E}, h, \gamma).$$

In the following, by *data* we mean the set  $\{N, p, \gamma\}$ . Moreover, given  $u \in [DG]_p^\pm(\bar{E}, h, \gamma)$ , for the values of  $k$  postulated in Definition 5.1.2, and for  $x_o \in \partial E$ , we always assume  $(u - k)_\pm$  to be defined as zero in  $B_\rho(x_o) \cap (\mathbb{R}^N \setminus E)$ ; in this way, the integrals can be extended to full balls; it is not hard to show that such extensions, which we continue to denote by  $(u - k)_\pm$  for simplicity, belong to  $[DG]_p^\pm(B_\rho(x_o), \gamma)$ .

**Definition 5.1.3** Take  $x_o \in \partial E$ ; we say that  $\partial E$  satisfies a geometric density condition at  $y$ , if there exist  $\theta \in (0, 1)$  and  $R > 0$  such that

$$|B_{\sigma R}(x_o) \cap (\mathbb{R}^N \setminus E)| > \theta |B_{\sigma R}|, \quad \forall \sigma \in (0, 1]. \quad (5.2)$$

In general,  $\theta$  and  $R$  depend on  $x_o$ . We assume that (5.2) holds uniformly on an open portion of  $\partial E$ , and in the following we work on such a portion, which we denote as  $(\partial E)_{\text{loc}}$ . In general, we have  $(\partial E)_{\text{loc}} \subseteq \partial E$ .

## Main Theorems

Let  $h \in W^{1,p}(\mathbb{R}^N) \cap C(\mathbb{R}^N)$  and  $u \in [DG]_p(\bar{E}, h, \gamma)$ ; take  $x_o \in \partial E$ , let  $2\rho < R$  with  $R$  as in Definition 5.1.3, and set

$$h^+(\rho) = \sup_{B_\rho(x_o) \cap \partial E} h, \quad h^-(\rho) = \inf_{B_\rho(x_o) \cap \partial E} h,$$

$$\omega_h(\rho) = h^+(\rho) - h^-(\rho) = \text{osc}_{B_\rho(x_o) \cap \partial E} h,$$

$$\mu^+(\rho) = \sup_{B_\rho(x_o) \cap E} u, \quad \mu^-(\rho) = \inf_{B_\rho(x_o) \cap E} u,$$

$$\omega(\rho) = \mu^+(\rho) - \mu^-(\rho) = \text{osc}_{B_\rho(x_o) \cap E} u.$$

We have the following result.



**Theorem 5.1.1** *Let  $\partial E$  satisfy the condition of positive geometric density (5.2), and let  $h \in W^{1,p}(\mathbb{R}^N) \cap C(\mathbb{R}^N)$ . Then every  $u \in [DG]_p(\bar{E}, h, \gamma)$  is continuous up to  $\partial E$ , and there exist constants  $C > 1$  and  $\alpha \in (0, 1)$ , depending only on the data, and the parameter  $\theta$ , and independent of  $h$  and  $u$ , such that for all  $y \in \partial E$  and all balls  $B_{2\rho}(y) \subset B_R(y)$*

$$\omega(\rho) \leq C \max \left\{ \omega(R) \left( \frac{\rho}{R} \right)^\alpha ; \omega_h(2\rho) \right\}. \quad (5.3)$$

A well-known straightforward consequence of the previous theorem is the following result.

**Corollary 5.1.1** *Let  $\partial E$  satisfy the condition of positive geometric density (5.2). A solution  $u \in [DG]_p(\bar{E}, h, \gamma)$  for a datum  $h \in W^{1,p}(\mathbb{R}^N) \cap C(\mathbb{R}^N)$  is continuous on  $\bar{E}$ . If  $h$  is Hölder continuous on  $\partial E$ , then  $u$  is Hölder continuous on  $\bar{E}$ .*

**Remark 5.1.1** Both Theorem 5.1.1 and Corollary 5.1.1 have been stated in a global way, but these results have a purely local thrust, as the proof will show; hence, we could assume  $(\partial E)_{\text{loc}} \subset \subset \partial E$ , and the previous continuity result would hold only on a proper subset of  $E$ .

## Novelty and Significance

We give a new proof of such a classical result, based on ideas that were originally introduced by Gariepy & Ziemer [42] when studying the sufficient part of the Wiener criterion for a wide class of quasilinear equation of  $p$ -Laplacian type. In their technique three auxiliary functions play a fundamental role, namely

$$\varphi(t) = \left( \frac{1}{1-t} \right)^\alpha, \quad \psi(t) = (1-t)^\beta, \quad \zeta(t) = \ln(1-t),$$

where  $\alpha > 0$ ,  $\beta \in (0, 1)$ , and  $t \in [0, 1)$ . The geometric density condition is needed to obtain proper estimates involving the composition of  $\varphi$ ,  $\psi$ ,  $\zeta$  with functions in De Giorgi classes; apart from that, it is immaterial. Whenever such estimates are available, the method works; therefore, this raises the natural question whether the same regularity of solutions continues to hold even when rougher boundaries are considered. This issue will not be dealt with here.

## Structure of the Chapter

In § 5.2.1 we prove a local upper bound for  $\varphi(u_k)$ , where  $u \in [DG]_p^+(E, \gamma)$ ,  $u_k := (u - k)_+$  with  $k \in \mathbb{R}$ , and  $\varphi = \varphi(t)$  is a proper convex function (namely, the one we mentioned above). § 5.2.2 is devoted to some further estimates for functions in  $[DG]_p^+(E, \gamma)$ . In § 5.2.3, relying on the estimates of § 5.2.2, and assuming that the boundary  $\partial E$  satisfies a geometric density condition, we show that a proper weak Harnack inequality holds. Finally, § 5.3 deals with the proof of Theorem 5.1.1.

## 5.2 Preliminaries

In the following we collect and prove some auxiliary results which are instrumental in proving the main theorem of this chapter.

### 5.2.1 A Local Upper Bound

It is known that convex, monotone, non-decreasing functions of sub-harmonic functions are sub-harmonic. Similarly, concave, monotone, non-decreasing functions of super-harmonic functions are super-harmonic. Similar statements hold for weak, sub(super)-solutions of linear elliptic equations with bounded and measurable coefficients [78].

Analogous properties were proven for functions  $u \in [DG]_p^\pm(E, \gamma)$  in [34]. A particular convex, monotone increasing function is  $\varphi(t) = \frac{1}{1-t}$ ; although we could rely directly on the above-mentioned result in order to provide a local estimate of the upper bound of  $\varphi((u - k)_+)$ , we give a direct proof of it; we have the following result.

**Lemma 5.2.1** *Consider  $u \in [DG]_p^+(E, \gamma)$ , assume  $B_{2R}(x_o) \subset\subset E$ , take  $k \in \mathbb{R}$ , let  $u_k := (u - k)_+$  and  $\mu := \sup_{B_{2R}(x_o)}(u - k)_+$ , and take  $\varphi(t) := \frac{\mu}{\mu - t}$  with  $t \in [0, \mu)$ . Then for all  $\sigma \in (0, 1)$ , and for all  $q > 0$  there exists a constant  $C_q$  depending only upon the data, and  $q$ , such that*

$$\sup_{B_{\sigma R}(x_o)} \frac{\mu}{\mu - u_k} \leq \frac{C_q}{(1 - \sigma)^{\frac{N}{q}}} \left( \int_{B_R(x_o)} \left( \frac{\mu}{\mu - u_k} \right)^q dx \right)^{\frac{1}{q}}. \quad (5.4)$$

**Proof.** Without loss of generality we may assume  $x_o = 0$ . For  $k \in \mathbb{R}$ , let  $\mu \stackrel{\text{def}}{=} \sup_{B_{2R}}(u - k)_+$ ; in order to simplify the notation, in the proof, instead of writing  $u_k$ ,

we let  $v \stackrel{\text{def}}{=} (u - k)_+$ . For  $\sigma \in (0, 1)$  inequality (5.1) can be rewritten as

$$\int_{B_{\sigma R}} |\nabla(u - h)_+|^p dx \leq \frac{\gamma}{(1 - \sigma)^p R^p} \int_{B_R} (u - h)_+^p dx.$$

Let us take  $h = k + t$  with  $t \geq 0$ . Then in terms of  $t$  we have,

$$\int_{B_{\sigma R}} |\nabla(v - t)_+|^p dx \leq \frac{\gamma}{(1 - \sigma)^p R^p} \int_{B_R} (v - t)_+^p dx.$$

We now multiply both terms by  $t^n$  with  $n \geq 0$  and integrate over  $t \in (0, \infty)$ . This yields

$$\int_0^\infty t^n dt \int_{B_{\sigma R}} |\nabla(v - t)_+|^p dx \leq \frac{\gamma}{(1 - \sigma)^p R^p} \int_0^\infty t^n dt \int_{B_R} (v - t)_+^p dx,$$

that is  $\mathbf{I}_{n+1} \leq \frac{\gamma}{(1 - \sigma)^p R^p} \mathbf{II}_{n+1}$ , where

$$\mathbf{I}_{n+1} = \int_{B_{\sigma R} \cap [v \geq 0]} |\nabla v|^p dx \int_0^v t^n dt = \int_{B_{\sigma R} \cap [v \geq 0]} \frac{v^{n+1}}{n+1} |\nabla v|^p dx.$$

This accounts for any  $n$  from 0 to  $\infty$ . On the other hand, for the zero-order power, we simply end up with

$$\mathbf{I}_0 = \int_{B_{\sigma R} \cap [v \geq 0]} |\nabla v|^p dx.$$

Similarly,

$$\mathbf{II}_{n+1} = \int_{B_R \cap [v \geq 0]} \int_0^v t^n (v - t)^p dt dx.$$

We have

$$\begin{aligned} \mathbf{II}_0 &= \int_{B_R \cap [v \geq 0]} (v - t)^p dx \leq \int_{B_R \cap [v \geq 0]} v^p dx; \\ \mathbf{II}_1 &= \int_{B_R \cap [v \geq 0]} \int_0^v (v - t)^p dx = - \int_{B_R \cap [v \geq 0]} \left. \frac{(v - t)^{p+1}}{p+1} \right|_0^v dx \\ &= \int_{B_R \cap [v \geq 0]} \frac{v^{p+1}}{p+1} dx, \end{aligned}$$

and if  $n \geq 2$

$$\begin{aligned} \mathbf{II}_{n+1} &= \int_{B_R \cap [v \geq 0]} \int_0^v t^n (v - t)_+^p dt dx \\ &= \frac{n}{p+1} \int_{B_R \cap [v \geq 0]} \int_0^v t^{n-1} (v - t)_+^{p+1} dt dx \\ &= \frac{n(n-1)}{(p+1)(p+2)} \int_{B_R \cap [v \geq 0]} \int_0^v t^{n-2} (v - t)_+^{p+2} dt dx \\ &= \frac{n(n-1) \cdots 1}{(p+1)(p+2) \cdots (p+n)} \int_{B_R \cap [v \geq 0]} \int_0^v (v - t)_+^{p+n} dt dx \end{aligned}$$

$$= \frac{n!}{(p+1)(p+2)\cdots(p+n)} \int_{B_R \cap [v \geq 0]} \frac{v^{p+n+1}}{p+n+1} dx.$$

Since

$$(n+p)\cdots(p+1) = (n+p)\cdots(n+p-(n-1)) = \binom{n+p}{n} n!$$

we have that  $\forall n \geq 1$

$$\int_0^v t^n (v-t)_+^p dx = \frac{n!}{\binom{n+p}{n} n!} \frac{v^{p+n+1}}{p+n+1}$$

and this contains the case  $n = 0$ . Hence,  $\forall n \geq 0$

$$\int_{B_R \cap [v \geq 0]} \int_0^v t^n (v-t)_+^p dt dx = \frac{1}{\binom{n+p}{n}} \int_{B_R \cap [v \geq 0]} \frac{v^{p+n+1}}{p+n+1} dx.$$

Collecting all terms, we have

$$\begin{aligned} \int_{B_{\sigma R} \cap [v \geq 0]} |\nabla v|^p dx &\leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} v^p dx \\ \int_{B_{\sigma R} \cap [v \geq 0]} v |\nabla v|^p dx &\leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{v^{p+1}}{p+1} dx \\ \int_{B_{\sigma R} \cap [v \geq 0]} \frac{v^2}{2} |\nabla v|^p dx &\leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{1}{p+1} \frac{v^{p+2}}{p+2} dx \\ \int_{B_{\sigma R} \cap [v \geq 0]} \frac{v^3}{3} |\nabla v|^p dx &\leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{1}{\binom{p+2}{2}} \frac{v^{p+3}}{p+3} dx \\ &= \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{2}{(p+1)(p+2)} \frac{v^{p+3}}{p+3} dx \\ &\vdots \\ \int_{B_{\sigma R} \cap [v \geq 0]} \frac{v^n}{n} |\nabla v|^p dx &\leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{1}{\binom{p+n-1}{n-1}} \frac{v^{p+n}}{p+n} dx \\ &= \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{(n-1)!}{(p+n-1)\cdots(p+1)} \frac{v^{p+n}}{p+n} dx. \end{aligned}$$

In general, we conclude that  $\forall n \geq 1$

$$\begin{aligned} \int_{B_{\sigma R} \cap [v \geq 0]} v^n |\nabla v|^p dx \\ \leq \frac{\gamma}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \frac{n!}{(n+p)(n+p-1)\cdots(p+1)} v^{p+n} dx. \end{aligned}$$

Let us now consider the function  $\varphi_1(t) = \left(\frac{\mu}{\mu-t}\right)^{\alpha p} = \mu^{\alpha p} (\mu-t)^{-\alpha p}$ . For  $k \geq 1$  we have

$$\varphi_1' = \mu^{\alpha p} \alpha p (\mu-t)^{-\alpha p-1},$$

$$\begin{aligned}
 & \vdots \\
 \varphi_1^{(k)} &= \mu^{\alpha p} \alpha p (\alpha p + 1) \cdots (\alpha p + k + 1) (\mu - t)^{-\alpha p - k}, \quad k \geq 1 \\
 \varphi_1(0) &= 1, \\
 & \vdots \\
 \varphi_1^{(k)}(0) &= \frac{\mu^{\alpha p} \alpha p (\alpha p + 1) \cdots (\alpha p + k - 1)}{\mu^{\alpha p + k}} = \frac{\alpha p (\alpha p + 1) \cdots (\alpha p + k - 1)}{\mu^k}.
 \end{aligned}$$

Hence,

$$\begin{aligned}
 & \int_{B_{\sigma R} \cap [v \geq 0]} \left[ 1 + \sum_{n=1}^{\infty} \frac{\varphi_1^{(n)}(0)}{n!} v^n \right] |\nabla v|^p dx \\
 & \leq \frac{\gamma}{(1 - \sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \left[ 1 + \sum_{n=1}^{\infty} \frac{\varphi_1^{(n)}(0)}{n!} \frac{n!}{(n + p) \cdots (p + 1)} v^n \right] v^p dx;
 \end{aligned}$$

we have

$$\mathbf{J}_1 \stackrel{\text{def}}{=} \int_{B_{\sigma R} \cap [v \geq 0]} \left( \frac{\mu}{\mu - v} \right)^{\alpha p} |\nabla v|^p dx = \int_{B_{\sigma R} \cap [v \geq 0]} \left[ \left( \frac{\mu}{\mu - v} \right)^{\alpha} |\nabla v| \right]^p dx.$$

Notice that

$$\begin{aligned}
 \nabla \left( \frac{\mu}{\mu - v} \right)^{\alpha - 1} &= \mu^{\alpha - 1} \nabla (\mu - v)^{-\alpha + 1} = \mu^{\alpha - 1} (-\alpha + 1) (-1) (\mu - v)^{-\alpha} \nabla v \\
 &= \mu^{\alpha - 1} (\alpha - 1) \frac{1}{(\mu - v)^{\alpha}} \nabla v = \frac{\alpha - 1}{\mu} \left( \frac{\mu}{\mu - v} \right)^{\alpha} \nabla v.
 \end{aligned}$$

Hence,

$$\left( \frac{\mu}{\mu - v} \right)^{\alpha} |\nabla v| = \frac{\mu}{\alpha - 1} \left| \nabla \left( \frac{\mu}{\mu - v} \right)^{\alpha - 1} \right|$$

and we obtain

$$\mathbf{J}_1 = \frac{\mu^p}{(\alpha - 1)^p} \int_{B_{\sigma R} \cap [v \geq 0]} \left| D \left( \frac{\mu}{\mu - v} \right)^{\alpha - 1} \right|^p dx,$$

and we can rewrite

$$\begin{aligned}
 & \int_{B_{\sigma R} \cap [v \geq 0]} \left| \nabla \left( \frac{\mu}{\mu - v} \right)^{\alpha - 1} \right|^p dx \\
 & \leq \frac{\gamma (\alpha - 1)^p}{(1 - \sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \left[ 1 + \sum_{n=1}^{\infty} \frac{\varphi_1^{(n)}(0)}{n!} \frac{n!}{(n + p) \cdots (p + 1)} v^n \right] \left( \frac{v}{\mu} \right)^p dx.
 \end{aligned}$$

We need to estimate the term on the right-hand side. Let us consider

$$\varphi_2(t) = \left( \frac{\mu}{\mu - t} \right)^{(\alpha - 1)p} = \mu^{(\alpha - 1)p} (\mu - t)^{-(\alpha - 1)p},$$

$$\begin{aligned}
 \varphi_2'(t) &= \mu^{(\alpha-1)p}(\alpha-1)p(\mu-t)^{-(\alpha-1)p-1}, \\
 &\vdots \\
 \varphi_2^{(n)}(t) &= \mu^{(\alpha-1)p}(\alpha-1)p \cdots [(\alpha-1)p+n-1](\mu-t)^{-(\alpha-1)p-n}, \quad n \geq 1, \\
 \varphi_2(0) &= 1, \\
 &\vdots \\
 \varphi_2^{(n)}(0) &= \frac{1}{\mu^n}(\alpha-1)p \cdot [(\alpha-1)p+1] \cdots [(\alpha-1)p+n-1], \quad n \geq 1.
 \end{aligned}$$

Hence, on the right-hand side the integrand is

$$\left[ 1 + \sum_{n=1}^{\infty} \frac{\alpha p(\alpha p+1) \cdots (\alpha p+n-1)}{n!} \frac{n!}{(n+p) \cdots (p+1)} \left( \frac{v}{\mu} \right)^n \right] \left( \frac{v}{\mu} \right)^p,$$

and we wonder whether this series can be upper bounded by the series of  $\varphi_2$ . This amounts to showing that for any  $n \geq 1$

$$\begin{aligned}
 &\frac{\alpha p(\alpha p+1) \cdots (\alpha p+n-1) \cdot n!}{(n+p) \cdots (p+1)} \\
 &\leq (\alpha-1)p [(\alpha-1)p+1] \cdots [(\alpha-1)p+n-1].
 \end{aligned} \tag{5.5}$$

Since (5.5) can be rewritten as

$$\begin{aligned}
 &\frac{\alpha p}{1+p} \cdot \frac{2(\alpha p+1)}{p+2} \cdots \frac{n(\alpha p+n-1)}{n+p} \\
 &\leq (\alpha-1)p \cdot [(\alpha-1)p+1] \cdots [(\alpha-1)p+n-1],
 \end{aligned}$$

it is enough to show that  $\forall k \geq 1$

$$\begin{aligned}
 &\frac{k(\alpha p+k-1)}{p+k} \leq (\alpha-1)p+k-1 \\
 &k(\alpha p+k-1) \leq (\alpha p-p)(p+k) + (k-1)(p+k) \\
 &\alpha p k + k^2 - k \leq \alpha p^2 - p^2 + \alpha p k - k p + k p - p + k^2 - k \\
 &0 \leq (\alpha-1)p^2 - p \\
 &\alpha \geq 1 + \frac{1}{p}.
 \end{aligned}$$

Hence, for  $\alpha \geq 1 + \frac{1}{p}$  we have

$$\begin{aligned}
 &\int_{B_R \cap [v \geq 0]} \left[ 1 + \sum_{n=1}^{\infty} \frac{\varphi_1^{(n)}(0)}{n!} \frac{n!}{(n+p) \cdots (1+p)} v^n \right] \left( \frac{v}{\mu} \right)^p dx \\
 &\leq \int_{B_R \cap [v \geq 0]} \sum_{n=0}^{\infty} \frac{\varphi_2^{(n)}(0)}{n!} v^n dx = \int_{B_R \cap [v \geq 0]} \left( \frac{\mu}{\mu-v} \right)^{(\alpha-1)p} dx,
 \end{aligned}$$

where we have also taken into account that  $\frac{v}{\mu} \leq 1$ . We have concluded that for any  $\alpha \geq 1 + \frac{1}{p}$

$$\int_{B_{\sigma R} \cap [v \geq 0]} \left| \nabla \left( \frac{\mu}{\mu - v} \right)^{\alpha-1} \right|^p dx \leq \frac{\gamma(\alpha-1)^p}{(1-\sigma)^p R^p} \int_{B_R \cap [v \geq 0]} \left( \frac{\mu}{\mu - v} \right)^{(\alpha-1)p} dx.$$

Inequality (5.4) with exponent  $q = p$  can now be derived by Moser's iteration [78]. By applying such a technique as described, for example, in [100], taking into account that  $v$  actually stands for  $u_k$ , for  $q = p$  we arrive at

$$\sup_{B_{\sigma R}} \frac{\mu}{\mu - u_k} \leq \frac{C}{(1-\sigma)^{\frac{N}{p}}} \left( \int_{B_R} \left( \frac{\mu}{\mu - u_k} \right)^p dx \right)^{\frac{1}{p}}. \quad (5.6)$$

In order to obtain the general result for any  $q > 0$ , once more we rely on a suitable interpolation. Set

$$R_o = \sigma R, \quad R_n = \sigma R + (1-\sigma)R \sum_{i=1}^n 2^{-i}, \quad n = 1, 2, \dots, \\ B_n = B_{R_n}, \quad M_n = \sup_{B_n} \frac{\mu}{\mu - u_k}.$$

Then for  $0 < q < p$  and  $\delta \in (0, 1)$  to be chosen, (5.6) yields

$$\begin{aligned} M_n &\leq \frac{C 2^{\frac{N}{p}n}}{(1-\sigma)^{\frac{N}{p}}} \left( \int_{B_{n+1}} \left( \frac{\mu}{\mu - u_k} \right)^p dx \right)^{\frac{1}{p}} \\ &\leq \frac{C 2^{\frac{N}{p}n}}{(1-\sigma)^{\frac{N}{p}}} M_{n+1}^{\frac{p-q}{p}} \left( \int_{B_{n+1}} \left( \frac{\mu}{\mu - u_k} \right)^q dx \right)^{\frac{1}{p}} \\ &\leq \delta M_{n+1} + \frac{C(\text{data}, q, \delta) 2^{\frac{N}{q}n}}{(1-\sigma)^{\frac{N}{q}}} \left( \int_{B_R} \left( \frac{\mu}{\mu - u_k} \right)^q dx \right)^{\frac{1}{q}}. \end{aligned}$$

By iteration

$$M_o \leq \delta^n M_n + \frac{C(\text{data}, q, \delta)}{(1-\sigma)^{\frac{N}{q}}} \left( \int_{B_R} \left( \frac{\mu}{\mu - u_k} \right)^q dx \right)^{\frac{1}{q}} \sum_{i=0}^n (\delta 2^{\frac{N}{q}})^i.$$

Now choose  $\delta$  in such a way that the sum is bounded above by a convergent sequence and let  $n \rightarrow \infty$ . ■

### 5.2.2 Estimates for Functions in $[DG]_p^+(E, \gamma)$ Classes

Let  $E \subset \mathbb{R}^N$  be open, and for some  $p > 1$  let  $u \in W_{\text{loc}}^{1,p}(E)$ . Having fixed  $B_R(x_o) \subset E$  and  $k \in \mathbb{R}$ , assume that

$$|[(u - k)_+ = 0] \cap B_R(x_o)| > \theta |B_R|, \quad \text{for some } \theta \in (0, 1). \quad (5.7)$$

**Proposition 5.2.1** *Assume  $p \geq 2$  and let (5.7) hold. There exists a constant  $\gamma = \gamma(N, p)$  such that*

$$\int_{B_R(x_o)} (u - k)_+^p dx \leq \frac{\gamma R^{p-1}}{\theta^{\frac{(N-1)(p-1)}{N}}} \int_{B_R(x_o)} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx \quad (5.8)$$

**Proof.** Without loss of generality, we may assume  $x_o = 0$ . If (5.7) is in force and  $p > 2$ , by the De Giorgi embedding argument (see, for example, [31, Proposition 2.1, Chapter I]) there is a constant  $\gamma = \gamma(N)$  such that

$$\begin{aligned} \int_{B_R} (u - k)_+^p dx &\leq \frac{\gamma R}{\theta^{\frac{N-1}{N}}} \int_{B_R} |\nabla(u - k)_+|^p dx \\ &= \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R} (u - k)_+^{p-1} |\nabla(u - k)_+| dx \\ &= \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R} (u - k)_+^{\frac{p-2}{p-1}} |\nabla(u - k)_+| (u - k)_+^{\frac{1}{p-1}} dx \\ &\leq \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \left( \int_{B_R} (u - k)_+^p dx \right)^{\frac{p-2}{p-1}} \\ &\quad \cdot \left( \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx \right)^{\frac{1}{p-1}}. \end{aligned}$$

If  $p = 2$ , by the same argument we have

$$\begin{aligned} \int_{B_R} (u - k)_+^2 dx &\leq \frac{\gamma R}{\theta^{\frac{N-1}{N}}} \int_{B_R} |\nabla(u - k)_+|^2 dx \\ &= \frac{2\gamma R}{\theta^{\frac{N-1}{N}}} \int_{B_R} (u - k)_+ |\nabla(u - k)_+| dx. \end{aligned}$$

■

**Proposition 5.2.2** *Assume  $1 < p < 2$  and let (5.7) hold. There exists a constant  $\gamma = \gamma(N, p)$  such that for all  $\eta \in (0, 1)$*

$$\begin{aligned} \int_{B_R(x_o)} (u - k)_+^p dx &\leq \eta R^p \int_{B_R(x_o)} |\nabla(u - k)_+|^p dx \\ &\quad + \left( \frac{\gamma p}{\theta^{\frac{N-1}{N}} \eta^{2-p}} \right)^{\frac{1}{p-1}} R^{p-1} \int_{B_R(x_o)} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx. \end{aligned} \quad (5.9)$$

**Proof.** Without loss of generality, we may assume  $x_o = 0$ . If (5.7) is in force, by the same De Giorgi embedding argument mentioned above, there is a constant  $\gamma = \gamma(N)$



such that

$$\begin{aligned}
 \int_{B_R} (u - k)_+^p dx &\leq \frac{\gamma R}{\theta^{\frac{N-1}{N}}} \int_{B_R} |\nabla(u - k)_+|^p dx \\
 &= \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R} (u - k)_+^{p-1} |\nabla(u - k)_+| dx \\
 &= \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R \cap [(u-k)_+ \leq \varepsilon |\nabla(u-k)_+|]} (u - k)_+^{p-1} |\nabla(u - k)_+| dx \\
 &\quad + \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R \cap [(u-k)_+ > \varepsilon |\nabla(u-k)_+|]} (u - k)_+^{p-1} |\nabla(u - k)_+| dx \\
 &= I_1 + I_2
 \end{aligned}$$

where  $\varepsilon > 0$  is to be chosen. The first term  $I_1$  is estimated as

$$\begin{aligned}
 I_1 &\leq \varepsilon^{p-1} \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R} |\nabla(u - k)_+|^p dx \\
 &= \eta R^p \int_{B_R} |\nabla(u - k)_+|^p dx
 \end{aligned}$$

for the choice  $\varepsilon^{p-1} = \eta \frac{\theta^{\frac{N-1}{N}}}{\gamma p} R^{p-1}$ . The choice of  $\varepsilon$  being made, estimate the second term  $I_2$  as

$$\begin{aligned}
 I_2 &\leq \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R \cap [(u-k)_+ > \varepsilon |\nabla(u-k)_+|]} \frac{1}{(u - k)_+^{2-p}} |\nabla(u - k)_+| (u - k)_+ dx \\
 &\leq \frac{1}{\varepsilon^{2-p}} \frac{\gamma p R}{\theta^{\frac{N-1}{N}}} \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx \\
 &= \eta^{\frac{p-2}{p-1}} \left( \frac{\gamma p}{\theta^{\frac{N-1}{N}}} \right)^{\frac{1}{p-1}} R^{p-1} \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx.
 \end{aligned}$$

Combining the estimates yields (5.9). ■

The previous two propositions concern general properties of functions in  $W_{\text{loc}}^{1,p}(E)$ , and have nothing to do with  $u$  belonging to a proper De Giorgi class. This is what we are going to assume next.

**Proposition 5.2.3** *Let  $p > 1$  and let  $u \in [DG]_p^+(E, \gamma)$ . If  $p \geq 2$  and (5.7) holds, then there exists a constant  $\gamma > 1$  which depends on the data and  $\theta$ , such that*

$$\begin{aligned}
 &\int_{B_{\sigma R}(x_o)} |\nabla(u - k)_+|^p dx \\
 &\leq \frac{\gamma}{(1 - \sigma)^p R} \int_{B_R(x_o)} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx.
 \end{aligned} \tag{5.10}$$

If  $1 < p < 2$ , and in addition

$$|[(u - k)_+ = 0] \cap B_{\sigma R}(x_o)| > \theta |B_{\sigma R}|,$$

for some  $\theta > 0$ , and for all  $\sigma \in \left(\frac{1}{2}, 1\right]$ , then, there exists a constant  $\gamma > 1$  which depends on the data and  $\theta$ , such that

$$\begin{aligned} \int_{B_{\sigma R}(x_o)} |\nabla(u - k)_+|^p dx \\ \leq \frac{\gamma}{(1 - \sigma)^{\frac{p}{p-1}}} \frac{1}{R} \int_{B_R(x_o)} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx. \end{aligned} \quad (5.11)$$

**Proof.** Without loss of generality, we may assume  $x_o = 0$ . Inequality (5.10) is a straightforward consequence of Proposition 5.2.1, whereas inequality (5.11) follows from Proposition 5.2.2 by interpolation.

Indeed, take  $\rho = R$  in (5.1), assume that  $B_R \subset\subset E$ , and then substitute it in the right side of (5.9); this yields

$$\begin{aligned} \int_{B_{\sigma R}} (u - k)_+^p dx &\leq \frac{\eta\gamma}{(1 - \sigma)^p} \int_{B_R} |(u - k)_+|^p dx \\ &\quad + \left( \frac{\gamma p}{\theta^{\frac{N-1}{N}} \eta^{2-p}} \right)^{\frac{1}{p-1}} (\sigma R)^{p-1} \int_{B_{\sigma R}} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx, \end{aligned}$$

for any  $\sigma \in (\frac{1}{2}, 1)$ . Let us consider the sequence of radii,

$$R_m = \sigma R + (1 - \sigma) R \sum_{i=1}^m \frac{1}{2^i}, \quad B_m \stackrel{\text{def}}{=} B_{R_m};$$

notice that  $R_{m+1} - R_m = \frac{(1 - \sigma)R}{2^{m+1}}$ . Hence, rewriting the previous inequality on  $B_m$  and  $B_{m+1}$  yields

$$\begin{aligned} \int_{B_m} (u - k)_+^p dx &\leq \left( \frac{\gamma p}{\theta^{\frac{N-1}{N}} \eta^{2-p}} \right)^{\frac{1}{p-1}} (R_m)^{p-1} \int_{B_m} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx \\ &\quad + \eta\gamma \frac{2^{mp}}{(1 - \sigma)^p} \int_{B_{m+1}} |(u - k)_+|^p dx. \end{aligned}$$

Now, choose  $\eta \in (0, 1)$ , such that  $\eta\gamma \frac{2^{mp}}{(1 - \sigma)^p} = \varepsilon_o$ , where  $\varepsilon_o \in (0, 1)$  is a constant to be determined below. Since  $\eta = \gamma^{-1}(1 - \sigma)^p 2^{-mp} \varepsilon_o$ , we have

$$\begin{aligned} \int_{B_m} (u - k)_+^p dx \\ \leq \left( \frac{Cp}{\theta^{\frac{N-1}{N}} \varepsilon_o^{2-p} (1 - \sigma)^{p(2-p)}} \right)^{\frac{1}{p-1}} b^m R^{p-1} \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx \end{aligned}$$

$$+ \varepsilon_o \int_{B_{m+1}} |(u - k)_+|^p dx,$$

where  $b = 2^{\frac{p(2-p)}{p-1}}$ . Iterating we obtain

$$\begin{aligned} \int_{B_{\frac{1+\sigma}{2}R}} (u - k)_+^p dx &\leq \varepsilon_o^m \int_{B_{m+1}} |(u - k)_+|^p dx \\ &+ \left( \frac{Cp}{\theta^{\frac{N-1}{N}} \varepsilon_o^{2-p} (1-\sigma)^{p(2-p)}} \right)^{\frac{1}{p-1}} b R^{p-1} \\ &\cdot \sum_{i=1}^{m-1} (\varepsilon_o b)^i \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx. \end{aligned}$$

Now, we choose  $\varepsilon_o$  such that  $\sum_{i=1}^{\infty} (\varepsilon_o b)^i$  converges, and let  $m \rightarrow +\infty$ . This gives

$$\int_{B_{\frac{1+\sigma}{2}R}} (u - k)_+^p dx \leq \frac{C(N, p, \theta)}{(1-\sigma)^{\frac{p}{p-1}}} R^{p-1} \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx.$$

Consequently, combining (5.1) with the inequality above yields

$$\int_{B_{\sigma R}} |\nabla(u - k)_+|^p dx \leq \frac{C(N, p, \theta)}{(1-\sigma)^{\frac{p}{p-1}}} \frac{1}{R} \int_{B_R} |\nabla(u - k)_+|^{p-1} (u - k)_+ dx.$$

■

## A Further Estimate

**Lemma 5.2.2** Consider  $u \in [DG]_p^+(E, \gamma)$ , assume  $B_{2R}(x_o) \subset\subset E$ , take  $k \in \mathbb{R}$ , let  $u_k := (u - k)_+$  and  $\mu := \sup_{B_{2R}(x_o)} (u - k)_+$ . If for  $\sigma \in (\frac{1}{2}, 1)$

$$\int_{B_{\sigma R}(x_o)} |\nabla u_k|^p dx \leq \frac{C_1}{(1-\sigma)^q R} \int_{B_R(x_o)} u_k |\nabla u_k|^{p-1} dx, \quad (5.12)$$

then

$$\int_{B_{\sigma R}(x_o)} v^{\beta+1} |\nabla u_k|^p dx \leq \frac{C_2 \mu}{(1-\sigma)^q R} \int_{B_R(x_o)} v^\beta |\nabla u_k|^{p-1} dx, \quad (5.13)$$

where  $v \stackrel{\text{def}}{=} \frac{\mu}{\mu - u_k}$ ,  $\beta > 0$ ,  $q = \begin{cases} p & \text{if } p \geq 2, \\ \frac{p}{p-1} & \text{if } 1 < p < 2, \end{cases}$  and  $C_2$  has the same functional dependence as  $C_1$ , but also depends on  $\beta$ .

**Remark 5.2.1** The method is very similar to the one used in the proof of Lemma 5.2.1. However, (5.12) is a further information, which allows to cover a wider range of exponents.

**Proof.** Without loss of generality, we may assume  $x_o = 0$  and  $\mu = 1$ . By writing  $k = h + t$ ,  $t \geq 0$  and setting  $(u - k)_+ = (w - t)_+$ , (5.12) can be rewritten as

$$\int_{B_{\sigma R}} |\nabla(w - t)_+|^p dx \leq \frac{C_1}{(1 - \sigma)^q R} \int_{B_R} (w - t)_+ |\nabla(w - t)_+|^{p-1} dx. \quad (5.14)$$

Let us now multiply both terms by  $t^\alpha$  with  $\alpha > -1$  and integrate over  $\mathbb{R}_+$ .

$$\begin{aligned} \int_{\mathbb{R}_+} t^\alpha dt \int_{B_{\sigma R}} |\nabla(w - t)_+|^p dx \\ \leq \frac{C_1}{(1 - \sigma)^q R} \int_{\mathbb{R}_+} t^\alpha dt \int_{B_R} (w - t)_+ |\nabla(w - t)_+|^{p-1} dx. \end{aligned}$$

Let us first deal with the left-hand term.

$$\begin{aligned} \int_{\mathbb{R}_+} t^\alpha dt \int_{B_{\sigma R}} |\nabla(w - t)_+|^p dx &= \int_{B_{\sigma R} \cap \{w \geq 0\}} |\nabla w|^p dx \int_0^w t^\alpha dt \\ &= \frac{1}{\alpha + 1} \int_{B_{\sigma R} \cap \{w \geq 0\}} w^{\alpha+1} |\nabla w|^p dx. \end{aligned}$$

As for the right-hand side

$$\begin{aligned} \int_{\mathbb{R}_+} t^\alpha dt \int_{B_R} (w - t)_+ |\nabla(w - t)_+|^{p-1} dx \\ = \int_{B_R \cap \{w \geq 0\}} |\nabla w|^{p-1} dx \int_0^w (w - t) t^\alpha dt \\ = \int_{B_R \cap \{w \geq 0\}} \frac{w^{\alpha+2}}{(\alpha + 1)(\alpha + 2)} |\nabla w|^{p-1} dx. \end{aligned}$$

Collecting the previous estimates yields

$$\int_{B_{\sigma R} \cap \{w \geq 0\}} w^{\alpha+1} |\nabla w|^p dx \leq \frac{C_1}{(\alpha + 2)(1 - \sigma)^q R} \int_{B_R \cap \{w \geq 0\}} w^{\alpha+2} |\nabla w|^{p-1} dx,$$

and it is easy to see that such an inequality holds also for  $\alpha = -1$  (we simply reduce to (5.14)). Let us now set  $\alpha + 1 = n$ ,  $n \in \mathbb{N}$ . The previous inequality becomes

$$\begin{aligned} \int_{B_{\sigma R} \cap \{w \geq 0\}} w^n |\nabla w|^p dx \\ \leq \frac{C_1}{(n + 1)(1 - \sigma)^q R} \int_{B_R \cap \{w \geq 0\}} w^{n+1} |\nabla w|^{p-1} dx. \end{aligned} \quad (5.15)$$

If we consider the function

$$\varphi(w) = \left( \frac{1}{1 - w} \right)^{\beta+1}, \quad \beta > 0,$$

it is easy to see that it is a convex, analytic function; hence,

$$\varphi(w) = \sum_{n=0}^{\infty} \frac{\varphi^{(n)}(0)}{n!} w^n, \quad \text{where } \forall n \in \mathbb{N} \quad \varphi^{(n)}(0) > 0. \quad (5.16)$$

We proceed as we did in the proof of Lemma 5.2.1. If we multiply both terms of (5.15) by  $\frac{\varphi^{(n)}(0)}{n!}$ , sum over  $n$ , and interchange sum with integral, we obtain

$$\begin{aligned} & \int_{B_{\sigma R} \cap \{w \geq 0\}} \sum_{n=0}^{\infty} \frac{\varphi^{(n)}(0)}{n!} w^n |\nabla w|^p dx \\ & \leq \frac{C_1}{(1-\sigma)^q R} \int_{B_R \cap \{w \geq 0\}} \sum_{n=0}^{\infty} \frac{\varphi^{(n)}(0)}{(n+1)!} w^{n+1} |\nabla w|^{p-1} dx. \end{aligned} \quad (5.17)$$

Since

$$\sum_{n=0}^{\infty} \frac{\varphi^{(n)}(0)}{(n+1)!} w^{n+1} = \int_0^w \varphi(t) dt = \int_0^w \frac{1}{(1-t)^{\beta+1}} dt = \frac{1}{\beta} \left[ \frac{1}{(1-w)^\beta} - 1 \right],$$

from (5.17) we conclude that for any  $\beta > 0$

$$\begin{aligned} & \int_{B_{\sigma R} \cap \{w \geq 0\}} \left( \frac{1}{1-w} \right)^{\beta+1} |\nabla w|^p dx \\ & \leq \frac{C_2}{(1-\sigma)^q R} \int_{B_R \cap \{w \geq 0\}} \left( \frac{1}{1-w} \right)^\beta |\nabla w|^{p-1} dx. \end{aligned}$$

Taking into account the previous normalization of  $\mu$ , we complete the proof.  $\blacksquare$

### A Consequence of Equation (5.13)

**Lemma 5.2.3** Consider  $u \in [DG]_p^+(E, \gamma)$ , assume  $B_{2R}(x_o) \subset\subset E$ , take  $k \in \mathbb{R}$ , let  $u_k := (u - k)_+$  and  $\mu := \sup_{B_{2R}(x_o)} (u - k)_+$ . If  $u_k$  satisfies (5.12) for  $\sigma \in (\frac{1}{2}, 1)$ , then

$$\int_{B_{\sigma R}(x_o)} v^{\beta+1} |\nabla u_k|^p dx \leq \frac{C_3 \mu^p}{(1-\sigma)^{pq} R^p} \int_{B_R(x_o)} v^{\beta+1-p} dx. \quad (5.18)$$

where  $v \stackrel{\text{def}}{=} \frac{\mu}{\mu - u_k}$ ,  $\beta > 0$ ,  $q = \begin{cases} p & \text{if } p \geq 2, \\ \frac{p}{p-1} & \text{if } 1 < p < 2, \end{cases}$  and  $C_3$  has the same functional dependence as  $C_1$ , but also depends on  $\beta$ .

**Proof.** Without loss of generality, we may assume  $x_o = 0$ . We start from (5.13) and set

$$R_o = \sigma R, \quad R_n = \sigma R + (1-\sigma)R \sum_{i=1}^n 2^{-i}, \quad n = 1, 2, \dots$$

Consequently, by Young's inequality

$$\int_{B_{R_n}} v^{\beta+1} |\nabla u_k|^p dx \leq \frac{C_2 \mu 2^{(n+1)q}}{(1-\sigma)^q R} \int_{B_{R_{n+1}}} v^\beta |\nabla u_k|^{p-1} dx$$

$$\begin{aligned}
 &= \frac{C_3 \mu 2^{nq}}{(1-\sigma)^q R} \int_{B_{R_{n+1}}} |\nabla u_k|^{p-1} v^{\frac{(\beta+1)(p-1)}{p}} v^{\frac{\beta+1-p}{p}} dx \\
 &\leq \eta \int_{B_{R_{n+1}}} |\nabla u_k|^p v^{\beta+1} dx + \frac{C_3^p \mu^p 2^{npq}}{\eta^{p-1} (1-\sigma)^{pq} R^p} \int_{B_{R_{n+1}}} v^{\beta+1-p} dx,
 \end{aligned}$$

where  $\eta \in (0, 1)$  is to be fixed. If we set

$$M_n = \int_{B_{R_n}} |\nabla u_k|^p v^{\beta+1} dx, \quad B = \frac{C_3^p \mu^p}{\eta^{p-1} (1-\sigma)^{pq} R^p} \int_{B_R} v^{\beta+1-p} dx, \quad d = 2^{pq},$$

we have the recursive inequality

$$M_n \leq \eta M_{n+1} + d^n B, \quad n = 0, 1, 2, \dots \quad (5.19)$$

By iteration

$$M_o \leq \eta^n M_{n+1} + B \sum_{i=0}^n (\eta d)^i.$$

We choose  $\eta = \frac{1}{2d}$  so that the sum on the right-hand side can be majorised by a convergent series and let  $n \rightarrow +\infty$  to conclude.  $\blacksquare$

### 5.2.3 A Weak Harnack Inequality at the Boundary

Assume that the geometric density condition holds, and choose  $k \in \mathbb{R}$  such that

$$u_k := (u - k)_+ = 0 \quad \text{in } B_R(x_o) \cap (\partial E)_{\text{loc}};$$

we have

$$|[(u - k)_+ = 0] \cap B_{\sigma R}(x_o)| > \theta |B_{\sigma R}| \quad \forall \sigma \in (0, 1).$$

Hence, Proposition 5.2.3 holds, and Lemma 5.2.3 can be applied. Therefore, for any  $\rho \in (0, R)$ , and for any  $\sigma \in (\frac{1}{2}, 1)$  we have

$$\int_{B_{\sigma\rho}(x_o)} v^{\beta+1} |\nabla u_k|^p dx \leq \frac{C_3 \mu^p}{(1-\sigma)^{pq} \rho^p} \int_{B_\rho(x_o)} v^{\beta+1-p} dx. \quad (5.20)$$

If  $\sigma \in (0, \frac{1}{2}]$  (5.20) continues to hold, so that we can assume without loss of generality that it holds for any  $\sigma \in (0, 1)$ .

Choosing  $\beta = p - 1$  in (5.20) yields

$$\int_{B_{\sigma\rho}(x_o)} |\nabla \ln(\mu - u_k)|^p dx \leq \frac{C_3}{(1-\sigma)^{pq}} \rho^{N-p}, \quad \forall \sigma \in (0, 1).$$

Working as in [100, § 1.2] we conclude that there exist  $C_4 > 1$  and  $\tau_o \in (0, 1)$  such that for all  $\tau \in (0, \tau_o)$  and for all  $\sigma \in (0, \frac{1}{2})$

$$\begin{aligned} \left( \int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{-\tau} dx \right) \left( \int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{\tau} dx \right) &\leq C_4 \\ \Leftrightarrow \int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{\tau} dx &\leq \frac{C_4}{\int_{B_{\sigma\rho}(x_o)} \frac{1}{(\mu - u_k)^{\tau}} dx}. \end{aligned}$$

On the other hand, (5.4) can be rewritten as

$$\frac{1}{\inf_{B_{\frac{\sigma}{2}\rho}(x_o)} (\mu - u_k)^{\tau}} \leq 2^N C_{\tau}^{\tau} \int_{B_{\sigma\rho}(x_o)} \frac{1}{(\mu - u_k)^{\tau}} dx.$$

Combining the previous two inequalities, we conclude that there exist  $C_5 > 1$  and  $\tau_o \in (0, 1)$ , both depending on the data and  $\theta$ , such that for any  $\tau \in (0, \tau_o)$  and for any  $\sigma \in (0, \frac{1}{2})$

$$\int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{\tau} dx \leq C_5 \inf_{B_{\frac{\sigma}{2}\rho}(x_o)} (\mu - u_k)^{\tau}, \quad (5.21)$$

which can be seen as a weak Harnack inequality at the boundary.

### 5.3 Proof of Theorem 5.1.1

Pick  $x_o \in \partial E$ ,  $2\rho < R$ , and with reference to the notation of Section 5.1, for simplicity let

$$\begin{aligned} \mu^+ &\stackrel{\text{def}}{=} \mu^+(2\rho), & \mu^- &\stackrel{\text{def}}{=} \mu^-(2\rho), & \omega &\stackrel{\text{def}}{=} \omega(2\rho), \\ h^+ &\stackrel{\text{def}}{=} h^+(2\rho), & h^- &\stackrel{\text{def}}{=} h^-(2\rho), & \omega_h &\stackrel{\text{def}}{=} \omega_h(2\rho). \end{aligned}$$

We remark that at least one of the two inequalities

$$\mu^+ - \frac{1}{4}\omega \leq h^+, \quad \mu^- + \frac{1}{4}\omega \geq h^-$$

cannot hold. Indeed, if both held, by subtracting from one another we would obtain

$$\omega \leq 2\omega_h,$$

and there would be nothing to prove. Hence, let us assume that

$$k := \mu^+ - \frac{1}{4}\omega > h^+; \quad (5.22)$$

at the end of the Section we will briefly discuss the changes that are needed in order to cover the case  $\mu^- + \frac{1}{4}\omega < h^-$ .

By (5.22), we have

$$u_k := (u - k)_+ = 0 \quad \text{in } B_\rho(x_o) \cap (\partial E)_{\text{loc}};$$

and by the geometric density condition holds, we have

$$|[(u - k)_+ = 0] \cap B_{\sigma\rho}(x_o)| > \theta |B_{\sigma\rho}| \quad \forall \sigma \in (0, 1).$$

Let  $\mu \stackrel{\text{def}}{=} \sup_{B_{2\rho}(x_o)} u_k$ ; once more, Proposition 5.2.3 holds, and Lemma 5.2.3 can be applied.

Therefore, for any  $\sigma \in (0, 1)$  we have

$$\int_{B_{\frac{3}{4}\sigma\rho}(x_o)} v^{\beta+1} |\nabla u_k|^p dx \leq \frac{C_3 \mu^p}{\sigma^{pq} \rho^p} \int_{B_{\sigma\rho}(x_o)} v^{\beta+1-p} dx. \quad (5.23)$$

Choosing  $\beta + 1 = s$  with  $0 < s < p$  in (5.23) yields

$$\int_{B_{\frac{3}{4}\sigma\rho}(x_o)} |\nabla (\mu - u_k)^{\frac{p-s}{p}}|^p dx \leq \frac{C_4}{\sigma^{pq} \rho^p} \int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{p-s} dx,$$

where  $C_4 > 0$  depends on the data,  $\theta$ , and  $s$ .

Let  $\varphi \in C_o^\infty(B_{\sigma\rho}(x_o))$ , such that  $0 \leq \varphi \leq 1$ , and

$$\varphi \equiv 1 \quad \text{in } B_{\frac{\sigma}{2}\rho}(x_o), \quad |\nabla \varphi| \leq \frac{5}{\sigma\rho}.$$

It is straightforward to see that

$$\begin{aligned} \int_{B_{\frac{3}{4}\sigma\rho}(x_o)} \left| \nabla \left[ \left( \frac{\mu - u_k}{\mu} \right)^{\frac{p-s}{p}} \varphi \right] \right|^p dx \\ \leq \frac{C_5}{\sigma^{pq} \sigma^p \rho^p} \int_{B_{\sigma\rho}(x_o)} \left( \frac{\mu - u_k}{\mu} \right)^{p-s} dx. \end{aligned}$$

Now we rely on an argument employed, for example, in [33], based on the  $p$ -capacity of a set. Based on its definition (see, for example, [40] for a simple introduction to the notion of  $p$ -capacity), it is not hard to see that

$$\int_{B_{\frac{3}{4}\sigma\rho}(x_o)} \left| \nabla \left[ \left( \frac{\mu - u_k}{\mu} \right)^{\frac{p-s}{p}} \varphi \right] \right|^p dx \geq \text{cap}_p(B_{\frac{\sigma}{2}\rho} \cap (\mathbb{R}^N \setminus E)),$$

and therefore,

$$\text{cap}_p(B_{\frac{\sigma}{2}\rho} \cap (\mathbb{R}^N \setminus E)) \leq \frac{C_5}{\sigma^{pq} \sigma^p \rho^p} \int_{B_{\sigma\rho}(x_o)} \left( \frac{\mu - u_k}{\mu} \right)^{p-s} dx$$



By the  $p$ -capacity properties and due to the geometric density condition we get

$$\text{cap}_p(B_{\frac{\sigma}{2}\rho} \cap (\mathbb{R}^N \setminus E)) \geq \theta \left( \frac{\sigma}{2}\rho \right)^{N-p},$$

whence we conclude

$$\mu^{p-s} \leq \frac{C_6}{\sigma^{pq}} \int_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{p-s} dx, \quad (5.24)$$

where  $C_6 > 0$  depends also on  $\theta$ . If we now choose  $s$  such that  $0 < p-s < \tau_o$ , by (5.21) we conclude that

$$\mu^{p-s} \leq \frac{C_7}{\sigma^{pq}} \inf_{B_{\sigma\rho}(x_o)} (\mu - u_k)^{p-s},$$

i.e., highlighting the explicit dependence on the radius of the ball where the supremum is taken,

$$\mu(2\rho) \leq C_8 [\mu(2\rho) - \mu(\frac{\sigma}{2}\rho)];$$

Notice that the constant  $C_8$  depends on  $\sigma$  as well. If we take  $\sigma = \frac{1}{4}$ , we have

$$\mu(2\rho) \leq C_8 \left[ \mu(2\rho) - \mu\left(\frac{\rho}{8}\right) \right]. \quad (5.25)$$

Since

$$\mu(2\rho) = \sup_{B_{2\rho}(x_o)} (u - k)_+, \quad \text{with} \quad k = \sup_{B_{2\rho}(x_o)} u - \frac{1}{4}\omega(2\rho),$$

it is apparent that  $\mu(2\rho) = \frac{1}{4}\omega(2\rho)$ . Hence (5.25) yields

$$\begin{aligned} \frac{\omega(2\rho)}{4} &\leq C_8 \left[ \mu^+(2\rho) - k - \left( \mu^+\left(\frac{\rho}{8}\right) - k \right) \right] \\ &\leq C_8 \left[ \mu^+(2\rho) - \mu^-(2\rho) - \left( \mu^+\left(\frac{\rho}{8}\right) - \mu^-\left(\frac{\rho}{8}\right) \right) \right] \\ &\leq C_8 \left[ \omega(2\rho) - \omega\left(\frac{\rho}{8}\right) \right], \end{aligned}$$

i.e.

$$\omega\left(\frac{\rho}{8}\right) \leq \left(1 - \frac{1}{4C_8}\right) \omega(2\rho).$$

From here on, a standard iteration allows to conclude the proof.

On the other hand, if  $\mu^- + \frac{1}{4}\omega < h^+$  holds, instead of working with  $u$ , we can consider  $v := -u$ , which takes the boundary value  $\tilde{h} \equiv -h$ . It is apparent that

$$\nu^+(2\rho) \stackrel{\text{def}}{=} \sup_{B_{2\rho}(x_o)} v = -\mu^-(2\rho), \quad \nu^-(2\rho) \stackrel{\text{def}}{=} \inf_{B_{2\rho}(x_o)} v = -\mu^+(2\rho),$$

$$\tilde{\omega}(2\rho) \stackrel{\text{def}}{=} \nu^+(2\rho) - \nu^-(2\rho) \equiv \omega(2\rho),$$

$$\tilde{h}^+(2\rho) = -h^-(2\rho), \quad \tilde{h}^-(2\rho) = -h^+(2\rho),$$

and also

$$\mu^- + \frac{1}{4}\omega < h^- \quad \Leftrightarrow \quad \nu^+ - \frac{1}{4}\tilde{\omega} > \tilde{h}^+,$$

and we finish exactly as in the former case. ■

## Chapter 6

# Non-local Uniformly Elliptic Equations

### 6.1 Main Results

We study the non-local integral–differential problem

$$\begin{cases} \mathcal{L}u = f, & x \in \Omega, \\ u = 0, & x \in \mathbb{R}^N \setminus \Omega, \end{cases} \quad (6.1)$$

where  $\Omega \subset \mathbb{R}^N$  is a bounded domain,  $N > 2s$  with  $s \in (0, 1)$ , and  $\mathcal{L}$  denotes the non-local operator

$$\mathcal{L}u(x) = \text{P. V.} \int_{\mathbb{R}^N} \frac{A(x, y)(u(x) - u(y))}{|x - y|^{N+2s}} dy,$$

with symmetric kernel  $A(x, y) = A(y, x)$  satisfying

$$0 < \lambda \leq A(x, y) \leq \Lambda, \quad \lambda, \Lambda > 0.$$

The datum is assumed to satisfy

$$f \in L^q(\Omega), \quad q > \frac{N}{2s}.$$

(The symbols  $q$  and  $s$  in this chapter are unrelated to those in previous chapters.)

**Theorem 6.1.1** *Let  $u$  be a weak subsolution of (6.1). Then there exists a constant  $C = C(\lambda, N, q, s)$  such that*

$$\|u_+\|_{L^\infty(\Omega)} \leq C \|f\|_{L^{\frac{N}{2s}}(\Omega)} \left[ \log \left( \frac{\|f\|_{L^q(\Omega)}}{\|f\|_{L^{\frac{N}{2s}}(\Omega)}} + 1 \right) + 1 \right], \quad q > \frac{N}{2s}. \quad (6.2)$$

**Remark 6.1.1** Notice that estimate (6.2) improves the classical  $L^\infty$  bound; see [37, Proposition 2.3.11]. The ratio  $\|f\|_{L^q(\Omega)}/\|f\|_{L^{\frac{N}{2s}}(\Omega)}$  quantifies how much the  $L^q$  norm exceeds the critical  $L^{\frac{N}{2s}}$  norm.

**Example 6.1.1 (Sharpness of the integrability exponent)** *For suitable  $f \in L^{\frac{N}{2s}}(\Omega)$ , problem (6.1) possesses an unbounded weak solution.*

## Organisation of the chapter

Section 6.2 recalls the definitions of fractional Sobolev spaces and weak solutions, and states several key lemmas required for the proof of Theorem 6.1.1. Section 6.3 constructs Example 6.1.1. Finally, Section 6.4 presents the proof of Theorem 6.1.1.

## 6.2 Preliminaries

### Fractional Sobolev spaces

Let  $0 < s < 1$  and  $N \geq 1$ . The fractional Sobolev space is defined by

$$H^s(\mathbb{R}^N) = \left\{ u \in L^2(\mathbb{R}^N) \mid [u]_{H^s(\mathbb{R}^N)} < \infty \right\},$$

where the Gagliardo semi-norm is

$$[u]_{H^s(\mathbb{R}^N)} = \left( \iint_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|u(x) - u(y)|^2}{|x - y|^{N+2s}} dx dy \right)^{1/2}.$$

The corresponding norm is

$$\|u\|_{H^s(\mathbb{R}^N)} = \left( \|u\|_{L^2(\mathbb{R}^N)}^2 + [u]_{H^s(\mathbb{R}^N)}^2 \right)^{1/2}.$$

### Weak subsolutions

**Definition 6.2.1** *A function  $u \in H^s(\mathbb{R}^N)$  with  $N > 2s$  and  $u \leq 0$  in  $\mathbb{R}^N \setminus \Omega$  is called a weak subsolution of (6.1) if*

$$\frac{1}{2} \iint_{\mathbb{R}^N \times \mathbb{R}^N} A(x, y) (u(x) - u(y)) (\phi(x) - \phi(y)) d\mu \leq \int_{\Omega} f \phi dx,$$

for every non-negative test function  $\phi \in H^s(\mathbb{R}^N)$  with  $\phi \equiv 0$  in  $\mathbb{R}^N \setminus \Omega$ , where

$$d\mu = \frac{1}{|x - y|^{N+2s}} dx dy.$$

**Remark 6.2.1** The integrability assumption on  $f$  ensures the right-hand side is well-defined; see [37, Chapter 2, Remark 2.2.17].

## Auxiliary lemmas for Moser iteration

we present several Lemmas that are essential for implementing the Moser iteration technique. The first Lemma provides a formula analogous to what one could obtain using the chain rule when applying a local differential operator to the exponential function  $e^x$ .

**Lemma 6.2.1** *For any  $s, t \in \mathbb{R}$ , and any fixed  $\alpha > 0$ , we have the following inequality*

$$(s - t)(e^{\alpha s} - e^{\alpha t}) \geq \frac{1}{\alpha} \left( e^{\frac{\alpha}{2}s} - e^{\frac{\alpha}{2}t} \right)^2.$$

**Proof.** First, for the case  $s > t$ , we have

$$e^{\alpha s} - e^{\alpha t} = \alpha \int_t^s e^{\alpha m} dm \leq \alpha e^{\alpha s} (s - t),$$

whence, we deduce

$$\begin{aligned} (s - t)(e^{\alpha s} - e^{\alpha t}) &\geq \frac{1}{\alpha e^{\alpha s}} (e^{\alpha s} - e^{\alpha t})^2 = \frac{1}{\alpha} (e^{\frac{\alpha}{2}s} - e^{\frac{\alpha}{2}t + \frac{\alpha}{2}(t-s)})^2 \\ &\geq \frac{1}{\alpha} \left( e^{\frac{\alpha}{2}s} - e^{\frac{\alpha}{2}t} \right)^2. \end{aligned}$$

For the case  $s < t$ , similarly, we have

$$e^{\alpha t} - e^{\alpha s} = \alpha \int_s^t e^{\alpha m} dm \leq \alpha e^{\alpha t} (t - s).$$

Consequently, we obtain

$$\begin{aligned} (s - t)(e^{\alpha s} - e^{\alpha t}) &= (t - s)(e^{\alpha t} - e^{\alpha s}) \geq \frac{1}{\alpha e^{\alpha t}} (e^{\alpha t} - e^{\alpha s})^2 \\ &= \frac{1}{\alpha} (e^{\frac{\alpha}{2}t} - e^{\frac{\alpha}{2}s + \frac{\alpha}{2}(s-t)})^2 \geq \frac{1}{\alpha} \left( e^{\frac{\alpha}{2}t} - e^{\frac{\alpha}{2}s} \right)^2. \end{aligned}$$

Therefore, the proof of the Lemma 6.2.1 is complete.  $\square$

The following result is similar to the previous lemma and serves as an alternative chain rule for polynomial, when working with the nonlocal differential operator.

**Lemma 6.2.2** *For any  $\beta \geq 1$ ,  $s, t \geq 0$ , we have*

$$(s - t)(s^\beta - t^\beta) \geq \frac{1}{\beta} \left( s^{\frac{\beta+1}{2}} - t^{\frac{\beta+1}{2}} \right)^2.$$

**Proof.** The proof is similar to that of Lemma 6.2.1; for the reader's convenience, we provide the detailed argument below.

First, we consider the case  $s \geq t$ : we have

$$s^\beta - t^\beta = \beta \int_t^s m^{\beta-1} dm \leq \beta s^{\beta-1}(s - t),$$

whence

$$\begin{aligned} (s - t)(s^\beta - t^\beta) &\geq \frac{1}{\beta s^{\beta-1}}(s^\beta - t^\beta)^2 = \frac{1}{\beta} \left( s^{\frac{\beta+1}{2}} - t^{\frac{\beta+1}{2}} \cdot \frac{t^{\frac{\beta-1}{2}}}{s^{\frac{\beta-1}{2}}} \right)^2 \\ &\geq \frac{1}{\beta} \left( s^{\frac{\beta+1}{2}} - t^{\frac{\beta+1}{2}} \right)^2. \end{aligned}$$

For the case  $s < t$ , we similarly get

$$t^\beta - s^\beta = \beta \int_s^t m^{\beta-1} dm \leq \beta t^{\beta-1}(t - s);$$

hence,

$$\begin{aligned} (s - t)(s^\beta - t^\beta) &= (t - s)(t^\beta - s^\beta) \geq \frac{1}{\beta t^{\beta-1}}(t^\beta - s^\beta)^2 \\ &= \frac{1}{\beta} \left( t^{\frac{\beta+1}{2}} - s^{\frac{\beta+1}{2}} \cdot \frac{s^{\frac{\beta-1}{2}}}{t^{\frac{\beta-1}{2}}} \right)^2 \geq \frac{1}{\beta} \left( t^{\frac{\beta+1}{2}} - s^{\frac{\beta+1}{2}} \right)^2. \end{aligned}$$

Therefore, the proof of the Lemma 6.2.2 is complete.  $\square$

We now present an auxiliary formula that resembles the change of variables technique commonly used in the local differential equations.

**Lemma 6.2.3** *For any  $s, t \in \mathbb{R}$  and  $r_1, r_2, \alpha > 0$ , we have*

$$(s - t)(r_1 e^{\alpha s} - r_2 e^{\alpha t}) \geq \frac{1}{\alpha} (e^{\alpha s} - e^{\alpha t})(r_1 - r_2).$$

**Proof.** First, when  $s \geq t$ , we have

$$s - t \geq \frac{1}{e^{\alpha s}} \int_t^s e^{\alpha m} dm = \frac{1}{\alpha e^{\alpha s}} (e^{\alpha s} - e^{\alpha t}),$$

and

$$s - t \leq \frac{1}{e^{\alpha t}} \int_t^s e^{\alpha m} dm = \frac{1}{\alpha e^{\alpha t}} (e^{\alpha s} - e^{\alpha t}).$$

Thus,

$$(s - t)(r_1 e^{\alpha s} - r_2 e^{\alpha t})$$

$$\begin{aligned}
 &= (s-t)r_1e^{\alpha s} - (s-t)r_2e^{\alpha t} \geq \frac{r_1}{\alpha}(e^{\alpha s} - e^{\alpha t}) - \frac{r_2}{\alpha}(e^{\alpha s} - e^{\alpha t}) \\
 &= \frac{(e^{\alpha s} - e^{\alpha t})}{\alpha}(r_1 - r_2).
 \end{aligned}$$

When  $s < t$ , we similarly have

$$\frac{1}{\alpha e^{\alpha t}}(e^{\alpha t} - e^{\alpha s}) \leq t - s \leq \frac{1}{\alpha e^{\alpha s}}(e^{\alpha t} - e^{\alpha s}).$$

Consequently,

$$\begin{aligned}
 (s-t)(r_1e^{\alpha s} - r_2e^{\alpha t}) &= (t-s)(r_2e^{\alpha t} - r_1e^{\alpha s}) = (t-s)r_2e^{\alpha t} - (t-s)r_1e^{\alpha s} \\
 &\geq \frac{r_2}{\alpha}(e^{\alpha t} - e^{\alpha s}) - \frac{r_1}{\alpha}(e^{\alpha t} - e^{\alpha s}) = \frac{(e^{\alpha t} - e^{\alpha s})}{\alpha}(r_2 - r_1) \\
 &= \frac{(e^{\alpha s} - e^{\alpha t})}{\alpha}(r_1 - r_2).
 \end{aligned}$$

Therefore, the proof of the Lemma 6.2.3 is complete.  $\square$

## Fractional Sobolev embedding

The following embedding is standard; see [85, Proposition 15.5] and [92].

**Lemma 6.2.4** *Let  $u \in H^s(\mathbb{R}^N)$  with  $0 < s < 1$ . Then*

$$\|u\|_{L^{\frac{2N}{N-2s}}(\mathbb{R}^N)} \leq C_* \left( \iint_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|u(x) - u(y)|^2}{|x - y|^{N+2s}} dx dy \right)^{1/2}.$$

## Exponential integrability

The next lemma shows that the positive part of any weak subsolution enjoys exponential integrability.

**Lemma 6.2.5** *Let  $u$  be a weak subsolution of (6.1). Then for every*

$$0 < \alpha < \frac{\lambda}{C_*^2 \|f\|_{L^{\frac{N}{2s}}(\Omega)}},$$

*we have*

$$\int_{\Omega} e^{\frac{2\alpha N}{N-2s} u_+(x)} dx \leq \left( \frac{2\lambda |\Omega|^{1-\frac{4s}{N+2s}} \|f\|_{L^{\frac{N}{2s}}(\Omega)}}{\lambda - \alpha C_*^2 \|f\|_{L^{\frac{N}{2s}}(\Omega)}} + |\Omega|^{\frac{N-2s}{2N}} \right)^{\frac{2N}{N-2s}},$$

*where  $C_*$  is the constant from Lemma 6.2.4.*

**Proof.** Taking  $\phi(x) = e^{2\alpha u_+(x)} - 1$  in Definition 6.2.1 yields,

$$\frac{\lambda}{2} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (u(x) - u(y))(e^{2\alpha u_+(x)} - e^{2\alpha u_+(y)}) d\mu \leq \int_{\Omega} f(x)(e^{2\alpha u_+(x)} - 1) dx. \quad (6.3)$$

Taking  $s = u_+(x)$ ,  $t = u_+(y)$  in Lemma 6.2.1 and then applying Lemma 6.2.4, we obtain

$$\begin{aligned} & \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (u(x) - u(y))(e^{2\alpha u_+(x)} - e^{2\alpha u_+(y)}) d\mu \\ & \geq \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (u_+(x) - u_+(y))(e^{2\alpha u_+(x)} - e^{2\alpha u_+(y)}) d\mu \\ & \geq \frac{2}{\alpha} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left( e^{\frac{2\alpha}{2} u_+(x)} - e^{\frac{2\alpha}{2} u_+(y)} \right)^2 d\mu \\ & = \frac{2}{\alpha} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left( e^{\frac{2\alpha}{2} u_+(x)} - 1 - (e^{\frac{2\alpha}{2} u_+(y)} - 1) \right)^2 d\mu \\ & \geq \frac{2}{\alpha C_*^2} \left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{N}}. \end{aligned}$$

Regarding the right side of (6.3), by the Hölder's inequality, we have

$$\begin{aligned} & \int_{\Omega} f(x)(e^{2\alpha u_+(x)} - 1) dx \\ & = \int_{\Omega} f(x) \left[ (e^{\frac{2\alpha}{2} u_+(x)} - 1)^2 + 2(e^{\frac{2\alpha}{2} u_+(x)} - 1) \right] dx \\ & \leq \|f\|_{L^{\frac{N}{2s}}(\Omega)} \left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{N}} \\ & \quad + 2 \left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} \|f\|_{L^{\frac{2N}{N+2s}}(\Omega)}. \end{aligned}$$

Combining this with the inequality above, and using the assumption  $\alpha \in (0, \frac{\lambda}{C_*^2 \|f\|_{L^{\frac{N}{2s}}(\Omega)}})$ , we have

$$\begin{aligned} & \left( 1 - \frac{\alpha C_*^2 \|f\|_{L^{\frac{N}{2s}}(\Omega)}}{\lambda} \right) \left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{N}} \\ & \leq 2 \left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} \|f\|_{L^{\frac{2N}{N+2s}}(\Omega)}. \end{aligned}$$

Without loss of generality, we assume  $\left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} \neq 0$ , then we divide both sides of the above equality by this quantity

$$\left( \int_{\Omega} \left| e^{\frac{2\alpha u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} \leq \frac{2\lambda \|f\|_{L^{\frac{2N}{N+2s}}(\Omega)}}{(\lambda - \alpha C_*^2 \|f\|_{L^{\frac{N}{2s}}(\Omega)})}$$



$$\leq \frac{2\lambda|\Omega|^{1-\frac{4s}{N+2s}}\|f\|_{L^{\frac{N}{2s}}(\Omega)}}{(\lambda - \alpha C_*^2\|f\|_{L^{\frac{N}{2s}}(\Omega)})}.$$

The proof is complete.  $\square$

### 6.3 Proof of Example 6.1.1

**Proof.** We set  $\Omega = B_{\frac{1}{e}}(0)$  and define  $v(x) = \begin{cases} \ln \ln \frac{1}{|x|}, & \text{in } B_{\frac{1}{e}}(0), \\ 0, & \text{in } \mathbb{R}^N \setminus B_{\frac{1}{e}}(0), \end{cases}$  and then take

$$u(x) = v(x)\eta(x),$$

where  $\eta(x) \in C_0^\infty(B_{\frac{1}{e}}(0))$  is a radial cut-off function with  $\eta \equiv 1$  in  $B_{\frac{1}{2e}}(0)$ ,  $\eta \equiv 0$  in  $B_{\frac{1}{e}}(0) \setminus B_{\frac{3}{4e}}(0)$ .

It is easy to verify that  $u \in H^s(\mathbb{R}^N)$ ; then we take

$$f(x) = P.V. \int_{\mathbb{R}^N} \frac{u(x) - u(y)}{|x - y|^{N+2s}} dy, \quad \text{for } x \in B_{\frac{1}{e}}(0).$$

Claim 1.  $f(x)$  is bounded on  $B_{\frac{1}{e}}(0) \setminus B_{c_o}(0)$ , where  $c_o = \frac{1}{6e}$ .

Indeed, for  $x \in B_{\frac{1}{e}}(0) \setminus B_{c_o}(0)$

$$\begin{aligned} f(x) &= P.V. \int_{\mathbb{R}^N} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\ &= P.V. \int_{B_{\frac{c_o}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz + P.V. \int_{\mathbb{R}^N \setminus B_{\frac{c_o}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz. \end{aligned}$$

For the first term on the right side of the previous equality, since  $|z \pm x| \geq |x| - \frac{c_o}{2} \geq \frac{c_o}{2}$ , and by applying Taylor's expansion, there exist parameters  $\theta_1, \theta_2 \in (0, 1)$  such that

$$\begin{aligned} &P.V. \int_{B_{\frac{c_o}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\ &= \frac{1}{2} P.V. \int_{B_{\frac{c_o}{2}}(0)} \frac{u(x) - u(x+z) + u(x) - u(x-z)}{|z|^{N+2s}} dz \\ &= -\frac{1}{2} \int_{B_{\frac{c_o}{2}}(0)} \frac{\sum_{i,j}^{N,N} u_{x_i, x_j}(x + \theta_1 z) z_i z_j + \sum_{i,j}^{N,N} u_{x_i, x_j}(x - \theta_2 z) z_i z_j}{|z|^{N+2s}} dz \\ &\leq \frac{N^2 \max_{|x| \geq \frac{c_o}{2}, i, j \in [1, N]} |D_{ij} u(x)|}{2} \int_{B_{\frac{c_o}{2}}(0)} \frac{1}{|z|^{N+2s-2}} dz \\ &= C(N) \max_{|x| \geq \frac{c_o}{2}, i, j \in [1, N]} |D_{ij} u(x)| \int_0^{\frac{c_o}{2}} \frac{1}{r^{2s-1}} dr \end{aligned}$$

$$= \frac{C(N)}{2-2s} \left(\frac{c_o}{2}\right)^{2-2s} \max_{|x| \geq \frac{c_o}{2}, i, j \in [1, N]} |D_{ij}u(x)| < +\infty.$$

As for the second term, we split it into two parts

$$\begin{aligned} P.V. \int_{\mathbb{R}^N \setminus B_{\frac{c_o}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\ = \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x| \geq |x+z|\}} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz + \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x| \leq |x+z|\}} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz. \end{aligned}$$

Regarding the second part, since  $u(x) = u(|x|)$  is a decreasing function with respect to  $|x|$ , and  $|x| \geq c_o$ , we derive

$$\begin{aligned} \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x| \leq |x+z|\}} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz &\leq 2u(|c_o|) \int_{\{|z| \geq \frac{c_o}{2}\}} \frac{1}{|z|^{N+2s}} dz \\ &= C(N)u(|c_o|) \int_{\frac{c_o}{2}}^{\infty} \frac{1}{r^{2s+1}} dr \\ &= \frac{C(N)v(|c_o|)}{s} \left(\frac{c_o}{2}\right)^{-2s} < +\infty; \end{aligned}$$

for the first part, by the Fubini's Theorem, we get

$$\begin{aligned} \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x| \geq |x+z|\}} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\ = \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x| \geq |x+z|\}} \frac{\int_{|z+x|}^{|z|} u'(t) dt}{|z|^{N+2s}} dz \\ = \int_0^{|x|} u'(t) \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\ = \int_0^{|x| - \frac{c_o}{2}} u'(t) \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\ + \int_{|x| - \frac{c_o}{2}}^{|x|} u'(t) \int_{\{|z| > \frac{c_o}{2}\} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt. \end{aligned}$$

Notice that when  $t \in [0, |x| - \frac{c_o}{2}]$ , we have

$$\{|z| > \frac{c_o}{2}\} \cap \{|x+z| \leq t\} = \{|x+z| \leq t\}.$$

Hence,

$$\begin{aligned} \int_0^{|x| - \frac{c_o}{2}} u'(t) \int_{\{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\ \leq C(N) \int_0^{|x| - \frac{c_o}{2}} |u'(t)| \int_{|x|-t}^{|x|+t} \frac{r^{N-1}}{r^{N+2s}} dr dt \end{aligned}$$

$$\begin{aligned}
 &= \frac{C(N)}{2s} \int_0^{|x|-\frac{c_o}{2}} |u'(t)| \left( \frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \right) dt \\
 &\leq \frac{C(N)}{2s} \int_0^{|x|-\frac{c_o}{2}} (|v'(t)| + |\eta'(t)||v(t)|) \left( \frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \right) dt \\
 &\leq C(N, s) + C(N, s) \int_0^{|x|-\frac{c_o}{2}} \frac{1}{\ln \frac{1}{t}} \frac{1}{t} \left( \frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \right) dt \\
 &\leq C(N, s).
 \end{aligned}$$

For the remaining part, we have

$$\begin{aligned}
 &\int_{|x|-\frac{c_o}{2}}^{|x|} u'(t) \int_{\{|z|>\frac{c_o}{2}\} \cap \{|x+z|\leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &\leq \int_{|x|-\frac{c_o}{2}}^{|x|} |u'(t)| \int_{\{|z|>\frac{c_o}{2}\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &\leq \frac{C(N)}{2s} \left( \frac{c_o}{2} \right)^{-2s} \int_{|x|-\frac{c_o}{2}}^{|x|} |u'(t)| dt \\
 &\leq \frac{C(N)}{2s} \left( \frac{c_o}{2} \right)^{-2s+1} \max_{t \in [\frac{c_o}{2}, \frac{3}{4e}]} |u'(t)| < +\infty.
 \end{aligned}$$

Then, combining all the inequalities above, we obtain

$$|f(x)| \leq C(N, s), \quad \text{for } |x| \in [c_o, \frac{1}{e}].$$

Claim 2. When  $|x| \leq c_o$ , we have  $|f(x)| \leq C(N, s)|x|^{-2s} \frac{1}{\ln \frac{1}{|x|}}$ .

First, we split  $f$  into three parts:

$$\begin{aligned}
 f(x) &= P.V. \int_{\mathbb{R}^N} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\
 &= P.V. \int_{B_{\frac{|x|}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz + \int_{B_{2|x|}(0) \setminus B_{\frac{|x|}{2}}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\
 &\quad + \int_{\mathbb{R}^N \setminus B_{2|x|}(0)} \frac{u(x) - u(x+z)}{|z|^{N+2s}} dz \\
 &= f_1(x) + f_2(x) + f_3(x).
 \end{aligned}$$

Then, we analyze  $f_3(x)$ , since  $\frac{3}{2}|z| \geq |x+z| \geq |x|$  for  $|z| \geq 2|x|$ , then applying Fubini's Theorem yields

$$\begin{aligned}
 f_3(x) &= \int_{\mathbb{R}^N \setminus B_{2|x|}(0)} \frac{\int_{|x|}^{|x+z|} -u'(t) dt}{|z|^{N+2s}} dz \\
 &= \int_{|x|}^{\frac{3}{4e}} -u'(t) \int_{\{|z+x|\geq t\} \cap \{|z|\geq 2|x|\}} \frac{1}{|z|^{N+2s}} dz dt
 \end{aligned}$$

$$\begin{aligned}
 &\leq \int_{|x|}^{\frac{3}{4e}} -u'(t) \int_{\{|z| \geq \frac{2}{3}t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &= \frac{C(N)}{2s} \left(\frac{2}{3}\right)^{-2s} \int_{|x|}^{\frac{3}{4e}} -u'(t) t^{-2s} dt \\
 &\leq C(N, s) \int_{|x|}^{\frac{3}{4e}} |v(t)| t^{-2s} dt + C(N, s) \int_{|x|}^{\frac{3}{4e}} |v'(t)| t^{-2s} dt.
 \end{aligned}$$

Notice that

$$\begin{aligned}
 &\int_{|x|}^{\frac{3}{4e}} |v(t)| t^{-2s} dt \\
 &\leq \ln \ln \frac{1}{|x|} \int_{|x|}^{\frac{3}{4e}} t^{-2s} dt \\
 &= \begin{cases} \frac{\ln \ln \frac{1}{|x|}}{1-2s} \left( \left(\frac{3}{4e}\right)^{1-2s} - |x|^{1-2s} \right), & s \neq \frac{1}{2}, \\ \left( \ln \ln \frac{1}{|x|} \right) \left( \ln \frac{3}{4e|x|} \right), & s = \frac{1}{2}, \end{cases}
 \end{aligned}$$

and

$$\int_{|x|}^{\frac{3}{4e}} |v'(t)| t^{-2s} dt \leq \int_{|x|}^{\frac{3}{4e}} \frac{1}{\ln \frac{1}{t}} t^{-2s-1} dt \leq C(s) \frac{1}{\ln \frac{1}{|x|}} |x|^{-2s}.$$

Next, we estimate  $f_2(x)$ ; since  $|x| \leq \frac{1}{6e} \leq \frac{1}{2e}$  and  $|z| \leq 2|x|$ , we have

$$u(x) = v(x) = \ln \ln \frac{1}{|x|}, \quad u(x+z) = v(x+z) = \ln \ln \frac{1}{|x+z|},$$

then, we split it into two parts,

$$\begin{aligned}
 f_2(x) &= \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \geq |x|\}} \frac{\ln \ln \frac{1}{|x|} - \ln \ln \frac{1}{|x+z|}}{|z|^{N+2s}} dz \\
 &\quad + \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| < |x|\}} \frac{\ln \ln \frac{1}{|x|} - \ln \ln \frac{1}{|x+z|}}{|z|^{N+2s}} dz \\
 &= f_{2,1}(x) + f_{2,2}(x).
 \end{aligned}$$

Regarding  $f_{2,1}(x)$ , from the Mean Value Theorem, we have

$$\begin{aligned}
 f_{2,1}(x) &\leq \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \geq |x|\}} \frac{\ln \ln \frac{1}{|x|} - \ln \ln \frac{1}{|x+|z||}}{|z|^{N+2s}} dz \\
 &\leq \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \}} \frac{\ln \ln \frac{1}{|x|} - \ln \ln \frac{1}{|x+|z||}}{|z|^{N+2s}} dz \\
 &\leq \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \}} \frac{2}{|x| \ln \frac{2}{|x|}} \frac{1}{|z|^{N+2s-1}} dz
 \end{aligned}$$

$$\begin{aligned}
 &\leq \frac{C(N)}{|x| \ln \frac{1}{|x|}} \int_{\frac{1}{2}|x|}^{2|x|} \frac{r^{N-1}}{r^{N+2s-1}} dr \\
 &\leq \begin{cases} \frac{C(N)}{|x| \ln \frac{1}{|x|}} & s = \frac{1}{2}, \\ \frac{C(N)}{(1-2s) \ln \frac{1}{|x|}} \left( |2x|^{-2s} - |\frac{1}{2}x|^{-2s} \right) & s \neq \frac{1}{2}. \end{cases}
 \end{aligned}$$

Next, we estimate  $f_{2,2}(x)$ , from Fubini's Theorem, we get

$$\begin{aligned}
 f_{2,2}(x) &= \int_0^{|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &= \int_0^{\frac{1}{2}|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &\quad + \int_{\frac{1}{2}|x|}^{|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt.
 \end{aligned}$$

Since  $\{|x+z| \leq t\} \subset \{\frac{1}{2}|x| \leq |z| \leq |x|\}$  for  $t \in [0, \frac{1}{2}|x|]$ , then

$$\begin{aligned}
 &\int_0^{\frac{1}{2}|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &= \int_0^{\frac{1}{2}|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt \\
 &\leq C(N) \int_0^{\frac{1}{2}|x|} \frac{1}{t \ln \frac{1}{t}} \int_{|x|-t}^{|x|+t} \frac{1}{|r|^{1+2s}} dr dt \\
 &= \frac{C(N)}{2s} \int_0^{\frac{1}{2}|x|} \frac{1}{t \ln \frac{1}{t}} \left( \frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \right) dt,
 \end{aligned}$$

as

$$\frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \leq \frac{4s}{|x|^{2s+1}} t + \frac{C(s)t^2}{|x|^{2s+2}},$$

then, we have

$$\begin{aligned}
 &\int_0^{\frac{1}{2}|x|} \frac{1}{t \ln \frac{1}{t}} \left( \frac{1}{(|x|-t)^{2s}} - \frac{1}{(|x|+t)^{2s}} \right) dt \\
 &\leq \frac{C(s)}{|x|^{2s+1}} \int_0^{\frac{|x|}{2}} \frac{1}{\ln \frac{1}{t}} dt + \frac{C(s)}{|x|^{2s+2}} \int_0^{\frac{|x|}{2}} \frac{t}{\ln \frac{1}{t}} dt \\
 &\leq C(s) |x|^{-2s} \frac{1}{\ln \frac{1}{|x|}}.
 \end{aligned}$$

As for  $t \in [\frac{|x|}{2}, |x|]$ , by direct computation, we have

$$\int_{\frac{1}{2}|x|}^{|x|} \frac{-1}{t \ln \frac{1}{t}} \int_{\{\frac{1}{2}|x| \leq |z| \leq 2|x| \} \cap \{|x+z| \leq t\}} \frac{1}{|z|^{N+2s}} dz dt$$

$$\begin{aligned}
 &\leq \frac{C(N)}{|x| \ln \frac{1}{|x|}} \cdot \frac{|x|}{2} \cdot \frac{2^{N+2s}}{|x|^{N+2s}} \cdot |x|^N \\
 &= C(N, s) |x|^{-2s} \frac{1}{\ln \frac{1}{|x|}}.
 \end{aligned}$$

Finally, we deal with  $f_1(x)$ , since  $|x \pm z| \in [\frac{|x|}{2}, \frac{3|x|}{2}] \subset [\frac{|x|}{2}, \frac{1}{4e}]$ , then applying Taylor's Expansions yields

$$\begin{aligned}
 f_1(x) &= \frac{1}{2} P.V. \int_{B_{\frac{|x|}{2}}(0)} \frac{u(x) - u(x+z) + u(x) - u(x-z)}{|z|^{N+2s}} dz \\
 &= \frac{1}{2} \int_{B_{\frac{|x|}{2}}(0)} \frac{-\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x+\theta_3 z)}{\partial x_i \partial x_j} z_i z_j - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x-\theta_4 z)}{\partial x_i \partial x_j} z_i z_j}{|z|^{N+2s}} dz.
 \end{aligned}$$

Here, if  $i \neq j$ , we have

$$\left| \frac{\partial^2 u(x)}{\partial x_i \partial x_j} \right| = \left| \frac{1}{(\ln \frac{1}{|x|})^2} \frac{x_i x_j}{|x|^4} + \frac{1}{\ln \frac{1}{|x|}} \frac{x_i x_j}{|x|^4} \right| \leq \frac{1}{|x|^2 \ln \frac{1}{|x|}},$$

and for  $i = j$

$$\left| \frac{\partial^2 u(x)}{\partial x_i^2} \right| = \left| \frac{3x_i^2}{|x|^4 \ln \frac{1}{|x|}} - \frac{1}{|x|^2 \ln \frac{1}{|x|}} \right| \leq \frac{4}{|x|^2 \ln \frac{1}{|x|}}.$$

Consequently,

$$\left| -\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x+\theta_3 z)}{\partial x_i \partial x_j} z_i z_j - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x-\theta_4 z)}{\partial x_i \partial x_j} z_i z_j \right| \leq \frac{4N^2 |z|^2}{|x|^2 \ln \frac{1}{|x|}}.$$

Thus,

$$\begin{aligned}
 &\int_{B_{\frac{|x|}{2}}(0)} \frac{-\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x+\theta_3 z)}{\partial x_i \partial x_j} z_i z_j - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u(x-\theta_4 z)}{\partial x_i \partial x_j} z_i z_j}{|z|^{N+2s}} dz \\
 &\leq \frac{4N^2}{|x|^2 \ln \frac{1}{|x|}} \int_{B_{\frac{|x|}{2}}(0)} \frac{1}{|z|^{N+2s-2}} dz \\
 &= \frac{C(N)}{|x|^2 \ln \frac{1}{|x|}} \int_0^{\frac{|x|}{2}} \frac{1}{r^{2s-1}} dr \\
 &= C(N) |x|^{-2s} \frac{1}{\ln \frac{1}{|x|}}.
 \end{aligned}$$

Combining the previous inequalities, we get

$$|f(x)| \leq C(N, s) \frac{1}{|x|^{2s} \ln \frac{1}{|x|}} \quad \text{for } x \in B_{\frac{1}{6e}}(0).$$

Finally, combining Claim 1 and Claim 2 immediately yields  $f \in L^{\frac{N}{2s}}(B_{\frac{1}{e}}(0))$ .  $\square$

## 6.4 Proof of Theorem 6.1.1

Due to the homogeneity of the problem (6.1), it is sufficient to assume  $\|f\|_{L^{\frac{N}{2s}}(\Omega)} = 1$ . First, we take the test function  $\phi = e^{\alpha u(x)}\eta(x)$ , where  $\eta(x) = e^{\alpha\beta u_+(x)} - 1$ ,  $\beta \geq 1$  and  $\alpha$  takes the values in the interval mentioned in Lemma 6.2.5, then we have

$$\frac{\lambda}{2} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (u(x) - u(y))(e^{\alpha u(x)}\eta(x) - e^{\alpha u(y)}\eta(y))d\mu \leq \int_{\Omega} f(x)e^{\alpha u(x)}\eta(x)dx. \quad (6.4)$$

By applying Lemma 6.2.3 with  $s = u(x)$ ,  $t = u(y)$ ,  $r_1 = \eta(x)$ , and  $r_2 = \eta(y)$ , we derive

$$\begin{aligned} & \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (u(x) - u(y))(e^{\alpha u(x)}\eta(x) - e^{\alpha u(y)}\eta(y))d\mu \\ & \geq \frac{1}{\alpha} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (e^{\alpha u(x)} - e^{\alpha u(y)})(\eta(x) - \eta(y))d\mu. \end{aligned}$$

Then, from Lemma 6.2.2 with  $s = e^{\alpha u_+(x)}$ ,  $t = e^{\alpha u_+(y)}$  and Lemma 6.2.4, we obtain

$$\begin{aligned} & \frac{1}{\alpha} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (e^{\alpha u(x)} - e^{\alpha u(y)})(\eta(x) - \eta(y))d\mu \\ & \geq \frac{1}{\alpha} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (e^{\alpha u_+(x)} - e^{\alpha u_+(y)})(e^{\alpha\beta u_+(x)} - e^{\alpha\beta u_+(y)})d\mu \\ & \geq \frac{1}{\alpha\beta} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (e^{\alpha(\frac{\beta+1}{2})u_+(x)} - e^{\alpha(\frac{\beta+1}{2})u_+(y)})^2 d\mu \\ & \geq \frac{1}{\alpha\beta C_*^2} \left( \int_{\Omega} \left| e^{\frac{\alpha(\beta+1)u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{N}}. \end{aligned}$$

On the other hand, for the right side of (6.4), Hölder's inequality yields

$$\begin{aligned} \int_{\Omega} f(x)e^{\alpha u(x)}\eta(x)dx & \leq \int_{\Omega} |f(x)|e^{\alpha(\beta+1)u_+(x)}dx \\ & \leq \|f\|_{L^q(\Omega)} \left( \int_{\Omega} \left| e^{\alpha(\frac{\beta+1}{2})u_+(x)} \right|^{2q'} dx \right)^{\frac{1}{q'}}, \end{aligned}$$

where  $q' = \frac{q}{q-1}$ . Combining the inequalities above, we obtain the reverse Hölder inequality

$$\begin{aligned} & \left( \int_{\Omega} \left| e^{\frac{\alpha(\beta+1)u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} \\ & \leq \left( \frac{2}{\lambda} \right)^{\frac{1}{2}} (\alpha\beta)^{\frac{1}{2}} C_* \|f\|_{L^q(\Omega)}^{\frac{1}{2}} \left( \int_{\Omega} \left| e^{\alpha(\frac{\beta+1}{2})u_+(x)} \right|^{2q'} dx \right)^{\frac{1}{2q'}}, \end{aligned}$$

then,

$$\left( \int_{\Omega} \left| e^{\frac{\alpha(\beta+1)u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}}$$

$$\begin{aligned}
 &\leq \left( \int_{\Omega} \left| e^{\frac{\alpha(\beta+1)u_+(x)}{2}} - 1 \right|^{\frac{2N}{N-2s}} dx \right)^{\frac{N-2s}{2N}} + |\Omega|^{\frac{N-2s}{2N}} \\
 &\leq \left[ \left( \frac{2}{\lambda} \right)^{\frac{1}{2}} (\alpha\beta)^{\frac{1}{2}} C_* \|f\|_{L^q(\Omega)}^{\frac{1}{2}} + |\Omega|^{\frac{N-2s}{2N} - \frac{1}{2q'}} \right] \left( \int_{\Omega} \left| e^{\alpha(\frac{\beta+1}{2})u_+(x)} \right|^{2q'} dx \right)^{\frac{1}{2q'}}.
 \end{aligned}$$

Since  $2q' < \frac{2N}{N-2s}$ , we define  $\chi = \frac{N}{(N-2s)q'} > 1$ , and set  $\frac{\beta+1}{2} = \chi^n$  with  $n \in \mathbb{N}_+$ , raising both sides of the previous inequality to the power  $\chi^{-n}$ , we obtain

$$\begin{aligned}
 \left( \int_{\Omega} \left| e^{\frac{2N\alpha u_+(x)}{N-2s}} \right|^{\chi^n} dx \right)^{\frac{N-2s}{2N\chi^n}} &\leq \left[ \left( \frac{4\alpha}{\lambda} \right)^{\frac{1}{2\chi^n}} \chi^{\frac{n}{2\chi^n}} C_*^{\frac{1}{\chi^n}} \|f\|_{L^q(\Omega)}^{\frac{1}{2\chi^n}} + |\Omega|^{\left(\frac{N-2s}{2N} - \frac{1}{2q'}\right)\chi^n} \right] \\
 &\cdot \left( \int_{\Omega} \left| e^{\frac{\alpha 2N u_+(x)}{N-2s}} \right|^{\chi^{n-1}} dx \right)^{\frac{N-2s}{2N\chi^{n-1}}}.
 \end{aligned}$$

By iteration, we get

$$\begin{aligned}
 \left\| e^{\frac{2N\alpha u_+(x)}{N-2s}} \right\|_{L^\infty(\Omega)} &\leq \left( \frac{4\alpha}{\lambda} \right)^{\frac{1}{2(\chi-1)}} \|f\|_{L^q(\Omega)}^{\frac{1}{2(\chi-1)}} C_*^{\frac{1}{\chi-1}} \chi^{\frac{\chi}{(\chi-1)^2}} \\
 &\cdot (|\Omega|^{\frac{N-2s}{2N} - \frac{1}{2q'}} + 1)^{\frac{1}{\chi-1}} \left( \int_{\Omega} \left| e^{\frac{\alpha 2N u_+(x)}{N-2s}} \right| dx \right)^{\frac{N-2s}{2N}},
 \end{aligned}$$

i.e.

$$\begin{aligned}
 \|u_+\|_{L^\infty(\Omega)} &\leq \frac{(N-2s)}{2N\alpha} \\
 &\cdot \log \left[ \left( \frac{4\alpha}{\lambda} \right)^{\frac{1}{2(\chi-1)}} \|f\|_{L^q(\Omega)}^{\frac{1}{2(\chi-1)}} C_*^{\frac{1}{\chi-1}} \chi^{\frac{\chi}{(\chi-1)^2}} (|\Omega|^{\frac{N-2s}{2N} - \frac{1}{2q'}} + 1)^{\frac{1}{\chi-1}} \right. \\
 &\cdot \left. \left( \int_{\Omega} \left| e^{\frac{2\alpha N u_+(x)}{N-2s}} \right| dx \right)^{\frac{N-2s}{2N}} \right].
 \end{aligned}$$

Consequently, applying Lemma 6.2.5 yields

$$\begin{aligned}
 &\|u_+\|_{L^\infty(\Omega)} \\
 &\leq \frac{(N-2s)}{2N\alpha} \log \left[ \left( \frac{4\alpha}{\lambda} \right)^{\frac{1}{2(\chi-1)}} \|f\|_{L^q(\Omega)}^{\frac{1}{2(\chi-1)}} C_*^{\frac{1}{\chi-1}} \chi^{\frac{\chi}{(\chi-1)^2}} (|\Omega|^{\frac{N-2s}{2N} - \frac{1}{2q'}} + 1)^{\frac{1}{\chi-1}} \right. \\
 &\quad \cdot \left. \left( \frac{2\lambda|\Omega|^{1-\frac{4s}{N+2s}}}{(\lambda - \alpha C_*^2)} + |\Omega|^{\frac{N-2s}{2N}} \right) \right] \\
 &\leq \frac{(N-2s)}{4N(\chi-1)\alpha} \left( \frac{2\lambda|\Omega|^{1-\frac{4s}{N+2s}}}{(\lambda - \alpha C_*^2)} + |\Omega|^{\frac{N-2s}{2N}} \right) \\
 &\quad \cdot \frac{\log \left( \|f\|_{L^q(\Omega)} + 1 \right) + 1}{\log \left( \left( \frac{4\alpha}{\lambda} \right)^{\frac{1}{2(\chi-1)}} C_*^{\frac{1}{\chi-1}} \chi^{\frac{\chi}{(\chi-1)^2}} (|\Omega|^{\frac{N-2s}{2N} - \frac{1}{2q'}} + 1)^{\frac{1}{\chi-1}} \right)},
 \end{aligned}$$



where  $\alpha \in (0, \frac{\lambda}{C_*^2})$ . This completes our proof.



# Conclusion and Outlook

In the preceding chapters, we have obtained regularity results for weak solutions of several classes of elliptic equations.

In Chapter 2 we studied the model without lower-order terms. Because the usual bounds on the ratio of the largest and smallest eigenvalues of the coefficient matrix are relaxed, the classical Moser-iteration scheme is no longer applicable. By a careful use of interpolation inequalities and the structural information in the equation, we removed the influence of the non-homogeneous term  $f$  on the iteration, and proved local  $L^\infty$ -estimates for weak solutions. Then counterexamples were constructed to demonstrate how the presence of  $f$  affects the regularity of both the solution and the coefficients. Subsequently, a local Harnack inequality was established by invoking the Trudinger-Moser theory. Concerning global regularity, we adopted the method of Xu [105] to derive a logarithmic upper bound for weak solutions of the Dirichlet problem. Inspired by Li and Wang [66], we further proved an ABP-type maximum principle. Finally, for uniformly elliptic equations with the source term  $f$  belonging to an Orlicz space, we exhibited a critical counterexample. It should be emphasized that, when the equation is non-uniformly elliptic and  $f$  merely lies in an Orlicz space, existing boundedness results are still incomplete.

Chapter 3 is devoted to non-homogeneous linear divergence-form non-uniformly elliptic equations *with* lower-order terms. We derived local  $L^\infty$ -estimates and Harnack inequalities for weak solutions, and supplied two elementary examples to illustrate how the presence of lower-order terms influences the regularity of both solutions and coefficients.

In Chapter 4 we investigated the possible growth at infinity of non-negative functions belonging to a homogeneous De Giorgi class. Owing to the richness of this class, the existence of a global comparison principle remains an open question, rendering the

classical methods usually employed for  $p$ -harmonic functions inapplicable. By introducing the  $p$ -capacity and exploiting the Harnack inequality intrinsic to the homogeneous De Giorgi class, we proved Phragmén-Lindelöf-type theorems. Counterexamples were constructed to show that the growth of non-negative functions in the De Giorgi classes we are interested in, can indeed be slower than that of  $p$ -harmonic functions, a phenomenon rooted in the generality of the class itself.

In Chapter 5 we gave a new proof of a classical result about the boundary regularity of functions in homogeneous De Giorgi classes. Our method is based on ideas that were originally introduced by Gariepy and Ziemer when they studied the sufficient part of the Wiener criterion for a wide class of quasilinear equation of  $p$ -Laplacian type. In their technique three auxiliary functions play a fundamental role, namely

$$\varphi(t) = \left( \frac{1}{1-t} \right)^\alpha, \quad \psi(t) = (1-t)^\beta, \quad \zeta(t) = \ln(1-t),$$

where  $\alpha > 0$ ,  $\beta \in (0, 1)$ , and  $t \in [0, 1)$ , and they satisfy proper estimates. We showed that such estimates are available even in our more general framework, so that the method eventually works; therefore, this raises the natural question whether the same regularity of solutions continues to hold even when rougher boundaries are considered. This issue was not dealt with in our work.

Chapter 6 focused on the Dirichlet problem for a class of non-local uniformly elliptic equations. As the fractional parameter  $s \rightarrow 1^-$ , the non-local operator converges to a local divergence-form uniformly elliptic operator. Building on earlier maximum-principle arguments, we obtained an improved, logarithmic-type upper bound for the weak solution. A sharp counterexample was provided to underscore the optimality of our result.

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1. **Zheng Li**, Bin Guo. A counter-example for Laplacian with data in Orlicz space. *Applied Mathematics Letters*, **158** (2024).

2. Ugo Gianazza, **Zheng Li**. Boundary regularity for functions in De Giorgi classes. *La Matematica*, **4**, 861–884 (2025).
3. **Zheng Li**. Logarithmic upper bound for weak subsolutions to the fractional Laplacian equation. To appear in *Annales Fennici Mathematici*.
4. **Zheng Li**, Bin Guo. The dependence of local regularity of solutions on the integrability of coefficients and nonhomogeneous term. arXiv:2402.07219.
5. Simone Ciani, Ugo Gianazza, **Zheng Li**. Phragmén–Lindelöf-type theorems for functions in homogeneous De Giorgi classes. arXiv:2505.17926.



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